

Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

12 September 2026

This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, the historical packet ghost, chain retractions, arithmetic moments, Gaussian resolvents, full kernel layers, symmetric tensor frontiers, measured support branches, and the original theta norm-volume identities. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

Contents

I	Integrated mathematical continuation	5
1	Exact Split-Zero carriers and the local spectral bridge	5
1.1	Scalar support and independent channel amplitudes	5
1.2	Tesseract sectors and a supported critical fixed locus	7
1.3	Exact sector, holonomy and positive-moment maps	8
1.4	The theta-packet form and its scalar comparison	10
1.5	Projective zeta at all zeros and at its pole	10
1.6	Matrix packets and their full relation module	11
2	The historical packet ghost and its full arithmetic jet quotient	12
3	Historical Hurwitz reconstruction and the complete Laurent jet bridge	16
3.1	Source receipts and exact locators	16
3.2	Every compact Hurwitz input is determined by one simple-zero germ	17
3.3	The complete zero cluster removes the simplicity restriction	20
3.4	The historical period map preserves every finite spectral jet	21
3.5	Exact conjugate-reciprocal involution on the jets and conormal	23
3.6	Period collisions: the exact unlabelled image and its missing fibres	24
3.7	Two prime periods recover the complete finite spectral algebra	25
3.8	Split-Zero support is preserved by the exact maps	27
3.9	All-order Hurwitz motion transported to the period and tesseract coordinates	28
3.10	Mathematical scope	29
4	Historical support branches, ideal completion, and measured observations	29
4.1	Exact ordinary and continuous Split-Zero branches	30
4.2	The exact frame comparison retaining all ordinary branches	32
4.3	A fully explicit nontrivial frame with algebraic branches and no frame points	34
4.4	Atomic observation and the exact support data it removes	35
4.5	The exact operator-algebraic branch map	36
5	Reconstruction with changes of support index	38
5.1	The internal quotient and the actual comparison kernel	39
6	The original theta complex and its arithmetic packet	40
6.1	Source spaces and fixed arithmetic factors	40
6.2	The exact operator range, including its inverse	42
6.3	The original extension supplies the boundary coefficient	43
6.4	Actual representative corrections and rank-two control	43
7	Certified moments from the original arithmetic theta source	44
7.1	Source polynomials and the exact reflection map	45
7.2	An absolutely convergent double series for every entry	45

7.3	A complete finite truncation certificate	47
7.4	Exact adjoint, parity, and finite-boundary checks	48
7.5	Reproducible certified finite arithmetic data	48
8	Exact source-metric optimization and aggregate arithmetic control	49
8.1	The fixed arithmetic maps and their boundary	50
8.2	A source-density calculation with its topology retained	51
8.3	Every positive packet metric has an exact finite source lift	51
8.4	Sharp control with all nilpotent jets present	53
8.5	The exact tensor cost of unrestricted metrics	54
8.6	A stronger bound on the actual orthogonal sequence	55
8.7	Original supported maps and the precise result delivered	56
9	The actual theta minimum as a confluent arithmetic kernel	56
9.1	Original functions, maps, coordinates, and measures	57
9.2	Exact constrained minimum over the admitted source	58
9.3	Monic arithmetic polynomials and their exact recurrence	59
9.4	Exact relative weight control from the same kernel	60
9.5	Exact change when one more theta direction is admitted	61
9.6	The original measure supplies a full tail recurrence	61
9.7	Metric dependence and the precise mathematical gain	63
10	The full jet kernel, finite resolvents, and the actual analytic tail	63
10.1	A finite compression in the original Mellin coordinate	64
10.2	Exact quadrature and the full confluent kernel	65
10.3	The infinite multiplication operator and its genuine tail	67
11	Quantitative convergence of the arithmetic Gaussian resolvents	68
11.1	The Gaussian rule inherits the original exponential moments	69
11.2	Exact moments give a quantitative analytic-strip estimate	69
11.3	Every confluent derivative and the original finite algebra	71
12	The exact terminal step and its retained boundary energy	73
12.1	A step written entirely in the original polynomial packet	73
12.2	The complete boundary-energy increment	74
12.3	The exact rational terminal step and its polynomial continuation	75
12.4	The extra jets required to recover the metric from the tail	76
12.5	Actual quartet packets close the energy in two volume ratios	77
12.6	What the recurrence proves about boundary energy	78
12.7	An exact positive-moment counterexample to consecutive energy monotonicity	78
13	The exact metric departure and the cost of complete jets	84
13.1	A quantitative constraint on every retained nilpotent chain	85
14	Coherent packets, joint tensor layers, and their exact spectral cost	86

14.1 The arithmetic unit and the coherent one-variable maps	86
14.2 The original reflection and the exact diagonal coefficient	87
14.3 The joint total-degree source and its relation layer	88
14.4 Nested packets on the joint source: the additional projection	89
14.5 Positive layer rank and the complete tensor spectral departure	89
15 Density of the original joint boundaries and determinant dissipation	91
15.1 Polynomial density with the original weighted measures	91
15.2 Strong limits of the actual source representatives	92
15.3 The accumulated determinant cost of the same boundary sequence	94
16 The actual kernel layer, its derivative image, and its quotient	95
16.1 The original arithmetic source and its constrained minimum	95
16.2 The complete next-layer frame and its theta primitives	96
16.3 The exact incidence map and the derivative-image rank	97
16.4 The whole quotient of the next layer by the derivative image	98
16.5 The derivative between consecutive relation quotients	99
16.6 The exact original-theta norm and volume bridge	99
16.7 The complete scalar dictionary for the boundary energy	101
17 Original orthogonal shells and the exact projective-ratio algebra	102
17.1 Two sections of the same polynomial filtration	102
17.2 The full correspondence between the two image spaces	104
17.3 The algebra generated by the original coordinate ratio	105
17.4 A complete finite witness for unequal shell ranks	107
18 The original frontier energy, determinant growth, and the entire border fibre	109
18.1 Original moment coordinates and the relation graph	109
18.2 Exact relative loss and the source derivative energy	111
18.3 The entire small border operator, including its zero extension	112
18.4 A quantified link between the frontier bound and determinant cost	113
19 The signed symmetric summand and its complete spectral cost	114
19.1 The chain projector and its completed and finite targets	114
19.2 Orbit coordinates, their Gram, and all local nilpotents	116
19.3 Complete symmetric spectral sums and their exact departure	117
19.4 The cost in the original symmetric relation layer	119
20 Arithmetic quartet parity and the exact positive frontier operator	120
20.1 The actual two frontier matrices	120
20.2 The original linear reflection separates the two frontiers	121
20.3 The complete positive operator and the sharper arithmetic bound	122
20.4 The same positive frontier on the actual symmetric summand	124
21 Which completion retains the arithmetic jets?	127

21.1	Logarithmic coordinates and the original measure	127
21.2	Exponential weights and exact jet norms	128
21.3	The energy term carried by the weighted topology	129
21.4	The tesserine coordinate in the actual weight form	129
22	The current arithmetic route and its next calculation	130
22.1	A single chain of exact comparison maps	130
22.2	Topology, duality and the two sides of a weight estimate	130
22.3	Provenance and mathematical status	132
22.4	The original theta norm and the positive arithmetic frontier	134
22.5	The symmetric summand, the actual relation layer, and support	135
II	Complete source proof notes and chronology	137
A	PR 8: Finite operator comparison	138
A.1	Exact finite operator quotients of the tau-base theta cohomology	138
B	PR 9: Chain descent	150
B.1	Chain-level arithmetic descent over the tau base	150
C	PR 10: Global retraction	160
C.1	A global retraction of the original theta complex over the tau base	160
D	PR 11: Homotopy control	170
D.1	Homotopy choices and source-preserving arithmetic control	170
E	PR 13: Orthogonal boundary control	176
E.1	Orthogonal theta boundaries and rank-two arithmetic control	176
F	Derived Split-Zero mathematics	182
F.1	SplitZero: reconstruction, internal homology, and balanced finite jets	182
G	PR 14: Coherent arithmetic interpolation and joint tensor boundary control	188
G.1	Coherent arithmetic interpolation and joint tensor boundary control	188
H	Full-jet interpolation kernels and the actual next relation layer	199
H.1	Full-jet interpolation kernels and the actual next relation layer	199
I	Arithmetic frontier estimates and symmetric tensor control	206
I.1	Arithmetic frontier estimates and symmetric tensor control	206

Part I

Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0, 1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R , including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ , and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection). *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad r \mapsto (r, 1).$$

The coordinate maps

$$\begin{aligned} p_R : G(R) &\rightarrow R, & p_R(\tau) &= 0, & p_R(r) &= r, \\ \chi_R : G(R) &\rightarrow \mathbb{B}, & \chi_R(\tau) &= 0, & \chi_R(r) &= 1 \end{aligned}$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r + s, 1)$ and product $(rs, 1)$. These identities retain support when the amplitude vanishes. The pair $(0, 0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R . If $f : G(R) \rightarrow A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \rightarrow A$. The equality $f = hp_R$ holds on R and on τ , and determines h . Conversely every such h gives hp_R . $\square$

The larger carrier $R \times \mathbb{B}$ contains $(1, 0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L , define

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), & (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu). \end{aligned}$$

Write $z_\lambda = (0, \lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring). *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \rightarrow R, \quad \chi_L : G_L(R) \rightarrow L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L . Both projections preserve operations, zero and unit. If $f : G_L(R) \rightarrow A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A . Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \rightarrow G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e , whereas the semiring zero is τ . Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_{\mathbb{C}}, p_{\mathbb{C}}) : M \rightarrow \mathbb{C}^2, \quad \chi^{(2)} = (\chi_{\mathbb{C}}, \chi_{\mathbb{C}}) : M \rightarrow \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau, u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \quad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c, (1, 1)) &= (j(c), j(c)), \\ b(0, (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a, b), (1, 1)) &= (j(a), j(b)), \\ B((0, 0), (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)}b = \Delta_{\mathbb{C}}p_{\mathbb{B}^2}, \quad \chi^{(2)}b = \chi_{\mathbb{B}^2}, \quad p^{(2)}B = p_{\mathbb{B}^2}, \quad \chi^{(2)}B = \chi_{\mathbb{B}^2},$$

where $\Delta_{\mathbb{C}}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\{((a, b), (\lambda_1, \lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \rightarrow G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2}r = p^{(2)}$ and $\chi_{\mathbb{B}^2}r = \chi^{(2)}$: it would send that element to $((1, 0), (1, 0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_{\mathbb{B}} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0, 0), (1, 1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M . Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only 0, 1. The positive relation between the original carriers is b , with diagonal amplitude projection.

1.2 Tesserine sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesserine algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \quad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \frac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \rightarrow \mathbb{C}^4, \quad x \mapsto (\text{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\kappa + (1 - \bar{s})\sigma\kappa, \quad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1), \quad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \rightarrow \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \rightarrow \mathbb{R} \times i\mathbb{R}, \quad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \quad s = \frac{1}{2} + \frac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon'\sigma)(1 + \delta'\kappa)$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned} c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\ c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\ c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\ c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma. \end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--} , retaining the constant character coordinate. The real tesserine algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \rightarrow M, \quad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s . The $G(\mathbb{C})$ -linear automorphisms

$$T_{\sigma}(x, y) = ((-1)x, (-1)y), \quad T_{\kappa}(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_{\sigma}\Psi(s), \quad \Psi(\kappa s) = T_{\kappa}\Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\text{Fix}(T_{\sigma}T_{\kappa}) = \{\tau, e\} \times G(\mathbb{C}), \quad \Psi^{-1} \text{Fix}(T_{\sigma}T_{\kappa}) = \{s : \Re s = \frac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ . It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e . The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. $\square$

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p , $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\text{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \text{Gal}(L/\mathbb{Q}) \rightarrow S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \rightarrow \mathbb{C}^\times, \quad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M . On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \frac{1}{2}, \quad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2}c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. $\square$

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^\top.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1 . For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^*T_2(z)x = |v_2(z)^*x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. $\square$

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \dots, z^{n-1})^\top, \quad M_n = \sum_{j=1}^r \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^r \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r . If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^r (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \dots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i < j} (z_j - z_i) \neq 0$, so V has rank r . The established kernel identity is $\ker M_n = \ker V^*$, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. $\square$

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1 . At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^r \alpha_j \left[\exp \left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \text{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel). *For every complex d -by- d matrix A , Hermitian matrix G , and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

*On the domain of these data where $v^*Gv \neq 0$, the quotient*

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \quad z_{p,u}(\rho) = u \exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω , and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. $\square$

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1, u = 1$,

$$z = \sqrt{p}, \quad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1), G = (1), v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}_{\text{um}}^1(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal). *The map*

$$p_* : \mathbb{P}_{\text{um}}^1(G(\mathbb{C})) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad x \mapsto [1 : x], \quad r : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.*

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. $\square$

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at 1, the full-support lift is*

$$Z_{\text{sp}} : \mathbb{C} \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad Z_{\text{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\hat{\zeta}$. At every zeta zero ρ its value is $[e : 1]$. Near the pole it has the chart

$$Z_{\text{sp}}(s) = [1 : j(h(s))], \quad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1. At a zero of order m , it has a pole of order m .

Proof. Put $t = s - 1$ and write $\zeta(1+t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1+t) = t/(1+tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m , write $\zeta(\rho+t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m} u(t)^{-1}$, proving its pole order; the first chart retains the original order m . $\square$

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_* . Hence J_*Z_{sp} is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι ; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let V be a complex Hilbert space of finite dimension $d > 0$ with fixed positive-definite Hermitian inner product. Let $g \in \text{GL}(V)$ and $m \geq 1$. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \rightarrow V^{m+1}, \quad v \mapsto (v, gv, \dots, g^m v), \quad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \rightarrow V^{m+1}, \quad \begin{cases} (D_m(g)y)_0 = -g^* y_1, \\ (D_m(g)y)_k = y_k - g^* y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ is positive semidefinite and

$$\ker \mathbb{T}_m(g) = \text{im } D_m(g), \quad \text{rank } \mathbb{T}_m(g) = d, \quad \text{corank } \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g is unitary for the fixed metric.

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v , so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \dots, v_m) = \sum_{k=0}^m (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} ((g^*)^k y_k - (g^*)^{k+1} y_{k+1}) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md , proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k} , hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0, 0), (1, 1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I , so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \mapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ with $t > 0$, the polynomial is $(w - 1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 The historical packet ghost and its full arithmetic jet quotient

The source of the packet construction in this section is the response dated 24 June 2026, 11:13:04 AM in the SSD export *Globalization Semiring Analysis*, whose exact path, hash and response-line locators are recorded in `logbook/historical_split_transcripts_20260912.md`. The source proves the permutation-packet calculation. We give that calculation completely, then construct its comparison with the full local algebras of the arithmetic packet. In particular, we retain the nilpotent jet operators and distinguish the source's complex-linear permutation from arithmetic conjugation by explicit maps.

Let Z be a finite set of complex numbers stable under $c(\rho) = \bar{\rho}$ and $\sigma(\rho) = 1 - \rho$. Assign a positive integer m_ρ constant on each orbit. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A[f] = [sf].$$

We use the original coordinate s , without replacing its multiplicative operator by its semisimple part. Put

$$L_Z = \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad e_{\rho,k} = t_\rho^k \quad (0 \leq k < m_\rho)$$

in the indicated direct summand. The coefficient inner product makes all $e_{\rho,k}$ orthonormal; it is an explicitly chosen reference inner product, not the theta-source metric.

Proposition 2.1 (Original-coordinate identification with all jets). *The map*

$$T : E_h \longrightarrow L_Z, \quad T[f]_\rho = \sum_{k=0}^{m_\rho-1} \frac{f^{(k)}(\rho)}{k!} t_\rho^k$$

is a complex-algebra isomorphism. In these coordinates

$$TAT^{-1} = A_{ss} + N, \quad A_{ss}e_{\rho,k} = \rho e_{\rho,k}, \quad Ne_{\rho,k} = \begin{cases} e_{\rho,k+1} & k+1 < m_\rho, \\ 0 & k+1 = m_\rho. \end{cases}$$

Proof. Taylor multiplication modulo $t_\rho^{m_\rho}$ proves multiplicativity. Its kernel consists exactly of polynomials divisible by every $(s - \rho)^{m_\rho}$ and hence by their product h , since the factors are pairwise coprime. For completeness an inverse is explicit. Set $h_\rho = h/(s - \rho)^{m_\rho}$, and let a_ρ be the Taylor polynomial of $1/h_\rho(s)$ at ρ through degree $m_\rho - 1$. Then $E_\rho = [a_\rho h_\rho]$ has residue 1 at the ρ factor and 0 at every other factor. Thus

$$T^{-1}e_{\rho,k} = E_\rho[(s - \rho)^k]$$

gives the inverse. Multiplication by $s = \rho + t_\rho$ gives the displayed operator, including its last nilpotent column. $\square$

In every subsequent operator formula on L_Z , the symbol A denotes the transported operator $TAT^{-1} : L_Z \rightarrow L_Z$ of Proposition 2.1. The polynomial-coordinate quotient at the end is equipped with the original multiplication-by- s operator.

Define complex-linear permutations C_0, S_0 and a complex-linear parity operator H on this complete basis by

$$C_0e_{\rho,k} = e_{\bar{\rho},k}, \quad S_0e_{\rho,k} = e_{1-\rho,k}, \quad He_{\rho,k} = (-1)^k e_{\rho,k}.$$

Also let K be the conjugate-linear map $K(\sum a_{\rho,k}e_{\rho,k}) = \sum \bar{a}_{\rho,k}e_{\rho,k}$. All four maps commute, square to the identity, and preserve products in L_Z , with K conjugate-linear on scalars.

Proposition 2.2 (Typed arithmetic involutions). *Under T , ordinary conjugation of polynomial coefficients is the conjugate-linear map $\mathcal{C} = C_0K$, and pullback by $s \mapsto 1 - s$ is the complex-linear map $\mathcal{S} = S_0H$. In particular*

$$C_0 = \mathcal{C}K, \quad S_0 = \mathcal{S}H, \quad \mathcal{C}A = A\mathcal{C}, \quad \mathcal{S}AS = I - A.$$

Here the equality involving $\mathcal{C}$ is an equality of conjugate-linear maps. Moreover $C_0N = NC_0$, $S_0N = NS_0$, and $HN = -NH$.

Proof. If $f^\#(s) = \overline{f(\bar{s})}$, then $(f^\#)^{(k)}(\rho)/k! = \overline{f^{(k)}(\bar{\rho})/k!}$, giving $\mathcal{C} = C_0K$. The k th derivative of $f(1 - s)$ at ρ is $(-1)^k f^{(k)}(1 - \rho)$, giving $\mathcal{S} = S_0H$. The ideal (h) is preserved by both operations: conjugation fixes the monic polynomial h , and $h(1 - s) = (-1)^{\deg h} h(s)$. Conjugating coefficients fixes the indeterminate s , whereas substituting $1 - s$ replaces multiplication by s by multiplication by $1 - s$. This proves the two identities for A . The assertions about N follow on each basis vector, including the vector of largest jet degree, which both sides send to zero. $\square$

Theorem 2.3 (Historical ghost with complete multiplicities). *Set*

$$R = C_0S_0, \quad \Gamma = C_0 - S_0, \quad P_- = \frac{I - R}{2}, \quad P_+ = \frac{I + R}{2}.$$

These satisfy

$$\Gamma^* = \Gamma, \quad \Gamma^2 = 4P_-, \quad \Gamma e_{\rho,k} = 0 \iff \Re \rho = \frac{1}{2}.$$

On a nonreal off-axis quartet of common multiplicity m , Γ has eigenvalues -2 and 2 each with multiplicity m , and 0 with multiplicity $2m$. Furthermore

$$\text{rank } P_- = 2m, \quad \text{Tr}(\Gamma^*\Gamma) = 8m.$$

On a critical-line orbit $\Gamma = P_- = 0$ at every jet degree.

Proof. The permutations C_0, S_0 are commuting selfadjoint involutions in the declared reference inner product. Consequently $(C_0 - S_0)^2 = 2I - 2C_0S_0$. The equality $C_0e_{\rho,k} = S_0e_{\rho,k}$ says exactly that the two labelled basis vectors have the same root, namely $\bar{\rho} = 1 - \rho$, which is equivalent to $2\Re\rho = 1$.

On an off-axis nonreal quartet order the roots as $(\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho})$. At every fixed jet degree the four joint (C_0, S_0) eigenspaces have characters $(1, 1), (1, -1), (-1, 1), (-1, -1)$. Explicit joint eigenvectors are

$$e_{\rho,k} + \epsilon e_{\bar{\rho},k} + \delta e_{1-\rho,k} + \epsilon \delta e_{1-\bar{\rho},k}, \quad \epsilon, \delta \in \{1, -1\}.$$

They form a basis: their pairwise inner products are zero and each has squared norm 4. The eigenvalue of Γ is $\epsilon - \delta$. Thus its squared eigenvalues are 0, 4, 4, 0, giving the rank and trace with every one of the m jet degrees included. On a critical orbit the two permutations agree on every basis vector. $\square$

For real off-axis orbits the same proof gives rank m and squared trace $4m$, since the orbit has two elements. Actual nontrivial zeta zeros in $0 < \Re s < 1$ are nonreal: for real $0 < s < 1$ the alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is strictly positive by grouping successive pairs, and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. For completeness the paired terms satisfy

$$(2n-1)^{-s} - (2n)^{-s} = s \int_{2n-1}^{2n} x^{-s-1} dx.$$

They converge uniformly on each compact subset of $\Re s > 0$, because on such a subset $|s| \leq M$, $\Re s \geq \delta > 0$, and the absolute value is at most $M(2n-1)^{-1-\delta}$. Thus η is analytic there. The product identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ follows first in $\Re s > 1$ by splitting even terms and then by analytic continuation. Division is used here only on the real interval $0 < s < 1$, where the denominator is strictly negative. Hence the quartet formula applies to off-axis arithmetic packets.

Proposition 2.4 (The exact group-algebra quotient). *For $W = \langle c, \sigma \mid c^2 = \sigma^2 = 1, c\sigma = \sigma c \rangle$ the homomorphism*

$$q : \mathbb{C}[W] \longrightarrow \mathbb{C}[u]/(u^2 - 1), \quad c \longmapsto u, \quad \sigma \longmapsto u$$

is surjective, with

$$I_q := \ker q = (c - \sigma)\mathbb{C}[W] = \mathbb{C}(c - \sigma) \oplus \mathbb{C}(1 - c\sigma).$$

Under the representation $c \mapsto C_0, \sigma \mapsto S_0, I_q L_Z = P_- L_Z$. Thus $L_Z/P_- L_Z$ is the largest quotient of the W -representation on which $C_0 = S_0$.

Proof. In the basis $(1, c, \sigma, c\sigma)$, q sends $a + bc + d\sigma + ec\sigma$ to $(a + e) + (b + d)u$. Its kernel is therefore spanned by $c - \sigma$ and $1 - c\sigma$. Multiplication by c sends $c - \sigma$ to $1 - c\sigma$, and multiplication by any generator stays in their span. This proves the ideal formula. Since $P_- = \Gamma^2/4$ and $\Gamma = \Gamma P_-$, their images agree and equal $I_q L_Z$. A quotient kills I_q precisely when it kills this image; this proves its universal property without any invariance assumption about the arithmetic generator. $\square$

The generator acts on the complete local rings. The next calculation constructs exactly the extra kernel needed to descend that action.

Theorem 2.5 (Arithmetic saturation of the historical ghost quotient). *Put*

$$Be_{\rho,k} = (\Re\rho - \tfrac{1}{2})e_{\rho,k}, \quad Z_{\text{crit}} = \{\rho \in Z : \Re\rho = \tfrac{1}{2}\}, \quad L_{\text{off}} = \bigoplus_{\rho \notin Z_{\text{crit}}} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}).$$

The smallest A -invariant subspace containing $I_q L_Z$ is exactly

$$I_q L_Z + A(I_q L_Z) = L_{\text{off}}.$$

On each off-axis orbit this sum is direct. The largest quotient on which both the original generator A descends and $C_0 = S_0$ is therefore the exact algebra and W -module quotient

$$L_Z \longrightarrow L_{Z_{\text{crit}}}, \quad (f_\rho)_\rho \longmapsto (f_\rho)_{\rho \in Z_{\text{crit}}}.$$

Its kernel is L_{off} , including every local multiplicity. In polynomial coordinates this is the quotient

$$\mathbb{C}[s]/(h) \longrightarrow \mathbb{C}[s]/(h_{\text{crit}}), \quad [f]_h \longmapsto [f]_{h_{\text{crit}}}, \quad h_{\text{crit}} = \prod_{\rho \in Z_{\text{crit}}} (s - \rho)^{m_\rho},$$

whose kernel is $(h_{\text{crit}})/(h)$; the target is zero when $h_{\text{crit}} = 1$.

Proof. The permutation R sends ρ to $1 - \bar{\rho}$, preserving its imaginary part and reversing $\Re \rho - 1/2$. It preserves the jet index and commutes with N . Accordingly $RB = -BR$ and

$$A = \frac{1}{2}I + B + iY + N, \quad Ye_{\rho,k} = \Im \rho e_{\rho,k}, \quad RY = YR, \quad RN = NR.$$

This is an exact identity for A , with N retained. If $v \in P_-L_Z$, then $(\frac{1}{2}I + iY + N)v \in P_-L_Z$ whereas $Bv \in P_+L_Z$. Therefore

$$P_+Av = Bv \quad (v \in P_-L_Z).$$

On each off-axis orbit all diagonal entries of B are nonzero, so $B : P_-L_Z \rightarrow P_+L_Z$ is an isomorphism on that orbit, its inverse being the restriction of the original diagonal operator B^{-1} . For $w = w_- + w_+$ on that orbit, take $v = B^{-1}w_+$; then $P_+Av = w_+$ and $w - Av \in P_-L_Z$. Thus every w lies in $P_-L_Z + A(P_-L_Z)$. If $Av \in P_-L_Z$ with $v \in P_-L_Z$, then $Bv = P_+Av = 0$ and $v = 0$, proving directness.

On a critical orbit $P_- = 0$. Both A and all permutations preserve the direct sum of off-axis factors, so the subspace just calculated is invariant and is precisely the smallest A -invariant subspace containing P_-L_Z . If a linear map F intertwines A with an operator on its target and satisfies $FC_0 = FS_0$, its kernel contains I_qL_Z and is A -invariant. It consequently contains L_{off} . Conversely the displayed coordinate projection intertwines A , preserves products and has $C_0 = S_0$ on its target. This proves the universal quotient assertion, also for targets carrying all the stated structures. The polynomial quotient and its kernel follow from $h_{\text{crit}} \mid h$ and Proposition 2.1. $\square$

Proposition 2.6 (Exact displacement and nilpotent contribution). *In the declared reference coefficient metric, let $W_{\text{ref}} = A^* + A - I$. Then*

$$W_{\text{ref}} = 2B + N + N^*, \quad \frac{W_{\text{ref}} - RW_{\text{ref}}R}{2} = 2B, \quad \frac{W_{\text{ref}} + RW_{\text{ref}}R}{2} = N + N^*.$$

For the original tessarine coordinate $c_{-+}(\rho) = 4(\Re \rho - 1/2)$, the first displayed component equals multiplication by $c_{-+}/2$.

Proof. The adjoint of A_{ss} is diagonal with entries $\bar{\rho}$, so $A_{\text{ss}}^* + A_{\text{ss}} - I = 2B$. Because R is a self-adjoint permutation commuting with N , it also commutes with N^* . Conjugation by R reverses B . Adding and subtracting the two resulting identities proves both formulas, including the entire nilpotent term. The tessarine identification uses $c_{-+} = 2(2\Re \rho - 1) = 4B$ with its original factor 4. $\square$

The precise map from this reference calculation to a theta-source metric retains the full metric matrix. Let G_E be that positive Hermitian matrix in the original polynomial basis of E_h ; then in the declared coefficient basis of L_Z it is

$$G_L = (T^{-1})^* G_E T^{-1}.$$

Writing the original polynomial-coordinate generator as A_E and the transported one as $A_L = TA_ET^{-1}$ for this formula, one obtains

$$\begin{aligned} W_E &= A_E^* G_E + G_E A_E - G_E, \\ W_L &= A_L^* G_L + G_L A_L - G_L = (T^{-1})^* W_E T^{-1}, \\ W_L &= G_L W_{\text{ref}} + [A_L^*, G_L], \quad [A_L^*, G_L] = A_L^* G_L - G_L A_L^*. \end{aligned}$$

The first two identities follow by inserting $A_L = TA_ET^{-1}$ and cancelling adjacent factors $T^{-1}T$ and $T^*(T^{-1})^*$. The last identity follows by expanding $G_L(A_L^* + A_L - I) + (A_L^* G_L - G_L A_L^*)$. Thus both nilpotent and metric-commutator contributions are present in the exact arithmetic weight; no zero-location estimate is supplied by discarding either term.

Finally form the split-zero semiring $G(\text{End}_{\mathbb{C}} L_Z)$, using the associative endomorphism ring. The original ghost has supported lift $\Gamma^\bullet$; when the comparison cancels it is $e = 0^\bullet$, whereas the absent comparison is τ . Conjugation by T extends to a semiring isomorphism

$$G(\text{End}_{\mathbb{C}} E_h) \longrightarrow G(\text{End}_{\mathbb{C}} L_Z), \quad X^\bullet \longmapsto (TXT^{-1})^\bullet, \quad \tau \longmapsto \tau.$$

It preserves addition, composition, the unit, e , and τ directly. Thus the complete jet comparison preserves supported cancellation. Theorem 2.5 states exactly which arithmetic factors are removed if that cancellation is imposed as a generator-compatible quotient. It does not assert that those factors are absent from the original arithmetic cohomology.

The quotient itself induces an endomorphism map on an exact specified domain. Define

$$\mathcal{D} = \{X \in \text{End}_{\mathbb{C}} L_Z : X(L_{\text{off}}) \subseteq L_{\text{off}}\}.$$

It is a unital subalgebra, and the induced quotient action is a surjective unital homomorphism

$$d : \mathcal{D} \longrightarrow \text{End}_{\mathbb{C}} L_{Z_{\text{crit}}}, \quad \ker d = \text{Hom}_{\mathbb{C}}(L_Z, L_{\text{off}}).$$

Indeed invariance of L_{off} is exactly what makes the quotient action well defined; the kernel condition says exactly that every value of X lies in L_{off} . Any endomorphism of the critical direct summand lifts by making it zero on the off-axis direct summand, proving surjectivity. Thus

$$G(d) : G(\mathcal{D}) \longrightarrow G(\text{End}_{\mathbb{C}} L_{Z_{\text{crit}}})$$

is the explicit split-zero descent map. Both A and Γ belong to $\mathcal{D}$; it sends $A^\bullet$ to the supported critical generator, $\Gamma^\bullet$ to the supported zero e , and τ to τ . This statement includes the zero target algebra when the critical summand is absent.

3 Historical Hurwitz reconstruction and the complete Laurent jet bridge

This section reconstructs a compact probability measure from the complete local Hurwitz zero cluster at any nontrivial zeta zero. It also proves the complete finite-jet relation with the marked Laurent period, including the involution coefficients and the exact collision quotient. Two distinct prime periods recover the entire finite spectral algebra with every multiplicity retained.

3.1 Source receipts and exact locators

1. The original *Split-Zero CUE Hurwitz phase space* source has SHA256 95CCA3275F2012E32F9E506EC0211273D5E7DA1FD72AD90F7167AA37094E94CC. Its exact excerpts accompany this edition in `sources/historical/hurwitz_laurent/`: `compact_hurwitz_and_residues.tex`

reproduces source lines 722–856; `zero_motion_and_bernstein.tex` reproduces lines 1110–1257; and `all_moments_claim.tex` reproduces lines 1907–1912. The statements below retain the original coordinate s , displacement t , support coordinate y , support bound L , and moments $m_n(\mu)$.

2. The *Mathematical Functor Construction* transcript has SHA256 8F94D3A8172F8585837A4E4F6060B5F8DAC38CF627C9689335F13792B7710D85. Its complete mathematical proof block, source lines 14142–15894, is retained as `sources/historical/hurwitz_laurent/marked_period_laurent_conormal_proofs.md`. Theorems 3, 7, 11 and 13 give the marked Laurent augmentation and period maps; Theorems 15–18 give the packet and Laurent involution; Theorems 20–25 give the marked conormal chart. Within that source, lines 15143–15227 prove $J_u^* = J_{-u}$, lines 15397–15502 give the $d \log |a_\infty|$ chart, and lines 15504–15631 give the augmentation-conormal diagram.
3. The corresponding standalone *Terzo-CC-CVS-JT deformation intertwiner* witness has SHA256 BAF7DB6506710CA9AA5B7705C7D5687B4E66536C08A9FE77549ABBC041BC9044. Its Laurent deformation appears at source lines 490–542 and its conormal calculation at lines 638–760. The complete proofs needed here are reproduced below and in the accompanying transcript excerpt.

The historical source gives the forward zero-motion recursion and the marked conormal map. The proofs below give the inverse moment recursion, the complete multiple-zero version, the full nilpotent Laurent isomorphism, and its collision quotient. Source line numbers and hashes identify the particular source editions; these comparisons make no claim of global mathematical priority.

3.2 Every compact Hurwitz input is determined by one simple-zero germ

Keep the original family

$$Z_{\mu,t}(s) = \int_{[0,L]} \zeta(s, 1+ty) d\mu(y), \quad m_n(\mu) = \int_{[0,L]} y^n d\mu(y), \quad m_0(\mu) = 1.$$

Here μ is a Borel probability measure, L is any finite support bound, and t is the original displacement; no rescaling of either variable is made. For $|t|L < 1$, put

$$b_n(s) = \frac{(-1)^n}{n!} (s)_n \zeta(s+n), \quad Z_{\mu,t}(s) = \sum_{n \geq 0} m_n(\mu) b_n(s) t^n. \quad (\text{H1})$$

The expansion is the source's compact Taylor formula. To verify its analytic scope directly, for $\text{Re } s > 1$ expand each $(k+ty)^{-s}$ in its binomial Taylor series. On compact s -sets and $|t|L \leq r < 1$, the ratios $|ty|/k \leq r$ give normally convergent double sums. Integration is termwise valid. The identity extends meromorphically in s ; near a nontrivial zero ρ all displayed coefficients are holomorphic. Joint holomorphicity and normal convergence in the strip can be checked directly: for a compact set K in $0 < \text{Re } s < 1$, the product

$$\left| \frac{(s)_n}{n!} \right| = \prod_{j=1}^n \left| 1 + \frac{s-1}{j} \right| \leq \prod_{j=1}^n \left(1 + \frac{C_K}{j} \right) \leq e^{C_K} (n+1)^{C_K}$$

is polynomially bounded uniformly in $s \in K$. The factor $\zeta(s+n)$ is uniformly bounded for $n \geq 1$ by the absolutely convergent Dirichlet series at $1 + \min_{s \in K} \text{Re } s$. Since $m_n \leq L^n$, the terms of (H1) are bounded by a constant times $(n+1)^{C_K} (L|t|)^n$. Thus the series is normally convergent when $L|t| < 1$, including on a surrounding compact annulus for any contour used below. Cauchy estimates give the same local derivative control. For $n \geq 1$, $\text{Re}(\rho+n) > 1$ and none of $\rho, \rho+1, \dots, \rho+n-1$ is zero, so

$$b_n(\rho) \neq 0. \quad (\text{H2})$$

The zeta factor in (H2) is nonzero because on $\operatorname{Re} z > 1$ the logarithmic Euler-product series

$$\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-kz}}{k}$$

converges absolutely, and its exponential is $\zeta(z)$ there. This conclusion uses the actual zero's strip $0 < \operatorname{Re} \rho < 1$; it does not assume RH.

Fix a simple nontrivial zero ρ , so $\zeta(\rho) = 0$ and $\zeta'(\rho) \neq 0$. The analytic implicit function theorem produces the unique local holomorphic germ

$$\rho_\mu(t) = \rho + w_\mu(t), \quad w_\mu(t) = \sum_{j \geq 1} c_j(\mu) t^j, \quad Z_{\mu,t}(\rho_\mu(t)) = 0. \quad (\text{H3})$$

This is the existing source theorem. Its simplicity hypothesis is retained explicitly, rather than silently extending a scalar branch through a multiple zero.

The inverse recurrence. For each integer $N \geq 1$, the input moment is recovered by the exact finite formula

$$m_N(\mu) = -\frac{1}{b_N(\rho)} \sum_{n=0}^{N-1} m_n(\mu) [t^{N-n}] b_n \left(\rho + \sum_{j=1}^N c_j(\mu) t^j \right). \quad (\text{H4})$$

Thus $(c_1, \dots, c_N)$ determines $(m_1, \dots, m_N)$ and conversely. In particular, two compact probability inputs have the same zero-trajectory jet through order N exactly when their first N moments agree.

Proof. Substitute (H3) in (H1). Its coefficient of t^N is

$$0 = \sum_{n=0}^N m_n(\mu) [t^{N-n}] b_n(\rho + w_\mu(t)).$$

The $n = N$ summand is exactly $m_N(\mu) b_N(\rho)$. It contains no positive-order coefficient of w . Every other summand uses only $m_0, \dots, m_{N-1}$ and $c_1, \dots, c_N$. Division is legitimate by (H2), proving (H4). In the same coefficient equation the only occurrence of c_N is $\zeta'(\rho) c_N$, from the $n = 0$ summand, because every other Taylor term has either a positive power of t already or at least two positive-order factors of w . This proves the forward triangular recurrence and both directions of the finite-jet assertion by induction. $\square$

The first two inverse formulas, with every sign retained, are

$$m_1 = \frac{\zeta'(\rho) c_1}{\rho \zeta(\rho + 1)}, \quad (\text{H5})$$

$$m_2 = -\frac{2}{\rho(\rho + 1)\zeta(\rho + 2)} \left[\zeta'(\rho) c_2 + \frac{\zeta''(\rho)}{2} c_1^2 - m_1(\zeta(\rho + 1) + \rho \zeta'(\rho + 1)) c_1 \right]. \quad (\text{H6})$$

These follow by $b_1(s) = -s\zeta(s + 1)$ and $b_2(s) = s(s + 1)\zeta(s + 2)/2$, not by changing the Hurwitz convention.

Exact measure reconstruction on the original interval. Assume $L > 0$. For every integer $M \geq 1$ and $0 \leq j \leq M$, set

$$\lambda_{M,j} = \binom{M}{j} \sum_{k=0}^{M-j} (-1)^k \binom{M-j}{k} \frac{m_{j+k}(\mu)}{L^{j+k}}, \quad \mu_M = \sum_{j=0}^M \lambda_{M,j} \delta_{Lj/M}. \quad (\text{H7})$$

Then every $\lambda_{M,j} \geq 0$, their sum is one, and μ_M converges weakly to μ . Thus (H4) and (H7) give an explicit reconstruction of the entire input from the zero germ, with the original support coordinate and bound present in every formula.

Proof. Expanding $(1 - y/L)^{M-j}$ proves

$$\lambda_{M,j} = \int_{[0,L]} \binom{M}{j} (y/L)^j (1 - y/L)^{M-j} d\mu(y).$$

This proves positivity, and the binomial theorem proves the sum is one. At fixed original y , let J have binomial law with M trials and success probability y/L . Its original-coordinate variable LJ/M has mean y and variance

$$\mathbb{E}[(LJ/M - y)^2] = \frac{y(L - y)}{M} \leq \frac{L^2}{4M}. \quad (\text{H8})$$

For continuous f on $[0, L]$ and any $\delta > 0$, split the expectation according to $|LJ/M - y| \leq \delta$. Uniform continuity and Chebyshev's inequality give

$$|\mathbb{E}f(LJ/M) - f(y)| \leq \omega_f(\delta) + 2\|f\|_\infty \frac{L^2}{4M\delta^2},$$

uniformly in y . First take $M \rightarrow \infty$, then $\delta \rightarrow 0$, and integrate against μ . This is weak convergence. For Lipschitz f , Cauchy-Schwarz gives the explicit bound $\text{Lip}(f)L/(2\sqrt{M})$. If $L = 0$, the input is exactly δ_0 , all positive moments vanish, and the original zero branch is constant; this exceptional fibre is retained separately. $\square$

Injectivity without a prespecified common support bound. If two compactly supported probability measures have the same zero germ at the chosen simple ρ , (H4) gives equal moments. Choose any common finite support bound for the two measures. Their measures (H7) are identical at every M , and their weak limits are therefore identical. Hence the map

$$\left\{ \begin{array}{c} \text{compact probability measures} \\ \text{on } [0, \infty) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{holomorphic zero germs} \\ \text{at the fixed simple } \rho \end{array} \right\},$$

$$\mu \longmapsto \rho_\mu$$

is injective onto its actual image. This statement asserts no surjectivity onto arbitrary holomorphic germs.

The exact support endpoint is also recovered:

$$\sup \text{supp } \mu = \lim_{n \rightarrow \infty} m_n(\mu)^{1/n}. \quad (\text{H9})$$

Indeed, if the endpoint is L_μ , then $m_n \leq L_\mu^n$. For any $0 < a < L_\mu$, positive mass lies in $(a, L_\mu]$, so $m_n \geq a^n \mu((a, L_\mu])$. Taking n th roots and then $a \uparrow L_\mu$ proves the claim. The zero endpoint gives δ_0 and both sides zero.

The completed residue comparison has its own exact factorization. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2} \Gamma(s/2) Z_{\mu,t}(s).$$

Then, for the same displacement domain $|t|L < 1$,

$$\text{Res}_{s=1} \Lambda_{\mu,t} = 1, \quad \text{Res}_{s=0} \Lambda_{\mu,t} = -1 - 2tm_1(\mu), \quad (\text{HRes1})$$

and consequently

$$\mathfrak{B}(\mu, t) := \text{Res}_{s=0} \Lambda_{\mu,t} + \text{Res}_{s=1} \Lambda_{\mu,t} = -2tm_1(\mu). \quad (\text{HRes2})$$

Proof. For $\operatorname{Re} \mathfrak{a} > 0$ and $\operatorname{Re} s > 1$, termwise integration of the absolutely convergent geometric series gives

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \int_0^\infty x^{s-1} \frac{e^{-\mathfrak{a}x}}{1 - e^{-x}} dx.$$

At zero the integrand's second factor is $x^{-1} + (1/2 - \mathfrak{a}) + O(x)$, uniformly on compact subsets of the indicated parameter half-plane. Subtract these two terms on $(0, 1)$. The remaining integrals define a function $H(s, \mathfrak{a})$ holomorphic for $\operatorname{Re} s > -1$, and

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \frac{1}{s-1} + \frac{1/2 - \mathfrak{a}}{s} + H(s, \mathfrak{a}).$$

The tail integral is holomorphic there because of exponential decay. Since $\Gamma(1) = 1$ and $\Gamma(s) = s^{-1} + O(1)$, the formula proves $\operatorname{Res}_{s=1} \zeta(s, \mathfrak{a}) = 1$ and $\zeta(0, \mathfrak{a}) = 1/2 - \mathfrak{a}$. For $\mathfrak{a} = 1 + ty$, the parameters stay in a compact subset of that half-plane as y ranges over the original support. Uniformity therefore permits integration of both coefficients against the probability measure. It gives residue one at $s = 1$ and $Z_{\mu,t}(0) = -1/2 - tm_1(\mu)$. Finally $\pi^{-1/2}\Gamma(1/2) = 1$, $\Gamma(s/2) = 2/s + O(1)$, and $\pi^{-s/2} = 1 + O(s)$ give (HRes1) and their sum (HRes2). $\square$

Thus the residue map factors through the explicitly recovered first moment:

$$\rho_\mu \mapsto \frac{\zeta'(\rho)c_1(\mu)}{\rho\zeta(\rho+1)} = m_1(\mu) \mapsto -2tm_1(\mu).$$

The full germ retains every higher moment through (H4), with measure reconstruction given by (H7).

3.3 The complete zero cluster removes the simplicity restriction

The preceding source theorem used a simple scalar branch. The full cluster gives an inverse for **every** nontrivial zero, with its original multiplicity retained. Fix a nontrivial zero ρ of exact order $r \geq 1$. Choose a closed disc about ρ entirely inside $0 < \operatorname{Re} s < 1$, whose boundary Γ contains no zero and whose interior contains no zero of ζ other than ρ . Thus the entire disc also excludes the pole at 1. Shrink the complex t -disc so $Z_{\mu,t}$ is nonzero on Γ and has exactly r zeros inside, with multiplicity. The normal-convergence estimate following (H1) supplies joint holomorphicity and uniform convergence to ζ on Γ ; Rouché's theorem then proves the exact zero count.

For $1 \leq j \leq r$ define the holomorphic power-sum germs

$$q_j(t) = \frac{1}{2\pi i} \oint_\Gamma (s - \rho)^j \frac{\partial_s Z_{\mu,t}(s)}{Z_{\mu,t}(s)} ds.$$

They are exactly the sums of the j th powers of the displacements of the r zeros, counted with multiplicity, by the residue theorem. Set $e_0(t) = 1$ and recursively

$$je_j(t) = \sum_{i=1}^j (-1)^{i-1} e_{j-i}(t) q_i(t).$$

Newton's identities show that the polynomial

$$P_\mu(s, t) = \sum_{j=0}^r (-1)^j e_j(t) (s - \rho)^{r-j} = (s - \rho)^r + \sum_{N \geq 1} P_N(s) t^N \quad (\text{H10})$$

has precisely the full zero cluster as its divisor, and every $P_N(s)$ has degree at most $r - 1$ in the original displacement $s - \rho$. The original point ρ and monic leading coefficient one are both retained. The polynomial is independent of any enumeration or Puiseux branch choices.

There is a jointly holomorphic unit $U_\mu(s, t)$ on a smaller disc such that

$$Z_{\mu,t}(s) = U_\mu(s, t)P_\mu(s, t), \quad U_\mu(s, t) = \sum_{N \geq 0} U_N(s)t^N, \quad U_0(s) = \frac{\zeta(s)}{(s - \rho)^r}. \quad (\text{H11})$$

For completeness, joint holomorphicity does not need to be assumed from the root parametrization. Choose an inner circle still enclosing all zeros for small t ; on that circle the quotient Z/P has nonzero denominator. Its Cauchy integral against $1/(\zeta - s)$ defines a jointly holomorphic function of (s, t) inside, since the integrand is jointly holomorphic on the fixed contour. For each fixed t , the zeros of Z and P agree with their exact multiplicities, so the quotient has removable singularities at the roots and the Cauchy integral equals Z/P . The quotient has no zero, because the two divisors already coincide. This proves (H11). The contour variable ζ here denotes a complex integration variable, not the Riemann zeta function.

Now the input moments have the exact recursive recovery

$$m_N(\mu) = \frac{1}{b_N(\rho)} \sum_{k=1}^N U_{N-k}(\rho)P_k(\rho), \quad (\text{H12})$$

followed by the exact analytic division

$$U_N(s) = \frac{m_N(\mu)b_N(s) - \sum_{k=1}^N U_{N-k}(s)P_k(s)}{(s - \rho)^r}. \quad (\text{H13})$$

In (H13) the numerator vanishes to order at least r ; its holomorphic extension at ρ has value equal to its r th derivative there divided by $r!$. Thus the formula retains every derivative needed by a multiple zero, rather than dividing a nonvanishing germ by a zero.

Proof. Comparing the coefficient of t^N in (H1) and (H11) gives

$$m_N(\mu)b_N(s) = U_N(s)(s - \rho)^r + \sum_{k=1}^N U_{N-k}(s)P_k(s).$$

Evaluation at $s = \rho$ and the nonvanishing (H2) prove (H12). Rearranging the same analytic equality proves (H13), including the divisibility and the value at the centre. Starting from the completely known U_0 , the pair (H12)–(H13) is triangular and recovers every moment from the polynomial-cluster germ.

Two measures with the same polynomial-cluster jet through order N therefore have the same moments through N , by induction. Conversely, equal moments through N give $Z_{\mu,t} - Z_{\nu,t} = O(t^{N+1})$ as holomorphic functions on the fixed circle. Since both denominators remain nonzero there, their logarithmic derivatives, contour power sums, Newton coefficients and hence polynomials (H10) agree through order N . This proves the converse without choosing scalar zero branches. Finally (H7) reconstructs the measure, so the cluster germ at any nontrivial zero determines the entire compact probability input. $\square$

For $r = 1$, $P_N(s) = -c_N$, so (H12)–(H13) reduce to the inverse branch recurrence. For $r > 1$, the contour construction keeps all r zeros and their multiplicities even when individual zeros have fractional-power trajectories. This is a fully proved extension of the historical source's simple-zero statement.

3.4 The historical period map preserves every finite spectral jet

The transcript's marked Frobenius character is

$$z_{p,\omega}(s) = \omega p^{s-1/2}, \quad p > 1, \quad \omega \in S^1, \quad a = \log p > 0. \quad (\text{J1})$$

For the original prime-circle construction, p is a rational prime and $\omega = \psi(\text{Frob}_p)$; the proof below only uses the displayed positive a and unit-modulus ω . It retains the mark and phase.

Fix $\rho \in \mathbb{C}$ and an integer multiplicity $m \geq 1$, and write

$$z = \omega p^{\rho-1/2}, \quad E_{\rho,m} = \mathbb{C}[X]/(X - \rho)^m, \quad L_{z,m} = \mathbb{C}[U, U^{-1}]/(U - z)^m.$$

The original spectral coordinate remains X , and the original Laurent period generator remains U .

Exact jet isomorphism. There is a complex-algebra isomorphism

$$\Phi_{\rho,m} : L_{z,m} \xrightarrow{\sim} E_{\rho,m}, \quad [U] \mapsto z \sum_{k=0}^{m-1} \frac{a^k (X - \rho)^k}{k!}. \quad (\text{J2})$$

Its inverse is

$$[X] \mapsto \rho + \frac{1}{a} \sum_{k=1}^{m-1} \frac{(-1)^{k+1}}{k} \left(\frac{U - z}{z} \right)^k. \quad (\text{J3})$$

For every $0 \leq k < m$, (J2) induces on the k th associated conormal grade the exact multiplier

$$[(U - z)^k] \mapsto (za)^k [(X - \rho)^k]. \quad (\text{J4})$$

Proof. The target of (J2) is a unit: it is z times a polynomial with constant term one and nilpotent remaining part. Its difference from z lies in $(X - \rho)$, so its m th power vanishes. The Laurent universal property and the quotient therefore define (J2). In $\mathbb{C}[t]/t^m$, truncated $\exp(t)$ and $\log(1 + t)$ are inverse operations. This identity can be proved without convergence: formal differentiation shows $(\log(\exp t))' = 1$ and its constant term is zero; likewise $\exp(\log(1 + t))/(1 + t)$ has derivative zero and constant term one. Reducing the resulting formal identities modulo t^m gives (J3) and both composites. Finally

$$\Phi(U - z) = za(X - \rho) + (X - \rho)^2 \left(z \sum_{j=2}^{m-1} \frac{a^j (X - \rho)^{j-2}}{j!} \right),$$

so its k th power modulo $(X - \rho)^{k+1}$ is exactly $(za)^k (X - \rho)^k$. For $m = 1$, the sums in (J3) are empty and both rings are the specified scalar fibres, as required. $\square$

For a finite primary spectral block $A = \rho I + N$ with $N^m = 0$, this means that its actual period operator is exactly

$$\omega \exp(a(A - \tfrac{1}{2}I)) = z \sum_{k=0}^{m-1} \frac{a^k N^k}{k!}. \quad (\text{J5})$$

There is no deletion of nilpotent multiplicity in this map. Indeed, $\exp(aN) - I = N(aI + NB(N))$; its second factor is an invertible polynomial in N , so its powers have exactly the same kernels as the corresponding powers of N . This proves preservation of all Jordan block lengths locally.

3.5 Exact conjugate-reciprocal involution on the jets and conormal

Define the original spectral involution

$$\rho^\dagger = 1 - \bar{\rho}, \quad z^\dagger = \omega p^{\rho^\dagger - 1/2} = 1/\bar{z}. \quad (\text{J6})$$

On polynomial classes define the conjugate-linear algebra isomorphism

$$\mathcal{S}_E : E_{\rho,m} \longrightarrow E_{\rho^\dagger,m}, \quad [f(X)] \longmapsto [\bar{f}(1 - X)]. \quad (\text{J7})$$

On Laurent classes define

$$\mathcal{S}_L : L_{z,m} \longrightarrow L_{z^\dagger,m}, \quad \left[\sum c_n U^n \right] \longmapsto \left[\sum \bar{c}_n U^{-n} \right]. \quad (\text{J8})$$

Both have square the identity after returning to the original labelled block, and the full finite-jet diagram commutes:

$$\boxed{\mathcal{S}_E \Phi_{\rho,m} = \Phi_{\rho^\dagger,m} \mathcal{S}_L.} \quad (\text{J9})$$

Proof. The polynomial generator transforms by

$$X - \rho \longmapsto 1 - X - \bar{\rho} = -(X - \rho^\dagger).$$

The Laurent generator transforms by the exact identity

$$U - z \longmapsto U^{-1} - \bar{z} = -\bar{z} U^{-1} (U - z^\dagger). \quad (\text{J10})$$

The factor $-\bar{z}U^{-1}$ is a unit, so the respective m th-power ideals map to one another and both maps are defined. Applying each twice fixes constants and the original generator. To check (J9), it suffices to check the Laurent generator and constants. Its left image is

$$\bar{z} \sum_{k=0}^{m-1} \frac{(-a)^k (X - \rho^\dagger)^k}{k!},$$

which is the inverse of $z^\dagger \exp(a(X - \rho^\dagger))$ modulo $(X - \rho^\dagger)^m$ and hence the right image. $\square$

For clarity, the entire nonlinear mixing of Laurent Taylor coefficients under (J8) is explicit. With $t' = U - z^\dagger$, for $1 \leq k < m$ one has

$$\boxed{\mathcal{S}_L((U - z)^k) = \sum_{\ell=k}^{m-1} (-1)^\ell \binom{\ell-1}{\ell-k} \bar{z}^{k+\ell} (t')^\ell.} \quad (\text{J11})$$

Proof. By (J10), the expression is

$$(-\bar{z})^k U^{-k} (t')^k = (-1)^k \bar{z}^{2k} (t')^k (1 + \bar{z}t')^{-k}.$$

The formal binomial expansion $(1 + v)^{-k} = \sum_{j \geq 0} (-1)^j \binom{k+j-1}{j} v^j$, truncated when the total degree reaches m , gives (J11). The constant basis vector maps to one separately. $\square$

In particular, the conormal maps are

$$[X - \rho] \longmapsto -[X - \rho^\dagger], \quad [U - z] \longmapsto -\bar{z}^2 [U - z^\dagger], \quad (\text{J12})$$

and their compatibility with (J4) is an exact equality of coefficients:

$$-\bar{z}^2 (z^\dagger a) = -\bar{z}a = \bar{z}\bar{a}(-1).$$

The higher grade k has factors $(-1)^k$ and $(-1)^k \bar{z}^{2k}$ respectively, with the corresponding $(za)^k$ comparison. This supplies the all-order anti-involution/conormal morphism requested by the historical construction; it does not erase the higher terms in (J11).

On the real analytic radial chart $s = \beta + i\gamma$, the same exact map gives

$$\nu = \log |z| = (\beta - \tfrac{1}{2}) \log p, \quad d\nu = (\log p) d\beta, \quad c_{-+} = 4(\beta - \tfrac{1}{2}), \quad \nu = \frac{\log p}{4} c_{-+}. \quad (\text{J13})$$

The source's marked transverse chart $j_S(\nu) = [((e^\nu, 1, \dots, 1), 0)]$ pulls its displayed form $\alpha = d \log |a_\infty|$ back to $d\nu$, since $\log(e^\nu) = \nu$. Thus the composite character→radial chart has pullback $\alpha = (\log p) d\beta$, with the original prime and tessarine factor retained. The local identity requires no conjecture about zeros. A global identification with any separately defined Weil quadratic form is not asserted by this local chart calculation; the exact algebraic and chart maps proved here are the morphisms actually available from this source block.

3.6 Period collisions: the exact unlabelled image and its missing fibres

Let Z be a finite set of distinct spectral points ρ , with original multiplicities $m_\rho \geq 1$, and

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[X]/h(X) \cong \prod_{\rho \in Z} E_{\rho, m_\rho}.$$

The last map is the CRT map $f \mapsto (f \bmod (X - \rho)^{m_\rho})$; pairwise coprimality gives its inverse by the Bezout identity. No multiplicity is discarded.

Set $z_\rho = \omega p^{\rho-1/2}$. The equality of two period values is exactly

$$z_\rho = z_\sigma \iff \rho - \sigma \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{J14})$$

The proof is the exact kernel $2\pi i \mathbb{Z}$ of the complex exponential. In particular, a fixed prime's character cannot automatically be treated as globally injective on a zero packet. No noncollision claim for actual zeta zeros is inserted here.

For each distinct period value z , put $M_z = \max_{\rho: z_\rho=z} m_\rho$ and

$$P(U) = \prod_z (U - z)^{M_z}, \quad L_P = \mathbb{C}[U, U^{-1}]/P(U).$$

There is an injective algebra map

$$\Phi_Z : L_P \hookrightarrow E_h \quad (\text{J15})$$

whose ρ th component is Laurent reduction modulo $(U - z_\rho)^{m_\rho}$ followed by (J2). Its image consists exactly of the tuples which, after (J3) on each labelled primary block, have the same Laurent Taylor coefficients at a common period value through every order visible in both blocks. More explicitly, if $a_\rho \in E_{\rho, m_\rho}$, write

$$\Phi_{\rho, m_\rho}^{-1}(a_\rho) = \sum_{k=0}^{m_\rho-1} b_{\rho, k} (U - z_\rho)^k.$$

Then (a_ρ) lies in the image exactly when

$$b_{\rho,k} = b_{\sigma,k} \quad \text{whenever } z_\rho = z_\sigma \text{ and } 0 \leq k < \min(m_\rho, m_\sigma). \quad (\text{J16})$$

The exact dimension and cokernel dimension as complex vector spaces are

$$\dim \operatorname{im} \Phi_Z = \sum_z M_z, \quad \dim \operatorname{coker} \Phi_Z = \sum_\rho m_\rho - \sum_z M_z. \quad (\text{J17})$$

Proof. Use CRT also on P . For a fixed common value z , a Laurent polynomial vanishes on every corresponding spectral block exactly when it is divisible by $(U - z)^{m_\rho}$ for every such ρ , which is equivalent to divisibility by $(U - z)^{M_z}$. Therefore the kernel of the original Laurent evaluation map is exactly (P) , giving (J15). Reduction of one degree- $< M_z$ polynomial to each smaller truncation gives (J16). Conversely choose any one block of maximal multiplicity M_z ; its Laurent coefficients define the common polynomial, and (J16) forces all other components to be its truncations. These constructions prove the image equality, while the dimensions follow from the explicit polynomial bases. $\square$

Keeping the labels restores a full exact isomorphism:

$$\prod_{\rho \in Z} L_{z_\rho, m_\rho} \xrightarrow{\Pi \Phi_{\rho, m_\rho}} E_h. \quad (\text{J18})$$

Thus the historical period map and the current finite arithmetic jet object have a proved relation at both the labelled and unlabelled levels. The collision compatibility space is exactly the image (J16). An explicit quotient records every lost independent coefficient: choose one maximal block ρ_z over each z , with $m_{\rho_z} = M_z$, and define the complex-linear map

$$\Delta_Z : E_h \longrightarrow \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho}, \quad (a_\rho) \longmapsto (b_{\rho,k} - b_{\rho_z,k})_{\substack{z, \rho \neq \rho_z \\ 0 \leq k < m_\rho}}. \quad (\text{J19})$$

It is surjective: set all maximal-block coefficients to zero and prescribe each remaining block's coefficients to be the desired output. Its kernel is exactly (J16). Consequently

$$0 \longrightarrow L_P \xrightarrow{\Phi_Z} E_h \xrightarrow{\Delta_Z} \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho} \longrightarrow 0 \quad (\text{J20})$$

is an exact sequence of complex vector spaces. The first map is an algebra homomorphism; the last map is explicitly linear and is not incorrectly promoted to a ring homomorphism. Its target has the cokernel dimension in (J17).

If Z is stable under $\rho \mapsto 1 - \bar{\rho}$ with the same multiplicities, the maps (J7) and (J8), together with permutation of the labelled factors, give conjugate-linear involutions on both sides of (J18). Formula (J9) proves that (J18) intertwines them. The unlabelled construction (J15) is also stable: $z \mapsto 1/\bar{z}$ permutes the common period values and preserves M_z ; (J10) carries the defining ideal of one value to that of the paired value.

3.7 Two prime periods recover the complete finite spectral algebra

Retain a finite nonempty set Z of distinct spectral points with the original orders $m_\rho \geq 1$, and $E_h = \mathbb{C}[X]/h(X)$, where $h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}$. Choose distinct rational primes p, q and fixed marks $\omega_p, \omega_q \in S^1$. Put

$$a = \log p, \quad b = \log q, \quad u_\rho = \omega_p e^{a(\rho-1/2)}, \quad v_\rho = \omega_q e^{b(\rho-1/2)}.$$

Define the units $\mathfrak{A}_p, \mathfrak{A}_q \in E_h$ by their complete original spectral CRT components:

$$\begin{aligned} (\mathfrak{A}_p)_\rho &= u_\rho \sum_{k=0}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \\ (\mathfrak{A}_q)_\rho &= v_\rho \sum_{k=0}^{m_\rho-1} \frac{b^k (X - \rho)^k}{k!}. \end{aligned} \tag{P1}$$

The period-algebra map therefore has the exact type

$$\mathcal{P}_{p,q} : \mathbb{C}[U^{\pm 1}, V^{\pm 1}] \longrightarrow E_h, \quad U \longmapsto \mathfrak{A}_p, \quad V \longmapsto \mathfrak{A}_q. \tag{P2}$$

This map uses the actual periods of the original generator; the notation $\mathcal{P}_{p,q}$ leaves the theta map Θ unchanged.

Theorem 3.1 (Complete two-prime Laurent presentation). *The map (P2) is surjective, with every local order retained. More precisely, in $\mathcal{L} = \mathbb{C}[U^{\pm 1}, V^{\pm 1}]$, put*

$$\begin{aligned} B_\rho(U) &= v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j, \\ I_\rho &= ((U - u_\rho)^{m_\rho}, V - B_\rho(U)). \end{aligned} \tag{P3}$$

Here $\binom{b/a}{0} = 1$, and for $j \geq 1$ the coefficient is $(b/a)(b/a - 1) \cdots (b/a - j + 1)/j!$. The ideals I_ρ are pairwise comaximal, and

$$\begin{aligned} \ker \mathcal{P}_{p,q} &= \bigcap_{\rho \in Z} I_\rho = \prod_{\rho \in Z} I_\rho, \\ \mathcal{L} / \bigcap_{\rho \in Z} I_\rho &\xrightarrow{\sim} E_h. \end{aligned} \tag{P4}$$

The map on each primary quotient is the original finite exponential map (J2), with the second generator determined by (P3).

Proof. First the two marked periods separate distinct complex points. If their values at ρ, σ agree, there exist integers k, l with

$$a(\rho - \sigma) = 2\pi i k, \quad b(\rho - \sigma) = 2\pi i l.$$

Eliminating $\rho - \sigma$ gives $k \log q = l \log p$, hence $q^k = p^l$. Prime valuations in the positive rationals give $k = l = 0$, including negative integer exponents. Thus $\rho = \sigma$. Each fixed phase cancels only within its own prime character; neither mark was changed.

For a fixed ρ , set $y = (U - u_\rho)/u_\rho$. The quotient $\mathcal{L}/I_\rho$ is exactly $\mathbb{C}[y]/y^{m_\rho}$: its generators satisfy

$$U = u_\rho(1 + y), \quad V = v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} y^j.$$

Both expressions are units, since their constant terms are nonzero and their remaining parts are nilpotent. The Laurent inverses therefore impose no extra relation. These expressions give a map from the Laurent quotient to $\mathbb{C}[y]/y^{m_\rho}$, whose inverse sends y to $(U - u_\rho)/u_\rho$; their composites fix the displayed generators.

The map to E_{ρ, m_ρ} is

$$y \longmapsto \sum_{k=1}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \quad X \longmapsto \rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} y^j \tag{P5}$$

for it and its inverse, respectively. The finite logarithm and exponential identities proved in (J2)–(J3) verify both composites. The identity $(1+y)^{b/a} = \exp((b/a) \log(1+y))$, reduced modulo y^{m_ρ} , verifies the prescribed V -image. For completeness its binomial coefficients are fixed by the formal differential equation $(1+y)F' = (b/a)F$, $F(0) = 1$: $(j+1)[y^{j+1}]F = (b/a-j)[y^j]F$. This recurrence gives exactly the coefficients in (P3). Thus I_ρ is precisely the kernel of the ρ -component of (P2), including order one.

For two different labels, either $u_\rho \neq u_\sigma$, or their v -values differ by the separation already proved. In the first case the powers of the two distinct linear U -factors have a polynomial Bezout identity, proving $I_\rho + I_\sigma = \mathcal{L}$. In the second case, $(V - v_\rho)^{m_\rho} \in I_\rho$: modulo I_ρ , $V - v_\rho = B_\rho(U) - v_\rho$ is a multiple of $U - u_\rho$. The corresponding powers of the two distinct linear V -factors again give a Bezout identity. Hence all primary kernels are pairwise comaximal.

The Chinese-remainder map now gives

$$\mathcal{L} / \bigcap_{\rho} I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} \mathcal{L} / I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} E_{\rho, m_{\rho}} \xleftarrow{\sim} E_h. \quad (\text{P6})$$

For clarity, the first map is onto because Bezout identities give classes ϵ_ρ equal to one in factor ρ and zero in every other factor; summing ϵ_ρ times prescribed representatives recovers any tuple. Its kernel is the intersection by definition. Pairwise comaximality also gives intersection equal to product, first for two ideals by multiplying a Bezout identity by an element of the intersection, and then by induction. These arguments prove every assertion in (P4).

The original coordinate itself has the explicit inverse image

$$[X] \mapsto \sum_{\rho \in Z} \epsilon_\rho \left(\rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j \right). \quad (\text{P7})$$

Each term is exactly (P5) in its own primary factor; the other factors vanish through ϵ_ρ . Thus the inverse retains the original generator and all of its nilpotent jets. $\square$

This presentation is an exact statement for each finite packet. The single-prime collision quotient (J20) and the two-prime map (P6) specify their respective information. The factorwise split lifts below retain their individual support labels.

3.8 Split-Zero support is preserved by the exact maps

For any ring R , keep the original $G(R) = \{\tau\} \sqcup R^\bullet$, global zero τ , and supported zero $e_R = 0_R^\bullet$. Every ring map $f : R \rightarrow S$ gives

$$G(f)(\tau) = \tau, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{S1})$$

This preserves addition and multiplication: if one input is τ , use its defining identity/absorber laws, and if both are supported, use the ring homomorphism identities with the supported tag retained. In particular $G(f)(e_R) = e_S \neq \tau$, even if f has a nontrivial kernel.

Apply (S1) separately to each map (J2), and then take the product of these lifted semiring isomorphisms. The resulting map is

$$\prod_{\rho \in Z} G(L_{z_\rho, m_\rho}) \xrightarrow{\sim} \prod_{\rho \in Z} G(E_{\rho, m_\rho}). \quad (\text{S2})$$

Every individual supported-zero coordinate is retained. The support map on both sides is the full vector of presence indicators in $\{0, 1\}^Z$; under (S2) it is unchanged.

For a finite nonempty index set $\mathcal{I}$, the comparison with applying G once to a product ring is the map

$$\begin{aligned} \iota_R : G \left(\prod_{i \in \mathcal{I}} R_i \right) &\longrightarrow \prod_{i \in \mathcal{I}} G(R_i), \\ \tau &\longmapsto (\tau)_i, \quad (r_i)_i^\bullet \longmapsto (r_i^\bullet)_i. \end{aligned} \quad (\text{S4})$$

It is an injective unital semiring homomorphism. For two supported inputs, the operations agree coordinate by coordinate by the ring laws. When a τ occurs, its additive-identity and multiplicative-absorber laws also agree coordinate by coordinate. The two sorts of image are disjoint because $\mathcal{I}$ is nonempty; equality of supported images recovers every r_i . Its support image is exactly the empty and full support vectors. For ring maps $f_i : R_i \rightarrow S_i$, direct evaluation proves

$$\left(\prod_i G(f_i) \right) \iota_R = \iota_S G \left(\prod_i f_i \right). \quad (\text{S5})$$

This is the exact natural comparison with the factorwise map (S2). If $\mathcal{I}$ is empty, the same comparison has the two-element semiring $G(0)$ as source and the one-element empty product as target, so it is not injective; its unique map identifies e and τ . The product map (S2) itself remains the identity between empty products in that case.

For the unlabelled map, a source factor $G(L_{z,M_z})$ maps to all spectral factors over z , using (J2) after the quotient $L_{z,M_z} \twoheadrightarrow L_{z,m_\rho}$. This is injective because one target block has multiplicity M_z . Its support image consists exactly of the vectors which are constant on every collision class $q^{-1}(z)$, where $q : Z \rightarrow \{z_\rho\}$ is $\rho \mapsto z_\rho$. A source τ gives τ in every such factor, while a supported element gives a supported element in every factor, including factors in which its amplitude becomes zero. The support embedding is exactly

$$q^{-1} : \mathcal{P}(\{z_\rho\}) \hookrightarrow \mathcal{P}(Z), \quad (\text{S3})$$

which preserves empty set, unions and intersections. The direct-image map $q_* : \mathcal{P}(Z) \rightarrow \mathcal{P}(\{z_\rho\})$ preserves unions and has $q_* q^{-1} = \text{id}$, but need not preserve intersections. For two distinct labels ρ, σ in the same collision class, $\{\rho\} \cap \{\sigma\} = \emptyset$ while $q_* \{\rho\} \cap q_* \{\sigma\} = \{z\}$. Thus (S3), rather than an unsupported lattice quotient isomorphism, is the exact support relation; the amplitude compatibility is (J16).

For a same-centre truncation $m' \leq m$, the quotient

$$L_{z,m} \twoheadrightarrow L_{z,m'}, \quad E_{\rho,m} \twoheadrightarrow E_{\rho,m'}$$

commutes with (J2) because the finite power series simply reduce to the corresponding lower nilpotence order. Its lifted map sends the supported nonzero class $(U - z)^{m'}$ to e when $m' < m$, and never to τ . This supplies a concrete example of the original distinction between a vanished jet amplitude and absent support, inside the current arithmetic finite-jet packet.

3.9 All-order Hurwitz motion transported to the period and tesserine coordinates

For the actual zero branch (H3), the exact period germ is

$$z_\mu(t) = \omega p^{\rho_\mu(t)-1/2} = z_\rho \exp \left((\log p) \sum_{n \geq 1} c_n(\mu) t^n \right). \quad (\text{C1})$$

Its N th coefficient is

$$[t^N]z_\mu(t) = z_\rho \sum_{k=1}^N \frac{(\log p)^k}{k!} \sum_{\substack{n_1+\dots+n_k=N \\ n_j \geq 1}} \prod_{j=1}^k c_{n_j}(\mu) \quad (N \geq 1). \quad (\text{C2})$$

The empty coefficient is z_ρ . Conversely, using the unique logarithm germ vanishing at $z_\mu/z_\rho = 1$,

$$c_N(\mu) = \frac{1}{\log p} [t^N] \log(z_\mu(t)/z_\rho). \quad (\text{C3})$$

Equations (C2)–(C3) are inverse because the same formal exponential/logarithm calculation used in (J2)–(J3) applies at every finite order. Combining (C3) with (H4) proves that the marked local period germ also determines the entire compact Hurwitz input.

For real t in the germ's domain,

$$\begin{aligned} \log |z_\mu(t)| &= (\log p) \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right), \\ c_{-+}(\rho_\mu(t)) &= 4 \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right). \end{aligned} \quad (\text{C4})$$

All coefficients, including tangential ones, are retained in (C1)–(C3). The real projection in (C4) is explicitly the radial/tessierine observable; its loss of imaginary coefficients is a specified map. For $t \neq 0$, the Hurwitz family generally lacks the original functional equation, so $1 - \rho_\mu(t)$ is not asserted to be another zero of that family. The tessierine coordinate formula is the proved coordinate map evaluated at the actual branch, with this spectral distinction preserved.

3.10 Mathematical scope

The maps (J2)–(J20) identify the full finite arithmetic algebra with its labelled Laurent jets and calculate the exact quotient of unlabelled single-prime observations. Equations (H12)–(H13) recover all moments from the actual zero cluster, and (H7) reconstructs the entire measure. These statements retain the original multiplicities, support coordinates and marked periods.

The constructions give exact algebraic and analytic comparison maps. They do not supply the uniform estimate on the arithmetic moment matrices required by the weight-control calculation elsewhere in this reader.

The accompanying source records retain the historical locators and independent proof reviews. The proofs in this section specify their own hypotheses and maps in full.

4 Historical support branches, ideal completion, and measured observations

This chapter gives the complete branch, ideal-completion, atomic-kernel, and operator-algebra proofs extracted and continued from the historical Split-Zero support program. Its mathematical report and independent audit are, respectively, `logbook/historical_next_responses_20260912.md` and `logbook/frame_branch_independent_audit_20260912.md`. The equation identifiers B, I, M, A and O are local to this chapter.

Source locators and comparison. Exact complete response bodies are staged at `sources/historical/next_responses/preboolean_prime_branches.md` (source lines 23137–23956; response

heading 23104; exported timestamp 7/6/2026, 12:23:35 PM), sources/historical/next_response/preboolean_arithmetic_branch_maps.md (source lines 25162–25715; heading 25058; timestamp 7/6/2026, 7:58:41 PM), and sources/historical/next_responses/pointfree_frame_arithmetic_claim.md (source lines 25754–26221; heading 25721; timestamp 7/6/2026, 10:01:27 PM). Their shared original source has SHA-256 8d1780a24df3bb31db981b5e437558de31e6d94bf21bb54103a0a336d6a73091. The respective excerpt SHA-256 values are f5f11dd6b26a3a9d6a4c77cf2f7b86898759a19fa740c993d5749c8d47c02618, e84873955876b2be5f7738f746b4e133fc479cfe726bcb85d477f9aec7d6d10e, and 57c6662732b723e451a495f0224507f61e3a969956178018af7633ae06797ac0. For each excerpt, its first line is the first indicated original source line, so every original locator has the exact excerpt locator obtained by subtracting that first source line and adding one.

The core corpus comparison is split_zero_projective_monads_surcomplex/split_support_geometry_arithmetic_curve_v11.tex, SHA-256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d. Its lines 1772–2054 already prove the ordinary prime-filter branches and their arithmetic maps; lines 2001–2018 explicitly impose arbitrary-join preservation for locale points. The support and measured realizations were also compared at lines 5006–5096 and 5346–5378. Accordingly the ordinary branch classification is a surviving historical construction, while the ideal quotient, explicit descent witnesses, atomic kernel and full L^∞ correspondence below supply the additional proved maps.

The final historical response, source lines 25982–26020 (excerpt lines 229–267), equates ordinary amplitude-fixed semiring branches with frame points without imposing arbitrary-join preservation. The exact correction is the classification (B1)–(B5), the frame quotient (I2), and the point-descent criterion (I3)–(I10). The measure-algebra construction then exhibits the kernel pairs that ordinary branches can distinguish. No external KMS classification or Morita equivalence is used as a theorem of this chapter.

4.1 Exact ordinary and continuous Split-Zero branches

The results below are stated in classical set theory. The ultrafilter construction explicitly uses Zorn’s lemma. No claim about existence or nonexistence of points in a constructive foundation is inferred from that use of choice; such an internal comparison would also have to specify its truth-value object and its available joins.

Let R be a nonzero commutative ring and let L be a nontrivial bounded distributive lattice. Retain the source carrier and all its exceptional coordinates:

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subseteq R \times L,$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu),$$

$$\tau_L = (0, 0_L), \quad e_L = (0, 1_L), \quad 1 = (1_R, 1_L).$$

Write $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $e = 0_R^\bullet$ distinct from global zero τ . The indicated subset is the carrier throughout.

Here $\mathbb{B} = \{0, 1\}$ has join and meet as its semiring operations. The supported copy in $G(R)$ has $r^\bullet + s^\bullet = (r + s)^\bullet$ and $r^\bullet s^\bullet = (rs)^\bullet$; its global zero satisfies $\tau + x = x + \tau = x$ and $\tau x = x\tau = \tau$. The map $G_{\mathbb{B}}(R) \rightarrow G(R)$ defined by $(0, 0) \mapsto \tau$, $(r, 1) \mapsto r^\bullet$ is a bijection and preserves these operations in each support case, hence is a unital semiring isomorphism. To verify the general carrier, the displayed operations are the restrictions of the product semiring $R \times L$. This subset is closed: a sum of nonzero amplitude has top support because at least one input has nonzero amplitude; a product of nonzero amplitude has two nonzero input amplitudes and hence top support. Every zero-amplitude

output belongs to the zero fibre. The zero and unit belong to the subset. Thus associativity, commutativity and distributivity restrict from the product, including all supported cancellations. All homomorphisms below preserve zero and unit. Write $z_\lambda = (0, \lambda)$.

Theorem B1 — all ordinary amplitude-fixed branches. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \xrightarrow{\sim} \{\beta : G_L(R) \rightarrow G(R) \text{ unital semiring map} : \beta(r, 1_L) = r^\bullet\}. \quad (\text{B1})$$

It sends ϵ to

$$\beta_\epsilon(r, 1_L) = r^\bullet, \quad \beta_\epsilon(0, \lambda) = \begin{cases} \tau & \epsilon(\lambda) = 0, \\ e & \epsilon(\lambda) = 1. \end{cases} \quad (\text{B2})$$

The map is surjective as a semiring map; its exact kernel congruence is equality of amplitudes together with equality of selected support values:

$$(r, \lambda) \sim_\epsilon (s, \mu) \iff r = s \text{ and } \epsilon(\lambda) = \epsilon(\mu). \quad (\text{B3})$$

Proof. The source formulas agree at $(0, 1_L)$ since $\epsilon(1_L) = 1$. They preserve zero and unit. On the zero fibre, preservation of addition and multiplication is precisely preservation of join and meet. If one argument has nonzero amplitude, its lattice coordinate is 1_L ; adding a zero-fibre argument leaves it unchanged and multiplying by that argument gives $(0, \lambda)$, matching respectively addition by τ or e , and multiplication by τ or e in $G(R)$. With both arguments at full support, even if their sum or product cancels to amplitude zero, both outputs are supported and ring identities prove preservation. This verifies all exceptional products and cancellations.

Conversely, $(0, \lambda)$ is additively idempotent. The only additive idempotents of $G(R)$ are τ and e : a supported $r^\bullet + r^\bullet = r^\bullet$ implies $2r = r$ in the ring, hence $r = 0$. Therefore every amplitude-fixed β sends the zero fibre into $\{\tau, e\}$. Identifying this two-element subsemiring with $\mathbb{B}$ gives a map ϵ preserving bottom, top, meet and join. Formula (B2) reconstructs β ; both constructions are inverse. Every amplitude and τ occurs in the image, and inspecting (B2) gives exactly (B3). $\square$

Now let L be a frame: all joins exist and finite meets distribute over arbitrary joins. Its zero fibre has specified infinitary joins

$$\bigvee_{i \in I}^{\text{fib}} (0, \lambda_i) = (0, \bigvee_{i \in I} \lambda_i).$$

Theorem B2 — exactly the continuous branches. Under (B1), the subset of branches preserving all these zero-fibre joins is exactly

$$\text{Hom}_{\text{Frm}}(L, \mathbb{B}). \quad (\text{B4})$$

Indeed the corresponding equality for (B2) says exactly $\epsilon(\bigvee_i \lambda_i) = \bigvee_i \epsilon(\lambda_i)$, and finite meets are already preserved. This is the precise additional structure required by the transcript's final frame formulation. Its ordinary algebraic maps retain a larger, explicitly identified category.

The classification extends to every map between original lattice-split carriers. For commutative unital rings R, A and nontrivial bounded distributive lattices L, M ,

$$\text{Hom}_{\text{Semi}}(G_L(R), G_M(A)) \cong \text{Hom}_{\text{Ring}}(R, A) \times \text{Hom}_{\text{DLat}_{0,1}}(L, M). \quad (\text{B5})$$

The pair (h, f) gives $(r, \lambda) \mapsto (h(r), f(\lambda))$. Conversely an arbitrary semiring map sends each additively idempotent z_λ to amplitude zero, say $z_{f(\lambda)}$; this preserves finite joins, meets and bottom. The identity $e_L = (1_R, 1_L) + (-1_R, 1_L)$ makes its image have top support, because one summand is the target unit; additive idempotence makes its amplitude zero. Thus $e_L \mapsto e_M$ and f preserves

top. Now $(r, 1_L) + e_L = (r, 1_L)$ forces its image to have top support. Write that image $(h(r), 1_M)$. The supported operations give a unital ring map h , and the formulas recover every source element, consistently at e_L . Coordinate composition proves naturality and both inverse constructions. In particular the entire two-factor datum is recoverable from the semiring map; scalar amplitude and support are not freely interchanged.

4.2 The exact frame comparison retaining all ordinary branches

For the underlying bounded distributive lattice of a frame L , let $\text{Idl}(L)$ be the set of lattice ideals. An ideal is a nonempty lower set closed under finite joins. Its bottom is $\{0_L\}$, its top is L , its finite meet is intersection, and the join of a family of ideals is

$$\bigvee_{\alpha} I_{\alpha} = \{x \in L : x \leq y_1 \vee \cdots \vee y_n \text{ for some finite } n \geq 0, y_j \in \bigcup_{\alpha} I_{\alpha}\}. \quad (\text{I1})$$

The empty finite join is 0_L ; this retains the bottom ideal in every formula.

Theorem I1 — frame quotient and principal-ideal section. The maps

$$q : \text{Idl}(L) \rightarrow L, \quad I \mapsto \bigvee I, \quad \eta : L \rightarrow \text{Idl}(L), \quad \lambda \mapsto \downarrow \lambda \quad (\text{I2})$$

have the following exact properties:

1. $\text{Idl}(L)$ is a frame and q is a surjective frame homomorphism.
2. η is an injective bounded-lattice homomorphism, $q\eta = \text{id}_L$, and $q(I) \leq \lambda \iff I \subseteq \eta(\lambda)$.
3. The kernel congruence of q is exactly $I \sim J \iff \bigvee I = \bigvee J$. No claim that η preserves arbitrary joins is made.

Proof. Formula (I1) is an ideal and is the least ideal containing every I_{α} ; intersections give infima. To prove frame distributivity, let $x \in I \cap \bigvee_{\alpha} J_{\alpha}$. Choose $x \leq y_1 \vee \cdots \vee y_n$ with $y_j \in J_{\alpha_j}$. In the distributive lattice,

$$x = x \wedge (y_1 \vee \cdots \vee y_n) = (x \wedge y_1) \vee \cdots \vee (x \wedge y_n).$$

Each $x \wedge y_j$ belongs to $I \cap J_{\alpha_j}$, proving $I \cap \bigvee_{\alpha} J_{\alpha} \subseteq \bigvee_{\alpha} (I \cap J_{\alpha})$; the reverse inclusion follows from containment. Thus $\text{Idl}(L)$ is a frame.

The supremum of the right side of (I1) equals $\bigvee_{\alpha} \bigvee I_{\alpha}$, proving arbitrary-join preservation by q , including the empty family. Since L is a frame,

$$(\bigvee I) \wedge (\bigvee J) = \bigvee_{i \in I, j \in J} (i \wedge j) = \bigvee (I \cap J).$$

The last equality holds because each $i \wedge j$ is in the intersection, and each $x \in I \cap J$ is the term $x \wedge x$. Hence q preserves finite meets and the top. Principal ideals satisfy $\downarrow \lambda \cap \downarrow \mu = \downarrow (\lambda \wedge \mu)$ and $\downarrow \lambda \vee \downarrow \mu = \downarrow (\lambda \vee \mu)$; they also preserve bottom/top. Their suprema are the original elements, proving $q\eta = \text{id}_L$, injectivity and surjectivity. The adjunction equivalence and the kernel description follow immediately from the definition of supremum. $\square$

Theorem I2 — all algebraic branches become frame points before the quotient. There are mutually inverse bijections

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \cong \text{Hom}_{\text{Frm}}(\text{Idl}(L), \mathbb{B}), \quad (\text{I3})$$

given by

$$\widehat{\epsilon}(I) = \bigvee_{\lambda \in I} \epsilon(\lambda) \quad \text{and} \quad \epsilon = \widehat{\epsilon}\eta. \quad (I4)$$

A frame point $\widehat{\epsilon}$ in (I3) factors through q if and only if its original ϵ is a frame point of L . Thus the exact inclusion of point sets is

$$q^* : \text{Pt}(L) \hookrightarrow \text{Pt}(\text{Idl}(L)), \quad \epsilon \mapsto \epsilon q, \quad (I5)$$

and its image is the points constant on the kernel congruence in Theorem I1.

Proof. In (I4), the value is 1 precisely when the ideal meets the prime filter $\epsilon^{-1}(1)$. For a join of ideals, a value 1 is witnessed by an element below a finite join of source elements; finite primeness then gives an element of value 1 in one of those source ideals. The reverse implication follows by containment, proving arbitrary-join preservation. For intersections, if I and J each have a value 1 element i, j , their meet $i \wedge j \in I \cap J$ has value 1; the reverse implication is immediate. Top and bottom follow from $\epsilon(1) = 1, \epsilon(0) = 0$. Therefore $\widehat{\epsilon}$ is a frame homomorphism.

Every ideal is the join in $\text{Idl}(L)$ of its principal ideals, so an arbitrary frame point is completely determined by its restriction along η , giving both inverse identities. If ϵ preserves arbitrary joins, $\widehat{\epsilon}(I) = \epsilon(\bigvee I) = (\epsilon q)(I)$. Conversely if $\widehat{\epsilon} = \chi q$ for a frame map χ , then $\epsilon = \widehat{\epsilon}\eta = \chi q\eta = \chi$, so ϵ is a frame map. A set map constant on the fibres of surjective q factors uniquely; if that map is a frame map, the induced factor preserves arbitrary joins and finite meets by lifting families through q . This proves the final image claim. $\square$

Theorem I3 — original Split-Zero comparison and exact congruence. Every bounded-lattice map $f : L \rightarrow M$ induces

$$G_f(R) : G_L(R) \rightarrow G_M(R), \quad (r, \lambda) \mapsto (r, f(\lambda)). \quad (I6)$$

For the nontrivial frames in Theorem I1 this gives a split pair of unital semiring maps

$$G_L(R) \xrightarrow{G_\eta(R)} G_{\text{Idl}(L)}(R) \xrightarrow{G_q(R)} G_L(R), \quad G_q G_\eta = \text{id}. \quad (I7)$$

They preserve τ and supported e as distinct elements. The exact kernel of $G_q(R)$ consists of equal amplitudes and equal suprema of zero-fibre ideals:

$$(r, I) \sim (s, J) \iff r = s \text{ and } \bigvee I = \bigvee J. \quad (I8)$$

Proof. A nonzero amplitude in $G_L(R)$ has full support and maps to full support because $f(1_L) = 1_M$; hence (I6) lands in the exact target carrier. Finite join and meet preservation establish the semiring identities coordinate by coordinate, including zero-amplitude cancellation. Formula (I7) follows from $q\eta = \text{id}_L$, and (I8) is equality of the two coordinates of the image. The bottom and top of the nontrivial source and target lattices differ; their images under these bounded maps are again bottom and top, proving the exact zero statements. $\square$

Thus an ordinary branch which fails zero-fibre join continuity is a fully explicit frame point **upstairs**, and the arithmetic map fails to descend through a proved frame quotient **downstairs**. The original support object and all its algebraic branches remain related by the maps (I2)–(I8).

The two available lifts have an exact comparison, which prevents a split algebra map from being mistaken for a split frame map. Given an ordinary $\epsilon : L \rightarrow \mathbb{B}$, both

$$\epsilon q : \text{Idl}(L) \rightarrow \mathbb{B}, \quad \widehat{\epsilon} : \text{Idl}(L) \rightarrow \mathbb{B} \quad (I9)$$

are bounded-lattice maps and agree on every principal ideal. The first descends through q by definition; the second preserves arbitrary joins by Theorem I2. They coincide on all ideals exactly when

ϵ preserves arbitrary joins. The forward implication follows because $\epsilon q = \widehat{\epsilon}$ and surjectivity of the frame map q force its factor ϵ to preserve joins; the converse is (I4). This also yields two amplitude-fixed Split-Zero branch maps out of $G_{\text{Idl}(L)}(R)$, with exactly the same criterion for equality.

There is a further exact localic structure. The map $j = \eta q$ is a nucleus on $\text{Idl}(L)$: it is monotone, $I \subseteq j(I)$ because each $i \in I$ lies below $\bigvee I$, it satisfies $j^2 = j$ because $q\eta = \text{id}_L$, and it preserves finite meets because both q and η do. Its fixed ideals are exactly the principal ideals. Their inherited meet and nucleus-applied join

$$\bigvee_{\alpha}^{\text{fix}} \downarrow \lambda_{\alpha} = j \left(\bigvee_{\alpha}^{\text{Idl}(L)} \downarrow \lambda_{\alpha} \right) = \downarrow \left(\bigvee_{\alpha}^L \lambda_{\alpha} \right) \quad (\text{I10})$$

identify the fixed-point frame exactly with the original L . This proves the quotient/sub-locale correspondence at the level of operations, while retaining the precise failure of η to preserve the ambient ideal-frame joins.

4.3 A fully explicit nontrivial frame with algebraic branches and no frame points

Let L be the Lebesgue probability measure algebra of $[0, 1]$: measurable subsets modulo null symmetric difference. Write $[A]$ for the class of A , keep the original measure $\mu(A)$, and use union, intersection and complement for finite Boolean operations. These operations are well defined modulo null sets, since changing their inputs changes any output only within the union of those null differences. The exact order is $[A] \leq [C]$ if and only if $\mu(A \setminus C) = 0$.

Lemma M1 — completeness and atomlessness. This L is a complete Boolean algebra, hence a frame, and has no atoms.

Proof. For an arbitrary family $([A_i])_{i \in I}$, let

$$b = \sup \{ \mu(A_{i_1} \cup \dots \cup A_{i_n}) : n \geq 0 \}.$$

Choose finite unions F_n with measures tending to b , and put $B = \bigcup_n F_n$. Replacing F_n by the union of the first n such finite unions makes them increasing without changing their defining kind, so $\mu(B) = b$. For each original A_i , the finite-union supremum bounds $\mu(F_n \cup A_i) \leq b$, and continuity from below gives $\mu(B \cup A_i) \leq b = \mu(B)$. Thus $A_i \setminus B$ is null. If $[C]$ is another upper bound, every one of the countably many constituent sets in B is contained in C modulo a null set; hence $B \setminus C$ is null. This proves that $[B]$ is the supremum. Infima follow by complements.

In a complete Boolean algebra finite meets distribute over arbitrary joins: if $v = \bigvee_i (a \wedge b_i)$, then every $b_i \leq \neg a \vee v$; taking joins and intersecting with a gives $a \wedge \bigvee_i b_i \leq v$. The reverse inequality is immediate. Thus L is a frame.

For a nonzero $[A]$, the function $F(t) = \mu(A \cap [0, t])$ is continuous because $|F(t) - F(u)| \leq |t - u|$. It takes the values 0 and $\mu(A) > 0$ at the endpoints. There is therefore t with $F(t) = \mu(A)/2$. The class of $A \cap [0, t]$ is strictly between 0 and $[A]$, so no nonzero class is an atom. $\square$

Theorem M2 — exact frame points of a complete Boolean algebra. If B is any complete Boolean algebra, its frame points correspond bijectively to its atoms:

$$\text{Hom}_{\text{Frm}}(B, \mathbb{B}) \cong \text{At}(B), \quad a \mapsto \epsilon_a(b) = \mathbf{1}_{\{a \leq b\}}. \quad (\text{M1})$$

Proof. An atom below $\bigvee_i b_i$ lies below some b_i : otherwise every $a \wedge b_i = 0$, while frame distributivity gives $a = a \wedge \bigvee_i b_i = \bigvee_i (a \wedge b_i) = 0$, a contradiction. Thus ϵ_a preserves arbitrary joins. It preserves finite meets, top and bottom directly.

Conversely, a frame map $\epsilon : B \rightarrow \mathbb{B}$ into the two-element Boolean algebra preserves complements, since complements are characterized by their finite meet 0 and join 1. By De Morgan it therefore also

preserves arbitrary meets. Put $a = \bigwedge \{b : \epsilon(b) = 1\}$. Its image is 1, so $a \neq 0$. If $0 < c \leq a$ and $\epsilon(c) = 0$, then $\epsilon(\neg c) = 1$, forcing $a \leq \neg c$ and hence $c = 0$, a contradiction. Consequently $\epsilon(c) = 1$, which by definition of a implies $a \leq c$. Thus a is an atom. Every element of the filter contains a ; conversely if $a \leq b$, then $\epsilon(b) = 1$. This proves the inverse formulas and their uniqueness. $\square$

In particular the measure algebra of Lemma M1 has **no frame points**.

It nevertheless has ordinary lattice characters. Here is an explicit witnessed failure of the required continuity. Let $D_n = [0, 2^{-n}]$ as original measurable subsets of $[0, 1]$; their classes are nonzero and any finite intersection is the smallest selected D_n . The filter generated by these classes is proper. Order proper filters containing it by inclusion; the union of every chain is proper, so Zorn's lemma gives a maximal proper filter F . In a Boolean algebra, maximal proper filters contain exactly one of $b, \neg b$: if neither lay in F , adjoining b would produce 0, so some $f \in F$ satisfies $f \wedge b = 0$ and hence $f \leq \neg b$, which already puts $\neg b \in F$. Its characteristic function is therefore a bounded-lattice homomorphism $\epsilon_F : L \rightarrow \mathbb{B}$.

Every $\epsilon_F([D_n]) = 1$, while every $\epsilon_F([D_n^c]) = 0$. But

$$\bigvee_{n \geq 1} [D_n^c] = [[0, 1] \setminus \{0\}] = 1_L, \quad \epsilon_F(1_L) = 1 \neq \bigvee_{n \geq 1} 0 = 0. \quad (\text{M2})$$

Thus (B2) gives an actual surjective semiring branch $\beta_{\epsilon_F} : G_L(R) \rightarrow G(R)$ preserving amplitudes, even though the frame has no point. Formula (M2) gives its exact failed zero-fibre join equation. The point $\hat{\epsilon}_F$ of $\text{Idl}(L)$ does not descend through q . More explicitly, set

$$J = \bigvee_{n \geq 1}^{\text{Idl}(L)} \downarrow [D_n^c].$$

Then $q(J) = 1_L = q(L)$, but $\hat{\epsilon}_F(J) = 0$ and $\hat{\epsilon}_F(L) = 1$. This is an explicit pair in the kernel congruence of q distinguished by the upstairs point; it proves the obstruction to descent at an exact source/target pair.

In fact **every** ordinary branch on this measure algebra fails already countable-join preservation. For its ultrafilter F , start at $a_0 = 1$. Split a representative of a_n into two equal-measure pieces by the continuous cumulative-measure construction in Lemma M1. Exactly one of their classes belongs to F ; call it a_{n+1} . The resulting decreasing sequence satisfies $a_n \in F$ and $\mu(a_n) = 2^{-n}$. Its infimum is 0 by continuity of measure. Consequently $b_n = \neg a_n$ increases to 1 and each $\epsilon_F(b_n) = 0$, again contradicting countable-join preservation. The ideal $I = \bigcup_n \downarrow b_n$ then satisfies $q(I) = q(L) = 1$, $\hat{\epsilon}_F(I) = 0$, and $\hat{\epsilon}_F(L) = 1$. This gives a kernel-congruence witness for each individual algebraic branch, not merely for the displayed dyadic choice.

4.4 Atomic observation and the exact support data it removes

For a complete Boolean algebra B , put

$$b_{\text{at}} = \bigvee_{a \in \text{At}(B)} a, \quad b_{\text{na}} = \neg b_{\text{at}}.$$

The frame-point evaluation map is

$$\text{ev} : B \rightarrow \mathcal{P}(\text{At}(B)), \quad x \mapsto \{a : a \leq x\}. \quad (\text{A1})$$

Theorem A1 — exact atomic quotient. This is a surjective complete Boolean-algebra homomorphism. Its kernel congruence is

$$\text{ev}(x) = \text{ev}(y) \iff x \wedge b_{\text{at}} = y \wedge b_{\text{at}}, \quad (\text{A2})$$

and its inverse image of the empty set is the full principal ideal $\downarrow b_{\text{na}}$. The complete decomposition retaining that ideal is

$$B \xrightarrow{\sim} (\downarrow b_{\text{at}}) \times (\downarrow b_{\text{na}}), \quad x \mapsto (x \wedge b_{\text{at}}, x \wedge b_{\text{na}}), \quad (\text{A3})$$

with inverse $(u, v) \mapsto u \vee v$ and respective local units $b_{\text{at}}, b_{\text{na}}$.

Proof. Each coordinate of (A1) is the frame point in (M1), so it preserves arbitrary joins and finite meets. Complements are preserved because an atom lies below exactly one of $x, \neg x$. For any subset S of atoms, the element $\bigvee_{a \in S} a$ maps exactly to S , by pairwise disjointness of distinct atoms and the arbitrary-join argument in (M1). Thus evaluation is surjective. Distributivity gives $x \wedge b_{\text{at}} = \bigvee_{a \in \text{At}(B)} (x \wedge a) = \bigvee_{a \leq x} a$, which proves (A2) and the empty-fibre claim. The two complementary components in (A3) reconstruct x by distributivity, and disjointness of $b_{\text{at}}, b_{\text{na}}$ proves both inverse identities and preservation of complete Boolean operations. $\square$

The complement on either local Boolean factor $\downarrow b$ in (A3) is explicitly $u \mapsto b \wedge \neg u$; its local top is b , not the global top of B unless $b = 1$.

This supplies an exact relation to the source's adelic measure-algebra support. In a finite measure-algebra realization, frame-point evaluation detects precisely its measure atoms; (A3) retains the entire nonatomic component rather than treating lack of points as lack of support. For a discrete countable measure with positive measure at each singleton, the singleton classes are the atoms and $b_{\text{at}} = 1$; evaluation is an isomorphism to the power set of the original places. For the Lebesgue example, $b_{\text{at}} = 0, b_{\text{na}} = 1$, and every frame-point evaluation is absent while the entire nontrivial Boolean carrier persists.

When applying $G_{\text{ev}}(R)$, retain the exceptional case: if there are no atoms, $\mathcal{P}(\text{At}(B))$ is the one-element lattice, so the exact carrier $G_{\mathcal{P}(\emptyset)}(R)$ is isomorphic to R , and this quotient identifies τ_B and e_B with ordinary zero. It is the explicit ring-reflection map on this observation, not evidence that $\tau_B = e_B$ in the original carrier. The split comparison (I7) has nontrivial lattices on both sides and preserves them as distinct; it remains available when the point-observation quotient collapses them.

Concretely, $\Psi : G_{\{\ast\}}(R) \rightarrow R, (r, \ast) \mapsto r$, is bijective and preserves both operations because the unique support coordinate is fixed. In the atomless case its composition is the exact identity

$$\Psi \circ G_{\text{ev}}(R) = p_B : G_B(R) \rightarrow R, \quad p_B(r, \lambda) = r. \quad (\text{A4})$$

4.5 The exact operator-algebraic branch map

The core manuscript's L^∞ realization permits an additional precise bridge to state spaces. Work with the same finite probability measure space, complete measure algebra B , and the complex algebra $M = L^\infty(\Omega, \mu; \mathbb{C})$ equipped with essential-supremum norm and complex conjugation. The original measure and every atomic mass are retained.

Theorem O1 — all Boolean branches extend exactly to multiplicative states. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(B, \mathbb{B}) \cong \{\chi : M \rightarrow \mathbb{C} : \chi \text{ is a unital complex-linear } \ast\text{-homomorphism}\}. \quad (\text{O1})$$

The correspondence is specified on every projection by

$$\chi \epsilon(\mathbf{1}_A) = \epsilon([A]), \quad (\text{O2})$$

and on arbitrary $f \in M$ by the uniform extension of this finite formula:

$$\chi_\epsilon \left(\sum_{j=1}^n c_j \mathbf{1}_{A_j} \right) = \sum_{j=1}^n c_j \epsilon([A_j]), \quad (O3)$$

where the A_j form the specified measurable partition of Ω modulo null sets. Every such χ is positive and has norm 1.

Proof. Exactly one member of a finite partition has ϵ -value 1: finite-join preservation gives at least one, and pairwise disjointness with meet preservation gives at most one. Thus (O3) selects one of the original values c_j . Refining two partitions by all intersections shows the formula is independent of the simple-function presentation. The unique selected intersection gives additivity and multiplicativity for two simple functions, while conjugating their values gives star preservation. The selected value has modulus at most the essential supremum of the simple function, so the map is contractive.

Every bounded measurable complex function has simple functions tending to it in essential-supremum norm: partition a square containing its essential range into finitely many squares of diameter at most 2^{-n} and use one original value in each nonempty square. The resulting evaluations are Cauchy by contractivity. Their limit defines a unique continuous extension independent of the approximating sequence. Addition, multiplication and conjugation commute with these uniform limits, proving the required homomorphism identities. Positivity follows from $f = g^*g$ with $g = \sqrt{f}$ for every nonnegative measurable f ; hence $\chi(f) = |\chi(g)|^2 \geq 0$. Contractivity and $\chi(1) = 1$ prove norm 1.

Conversely a unital star homomorphism sends projections to real idempotents 0 or 1. Since $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$ and $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A \cap B}$, its projection values form the required Boolean lattice map. It is contractive because $\|f\|_\infty^2 \mathbf{1} - f^*f$ is the square of a bounded measurable function, so positivity gives $|\chi(f)|^2 \leq \|f\|_\infty^2$. It agrees with (O3) on simple functions, and continuity forces equality on all of M . This proves both inverse constructions. $\square$

Theorem O2 — normal branches, exact atomic masses, and failure of normality. For the correspondence (O1), the following are equivalent:

1. χ preserves suprema of bounded increasing nets of nonnegative elements of M (normality).
2. ϵ is a frame point of the complete measure algebra.
3. There is a measure atom a such that

$$\chi(f) = \frac{1}{\mu(a)} \int_a f d\mu \quad (f \in M). \quad (O4)$$

In particular the atomless Lebesgue example has multiplicative states for all of its ordinary branches, and none is normal; its exact countable failure is exhibited by (M2).

All bounded increasing nets used here have a supremum in the specified M . An explicit proof for nonnegative nets retains the finite measure: put $b = \sup_i \int_\Omega f_i d\mu$. Choose indices with integrals tending to b , and use directedness to replace them by an increasing dominating sequence. Choose representatives outside one common null set for the countably many sequence comparisons, so this selected sequence is pointwise increasing there. Its pointwise supremum g has integral b . For every fixed original index i , each finite maximum with a sequence member lies below a net member, so $\int_\Omega \max(g, f_i) d\mu \leq b = \int_\Omega g d\mu$ by monotone convergence. Thus $f_i \leq g$ almost everywhere. Every upper bound of the original net bounds the selected sequence and hence g , proving that g is the required least upper bound.

Proof. The supremum in M of the finite-union projections from an arbitrary family $([A_i])$ is the projection $\mathbf{1}_{\bigvee_i [A_i]}$: Lemma M1's countable essential union is an upper bound, and its increasing finite selected unions show it is the least upper bound also among arbitrary measurable nonnegative functions. Normality applied to this increasing net therefore gives preservation of its full projection

join. Together with (O2), this is precisely frame continuity of ϵ , proving $1 \Rightarrow 2$. By Theorem M2, a frame point corresponds to a unique atom a .

Every bounded measurable f is essentially constant on a . To see this, use the finite measurable partitions of its essential range from the proof of Theorem O1. Each partition has exactly one cell whose intersection with a is nonzero, and by atomhood that intersection equals a modulo null sets. Its diameter tends to 0, so the simple approximants restrict to constants c_n on a , with $|c_n - c_m|$ bounded by their uniform difference. Their limit c is the almost-everywhere value of f on a . Formula (O3) selects exactly these constants and gives $\chi(f) = c$; integration over the original atom gives precisely (O4), with the factor $1/\mu(a)$ retained. Thus $2 \Rightarrow 3$.

Finally let a bounded increasing net $f_i \geq 0$ have supremum f in M , and let c_i be its respective constants on a . These constants increase; put $c = \sup_i c_i$. Every f_i is bounded above by the function equal to c on a and to f off a ; hence the least-upper-bound property of f forces its constant on a to be at most c . Conversely each $f_i \leq f$ forces $c_i \leq f|_a$, and taking their supremum gives the reverse inequality. Thus $\chi(f) = \sup_i \chi(f_i)$. This proves $3 \Rightarrow 1$, including every original coefficient and atomic mass. $\square$

Equations (B1),(O1),(O2) form the promised typed connection between algebraic Split-Zero branches and operator-algebraic states. They identify exactly which branches can be normal observations of the measured support.

5 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (\text{D1})$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 5.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (\text{D2})$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (\text{D3})$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (\text{D4})$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \mapsto (l, x). \quad (\text{D5})$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

5.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

6 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

6.1 Source spaces and fixed arithmetic factors

Let

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \end{aligned}$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 6.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \operatorname{Re} s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k) |\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\operatorname{Re} s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2} \{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\operatorname{im} d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

6.2 The exact operator range, including its inverse

Proposition 6.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \operatorname{Re} \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \operatorname{Re} \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho_i y}/k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 6.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}}u = S_h^{\mathcal{B}}\Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad \sigma = qR_{\text{ref}}.$$

Then

$$\mathcal{M}R_{\text{ref}}u = (g/h)R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

6.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$DR_{\text{ref}} - R_{\text{ref}}A = f_0 \ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $tR_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}}E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\sim} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

6.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below

uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy-Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

7 Certified moments from the original arithmetic theta source

The moment matrices in the arithmetic kernel formulas can be calculated directly from the specified theta source. This section gives an exact incomplete-gamma series, proves an explicit bound for the entire omitted double tail, and applies directed interval arithmetic to the retained terms. The construction retains the original factor $g = 2\xi$, the measure dx , the Euler operator $D = -x\partial_x$, and every polynomial coefficient. No information about the location of a zeta zero enters this calculation.

7.1 Source polynomials and the exact reflection map

Keep

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_j = D^j \Theta \phi_*, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx). \quad (\text{AM.1})$$

Introduce polynomials with their original integer coefficients by

$$P_0(z) = 4z^2 - 6z, \quad P_{j+1}(z) = 2z(P_j(z) - P'_j(z)), \quad Q_j(z) = \sum_{k=0}^j (-1)^k \binom{j}{k} P_k(z). \quad (\text{AM.2})$$

Write p_{ja} and q_{ja} for the coefficients of z^a . Both polynomials have degree $d_j = j + 2$, with leading coefficients 2^{j+2} and $(-1)^j 2^{j+2}$, respectively, and both have zero constant coefficient. The formula defining Q_j is an explicit change of polynomial coordinates; neither family is discarded.

Define the linear involution on the actual Hilbert observation by

$$\mathcal{T} : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}_{>0}, dx), \quad (\mathcal{T}F)(x) = x^{-1}F(1/x). \quad (\text{AM.3})$$

It is unitary and self-adjoint, satisfies $\mathcal{T}^2 = 1$, and, on the strong Schwartz space $\mathcal{B}$,

$$D\mathcal{T} = \mathcal{T}(1 - D), \quad \mathcal{T}f_0 = f_0, \quad \mathcal{T}f_j = (1 - D)^j f_0. \quad (\text{AM.4})$$

Consequently, the complete source series and its reflected series are

$$f_j(x) = 2 \sum_{n \geq 1} P_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \tilde{f}_j(x) := (1 - D)^j f_0(x) = 2 \sum_{n \geq 1} Q_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad (\text{AM.5})$$

and $f_j(1/x) = x \tilde{f}_j(x)$.

Proof. For $z = \pi x^2$, applying $-x\partial_x$ to $P(z)e^{-z}$ gives $2z(P - P')e^{-z}$. This proves the recurrence and (AM.5). The leading coefficients and zero constant coefficients follow by induction. Under the Fourier convention $\hat{\phi}(\xi) = \int \phi(x)e^{-2\pi i x \xi} dx$, differentiating the Fourier transform of the Gaussian gives

$$\widehat{x^2 e^{-\pi x^2}}(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}}(\xi) = \left(\xi^4 - \frac{3\xi^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi \xi^2}.$$

Multiplication by the exact coefficients $4\pi^2, -6\pi$ therefore gives $\hat{\phi}_* = \phi_*$. Its value at zero is zero, and its integral is $\hat{\phi}_*(0) = 0$. The theta-Fourier identity already proved for the source V then gives $\mathcal{T}f_0 = f_0$.

Substitution $u = 1/x$ gives $\int |x^{-1}F(1/x)|^2 dx = \int |F(u)|^2 du$, so $\mathcal{T}$ is unitary. It is an involution and hence self-adjoint. Direct differentiation proves $D\mathcal{T} = \mathcal{T}(1 - D)$, including the constant 1 arising from the original factor x^{-1} . Induction proves the other identities in (AM.4). Local uniform convergence of each differentiated Gaussian series follows on $x \geq \delta > 0$ by a polynomial times Gaussian majorant. The series gives decay of all Euler derivatives at infinity, and its exact reflected expression gives the corresponding decay at zero. Thus these operations apply to the original functions in $\mathcal{B}$. $\square$

7.2 An absolutely convergent double series for every entry

For integers $i, j \geq 0$, retain the raw moment entry

$$H_{ij} = \langle D^i f_0, D^j f_0 \rangle_{L^2(dx)} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j |2\xi(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{AM.6})$$

The equality is the original Mellin Plancherel identity. In particular H_{ij} is real because the source functions are real. For $L \geq 0$ and $\lambda > 0$, set

$$I_L(\lambda) = \int_1^\infty x^{2L} e^{-\lambda x^2} dx = \frac{\Gamma(L + 1/2, \lambda)}{2\lambda^{L+1/2}}, \quad \Gamma(a, \lambda) = \int_\lambda^\infty u^{a-1} e^{-u} du. \quad (\text{AM.7})$$

Theorem 7.1 (Exact original arithmetic moment series). *With $\lambda_{nm} = \pi(n^2 + m^2)$, the following double series converges absolutely, including after the finite coefficient sums are replaced by their absolute values:*

$$H_{ij} = 2 \sum_{n,m \geq 1} \sum_{a=1}^{d_i} \sum_{b=1}^{d_j} (p_{ia} p_{jb} + q_{ia} q_{jb}) \pi^{a+b} n^{2a} m^{2b} \cdot \frac{\Gamma(a+b+1/2, \lambda_{nm})}{\lambda_{nm}^{a+b+1/2}}. \quad (\text{AM.8})$$

Equivalently, it is the Gram integral on the positive half-domain

$$H_{ij} = \int_1^\infty (f_i(x) f_j(x) + \tilde{f}_i(x) \tilde{f}_j(x)) dx. \quad (\text{AM.9})$$

Thus the precise map to the positive Gram presentation is

$$\mathbb{C}[s] \longrightarrow L^2([1, \infty), dx)^{\oplus 2}, \quad P \longmapsto (P(D)f_0, P(1-D)f_0)|_{[1, \infty)}. \quad (\text{AM.10})$$

It preserves the original polynomial inner product (AM.6) and is injective. The signs in (AM.8) are retained; the positivity is the Gram positivity of this specified map.

Proof. Split the integral in (AM.6) at $x = 1$. On $(0, 1)$, set $x = 1/u$. The two factors u supplied by (AM.5) cancel the Jacobian u^{-2} exactly. This gives (AM.9). For absolute convergence, write

$$C_j = \sum_{a=1}^{d_j} |p_{ja}| \pi^a, \quad E_j = \sum_{a=1}^{d_j} |q_{ja}| \pi^a. \quad (\text{AM.11})$$

For $n, x \geq 1$, the polynomial bounds are

$$|P_j(\pi n^2 x^2)| \leq C_j n^{2d_j} x^{2d_j}, \quad |Q_j(\pi n^2 x^2)| \leq E_j n^{2d_j} x^{2d_j}.$$

Since $n^2 + m^2 \geq 2$ and $x^2 \geq 1$,

$$(n^2 + m^2)x^2 \geq \frac{1}{2}(n^2 + m^2) + x^2.$$

The absolute integrand series is therefore bounded by

$$4(C_i C_j + E_i E_j) x^{2(d_i + d_j)} e^{-\pi x^2} \left(\sum_{n \geq 1} n^{2d_i} e^{-\pi n^2/2} \right) \left(\sum_{m \geq 1} m^{2d_j} e^{-\pi m^2/2} \right),$$

which is integrable on $[1, \infty)$. This proves all required sum-integral interchanges. Substitution $u = \lambda x^2$ gives (AM.7). The two theta factors contribute 4, and (AM.7) contributes $1/2$, leaving exactly the factor 2 in (AM.8).

For complex polynomial coefficients, conjugate the coefficients of the first factor in the Gram integral. Equation (AM.9) then proves the inner-product assertion in (AM.10). If a nonzero polynomial P were in its kernel, its first component would vanish on $[1, \infty)$. Put $U(z) = \sum_j P_j(z) c_j$ when $P(s) = \sum_j c_j s^j$. The distinct degrees and nonzero leading coefficients of the P_j imply $U \neq 0$. Multiplying the vanishing theta series by $e^{\pi x^2}/2$ gives

$$U(\pi x^2) = - \sum_{n \geq 2} U(\pi n^2 x^2) e^{-\pi(n^2-1)x^2}.$$

The right side decays exponentially times a fixed polynomial as $x \rightarrow \infty$, by the same Gaussian bound. A nonzero polynomial on the left cannot have that decay. This contradiction proves injectivity. $\square$

7.3 A complete finite truncation certificate

Let $H_{ij}^{[N]}$ be (AM.8) with both theta indices restricted to $1 \leq n, m \leq N$, keeping all coefficients p_{ia}, q_{ia} . For the calculation of this entry choose an integer $N \geq \max(d_i, d_j)$. Define, for $k = i, j$,

$$\begin{aligned} r_{k,N} &= \left(\frac{N+2}{N+1} \right)^{2d_k} e^{-\pi(2N+3)}, & T_{k,N} &= \frac{(N+1)^{2d_k}}{1 - r_{k,N}}, \\ A_{k,N} &= \sum_{n=1}^N n^{2d_k} e^{-\pi(n^2-1)} + T_{k,N} e^{-\pi((N+1)^2-1)}. \end{aligned} \quad (\text{AM.12})$$

All these numbers are specified positive real constants. The choice of N ensures $r_{k,N} < 1$, since

$$\log r_{k,N} < \frac{2d_k}{N+1} - \pi(2N+3) < 2 - 5\pi < 0.$$

Theorem 7.2 (Rigorous complete double-tail bound). *For every $i, j \geq 0$ and $N \geq \max(i, j) + 2$,*

$$\boxed{|H_{ij} - H_{ij}^{[N]}| \leq \mathcal{E}_{ij,N} := 4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) \cdot I_{d_i+d_j}(\pi((N+1)^2 + 1))}. \quad (\text{AM.13})$$

In particular the bound tends to zero as $N \rightarrow \infty$, for each fixed pair i, j , and provides an explicit finite certificate for every raw arithmetic moment.

Proof. For $x \geq 1$, the ratio of successive terms of $n^{2d_k} e^{-\pi n^2 x^2}$ is at most

$$\left(\frac{n+1}{n} \right)^{2d_k} e^{-\pi(2n+1)}.$$

The logarithm of this expression has derivative $-2d_k/(n(n+1)) - 2\pi < 0$ for positive real n . For $n \geq N+1$ every ratio is consequently at most $r_{k,N}$. Summing the geometric majorant yields

$$\sum_{n \geq N} n^{2d_k} e^{-\pi n^2 x^2} \leq T_{k,N} e^{-\pi(N+1)^2 x^2}, \quad \sum_{n \geq 1} n^{2d_k} e^{-\pi n^2 x^2} \leq A_{k,N} e^{-\pi x^2}. \quad (\text{AM.14})$$

For the second inequality, divide by $e^{-\pi x^2}$, bound each retained factor $e^{-\pi(n^2-1)x^2}$ at $x = 1$, and apply the first inequality to the tail.

Write $F_{k,N}$ for the finite P_k -series and $\tilde{F}_{k,N}$ for the finite Q_k -series. Equations (AM.11) and (AM.14) give

$$\begin{aligned} |f_k(x)|, |F_{k,N}(x)| &\leq 2C_k x^{2d_k} A_{k,N} e^{-\pi x^2}, \\ |f_k(x) - F_{k,N}(x)| &\leq 2C_k x^{2d_k} T_{k,N} e^{-\pi(N+1)^2 x^2}. \end{aligned}$$

The reflected inequalities have exactly E_k in place of C_k . In each product of (AM.9), use the exact decomposition $ab - a_N b_N = (a - a_N)b + a_N(b - b_N)$. The resulting bound on the absolute difference of integrands is

$$4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) x^{2(d_i+d_j)} e^{-\pi((N+1)^2+1)x^2}.$$

Its integral is exactly the right side of (AM.13). Finally, the $A_{k,N}$ remain bounded for fixed k , the $T_{k,N}$ grow at most polynomially, and the integral decays exponentially in $(N+1)^2$. For completeness, with $L = d_i + d_j$ fixed and $\lambda \geq 2L$, the inequality $\log x \leq (x^2 - 1)/2$ for $x \geq 1$ gives

$$I_L(\lambda) \leq e^{-\lambda} \int_1^\infty e^{-(\lambda-L)(x^2-1)} dx \leq \frac{e^{-\lambda}}{2(\lambda-L)}.$$

Here the last step uses $x^2 - 1 \geq 2(x-1)$. This proves the asserted limit with an additional elementary upper bound when desired. $\square$

The finite matrix $(H_{ij}^{[N]})_{0 \leq i, j \leq m}$ is itself positive definite for every $N \geq 1$. Indeed it is the Gram matrix of $(F_{j,N}, \tilde{F}_{j,N})$ on the same half-domain; the injectivity proof in (AM.10) applies verbatim to the finite sum. This fact does not assign an ordering to its difference from the full matrix. The relation to the full matrix is the entrywise enclosure (AM.13), whose signs and error bounds are explicit.

7.4 Exact adjoint, parity, and finite-boundary checks

On $\mathcal{B}$ in $L^2(dx)$, integration by parts gives $\langle DF, H \rangle = \langle F, (1 - D)H \rangle$. Therefore the complete moments satisfy

$$H_{i+1,j} + H_{i,j+1} = H_{ij}, \quad H_{ij} = \sum_{k=0}^i (-1)^k \binom{i}{k} H_{0,j+k}, \quad H_{01} = \frac{1}{2} H_{00}. \quad (\text{AM.15})$$

All boundary terms vanish because the original functions have rapid decay at both ends. More explicitly,

$$(Df_i)f_j + f_i(Df_j) - f_i f_j = -\partial_x(x f_i f_j),$$

whose integral on $(0, \infty)$ is zero. Moving each of the i copies of D in the first argument to the second gives the middle identity; symmetry gives the last one.

The finite sums retain an exact boundary correction at $x = 1$:

$$H_{i+1,j}^{[N]} + H_{i,j+1}^{[N]} - H_{ij}^{[N]} = F_{i,N}(1)F_{j,N}(1) - \tilde{F}_{i,N}(1)\tilde{F}_{j,N}(1). \quad (\text{AM.16})$$

To prove this, use $DF_{j,N} = F_{j+1,N}$ for the first component and $D\tilde{F}_{j,N} = \tilde{F}_{j,N} - \tilde{F}_{j+1,N}$ for the second. Applying the preceding integration-by-parts calculation on $[1, \infty)$ gives the first displayed endpoint product for the first component and its negative for the second. The endpoints at infinity vanish term by term. In particular the exact finite identity $H_{01}^{[N]} = H_{00}^{[N]}/2$ follows even before the omitted tail is enclosed, since $F_{0,N} = \tilde{F}_{0,N}$.

One may also retain the exact triangular coordinate map

$$B_{ja} = \begin{cases} \binom{j}{a} (-1/2)^{j-a}, & a \leq j, \\ 0, & a > j, \end{cases} \quad C = BHB^T. \quad (\text{AM.17})$$

This is the Gram matrix of $u_j = (D - 1/2)^j f_0$, with inverse coordinate matrix $(B^{-1})_{ja} = \binom{j}{a} (1/2)^{j-a}$. No original moment is lost. Since $\mathcal{T}u_j = (-1)^j u_j$, unitarity proves $C_{ij} = 0$ for odd $i + j$. The operator $D - 1/2$ is skew-symmetric on this displayed strong Schwartz domain, so for $i + j = 2q$,

$$C_{ij} = (-1)^{i+q} \|u_q\|^2, \quad \|u_q\|^2 = \int_{\mathbb{R}} t^{2q} |2\xi(1/2 + it)|^2 \frac{dt}{2\pi} > 0. \quad (\text{AM.18})$$

For the first equality move all i powers to the second argument, and compare with moving the q powers in $\langle u_q, u_q \rangle$. Mellin Plancherel gives the second equality. Strict positivity follows from the nonzero original function and the polynomial injectivity already proved.

7.5 Reproducible certified finite arithmetic data

The companion program `scripts/check_arithmetic_moments.py` implements the exact integer recurrence (AM.2), evaluates every retained term of (AM.8) by Arb directed ball arithmetic, and enlarges its radius by an upward-rounded value of (AM.13). Thus the final balls enclose the infinite-source entries, including rounding error and both omitted theta tails. The default run uses $0 \leq i, j \leq 5, N = 7$, and 110 decimal working digits; its machine-readable output is `checks/arithmetic_moments.json`.

Let $\Delta_k = \det(H_{ij})_{0 \leq i,j < k}$ and $\Delta_0 = 1$. The squared norm h_k of the monic degree- k polynomial for the original measure μ is

$$h_k = \frac{\Delta_{k+1}}{\Delta_k}. \quad (\text{AM.19})$$

Indeed the coefficients below the leading term solve the Gram system making that monic polynomial orthogonal to degrees below k . Substitution of that unique solution into its squared norm gives the Schur complement of the leading k -by- k Gram block. The block determinant identity gives (AM.19). The interval calculation certifies strict positivity of all six leading principal determinants recorded in the companion certificate and all six corresponding h_k ; those are the actual arithmetic norms used by the orthogonal recurrence, through the finite degrees stated.

Here are explicit certified intervals from that run. Each row means the entire interval $[c - r, c + r]$, with the printed decimal centre c and radius r interpreted exactly. Rounding the stored Arb balls to the displayed decimals was performed outwards.

raw moment	c	r
H_{00}	1.279007247846485140480	$4.67 \cdot 10^{-22}$
H_{01}	0.6395036239232425702398	$3.33 \cdot 10^{-23}$
H_{02}	-12.73579749060893715027	$2.81 \cdot 10^{-21}$
H_{11}	13.37530111453217972051	$2.57 \cdot 10^{-21}$
H_{12}	6.687650557266089860256	$2.85 \cdot 10^{-22}$
H_{22}	389.8820543216399270934	$1.65 \cdot 10^{-20}$

Symmetry supplies the other entries of this original three-by-three moment block. The six original monic squared norms are enclosed by

k	c for h_k	r
0	1.279007247846485140480	$4.67 \cdot 10^{-22}$
1	13.05554930257055843539	$2.69 \cdot 10^{-21}$
2	250.0089768190499145175	$4.88 \cdot 10^{-20}$
3	6694.666685525627506773	$4.76 \cdot 10^{-19}$
4	224298.1953314520556463	$9.95 \cdot 10^{-18}$
5	9383436.471794727053469	$2.54 \cdot 10^{-16}$

The analytic double-tail bound is less than $2.961369 \cdot 10^{-63}$ for every entry of the six-by-six matrix, and is less than $1.150030 \cdot 10^{-83}$ for H_{00} . These bounds account for omitted source terms; the displayed intervals additionally include finite-term arithmetic and decimal conversion errors.

The checker also evaluates (AM.15) and the independent exact finite-boundary identity (AM.16), and detects removal of either the reflected half-domain, the theta factor 2, or the adjoint constant 1. Its checks execute identically under ordinary Python and Python's optimization flag. Finite interval certificates do not assert an infinite-degree weight estimate; (AM.8)–(AM.14) give the exact arithmetic input and its complete error control for calculating further degrees.

8 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

8.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$\begin{aligned} R_{\text{ref}} u &= S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = q R_{\text{ref}}, \\ JR_{\text{ref}} &= 1_E, \quad J\Theta = 0, \quad DR_{\text{ref}} - R_{\text{ref}} A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}] R_Z(u). \end{aligned} \quad (\text{TC1})$$

Here $S_\rho^{\mathcal{B}} F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^* R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x \overline{F(x)} H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^* G_R + G_R A - G_R = -(R^* \Theta K_R + (\Theta K_R)^* R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

8.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2} F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\operatorname{Im} z| < \pi/4$. On every closed narrower strip, $\operatorname{Re}(e^{2z}) \geq c_a e^{2\operatorname{Re} z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\operatorname{Re} z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1} f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2} f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\operatorname{Im} z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

8.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\operatorname{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \operatorname{rank} \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m + 1$ has a factorization $\Delta = T^*T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m + 1$ rows. Taking $C = B_m + H_m^{-1/2}T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^*R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d - 1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^*R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\text{Re } s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h \iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho \psi(x) = x^{-\rho} \int_x^\infty \psi(y) y^{\rho-1} dy = - \int_0^1 \psi(tx) t^{\rho-1} dt.$$

The second expression and $0 < \text{Re } \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\text{Re } \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho \psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho \psi = \int \psi = 0$, so $S_\rho \psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h \mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$,

where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

8.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^*W_A(G)u|}{u^*Gu}, \quad \eta_Z = \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2 \operatorname{Re} \rho - 1)u^*Gu. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA}, \\ G_\epsilon &= \int_{-\infty}^{\infty} e^{-\epsilon|t|}F(t)dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^{\infty} \operatorname{sign}(t)e^{-\epsilon|t|}F(t)dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i!j!} \cdot (N_\rho^i P_\rho)^* G_0(N_\sigma^j P_\sigma). \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)) = \eta_Z. \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA}u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA}u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

8.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \cdots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \operatorname{Spec} A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\inf_{G>0} \sup_{v \neq 0} \frac{|v^* W_{A,n}(G) v|}{v^* G v} = n\eta_Z. \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

8.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \text{rank } W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\text{Tr}(G_m^{-1} W_m) = 2 \text{Re Tr } A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \text{Re } \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\text{Re } \rho - \tfrac{1}{2}|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal entries $\delta_j = 2 \text{Re } \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \text{Tr}(PH) \leq \text{Tr}(PH_+) \leq \text{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\operatorname{Im} c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\operatorname{Re} c_m = 0$. If $a_{m+1} = (1 - \Pi_m)f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\operatorname{Im} c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

8.7 Original supported maps and the precise result delivered

The passage from a finite source correction C to its representative, Gram, and boundary is the explicit chain of maps

$$C \mapsto R_{\text{ref}} + \Theta \Phi_L C \mapsto (qR, JR, R^*R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

For every active support label λ , the linear maps lift as $(\lambda, u) \mapsto (\lambda, Ru)$ and $(\lambda, u) \mapsto (\lambda, K_R u)$. The quotient of the boundary is $(\lambda, \Theta K_R u) \mapsto (\lambda, 0_Q)$; external absence remains the external zero τ . For the two-leg differential $d(\phi, \psi) = \Theta(\phi - \widehat{\psi})$, the two primitives are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$. Their difference $(K_R u, \widehat{K_R u})$ is the retained degree-zero cycle. Thus all source changes in the optimization have actual typed cochain lifts with the same support synchronization as the input programme.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

9 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

9.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2} F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

9.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j(Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 9.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad \mathcal{M} R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$\boxed{G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon.} \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $q R_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

9.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 9.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 9.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1)\mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_hq_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n)v_n^* + v_n(Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

9.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^*(DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x\bar{F}H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon.} \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\boxed{\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.} \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \bar{\gamma}_N \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_{\pm} = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

9.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \operatorname{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The quotient map $\mathcal{B}/\Theta W_m \rightarrow \mathcal{B}/\Theta W_{m+1}$ has kernel $\Theta W_{m+1}/\Theta W_m$; injectivity of Θ identifies this kernel with the original one-dimensional space W_{m+1}/W_m . The scalar e sends a class to the zero in its same support fibre, while τ sends it to external absence. None of (TB.24)–(TB.26) changes those maps.

9.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \frac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n.} \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n(\tfrac{1}{2}I - A) - \delta_{n0}\ell_Z.} \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1}).} \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 9.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m.} \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^m (sQ_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1} d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed

primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D)f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

9.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section 22. The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \rightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \rightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

10 The full jet kernel, finite resolvents, and the actual analytic tail

Retain the arithmetic measure, coordinates and maps of Section 9:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = |g(s)/h(s)|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

In particular, every selected zero has its complete multiplicity in h , and c is the coordinate column of the unit in the retained monomial basis of E . Write q_k for the monic orthogonal polynomials of ν_Z , $\kappa_k = \|q_k\|_{\nu_Z}^2$, and $a_k = \kappa_k/\kappa_{k-1} > 0$ for $k \geq 1$. The exact recurrence is

$$sq_k = q_{k+1} + b_k q_k - a_k q_{k-1}, \quad b_k + \bar{b}_k = 1, \quad (\text{KR.1})$$

where the last term is omitted at $k = 0$.

10.1 A finite compression in the original Mellin coordinate

Fix $n \geq d$, so $N = n - 1$ is the degree bound in the preceding kernel construction. On the coefficient space with basis $q_0, \dots, q_{n-1}$, put

$$H_n = \text{diag}(\kappa_0, \dots, \kappa_{n-1}), \quad (J_n)_{k,k} = b_k, \quad (J_n)_{k+1,k} = 1, \quad (J_n)_{k,k+1} = -a_{k+1}. \quad (\text{KR.2})$$

All unspecified entries are zero, and indices start at zero. Multiplication of the displayed matrices proves

$$J_n^* H_n + H_n J_n = H_n. \quad (\text{KR.3})$$

The isometry from this coefficient space to its polynomial image is $u \mapsto \sum_k q_k u_k$. For a standard coefficient Hermitian form, the corresponding isometry is

$$\mathcal{I}_n u = \sum_{k=0}^{n-1} \frac{q_k}{\sqrt{\kappa_k}} u_k, \quad B_n = H_n^{1/2} J_n H_n^{-1/2}. \quad (\text{KR.4})$$

This explicitly stated auxiliary coordinate map leaves A , the packet coordinates, q_k , and κ_k unchanged. The matrix B_n has lower off-diagonal entry $\sqrt{a_{k+1}}$, upper off-diagonal entry $-\sqrt{a_{k+1}}$, and satisfies $B_n^* + B_n = I$.

Lemma 10.1 (Nodes and endpoint resolvents). *The characteristic polynomial of J_n , and hence of B_n , is $q_n(z)$. Its roots $\lambda_{n,1}, \dots, \lambda_{n,n}$ are distinct and lie on $\text{Re } z = 1/2$. There are positive numbers $\omega_{n,j}$ and $\sigma_{n,j}$ such that*

$$\begin{aligned} m_n(z) &= \kappa_0 e_0^* (zI - B_n)^{-1} e_0 = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \omega_{n,j} &= \kappa_0, \\ F_n(z) &= e_{n-1}^* (zI - B_n)^{-1} e_{n-1} = \frac{q_{n-1}(z)}{q_n(z)} = \sum_{j=1}^n \frac{\sigma_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \sigma_{n,j} &= 1. \end{aligned} \quad (\text{KR.5})$$

Here m_n and F_n are respectively the initial and terminal observations of the same finite compression.

Proof. Expanding the tridiagonal determinant along its last row gives $D_{k+1}(z) = (z - b_k)D_k(z) + a_k D_{k-1}(z)$, with $D_0 = 1$, $D_1 = z - b_0$. This is (KR.1), proving $D_n = q_n$. The matrix $(B_n - \frac{1}{2}I)/i$ is Hermitian. If an eigenvector has first coordinate zero, its first row equation forces its second coordinate to be zero; recurrence through the rows forces every coordinate to vanish. Thus each eigenspace has dimension at most one. Hermitian diagonalization gives a complete eigenbasis, hence all eigenvalues are simple. The same argument beginning at the last row shows every eigenvector has nonzero last coordinate. If P_j are its rank-one orthogonal eigenprojections, set $\omega_{n,j} = \kappa_0 e_0^* P_j e_0 > 0$ and $\sigma_{n,j} = e_{n-1}^* P_j e_{n-1} > 0$. Their sums and the two spectral expansions follow from $\sum_j P_j = I$. The last diagonal cofactor of $zI - B_n$ is $q_{n-1}(z)$, proving the ratio in (KR.5). $\square$

Define $\delta(z) = |\operatorname{Re} z - 1/2|$. For $\delta(z) > 0$, (KR.3)–(KR.4) give $\|(zI - B_n)^{-1}\| \leq \delta(z)^{-1}$: for a vector u , the real part of $u^*(zI - B_n)u$ is $(\operatorname{Re} z - 1/2)\|u\|^2$, and Cauchy–Schwarz then gives the lower singular-value bound. In particular $|F_n(z)| \leq \delta(z)^{-1}$, and

$$\operatorname{Re} F_n(z) = (\operatorname{Re} z - \tfrac{1}{2}) \sum_j \frac{\sigma_{n,j}}{|z - \lambda_{n,j}|^2}. \quad (\text{KR.6})$$

10.2 Exact quadrature and the full confluent kernel

Theorem 10.2 (A polynomial formula valid at every packet centre). *Put*

$$L_{n,j}(z) = \frac{q_n(z)}{(z - \lambda_{n,j})q'_n(\lambda_{n,j})}. \quad (\text{KR.7})$$

This denotes the polynomial obtained by cancelling the displayed linear factor. For every polynomial P of degree at most $2n - 1$,

$$\int P(s) d\nu_Z = \sum_j \omega_{n,j} P(\lambda_{n,j}). \quad (\text{KR.8})$$

Consequently the complete arithmetic kernel is

$$K_{n-1} = \sum_{j=1}^n \frac{L_{n,j}(A) c c^* L_{n,j}(A)^*}{\omega_{n,j}}. \quad (\text{KR.9})$$

Formula (KR.9) includes all Taylor derivatives at all selected zeros, even when a selected critical zero equals a quadrature node.

Proof. For degree below n , the vector $P(J_n)e_0$ records exactly the coefficients of P in the monic orthogonal basis: in the successive multiplications no term of degree n occurs. Its initial coefficient is $\kappa_0^{-1} \int P d\nu_Z$. Thus the spectral expansion in (KR.5) proves (KR.8) in that degree range. For a general P in the asserted range, polynomial division gives $P = q_n U + R$, where $\deg U \leq n - 1$ and $\deg R < n$. On the line, the conjugate of $U(s)$ is again a polynomial in s , namely $\overline{U(1 - \bar{s})}$. Orthogonality therefore gives $\int q_n U d\nu_Z = 0$. The same multiple vanishes at all the nodes, proving (KR.8).

The polynomials $L_{n,j}$ form the interpolation basis of $\mathcal{P}_{n-1}$. Apply (KR.8) to $\overline{L_{n,j}(s)} L_{n,k}(s)$, interpreted as a polynomial on the line; its degree is at most $2n - 2$. Their Gram matrix is $\operatorname{diag}(\omega_{n,j})$. In this basis the residue map has column $j_h L_{n,j} = L_{n,j}(A)c$. The definition $K = CH^{-1}C^*$ now proves (KR.9). This is a basis change in the polynomial source only. Polynomial evaluation at A retains its nilpotent powers, so the formula gives the complete jets without any distinct-centre or simple-zero assumption. $\square$

To identify the quadrature endpoints exactly, one has

$$\sigma_{n,j} = \frac{\omega_{n,j} |q_{n-1}(\lambda_{n,j})|^2}{\kappa_{n-1}} = \frac{\kappa_{n-1}}{\omega_{n,j} |q'_n(\lambda_{n,j})|^2}. \quad (\text{KR.10})$$

Indeed an eigenvector of B_n at λ has coordinates proportional to $\overline{q_k(\lambda)}/\sqrt{\kappa_k}$: substitution into the recurrence proves this assertion, including the endpoint row. The squared ratio of its last and first coordinates proves the first equality. The preceding polynomial kernel formula gives $\mathcal{K}_{n-1}(\lambda_{n,j}, \lambda_{n,j}) = 1/\omega_{n,j}$. In the original Christoffel–Darboux polynomial identity, set $w = \lambda_{n,j}$, divide its common polynomial factor $z - \lambda_{n,j}$, then evaluate at that point. This gives

$$\frac{1}{\omega_{n,j}} = \frac{q'_n(\lambda_{n,j}) \overline{q_{n-1}(\lambda_{n,j})}}{\kappa_{n-1}}.$$

Its squared modulus gives the second equality in (KR.10).

Theorem 10.3 (The exact resolvent metric for the packet). *Suppose, for this formula, that no selected centre equals one of the specified n quadrature nodes. Define maps on the original space E by*

$$Q_n = q_n(A), \quad X_{n,j} = (A - \lambda_{n,j}I)^{-1}c, \quad S_n = \sum_j \sigma_{n,j} X_{n,j} X_{n,j}^*, \quad r_n = F_n(A)c. \quad (\text{KR.11})$$

Then Q_n is invertible, $S_n > 0$, and

$$\boxed{K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*}. \quad (\text{KR.12})$$

In particular, the original relative-control quantity at $m = n - 1 - d$ is exactly

$$\epsilon_m = \sup_{y \neq 0} \frac{|y^*(cr_n^* + r_n c^*)y|}{y^* S_n y}. \quad (\text{KR.13})$$

The identity is valid whenever that index belongs to the original source sequence; $m = -1$ is its already specified fixed-section extension. Every off-line packet obeys the nonintersection condition for every n . At an intersecting critical centre, (KR.9) gives the exact polynomial continuation, with no deletion of that centre or its jets.

Proof. The finite algebra E has an inverse for a polynomial precisely when the polynomial is nonzero at every centre: the Euclidean algorithm gives an inverse modulo h , or, equivalently, the finite Taylor inverse at each centre gives the inverse under the complete Chinese-remainder map. This proves the asserted invertibilities. Substitution of (KR.7) and (KR.10) into (KR.9) proves the first identity in (KR.12). It proves $S_n > 0$, since the residue map is onto at $n \geq d$, hence $K_{n-1} > 0$. The identity $AX_{n,j} = \lambda_{n,j} X_{n,j} + c$, together with $\lambda_{n,j} + \bar{\lambda}_{n,j} = 1$, proves

$$AS_n + S_n A^* - S_n = c \left(\sum_j \sigma_{n,j} X_{n,j} \right)^* + \left(\sum_j \sigma_{n,j} X_{n,j} \right) c^*.$$

The partial fractions in (KR.5), evaluated in the finite algebra, identify the sum with $F_n(A)c$. Because Q_n commutes with A , both kernel and its weight form are congruent, with the same factor κ_{n-1}^{-1} , to S_n and its displayed weight form. The invertible change $y = Q_n^* x$ proves equality of their Rayleigh suprema. The previous exact map from K_{n-1} to G_m then proves (KR.13). All of these maps are stated in the original quotient coordinates; the Taylor expansion of each rational factor includes every power through $m_\rho - 1$. $\square$

The scalar form of (KR.13), with every inverse taken on the same retained packet coordinates, is

$$\alpha_n = r_n^* S_n^{-1} r_n, \quad \beta_n = c^* S_n^{-1} c, \quad \gamma_n = c^* S_n^{-1} r_n,$$

$$\boxed{\epsilon_m = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}}. \quad (\text{KR.13a})$$

Indeed the nonzero eigenvalues of $S_n^{-1/2}(cr_n^* + r_n c^*)S_n^{-1/2}$ coincide, with multiplicities, with the nonzero eigenvalues of $\begin{pmatrix} \gamma_n & \beta_n \\ \alpha_n & \bar{\gamma}_n \end{pmatrix}$: factor the former through its two displayed vectors and interchange the two rectangular factors. Its characteristic polynomial gives $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Cauchy-Schwarz gives $|\gamma_n|^2 \leq \alpha_n \beta_n$, so their greatest absolute value is (KR.13a), including zero or one-dimensional rank. The full congruence (KR.12) specifies exactly how the original q_n , κ_{n-1} and local unit enter the metric whose relative form is evaluated here.

10.3 The infinite multiplication operator and its genuine tail

Let $\mathcal{L} = L^2(\nu_Z)$, and let M be multiplication by s , with domain $\{f : sf \in \mathcal{L}\}$. It is a closed operator and $M^* = I - M$. These statements follow directly from the multiplication formula; its resolvent off the line is multiplication by $(z - s)^{-1}$, of norm at most $\delta(z)^{-1}$. The auxiliary vectors $\psi_k = q_k / \sqrt{\kappa_k}$ form a complete orthonormal basis. Completeness follows from the positive exponential moment of the actual arithmetic measure by the Fourier-annihilator proof already given for the theta measures.

Let P_n project onto $\psi_0, \dots, \psi_{n-1}$, and put $Q_n^{\text{tail}} = I - P_n$; this projection is distinct from the polynomial map $Q_n = q_n(A)$ in (KR.11). Define

$$M_n^{\text{tail}} = Q_n^{\text{tail}} M Q_n^{\text{tail}} \quad \text{on } Q_n^{\text{tail}} \text{Dom } M, \quad \zeta_n(z) = \langle \psi_n, (z - M_n^{\text{tail}})^{-1} \psi_n \rangle. \quad (\text{KR.14})$$

This tail resolvent exists at every off-line z , with norm at most $\delta(z)^{-1}$. Here is the domain justification. Every polynomial belongs to $\text{Dom } M$, so P_n preserves that domain and $Q_n^{\text{tail}} \text{Dom } M = \text{Dom } M \cap Q_n^{\text{tail}} \mathcal{L}$. The operator $(M - \frac{1}{2}I)/i$ is self-adjoint multiplication by t . Its off-diagonal blocks for $P_n \oplus Q_n^{\text{tail}}$ extend to a bounded, finite-rank self-adjoint operator, by (KR.1). Subtracting this operator preserves self-adjointness: for a self-adjoint T and bounded self-adjoint C , the definition of the adjoint gives $(T - C)^* = T^* - C$ on exactly $\text{Dom } T^*$. The resulting diagonal operator reduces the two projections. Its tail block is therefore self-adjoint on the displayed tail domain. Multiplication by i and addition of $1/2$ gives the claimed tail resolvent.

Theorem 10.4 (Exact finite-section, terminal, and tail comparison). *For $\delta(z) > 0$, let*

$$m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t).$$

Then

$$H_n^{\text{obs}}(z) = \frac{\zeta_n(z)}{1 + a_n F_n(z) \zeta_n(z)}, \quad (\text{KR.15})$$

$$m(z) - m_n(z) = \frac{(-1)^n \kappa_n \zeta_n(z)}{q_n(z)^2 [1 + a_n F_n(z) \zeta_n(z)]} = \frac{(-1)^n}{q_n(z)^2} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t). \quad (\text{KR.16})$$

Every denominator in these formulas is nonzero off the original Mellin line. The terminal finite observations additionally obey the exact Schur recurrence

$$F_1(z) = \frac{1}{z - b_0}, \quad F_{n+1}(z) = \frac{1}{z - b_n + a_n F_n(z)}. \quad (\text{KR.17})$$

Proof. Under $\mathcal{I}_n \oplus I$, the block operator M is

$$\begin{pmatrix} B_n & -\sqrt{a_n} e_{n-1} \langle \psi_n, \cdot \rangle \\ \sqrt{a_n} \psi_n e_{n-1}^* & M_n^{\text{tail}} \end{pmatrix}. \quad (\text{KR.18})$$

The signs follow directly from (KR.1); both coupling maps are bounded on the displayed summands. Eliminate the tail component in $z - M$. The finite Schur complement is $zI - B_n + a_n \zeta_n(z) e_{n-1} e_{n-1}^*$. For $\text{Re } z > 1/2$, the real part of its quadratic form is at least $\delta(z) \|u\|^2$, because $\text{Re } \zeta_n(z) > 0$; on the other side all signs reverse. The sign of this scalar real part follows either from the self-adjoint tail resolvent or from the resolvent identity $\text{Re} \langle v, (z - M_n^{\text{tail}})^{-1} v \rangle = (\text{Re } z - 1/2) \|(z - M_n^{\text{tail}})^{-1} v\|^2$. Thus the Schur complement is invertible and $1 + a_n \zeta_n F_n \neq 0$. Direct multiplication verifies the rank-one inverse formula

$$R_n - \frac{a_n \zeta_n}{1 + a_n \zeta_n F_n} R_n e_{n-1} e_{n-1}^* R_n, \quad R_n = (zI - B_n)^{-1}. \quad (\text{KR.19})$$

The analogous bottom Schur calculation, paired against ψ_n , gives (KR.15), since $\langle \psi_n, (z - M)^{-1} \psi_n \rangle = H_n^{\text{obs}}(z)$.

The two endpoint cofactors of $zI - B_n$ give

$$e_0^* R_n e_{n-1} = \frac{(-1)^{n-1} \sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}, \quad e_{n-1}^* R_n e_0 = \frac{\sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}. \quad (\text{KR.20})$$

These cofactors are the products of the successive upper and lower entries, respectively; the formula also covers $n = 1$. Pair (KR.19) against $\sqrt{\kappa_0} e_0$, the coefficient vector of the unchanged function 1. Substitute (KR.20) and $a_n \kappa_{n-1} = \kappa_n$. This proves the first equality in (KR.16); (KR.15) proves the second. Finally (KR.17) is (KR.1) divided by q_{n+1} , equivalently the last-coordinate finite Schur complement. $\square$

There is also a direct polynomial check on every factor in (KR.16). The polynomial

$$p_n(z) = \int \frac{q_n(z) - q_n(s)}{z - s} d\nu_Z(t)$$

has degree below n . At a quadrature node its ratio residue is $p_n(\lambda_{n,j})/q'_n(\lambda_{n,j}) = \int L_{n,j} d\nu_Z = \omega_{n,j}$, so $m_n = p_n/q_n$. Orthogonality to the polynomial $(q_n(z) - q_n(s))/(z - s)$ gives

$$m - m_n = \frac{1}{q_n(z)^2} \int \frac{q_n(s)^2}{z - s} d\nu_Z.$$

The polynomial $(-1)^n \overline{q_n(1 - \bar{s})}$ is monic and has the same orthogonality conditions as q_n ; uniqueness makes it equal to q_n . On the line this says $q_n(s)^2 = (-1)^n |q_n(s)|^2$, giving (KR.16) again. In particular the explicit error inequality is

$$|m(z) - m_n(z)| \leq \frac{\kappa_n}{\delta(z) |q_n(z)|^2}. \quad (\text{KR.21})$$

The initial and terminal observations have now been related by the actual block maps (KR.18)–(KR.20), the tail (KR.14), and the exact remainder (KR.16). The relative form uses the terminal observation through (KR.11)–(KR.13). An estimate for the initial observation can be propagated through these equalities while retaining the factors q_n , κ_n , a_n , and the tail denominator. None of those arithmetic quantities is discarded in passing from one observation to the other.

11 Quantitative convergence of the arithmetic Gaussian resolvents

Retain the actual measure and every coordinate of Section 10:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

The polynomial h contains the complete multiplicities of its selected actual zeros. The quotient g/h is therefore entire, including at the selected critical zeros. Use the initial Gaussian observation of (KR.5) and (KR.8),

$$m_n(z) = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \quad m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad \mathcal{E}_n(z) = m(z) - m_n(z), \quad \delta(z) = |\operatorname{Re} z - \tfrac{1}{2}|. \quad (\text{GC.1})$$

The Gaussian weights are positive and the distinct nodes lie on the original Mellin line, as proved in Section 10. Its exact quadrature (KR.8) and exact error (KR.16) are the inputs below. No asymptotic for q_n or κ_n is assumed.

For each fixed $0 < a < \pi/2$, the number

$$E_a = \max \left\{ \int e^{at} d\nu_Z(t), \int e^{-at} d\nu_Z(t) \right\} \quad (\text{GC.2})$$

is finite. Here is the connection to the proved arithmetic decay. The actual theta calculation in Section 21 gives $|g(1/2 + it)| \leq C_b e^{-b|t|}$ for every $0 < b < \pi/4$. Choose $a/2 < b < \pi/4$. Outside a sufficiently large compact interval, the monic polynomial h of degree d satisfies $|h(1/2 + it)| \geq c_h (1 + |t|)^d$. Thus the density multiplied by $e^{a|t|}$ is bounded there by a constant times $(1 + |t|)^{-2d} e^{-(2b-a)|t|}$. On the compact interval the entire quotient is bounded. Both parts are integrable, proving (GC.2) with the original factor $dt/(2\pi)$ retained.

11.1 The Gaussian rule inherits the original exponential moments

Lemma 11.1 (Exponential moment domination). *For every real v with $|v| \leq a$, write $\lambda_{n,j} = 1/2 + it_{n,j}$. Then*

$$\sum_j \omega_{n,j} e^{vt_{n,j}} \leq \int e^{vt} d\nu_Z(t) \leq E_a. \quad (\text{GC.3})$$

Proof. Let H be the real Hermite interpolation polynomial of degree at most $2n - 1$ whose values and first derivatives at the distinct real nodes $t_{n,j}$ equal those of $f(t) = e^{vt}$. Its existence and uniqueness follow from the invertibility of the interpolation map: a polynomial in its kernel has a double zero at each of the n nodes, hence is zero in degree below $2n$; domain and target both have dimension $2n$. For a real t outside the nodes the interpolation remainder is

$$f(t) - H(t) = \frac{f^{(2n)}(\xi)}{(2n)!} \prod_j (t - t_{n,j})^2 \geq 0 \quad (\text{GC.4})$$

for some real ξ between t and the nodes. To prove the remainder, subtract $C \prod_j (x - t_{n,j})^2$ from $f(x) - H(x)$, choosing C to produce an additional zero at $x = t$. The resulting function has the n double zeros and the additional zero. Repeated Rolle's theorem gives a zero of its $2n$ -th derivative; solving at that zero gives (GC.4). The last sign follows from $f^{(2n)}(x) = v^{2n} e^{vx} \geq 0$. At a node the remainder is zero.

Integrate the pointwise inequality. The exponential and every polynomial are integrable by (GC.2). Since H is a polynomial in $t = (s - 1/2)/i$ of degree at most $2n - 1$, (KR.8) yields

$$\int H(t) d\nu_Z(t) = \sum_j \omega_{n,j} H(t_{n,j}) = \sum_j \omega_{n,j} e^{vt_{n,j}}.$$

This proves the first inequality in (GC.3). Pointwise convexity of e^{vt} in $v \in [-a, a]$, followed by integration, proves the second. No bound on the largest quadrature node has been inserted. $\square$

11.2 Exact moments give a quantitative analytic-strip estimate

Define

$$\Phi(u) = \int e^{iut} d\nu_Z(t), \quad \Phi_n(u) = \sum_j \omega_{n,j} e^{iut_{n,j}}, \quad \Delta_n(u) = \Phi(u) - \Phi_n(u). \quad (\text{GC.5})$$

These functions are holomorphic on $|\operatorname{Im} u| < a$. Indeed, on each smaller closed strip, every differentiated integrand is dominated by a constant times $e^{at} + e^{-at}$, because $|t|^k e^{b|t|} \leq C_{k,b,a} (e^{at} + e^{-at})$ when $b < a$. Dominated convergence proves all the derivatives of Φ ; the finite sum is entire. Equation (GC.3), applied at $v = -\operatorname{Im} u$, gives

$$|\Delta_n(u)| \leq 2E_a \quad (|\operatorname{Im} u| < a), \quad \Delta_n^{(k)}(0) = 0 \quad (0 \leq k < 2n). \quad (\text{GC.6})$$

The derivative identities are exactly the moments in (KR.8).

Lemma 11.2 (The complete strip bound). *For $|\operatorname{Im} u| < a$,*

$$|\Delta_n(u)| \leq 2E_a \left| \tanh\left(\frac{\pi u}{4a}\right) \right|^{2n}. \quad (\text{GC.7})$$

Proof. The explicit holomorphic coordinate maps

$$w = \tanh\left(\frac{\pi u}{4a}\right), \quad u = \frac{2a}{\pi} \operatorname{Log} \frac{1+w}{1-w}$$

are inverse bijections between the strip and the unit disk. For $|w| < 1$, the fraction lies in the right half-plane, so its principal logarithm has imaginary part in $(-\pi/2, \pi/2)$, which proves the asserted strip range. Compose $\Delta_n/(2E_a)$ with this inverse. By (GC.6), the resulting disk function is bounded by one and has a zero of order at least $2n$ at zero. Divide by w^{2n} , filling in its removable value at zero. On each circle $|w| = r < 1$, its modulus is at most r^{-2n} ; the maximum principle gives this bound inside the circle. Letting r increase to one proves the required bound by $|w|^{2n}$, hence (GC.7). $\square$

Theorem 11.3 (Direct off-line convergence with an explicit rate). *For every $0 < a < \pi/2$, every $n \geq 1$, and every z off the original Mellin line,*

$$|m(z) - m_n(z)| \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\delta(z)}{\pi}\right) (4n)^{-2a\delta(z)/\pi}. \quad (\text{GC.8})$$

On a compact set K disjoint from that line the same bound holds uniformly after replacing $\delta(z)$ everywhere by $\delta_K = \min_{z \in K} \delta(z) > 0$.

Proof. Put $w = z - 1/2$. On $\operatorname{Re} w > 0$, the elementary Laplace identity and Fubini's theorem give

$$\mathcal{E}_n(z) = \int_0^\infty e^{-wu} \Delta_n(u) du. \quad (\text{GC.9})$$

For example the absolute integral for the original measure before subtraction is at most $\nu_Z(\mathbb{R})/\operatorname{Re} w$, so the interchange is justified. On $\operatorname{Re} w < 0$, the exact orientation is

$$\mathcal{E}_n(z) = - \int_0^\infty e^{wu} \Delta_n(-u) du. \quad (\text{GC.10})$$

These follow respectively from integrating $e^{-(w-it)u}$ and $-e^{(w-it)u}$. Set $b = \pi/(4a)$. Equation (GC.7) gives

$$|\mathcal{E}_n(z)| \leq 2E_a I_n(\delta(z)), \quad I_n(\delta) = \int_0^\infty e^{-\delta u} \tanh(bu)^{2n} du. \quad (\text{GC.11})$$

For $0 < x < 1$, direct integration gives

$$\log \frac{1-x}{1+x} = -2 \int_0^x \frac{dv}{1-v^2} \leq -2x.$$

Taking $x = e^{-2bu}$ yields $\tanh(bu)^{2n} \leq e^{-4ne^{-2bu}}$. With $\beta = \delta/(2b) = 2a\delta/\pi$, the same variable change gives

$$\begin{aligned} I_n(\delta) &\leq \frac{1}{2b} \int_0^1 x^{\beta-1} e^{-4nx} dx \\ &\leq \frac{1}{2b} (4n)^{-\beta} \Gamma(\beta), \quad \Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy. \end{aligned} \quad (\text{GC.12})$$

Substitution in (GC.11) proves (GC.8). On the compact set one uses $e^{-\delta(z)u} \leq e^{-\delta_K u}$ in (GC.11) and then (GC.12) at δ_K ; no monotonicity of the gamma factor is assumed. This proves locally uniform convergence on both components of the off-line domain. $\square$

11.3 Every confluent derivative and the original finite algebra

Write $F^{[k]} = F^{(k)}/k!$. The exact error (KR.16), with its vertical-line sign, gives for all $k \geq 0$

$$\mathcal{E}_n^{[k]}(z) = (-1)^n \sum_{j=0}^k (q_n^{-2})^{[k-j]}(z) (-1)^j \int \frac{|q_n(s)|^2}{(z-s)^{j+1}} d\nu_Z(t). \quad (\text{GC.13})$$

Every derivative under this integral is justified on a compact off-line neighborhood by its positive distance from the line and the finite measure $|q_n|^2 \nu_Z$. The product rule for Taylor coefficients has no binomial factors; those are precisely cancelled by the convention $F^{[k]} = F^{(k)}/k!$. For completeness its polynomial factor is explicitly

$$(q_n^{-2})^{[r]}(z) = \frac{(-1)^r}{q_n(z)^2} \sum_{r_1+\dots+r_n=r} \prod_{j=1}^n \frac{r_j+1}{(z-\lambda_{n,j})^{r_j}}. \quad (\text{GC.14})$$

This follows by multiplying the Taylor series of the n original factors $(z - \lambda_{n,j})^{-2}$, retaining every cross term.

Differentiating (GC.9)–(GC.10) gives the independent estimate

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{2E_a}{k!} \int_0^\infty u^k e^{-\delta(z)u} \tanh(bu)^{2n} du. \quad (\text{GC.15})$$

The integrable majorant $u^k e^{-\delta_K u}$ on a smaller off-line compact neighborhood justifies this differentiation. If $0 < \sigma < \delta$, the exponential series gives $u^k e^{-(\delta-\sigma)u} \leq k!/(\delta-\sigma)^k$. Equations (GC.11)–(GC.12) prove

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{4aE_a}{\pi(\delta-\sigma)^k} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi}, \quad 0 < \sigma < \delta \leq \delta(z). \quad (\text{GC.16})$$

Taking $\delta = \delta_K$ makes this a simultaneous compact-uniform bound on every fixed finite list of derivatives.

Here is the complete algebra transport, including the part of the original packet outside this resolvent domain. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{d_\rho}, \quad H_\rho(s) = \frac{h(s)}{(s - \rho)^{d_\rho}}, \quad U_\rho(s) = \sum_{k=0}^{d_\rho-1} (H_\rho^{-1})^{[k]}(\rho) (s - \rho)^k,$$

and define the actual polynomial remainder

$$e_\rho = j_h(H_\rho U_\rho) \in E. \quad (\text{GC.17})$$

Its remainder is one modulo $(s - \rho)^{d_\rho}$ and zero modulo every other factor of h . The distinct factors are coprime, and a polynomial divisible by each is divisible by their product: successive use of the Bezout identity proves this last statement. It follows directly that

$$e_\rho e_\eta = 0 \ (\rho \neq \eta), \quad e_\rho^2 = e_\rho, \quad \sum_{\rho \in Z} e_\rho = 1. \quad (\text{GC.18})$$

Restriction of a polynomial to its truncated Taylor coefficients therefore gives the exact algebra isomorphism

$$E \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{d_\rho}), \quad j_h P \longmapsto \left(\sum_{k < d_\rho} P^{[k]}(\rho) \epsilon_\rho^k \right)_\rho. \quad (\text{GC.19})$$

Its inverse sends the coordinate ϵ_ρ^k to $e_\rho(s-\rho)^k$; composition is the identity by the just proved congruences. Multiplication is preserved because the Taylor coefficient of a product is the exact convolution of the two coefficient lists, truncated only by the displayed ideal.

Let $Z_{\text{off}} = \{\rho \in Z : \text{Re } \rho \neq 1/2\}$ and $Z_{\text{line}} = Z \setminus Z_{\text{off}}$. Define

$$e_{\text{off}} = \sum_{\rho \in Z_{\text{off}}} e_\rho, \quad e_{\text{line}} = 1 - e_{\text{off}}, \quad E_{\text{off}} = e_{\text{off}}E, \quad E_{\text{line}} = e_{\text{line}}E. \quad (\text{GC.20})$$

Then $E = E_{\text{off}} \oplus E_{\text{line}}$ as algebras with their respective idempotent units. The projection $u \mapsto e_{\text{off}}u$ has exactly kernel E_{line} and commutes with the original multiplication operator A . Under (GC.19), its restriction at every retained centre is $A_\rho = \rho I + N_\rho$, where multiplication by ϵ_ρ gives $N_\rho^{d_\rho} = 0$ and retains all lower powers. This constructs the precise map from the full packet to the off-line resolvent domain, together with its full kernel and the operator on that kernel. An empty off-line set gives the zero summand, with the full original packet in the kernel.

For a function f holomorphic near Z_{off} , define

$$f(A_{\text{off}}) = \sum_{\rho \in Z_{\text{off}}} \sum_{k=0}^{d_\rho-1} f^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \quad (\text{GC.21})$$

Here $e_\rho(A)$ means multiplication by the explicitly constructed remainder (GC.17). Equation (GC.19) proves that (GC.21) is a unital algebra homomorphism on the off-line summand. No diagonalizability of A has entered this construction.

In particular, finite geometric expansion in each nilpotent block proves

$$\begin{aligned} m(A_{\text{off}}) &= \int (A_{\text{off}} - sI)^{-1} d\nu_Z(t), \\ m_n(A_{\text{off}}) &= \sum_j \omega_{n,j} (A_{\text{off}} - \lambda_{n,j}I)^{-1}, \\ \mathcal{E}_n(A_{\text{off}}) &= \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \mathcal{E}_n^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \end{aligned} \quad (\text{GC.22})$$

For the first equality the expansion is

$$(A - sI)^{-1} e_\rho(A) = \sum_{k < d_\rho} \frac{(-1)^k (A - \rho I)^k e_\rho(A)}{(\rho - s)^{k+1}}.$$

Each coefficient is integrable since $\delta(\rho) > 0$, and its integral is exactly $m^{[k]}(\rho)$. This proves the integral identity, rather than assigning a formal meaning to it.

For a concrete bound keep any fixed positive Hermitian form G_* on the original coefficient space and restrict it to E_{off} . When that summand is nonzero, put

$$\delta_* = \min_{\rho \in Z_{\text{off}}} \delta(\rho) > 0, \quad C_{\rho,k} = \left\| (A - \rho I)^k e_\rho(A)|_{E_{\text{off}}} \right\|_{G_*}.$$

The triangle inequality and (GC.16) give, for every $0 < \sigma < \delta_*$,

$$\|m(A_{\text{off}}) - m_n(A_{\text{off}})\|_{G_*} \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi} \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \frac{C_{\rho,k}}{(\delta_* - \sigma)^k}. \quad (\text{GC.23})$$

Thus the full initial resolvent converges in the original finite coefficient space, with every nilpotent contribution controlled. Applying the displayed operators to the retained vector $e_{\text{off}}c$ gives the corresponding vector statement with the additional factor $\|e_{\text{off}}c\|_{G_*}$.

The polynomial kernel (KR.9) continues to act on all of E , including E_{line} and collisions with quadrature nodes. The off-line estimates above concern the initial observation m_n . Its exact map to the terminal observation F_n is the block and tail comparison (KR.14)–(KR.20), with the factors q_n, κ_n, a_n and the tail denominator still present. The estimate (GC.23) may be inserted into those identities; it does not remove their arithmetic factors or the critical-line summand of (GC.20).

12 The exact terminal step and its retained boundary energy

All measures, packet coordinates and maps in this section are the original arithmetic ones. Thus $s = 1/2 + it$, $d\nu_Z = |g(s)/h(s)|^2 dt/(2\pi)$, $E = \mathbb{C}[s]/(h)$, $A = M_s$, and $c = j_h 1$. Every selected zero occurs with its complete multiplicity in h . Retain the monic q_n , their actual norms κ_n , and

$$a_n = \frac{\kappa_n}{\kappa_{n-1}} > 0, \quad q_{n+1}(s) = (s - b_n)q_n(s) + a_n q_{n-1}(s), \quad b_n + \bar{b}_n = 1. \quad (\text{KT.1})$$

No recurrence coefficient is replaced by a limiting value.

12.1 A step written entirely in the original polynomial packet

For $n \geq d = \deg h$, put

$$v_j = q_j(A)c, \quad K = K_{n-1} = \sum_{j=0}^{n-1} \frac{v_j v_j^*}{\kappa_j} > 0, \quad M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}. \quad (\text{KT.2})$$

Use the positive form $\langle x, y \rangle_n = x^* M_n y$, conjugate-linear in its first argument, and define

$$u_n = v_n, \quad w_n = v_{n-1}, \quad \alpha_n = \langle w_n, w_n \rangle_n, \quad \beta_n = \langle u_n, u_n \rangle_n, \quad \gamma_n = \langle u_n, w_n \rangle_n. \quad (\text{KT.3})$$

The arithmetic relative allowance at source level $m = n - d - 1$ will be denoted by $\epsilon^{(n)}$. The fixed-section extension at $m = -1$ has already been specified; the original corrected sequence has $n \geq d + 1$. The exact kernel boundary identity gives

$$\epsilon^{(n)} = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}, \quad \operatorname{Re} \gamma_n = \operatorname{Re} \operatorname{tr} A - \frac{d}{2} =: g_Z. \quad (\text{KT.4})$$

Indeed the kernel weight form is $(u_n w_n^* + w_n u_n^*)/\kappa_{n-1}$. Its possible nonzero generalized eigenvalues are obtained by reversing the two rectangular factors, giving $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Taking the trace after multiplication by K^{-1} gives the second identity. Cauchy–Schwarz proves the nonnegativity of the radicand and justifies the maximum absolute value in (KT.4).

Theorem 12.1 (Universal scalar and metric step). *Write $a = a_n$, and retain the exact next vector*

$$t_n = (A - b_n I)u_n + a_n w_n = v_{n+1}. \quad (\text{KT.5})$$

Then

$$M_{n+1} = \frac{1}{a} \left(M_n - \frac{M_n u_n u_n^* M_n}{a + \beta_n} \right), \quad (\text{KT.6})$$

$$\begin{aligned} \alpha_{n+1} &= \frac{\beta_n}{a + \beta_n}, \\ \gamma_{n+1} &= \frac{\langle t_n, u_n \rangle_n}{a + \beta_n}, \\ \beta_{n+1} &= \frac{1}{a} \left(\langle t_n, t_n \rangle_n - \frac{|\langle u_n, t_n \rangle_n|^2}{a + \beta_n} \right). \end{aligned} \quad (\text{KT.7})$$

Every denominator is strictly positive. These identities hold also when any selected critical centre equals a quadrature node, and when $q_n(A)$ is singular.

The exact determinant changes of the kernel and original theta metric are

$$\boxed{\frac{\det K_n}{\det K_{n-1}} = 1 + \frac{\beta_n}{a_n}, \quad \frac{\det G_{m+1}}{\det G_m} = \frac{a_n}{a_n + \beta_n} = 1 - \alpha_{n+1}.} \quad (\text{KT.8})$$

Proof. The unchanged kernel update is $K_n = K_{n-1} + u_n u_n^* / \kappa_n$. Its inverse is verified by multiplication:

$$K_n^{-1} = K_{n-1}^{-1} - \frac{K_{n-1}^{-1} u_n u_n^* K_{n-1}^{-1}}{\kappa_n + u_n^* K_{n-1}^{-1} u_n}.$$

Insert $\kappa_n = a \kappa_{n-1}$ and the definition of M_n ; this gives (KT.6). At the next step the ordered vectors in (KT.3) are $u_{n+1} = t_n$, $w_{n+1} = u_n$. Pair each ordered pair with (KT.6). The three resulting expressions are precisely (KT.7), including their conjugations. The denominator is positive because $a > 0$, $\beta_n \geq 0$. The determinant of the rank-one update is $1 + u_n^* K_{n-1}^{-1} u_n / \kappa_n = 1 + \beta_n / a$; this follows directly by expanding the determinant multilinearly in its columns. The original metric is $G_m = U_\varepsilon^* K_{n-1}^{-1} U_\varepsilon$, with the same fixed invertible local unit at adjacent levels. Its determinant ratio is therefore the inverse kernel ratio. Equation (KT.7) gives the last equality in (KT.8). All calculations use polynomial residues at A ; no inverse of $q_n(A)$ has been used. $\square$

12.2 The complete boundary-energy increment

Suppose first that $\beta_n > 0$. The exact orthogonal projection for $\langle \cdot, \cdot \rangle_n$ is

$$\begin{aligned} P_n x &= x - u_n \frac{\langle u_n, x \rangle_n}{\beta_n}, & p_n &= P_n w_n, & z_n &= P_n A u_n, \\ \Delta_n &= \alpha_n \beta_n - |\gamma_n|^2 = \beta_n \|p_n\|_n^2, & \Omega_n &= \beta_n \|z_n\|_n^2. \end{aligned} \quad (\text{KT.9})$$

The projection records a specified map inside the original packet; $A u_n$, its removed coefficient, and its remaining vector z_n are all retained in (KT.9).

Theorem 12.2 (Exact transverse energy law). *At every such step,*

$$\boxed{\Delta_{n+1} = \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}}. \quad (\text{KT.10})$$

Consequently

$$\boxed{\Delta_{n+1} - \Delta_n = \frac{\beta_n}{a_n(a_n + \beta_n)} (\|z_n\|_n^2 + 2a_n \operatorname{Re} \langle z_n, p_n \rangle_n - a_n \beta_n \|p_n\|_n^2)}. \quad (\text{KT.11})$$

The exact energy law gives the following two-sided bounds by the sharp triangle inequalities:

$$\frac{|a_n \sqrt{\Delta_n} - \sqrt{\Omega_n}|}{\sqrt{a_n(a_n + \beta_n)}} \leq \sqrt{\Delta_{n+1}} \leq \frac{a_n \sqrt{\Delta_n} + \sqrt{\Omega_n}}{\sqrt{a_n(a_n + \beta_n)}}. \quad (\text{KT.12})$$

The actual relative allowance is recovered exactly by

$$\boxed{\epsilon^{(n)} = |g_Z| + \sqrt{g_Z^2 + \Delta_n}}. \quad (\text{KT.13})$$

For a reflection-stable packet with its multiplicities, $g_Z = 0$, so the energy law calculates the square of the actual allowance without a trace term.

Proof. Let $T = \langle t_n, t_n \rangle_n$ and $v = \langle u_n, t_n \rangle_n$. Substitution of (KT.7) gives

$$\alpha_{n+1}\beta_{n+1} - |\gamma_{n+1}|^2 = \frac{\beta_n T - |v|^2}{a_n(a_n + \beta_n)} = \frac{\beta_n \|P_n t_n\|_n^2}{a_n(a_n + \beta_n)}.$$

Since $P_n u_n = 0$, the exact recurrence (KT.5) gives $P_n t_n = P_n A u_n + a_n P_n w_n = z_n + a_n p_n$. This proves (KT.10). Expand the squared norm and subtract $\Delta_n = \beta_n \|p_n\|_n^2$ to prove (KT.11). The triangle inequality and its reverse, applied to these two specific vectors, prove (KT.12). Finally (KT.4) gives (KT.13). Its real trace term is unchanged with n ; equivalently the same trace computation can be performed at each adjacent positive kernel. Reflection permutes all roots and multiplicities, so $2 \operatorname{Re} \operatorname{tr} A = d$ in that case. $\square$

If $\beta_n = 0$, positivity of M_n forces $u_n = v_n = 0$. Then $\gamma_n = 0$, the current kernel weight is zero, $K_n = K_{n-1}$, and (KT.5)–(KT.7) give the entire exceptional step:

$$t_n = a_n w_n, \quad \alpha_{n+1} = \gamma_{n+1} = 0, \quad \beta_{n+1} = a_n \alpha_n, \quad \Delta_n = \Delta_{n+1} = 0, \quad \epsilon^{(n)} = \epsilon^{(n+1)} = 0. \quad (\text{KT.14})$$

Moreover $v_n = 0$ is exactly the divisibility $h \mid q_n$. Since every root of q_n is simple and lies on the Mellin line, this particular event forces the selected packet to consist of simple line roots. It does not remove the resulting packet or the next vectors in (KT.14).

12.3 The exact rational terminal step and its polynomial continuation

Where $q_n(A)$ and $q_{n+1}(A)$ are invertible, use the actual terminal maps and Gram from Section 10:

$$T_n = F_n(A), \quad r_n = T_n c, \quad Q_n = q_n(A), \quad K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*.$$

The map Q_n is the exact isometry

$$Q_n : (E, S_n^{-1}) \longrightarrow (E, M_n), \quad Q_n c = u_n, \quad Q_n r_n = w_n. \quad (\text{KT.15})$$

Indeed $Q_n^* M_n Q_n = S_n^{-1}$ follows by inverting the displayed Gram congruence, and the vector identity uses $F_n = q_{n-1}/q_n$. This proves the precise relation between the rational and polynomial forms of (KT.3)–(KT.13).

The terminal recurrence gives

$$T_{n+1} = (A - b_n I + a_n T_n)^{-1}, \quad \boxed{S_{n+1} = T_{n+1} (c c^* + a_n S_n) T_{n+1}^*}. \quad (\text{KT.16})$$

For a direct full-jet proof, the scalar kernel of a terminal observation is

$$H_{F_n}(z, w) = \frac{F_n(z) + \overline{F_n(w)}}{z + \overline{w} - 1}.$$

The quotient denotes its removable polynomial/rational continuation where numerator and denominator both vanish. It is identically the positive resolvent Gram $\sum_j \sigma_{n,j} / [(z - \lambda_{n,j})(\overline{w} - \overline{\lambda}_{n,j})]$. Substitution of $F_{n+1} = (z - b_n + a_n F_n)^{-1}$ gives

$$H_{F_{n+1}}(z, w) = F_{n+1}(z) \overline{F_{n+1}(w)} [1 + a_n H_{F_n}(z, w)]. \quad (\text{KT.17})$$

This identity holds first away from its removable sets and hence through them by continuation. Taking the complete residue on both variables proves (KT.16). It also follows from the original kernel update and $Q_{n+1}^{-1} Q_n = T_{n+1}$. Thus a quadrature collision is handled by the polynomial formulas (KT.2)–(KT.14), which retain all coordinates; (KT.15) states exactly the isometry on the open set where the rational presentation is defined.

In this presentation the determinant still retains the entire polynomial factor:

$$\frac{\det S_{n+1}}{\det S_n} = \frac{a_n^{d-1}(a_n + \beta_n)}{|\det(A - b_n I + a_n T_n)|^2}, \quad \det(A - b_n I + a_n T_n) = \prod_{\rho \in Z} \left(\frac{q_{n+1}(\rho)}{q_n(\rho)} \right)^{m_\rho}. \quad (\text{KT.18})$$

The determinant of $a_n S_n + cc^*$ gives the first formula. The second follows by evaluating the polynomial ratio on each full triangular Taylor block $\rho I + N_\rho$. The determinant has the stated multiplicity even when the block is not semisimple.

12.4 The extra jets required to recover the metric from the tail

There is an exact map from the scalar tail identity to the matrix energy, with a specified number of Taylor coefficients. For (KT.19)–(KT.22), take centres at which q_n is nonzero, so F_n is holomorphic there. Every off-line centre meets this condition. At a critical quadrature intersection the universal polynomial formulas (KT.2)–(KT.14) retain the complete packet. The terminal function satisfies $\overline{F_n(1 - \bar{z})} = -F_n(z)$, by its positive spectral sum on the line. Thus

$$H_{F_n}(z, w) = \frac{F_n(z) - F_n(1 - \bar{w})}{z - (1 - \bar{w})}. \quad (\text{KT.19})$$

For centres ρ, σ , and Taylor orders $0 \leq k < m_\rho, 0 \leq l < m_\sigma$, let $S_n^{\text{jet}}(\rho, k; \sigma, l)$ denote the coefficient of $(z - \rho)^k (\bar{w} - \bar{\sigma})^l$ in this kernel. In particular, at a reflection pair,

$$\boxed{S_n^{\text{jet}}(\rho, k; \sigma, l) = (-1)^l F_n^{[k+l+1]}(\rho) \quad \text{if } \rho = 1 - \bar{\sigma}.} \quad (\text{KT.20})$$

Here $F^{[j]} = F^{(j)}/j!$. To prove the formula, put $z = \rho + x, 1 - \bar{w} = \rho - y$ in (KT.19), and expand

$$\frac{x^r - (-y)^r}{x + y} = \sum_{j=0}^{r-1} x^{r-1-j} (-y)^j.$$

The coefficient of $x^k y^l$ is supplied exactly by $r = k + l + 1, j = l$, proving (KT.20) without dividing by zero.

For $D_{\rho\sigma} = \rho + \bar{\sigma} - 1 \neq 0$, the corresponding complete coefficient is

$$\begin{aligned} S_n^{\text{jet}}(\rho, k; \sigma, l) &= \sum_{i=0}^k F_n^{[i]}(\rho) \frac{(-1)^{k-i+l} \binom{k-i+l}{k-i}}{D_{\rho\sigma}^{k-i+l+1}} \\ &\quad + \sum_{j=0}^l \overline{F_n^{[j]}(\sigma)} \frac{(-1)^{k+l-j} \binom{k+l-j}{k}}{D_{\rho\sigma}^{k+l-j+1}}. \end{aligned} \quad (\text{KT.21})$$

This is the full product of the numerator Taylor series and the geometric expansion of $(D_{\rho\sigma} + x + y)^{-1}$. Both constant terms in the numerator are retained. If $V_Z : E \rightarrow \bigoplus_\rho \mathbb{C}^{m_\rho}$ is the fixed complete Taylor-coordinate map, then

$$S_n^{\text{jet}} = V_Z S_n V_Z^*, \quad S_n = V_Z^{-1} S_n^{\text{jet}} (V_Z^{-1})^*. \quad (\text{KT.22})$$

This follows by applying V_Z to each resolvent column in the positive Gram representation of S_n .

To specify the domain of the analytic extraction, let $h_{\text{off}} = \prod_{\text{Re } \rho \neq 1/2} (s - \rho)^{m_\rho}$. The full original algebra projects onto $E_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}})$; the kernel is the retained critical-line CRT summand, with its original action of A . Above the off-line summand use the doubled algebra $\widehat{E}_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}}^2)$. Its exact reduction sequence of $\mathbb{C}[s]$ -modules is

$$0 \longrightarrow E_{\text{off}} \xrightarrow{[p] \mapsto [h_{\text{off}} p]} \widehat{E}_{\text{off}} \xrightarrow{[p] \mapsto [p]} E_{\text{off}} \longrightarrow 0. \quad (\text{KT.23})$$

The first map is well-defined and injective because $h_{\text{off}}^2 \mid h_{\text{off}}p$ holds exactly when $h_{\text{off}} \mid p$; its image is precisely the kernel of the reduction. All maps commute with multiplication by s . The reduction is also a unital algebra map. The image of the injection is the square-zero ideal $(h_{\text{off}})/(h_{\text{off}}^2)$: the product of the images of $[p]$ and $[q]$ is $[h_{\text{off}}^2 pq] = 0$. This records its complete inherited multiplication as well as its module isomorphism with E_{off} . The nilpotent lengths in the middle term are exactly $2m_\rho$, and reduction retains the lower m_ρ coefficients. At reflected actual zeros, the full multiplicities are equal, by the functional equation and conjugation of g . Thus (KT.20) uses at most order $2m_\rho - 1$, exactly a coefficient of this doubled algebra. Formula (KT.21) uses only the stated lower coefficients. Equations (KT.20)–(KT.23) prove the precise passage from terminal Taylor data to the full metric.

Write $\mathcal{E}_n = m - m_n$ for the initial observation error, and retain the genuine tail ζ_n from (KR.14). For every off-line point its real part has the sign of $\text{Re } z - 1/2$, so $\zeta_n \neq 0$. The exact identity (KR.16) also gives

$$\mathcal{E}_n(z) = \frac{(-1)^n \kappa_n}{q_n(z)^2} H_n^{\text{obs}}(z), \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z.$$

The real part of H_n^{obs} has that same strict sign: its defining measure is positive of total mass one. Hence $\mathcal{E}_n(z) \neq 0$ off the line as well. Solving the actual tail equation, before taking any residues, therefore gives the exact extraction

$$F_n(z) = \frac{1}{a_n} \left(\frac{(-1)^n \kappa_n}{q_n(z)^2 \mathcal{E}_n(z)} - \frac{1}{\zeta_n(z)} \right). \quad (\text{KT.24})$$

It holds in the complete doubled algebra at the off-line centres. Every reciprocal there is explicit: if $f(\rho + x) = \sum_{j=0}^{2m_\rho-1} f_j x^j$ modulo x^{2m_ρ} , with $f_0 \neq 0$, the inverse coefficients are

$$b_0 = f_0^{-1}, \quad b_k = -f_0^{-1} \sum_{j=1}^k f_j b_{k-j} \quad (1 \leq k < 2m_\rho). \quad (\text{KT.25})$$

Multiplication of the two finite series proves all coefficients. The already proved bounds (GC.13)–(GC.16) apply to every one of the error coefficients required here. Equations (KT.24)–(KT.25), then (KT.20)–(KT.22), give a complete finite extraction of the terminal metric on E_{off} from the initial error and genuine tail data. In the retained monomial quotient coordinates, set $A_{\text{off}} = M_s$, $\pi_{\text{off}}([p]_h) = [p]_{h_{\text{off}}}$, and $c_{\text{off}} = [1] = \pi_{\text{off}}c$. Explicitly, retain the same ν_Z , q_n , nodes and terminal weights, and put

$$S_n^{\text{off}} = \sum_j \sigma_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1} c_{\text{off}} c_{\text{off}}^* (A_{\text{off}}^* - \bar{\lambda}_{n,j} I)^{-1}. \quad (\text{KT.26})$$

For the original quotient map $\pi_{\text{off}} : E \rightarrow E_{\text{off}}$, the resolvent intertwining identity gives $S_n^{\text{off}} = \pi_{\text{off}} S_n \pi_{\text{off}}^*$ whenever the full rational Gram is defined. The right side of (KT.26) remains defined at every off-line centre even when the full packet has a critical quadrature collision. No replacement of the measure by $|g/h_{\text{off}}|^2 dt / (2\pi)$ occurs in this projection. The full polynomial quadrature formula, applied to these projected residue columns, proves even at such a collision that

$$\pi_{\text{off}} K_{n-1} \pi_{\text{off}}^* = \kappa_{n-1}^{-1} q_n (A_{\text{off}}) S_n^{\text{off}} q_n (A_{\text{off}})^*. \quad (\text{KT.27})$$

These extraction formulas retain the inverse error, the inverse tail, q_n , κ_n , and a_n , including their cancellations. The initial error bound alone is not used to replace any of those terms. The polynomial metric step (KT.6) continues to include the critical-line summand, where the off-line resolvent domain ends.

12.5 Actual quartet packets close the energy in two volume ratios

Assume here that the selected actual packet is closed under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \bar{\rho}$, with the complete multiplicities. These are exact invariances of the specified divisor; no value of $\text{Re } \rho$ has been imposed.

Theorem 12.3 (Two consecutive arithmetic volume contractions). *Let G_{-1} be the fixed-section metric already identified with $U_\varepsilon^* K_{d-1}^{-1} U_\varepsilon$, and define the actual positive determinant ratios*

$$r_m = \frac{\det G_m}{\det G_{m-1}} \quad (m \geq 0).$$

Then $0 < r_m \leq 1$, and for every $m \geq 0$, with $n = d + m + 1$,

$$\epsilon_m^2 = \frac{\kappa_{d+m+1}}{\kappa_{d+m}} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{KT.28})$$

This is the original relative allowance and the original theta Gram sequence, including their local unit and all multiplicities.

Proof. Conjugation invariance of the divisor makes every coefficient of the monic h real. The entire function g obeys $g(\bar{s}) = \overline{g(s)}$, so the unchanged density $|g(1/2 + it)/h(1/2 + it)|^2$ is even in t . For a monic orthogonal polynomial q , form $q^c(s) = \overline{q(\bar{s})}$. Substitution $t \mapsto -t$ and conjugation of each orthogonality equation show that q^c is monic with the same degree and the same lower-degree orthogonality. Positivity of the polynomial Gram proves uniqueness, so $q^c = q$. Thus every q_n has real coefficients in the original variable s . The coefficient of q_n in sq_n is therefore real as well; (KT.1) proves its exact value $b_n = 1/2$. This is a consequence of the given packet symmetries, with every other recurrence coefficient retained.

The original companion matrix A , vector c , residue columns v_j , and Gram matrices K_{n-1}, M_n are real. Therefore the scalar γ_n in (KT.3) is real. The second specified symmetry gives $g_Z = 0$ in (KT.4), so $\gamma_n = 0$ exactly. Hence $\epsilon_m^2 = \alpha_n \beta_n$. At the preceding and following steps (KT.7)–(KT.8) give

$$\alpha_n = 1 - r_m, \quad \alpha_{n+1} = 1 - r_{m+1}, \quad \beta_n = \frac{a_n \alpha_{n+1}}{1 - \alpha_{n+1}}.$$

All these denominators are positive: $G_m > 0$ and $a_n/(a_n + \beta_n) > 0$. Substitution proves (KT.28), including the exceptional zero-column steps. The inequality $0 < r_m \leq 1$ follows directly from the same determinant formula. At $m = 0$ the preceding metric is precisely the stated G_{-1} , so no earlier undefined source level is used. $\square$

12.6 What the recurrence proves about boundary energy

The unchanged arithmetic source has supplied the explicit energy increment (KT.11). Its three terms are the squared transverse A -image, its real pairing with the retained terminal vector, and the specified negative term $-a_n \beta_n \|p_n\|_n^2$. Thus the source recurrence produces an exactly computable boundary energy, including its sign-determining cross term. No sign for their sum has been assumed. Every term is a polynomial residue, an actual moment Gram inverse, or a retained recurrence coefficient. The rational route to those same quantities is (KT.15)–(KT.25). The original theta representative and its boundary are recovered from the preceding kernel-minimum formula and its explicit division by h . Consequently all updates are maps between the original source representatives, with the same quotient class and supported zero. For each support label the displayed linear maps lift by $(\lambda, x) \mapsto (\lambda, Tx)$; external absence remains τ , and a zero vector in that fibre remains its supported zero throughout.

12.7 An exact positive-moment counterexample to consecutive energy monotonicity

This section concerns the moment-to-packet construction for the explicitly specified positive measure below. It proves that positivity, the vertical orthogonal-polynomial recurrence, reflection stability of the packet, and the terminal rank-two identity together allow an increase of the relative weight

energy at consecutive degrees. The measure is specified by its nine atoms; no assertion identifies it with the arithmetic measure $d\nu_Z(t) = |g(\frac{1}{2} + it)/h(\frac{1}{2} + it)|^2 dt/(2\pi)$. The exact correspondence with that construction will be the same polynomial residue map, multiplication operator, kernel sum and Hermitian defect, with the measure retained as part of the data throughout.

12.7.1 The measure and every moment needed below

Keep the original coordinate $s = \frac{1}{2} + it$. Define a probability measure ν on this line by the following table. In a row containing two signs, the displayed weight is assigned to each of the two atoms.

t	weight at each atom	
0	115/288	
$\pm 1/4$	29/120	
$\pm 1/2$	13/240	
$\pm 3/4$	11/2520	
± 1	1/6720	(TEC.1)

All nine weights are strictly positive, and

$$\frac{115}{288} + \frac{29}{60} + \frac{13}{120} + \frac{11}{1260} + \frac{1}{3360} = 1. \quad (\text{TEC.2})$$

Write $\mu_k = \int t^k d\nu$ and $v = 1/16$. The paired atoms give $\mu_k = 0$ for every odd integer $k \geq 1$, including $k = 1, 3, 5, 7, 9$. For positive integers r , the even-moment formula, with all original atom weights retained, is

$$\mu_{2r} = 4^{-2r} \left(\frac{29}{60} 1^{2r} + \frac{13}{120} 2^{2r} + \frac{11}{1260} 3^{2r} + \frac{1}{3360} 4^{2r} \right). \quad (\text{TEC.3})$$

Here the factor 4^{-2r} comes from the explicitly displayed atom locations in (TEC.1). The four contributions and their sums are

r	$\frac{29}{60} 1^{2r}$	$\frac{13}{120} 2^{2r}$	$\frac{11}{1260} 3^{2r}$	$\frac{1}{3360} 4^{2r}$	sum	
1	29/60	13/30	11/140	1/210	1	
2	29/60	26/15	99/140	8/105	3	
3	29/60	104/15	891/140	128/105	15	
4	29/60	416/15	8019/140	2048/105	105	
5	29/60	1664/15	72171/140	32768/105	939	(TEC.4)

Consequently

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8, \mu_{10}) = (1, v, 3v^2, 15v^3, 105v^4, 939v^5). \quad (\text{TEC.5})$$

Equations (TEC.2)–(TEC.5), together with cancellation of each odd moment between the two atoms, verify all moments through degree ten.

12.7.2 The actual monic orthogonal polynomials of this measure

Use the inner product

$$\langle P, Q \rangle_\nu = \int \overline{P(\frac{1}{2} + it)} Q(\frac{1}{2} + it) d\nu(t), \quad x = s - \frac{1}{2}. \quad (\text{TEC.6})$$

The following table retains the monic polynomials in the original s -coordinate and gives their actual squared norms.

k	$q_k(s)$	$\kappa_k = \langle q_k, q_k \rangle_\nu$
0	1	1
1	x	v
2	$x^2 + v$	$2v^2$
3	$x^3 + 3vx$	$6v^3$
4	$x^4 + 6vx^2 + 3v^2$	$24v^4$
5	$x^5 + 10vx^3 + 15v^2x$	$114v^5$

(TEC.7)

Proof. For an explicit verification, define real polynomials

$$\begin{aligned}
 P_0(t) &= 1, & P_1(t) &= t, & P_2(t) &= t^2 - v, \\
 P_3(t) &= t^3 - 3vt, & P_4(t) &= t^4 - 6vt^2 + 3v^2, \\
 P_5(t) &= t^5 - 10vt^3 + 15v^2t.
 \end{aligned}
 \tag{TEC.8}$$

Direct substitution gives $q_k(\frac{1}{2} + it) = i^k P_k(t)$. Thus

$$\langle q_j, q_k \rangle_\nu = (-i)^j i^k \int P_j(t) P_k(t) d\nu(t). \tag{TEC.9}$$

If j, k have different parity, the integrand on the right is odd and its integral is zero. The remaining six off-diagonal integrals, using (TEC.5), are precisely

$$\begin{aligned}
 \int P_0 P_2 d\nu &= v(1 - 1) = 0, \\
 \int P_0 P_4 d\nu &= v^2(3 - 6 + 3) = 0, \\
 \int P_2 P_4 d\nu &= v^3(15 - 21 + 9 - 3) = 0, \\
 \int P_1 P_3 d\nu &= v^2(3 - 3) = 0, \\
 \int P_1 P_5 d\nu &= v^3(15 - 30 + 15) = 0, \\
 \int P_3 P_5 d\nu &= v^4(105 - 195 + 135 - 45) = 0.
 \end{aligned}
 \tag{TEC.10}$$

For the diagonal integrals, direct polynomial multiplication gives

$$\begin{aligned}
 \int P_0^2 d\nu &= 1, & \int P_1^2 d\nu &= v, \\
 \int P_2^2 d\nu &= v^2(3 - 2 + 1) = 2v^2, \\
 \int P_3^2 d\nu &= v^3(15 - 18 + 9) = 6v^3, \\
 \int P_4^2 d\nu &= v^4(105 - 180 + 126 - 36 + 9) = 24v^4, \\
 \int P_5^2 d\nu &= v^5(939 - 2100 + 1950 - 900 + 225) = 114v^5.
 \end{aligned}
 \tag{TEC.11}$$

In (TEC.9), the diagonal phase is $(-i)^k i^k = 1$. These equations therefore prove every entry of the Gram matrix asserted in (TEC.7). Each polynomial in that table is monic. A nonzero polynomial of degree at most five has at most five distinct roots, so it cannot vanish at all nine atoms in (TEC.1). Positivity of the weights makes the inner product positive definite on polynomials of degree at most five.

If two monic polynomials of the same degree were both orthogonal to every lower degree polynomial, their difference would have lower degree and zero squared norm. Positive definiteness forces that difference to vanish. Thus (TEC.7) consists of the actual monic orthogonal polynomials and norms of the stated measure, including the fifth norm. $\square$

Polynomial multiplication in (TEC.7) also proves

$$sq_k = q_{k+1} + \frac{1}{2}q_k - kvq_{k-1} \quad (0 \leq k \leq 4), \quad (\text{TEC.12})$$

where the last term is omitted for $k = 0$. For $1 \leq k \leq 4$, the displayed coefficient is exactly $a_k = \kappa_k/\kappa_{k-1} = kv = k/16 > 0$. Thus (TEC.12) has the required vertical sign and $b_k + \overline{b_k} = 1$, with $b_k = 1/2$. The fifth norm in (TEC.7) is $\kappa_5 = 114/16^5 = 57/524288$; its value was derived from the actual tenth moment $939/16^5$.

12.7.3 The retained packet and its exact evaluation isomorphism

Set

$$h(s) = (s - \frac{3}{4})(s - \frac{1}{4}) = s^2 - s + \frac{3}{16}, \quad E = \mathbb{C}[s]/(h), \quad j_h : \mathbb{C}[s] \longrightarrow E. \quad (\text{TEC.13})$$

Both packet centres lie in $0 < \operatorname{Re} s < 1$, and the reflection $\rho \mapsto 1 - \overline{\rho}$ exchanges them. They are distinct, and each has its complete multiplicity one. In the retained monomial basis $(1, s)$ of E , the unit and multiplication operator are

$$c = j_h 1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = M_s = \begin{pmatrix} 0 & -3/16 \\ 1 & 1 \end{pmatrix}. \quad (\text{TEC.14})$$

Indeed $s^2 = s - 3/16$ in this quotient, which proves both columns of the multiplication matrix.

The complete evaluation map is

$$\begin{aligned} T : E &\longrightarrow \mathbb{C} \oplus \mathbb{C}, & [p] &\longmapsto (p(3/4), p(1/4)), \\ T &= \begin{pmatrix} 1 & 3/4 \\ 1 & 1/4 \end{pmatrix}, & T^{-1} &= \begin{pmatrix} -1/2 & 3/2 \\ 2 & -2 \end{pmatrix}. \end{aligned} \quad (\text{TEC.15})$$

Both evaluations vanish on the ideal (h) , so the map is well defined. Evaluations preserve addition, multiplication and the unit. For any pair (u_+, u_-) , the polynomial

$$2u_+(s - \frac{1}{4}) + 2u_-(\frac{3}{4} - s) \quad (\text{TEC.16})$$

evaluates to that pair. It has the coefficients given by T^{-1} , and multiplication of the two matrices in (TEC.15) gives the identity. This proves the typed algebra isomorphism, including its inverse. Evaluation of sp also proves

$$TA = \tilde{A}T, \quad \tilde{A} = \operatorname{diag}(3/4, 1/4), \quad Tc = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{TEC.17})$$

In the original quotient E one has $x^2 = v$. Reducing the explicit polynomials of (TEC.7), with no deletion of their supported residue classes, gives

k	$v_k := j_h q_k$ in the monomial basis $(1, s)$	
0	$(1, 0)^T$	
1	$(-1/2, 1)^T$	
2	$(1/8, 0)^T$	
3	$(-1/8, 1/4)^T$	
4	$(5/128, 0)^T$	
5	$(-13/256, 13/128)^T$	(TEC.18)

For example $q_3 = 4vx = x/4$, $q_4 = 10v^2 = 5/128$, and $q_5 = 26v^2x = (13/128)x$ in E . These calculations establish every entry of (TEC.18) directly from the original ideal relation.

12.7.4 The original kernels, terminal defects and exact energies

For $1 \leq N \leq 4$, define

$$\begin{aligned} K_N &= \sum_{k=0}^N \frac{v_k v_k^*}{\kappa_k}, \\ L_N &= AK_N + K_N A^* - K_N, \\ \epsilon_N &= \left\| K_N^{-1/2} L_N K_N^{-1/2} \right\|. \end{aligned} \tag{TEC.19}$$

All adjoints in (TEC.19) are conjugate transposes in the explicitly retained monomial coordinates. The columns v_0, v_1 are linearly independent. If a nonzero column u had $u^* K_N u = 0$, the positive sum in (TEC.19) would force both $v_0^* u = 0$ and $v_1^* u = 0$, a contradiction. Thus each K_N is positive definite, so its positive inverse square root and the displayed Hermitian operator norm are well defined.

Using (TEC.7) and (TEC.18), the successive kernels are

$$\begin{aligned} K_1 &= \begin{pmatrix} 5 & -8 \\ -8 & 16 \end{pmatrix}, & K_2 &= \begin{pmatrix} 7 & -8 \\ -8 & 16 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 53/3 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}, & K_4 &= \begin{pmatrix} 131/6 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}. \end{aligned} \tag{TEC.20}$$

In particular the last step is the exact positive rank-one update

$$K_4 - K_3 = \frac{v_4 v_4^*}{\kappa_4} = \frac{25}{6} cc^*. \tag{TEC.21}$$

Multiplication by the matrix in (TEC.14) gives

$$L_3 = \begin{pmatrix} -20/3 & 20/3 \\ 20/3 & 0 \end{pmatrix}, \quad L_4 = \begin{pmatrix} -65/6 & 65/6 \\ 65/6 & 0 \end{pmatrix}. \tag{TEC.22}$$

These defects satisfy the original terminal identity, with its exact coefficients and signs:

$$L_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N} \quad (N = 3, 4). \tag{TEC.23}$$

For $N = 3$, the numerator calculated from (TEC.18) is

$$\begin{pmatrix} -5/512 & 5/512 \\ 5/512 & 0 \end{pmatrix}, \quad \kappa_3 = 3/2048.$$

For $N = 4$, it is

$$\begin{pmatrix} -65/16384 & 65/16384 \\ 65/16384 & 0 \end{pmatrix}, \quad \kappa_4 = 3/8192.$$

Division gives exactly the two matrices in (TEC.22), proving (TEC.23) without an appeal to a numerical rank calculation.

Theorem 12.4 (Strict increase at two consecutive permitted levels). *For the positive measure (TEC.1) and the complete reflection-stable packet (TEC.13), the energies in (TEC.19) satisfy*

$$\epsilon_3^2 = \frac{25}{99}, \quad \epsilon_4^2 = \frac{4225}{15136}, \quad \epsilon_4^2 - \epsilon_3^2 = \frac{39875}{1498464} > 0. \tag{TEC.24}$$

In the indexing $N = d + m$, with $d = \dim_{\mathbb{C}} E = 2$, these are the consecutive levels $m = 1$ and $m = 2$, both with $m \geq 0$.

Proof. Transport the two forms by the proved isomorphism (TEC.15):

$$\begin{aligned}\tilde{K}_3 &= TK_3T^* = \begin{pmatrix} 20/3 & -2/3 \\ -2/3 & 20/3 \end{pmatrix}, \\ \tilde{K}_4 &= TK_4T^* = \begin{pmatrix} 65/6 & 7/2 \\ 7/2 & 65/6 \end{pmatrix}.\end{aligned}\tag{TEC.25}$$

By (TEC.17), the transported defect is

$$\tilde{L}_N = TL_NT^* = \tilde{A}\tilde{K}_N + \tilde{K}_N\tilde{A}^* - \tilde{K}_N.\tag{TEC.26}$$

For either matrix $\tilde{K} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ in (TEC.25), one has

$$\tilde{L} = \text{diag}(a/2, -a/2), \quad \det(\tilde{L} - \lambda\tilde{K}) = \lambda^2(a^2 - b^2) - a^2/4.\tag{TEC.27}$$

To check that this calculation gives the norm defined in the original coordinates, set $H = K^{-1/2}LK^{-1/2}$. Then

$$\det(H - \lambda I) = \det(K)^{-1} \det(L - \lambda K), \quad \det(T(L - \lambda K)T^*) = |\det T|^2 \det(L - \lambda K).\tag{TEC.28}$$

The two determinant factors are nonzero and independent of λ . Consequently the eigenvalues of the Hermitian matrix H equal the roots of (TEC.27), with their multiplicities. The roots are the opposite real numbers $\pm a/(2\sqrt{a^2 - b^2})$. A Hermitian matrix has operator norm equal to the greatest absolute value of its eigenvalues, so

$$\epsilon^2 = \frac{a^2}{4(a^2 - b^2)}.\tag{TEC.29}$$

For $N = 3$, substitution of $a = 20/3, b = -2/3$ gives $\epsilon_3^2 = (400/9)/(4 \cdot 396/9) = 25/99$. For $N = 4$, substitution of $a = 65/6, b = 7/2$ gives $\epsilon_4^2 = (4225/36)/(4 \cdot 3784/36) = 4225/15136$. Their difference, retaining the two denominators, is

$$\frac{4225 \cdot 99 - 25 \cdot 15136}{15136 \cdot 99} = \frac{418275 - 378400}{1498464} = \frac{39875}{1498464} > 0.\tag{TEC.30}$$

Both energies are nonnegative, and their squared values have strictly positive difference, so $\epsilon_4 > \epsilon_3$. The original coordinate characteristic polynomials provide the same calculation:

$$\begin{aligned}\det(L_3 - \lambda K_3) &= 176\lambda^2 - 400/9, \\ \det(L_4 - \lambda K_4) &= (3784/9)\lambda^2 - 4225/36.\end{aligned}\tag{TEC.31}$$

Finally $N = 3 = 2 + 1$ and $N = 4 = 2 + 2$, proving the stated degree range without using the lower degree $N = 1$. $\square$

12.7.5 The exact scope of the obstruction

For this measure the construction is the composite of the polynomial source with the original residue quotient

$$\mathcal{P}_N \hookrightarrow \mathbb{C}[s] \xrightarrow{j_h} E, \quad E \xrightarrow{T} \mathbb{C} \oplus \mathbb{C},\tag{TEC.32}$$

and multiplication is intertwined exactly by (TEC.17). Its source inner product is (TEC.6); its monic source Gram matrix is $\text{diag}(\kappa_0, \dots, \kappa_N)$; its kernel is the full sum (TEC.19). Thus it has precisely the vertical recurrence, positive kernel update, complete packet quotient and terminal defect maps appearing in the arithmetic moment construction. Equations (TEC.24) and (TEC.30) prove that these data alone do not imply nonincreasing consecutive energy. An argument for that property for ν_Z must therefore establish and use additional identities or estimates of its stated arithmetic density.

The present example establishes the indicated obstruction to a universal recurrence-based argument; it asserts no consecutive increase for ν_Z .

The companion exact checker verifies 88 rational and complex-rational identities in both ordinary Python and optimized Python. It checks the atom weights and relevant moments, source Gram matrix, packet isomorphism, residue columns, kernel updates, terminal identities and generalized characteristic polynomials. Those finite checks supplement the complete computations above.

13 The exact metric departure and the cost of complete jets

Retain the original packet $E = \mathbb{C}[s]/h$, its polynomial coordinates and $A = M_s$. Write its eigenvalues with their algebraic multiplicities as $\rho_1, \dots, \rho_d$, and put $\delta_j = \operatorname{Re} \rho_j - 1/2$. No eigenvalue or repeated factor is discarded. For a positive Hermitian matrix G in these coordinates define

$$W_G = A^*G + GA - G, \quad S_G = G^{-1/2}W_GG^{-1/2}, \quad \epsilon_G = \|S_G\|_{\text{op}},$$

$$\mathfrak{d}_G(A) = \operatorname{Tr}(G^{-1}A^*GA) - \sum_{j=1}^d |\rho_j|^2. \quad (\text{MD1})$$

The positive square root is used only for the explicit isometry $x \mapsto G^{1/2}x$ from the specified metric space (E, G) to the Euclidean coordinate space. The original A, G , their entries and their maps remain those in (MD1).

Theorem 13.1 (Exact departure identity). *The number $\mathfrak{d}_G(A)$ is real and nonnegative, and*

$$\operatorname{Tr} S_G = 2 \sum_j \delta_j, \quad \operatorname{Tr}(S_G^2) = 4 \sum_j \delta_j^2 + 2\mathfrak{d}_G(A). \quad (\text{MD2})$$

*It vanishes precisely when A is normal for the specified metric, that is, when $A(G^{-1}A^*G) = (G^{-1}A^*G)A$. For the original orthogonal representatives R_m , their matrices G_m and rank-two forms W_m , one has exactly*

$$\epsilon_m = \left| \sum_j \delta_j \right| + \sqrt{2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j \right)^2}. \quad (\text{MD3})$$

For a reflection-stable packet this becomes

$$\boxed{\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).} \quad (\text{MD4})$$

Proof. Let $C = G^{1/2}AG^{-1/2}$. Then $S_G = C + C^* - I$ and $\operatorname{Tr}(G^{-1}A^*GA) = \operatorname{Tr}(C^*C)$. For completeness, choose a unit eigenvector of C , extend it to an orthonormal basis, and repeat on the lower-right compression. Induction gives a unitary matrix U with upper-triangular $T = U^*CU$ whose diagonal lists the same eigenvalues with their algebraic multiplicities. Unitary invariance of the trace gives

$$\mathfrak{d}_G(A) = \sum_{i < j} |T_{ij}|^2 \geq 0. \quad (\text{MD5})$$

If it is zero, T is diagonal and C is normal. Conversely, an upper-triangular normal matrix is diagonal: its first row norm equals its first column norm, eliminating the strict first-row entries, and induction eliminates every remaining such entry. Thus normality of C is equivalent to the stated equality for A , through the specified isometry.

Expanding $(C + C^* - I)^2$ under the trace gives

$$\mathrm{Tr}(S_G^2) = 2 \mathrm{Tr}(C^* C) + 2 \mathrm{Re} \mathrm{Tr}(C^2) - 4 \mathrm{Re} \mathrm{Tr} C + d.$$

The traces of C and C^2 are respectively $\sum_j \rho_j$ and $\sum_j \rho_j^2$, even in the presence of nilpotents. Since $2|\rho|^2 + 2 \mathrm{Re} \rho^2 - 4 \mathrm{Re} \rho + 1 = 4(\mathrm{Re} \rho - 1/2)^2$, this proves (MD2). The trace formula follows directly from $S_G = C + C^* - I$.

For G_m , Section 6 proved $\mathrm{rank} S_{G_m} \leq 2$. List its two possibly nonzero real eigenvalues a, b , appending zero if necessary (also when $d = 1$). Equations (MD2) give

$$a + b = 2 \sum_j \delta_j, \quad (a - b)^2/4 = 2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j\right)^2.$$

Thus the radicand is nonnegative, and $\max(|a|, |b|) = |(a + b)/2| + |a - b|/2$, proving (MD3). Reflection-stability pairs δ with $-\delta$, with the same multiplicities, proving (MD4). $\square$

The actual arithmetic kernel of Section 9 also gives this quantity without changing the local unit. Indeed $G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon$, $N = d + m$, and $U_\varepsilon A = A U_\varepsilon$. Cyclicity of trace gives

$$\mathfrak{d}_{G_m}(A) = \mathrm{Tr}(K_N A^* K_N^{-1} A) - \sum_j |\rho_j|^2. \quad (\text{MD6})$$

To verify the cancellation, substitute $G_m^{-1} = U_\varepsilon^{-1} K_N (U_\varepsilon^*)^{-1}$ in (MD1), commute A with U_ε and A^* with U_ε^* , and cancel the outer similar factors inside the trace. This calculation removes no source representative: their exact recovery remains (TB.7)–(TB.9).

13.1 A quantitative constraint on every retained nilpotent chain

For a selected root ρ of full order r , let

$$I_\rho : \mathbb{C}[z]/z^r \longrightarrow E$$

be the original Chinese-remainder inclusion: the ρ -component is the given Taylor polynomial in $z = s - \rho$, and every other component is zero. Its Gram matrix is $G_\rho = I_\rho^* G I_\rho$, in the basis $1, z, \dots, z^{r-1}$. Put $N_\rho z^k = z^{k+1}$, with $z^r = 0$. The exact intertwining relation is $A I_\rho = I_\rho(\rho I + N_\rho)$.

Theorem 13.2 (Full-jet metric inequalities). *For every positive original metric G , write $\delta = \mathrm{Re} \rho - 1/2$. Then $\epsilon_G \geq 2|\delta|$, and for all $0 \leq k < k + \ell < r$,*

$$\frac{(G_\rho)_{k+\ell, k+\ell}}{(G_\rho)_{k, k}} \leq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^\ell. \quad (\text{MD7})$$

In particular, for $r \geq 2$,

$$\kappa_2(G_\rho) \geq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^{-(r-1)}. \quad (\text{MD8})$$

The bracket is strictly positive when $r \geq 2$. Here κ_2 is the ratio of the largest and smallest eigenvalues in the specified Taylor coordinates.

Proof. The definition of ϵ_G is the two-sided form inequality $-\epsilon_G G \leq W_G \leq \epsilon_G G$. On an eigenvector it gives $\epsilon_G \geq 2|\delta|$. Pull this same inequality back by I_ρ . Let $L = G_\rho^{1/2} N_\rho G_\rho^{-1/2}$ and $T = 2\delta I + L + L^*$. The pulled-back inequality gives $\|T\| \leq \epsilon_G$. Nilpotence retains $\mathrm{Tr} L = \mathrm{Tr}(L^2) = 0$, whence

$$\mathrm{Tr} T = 2r\delta, \quad \|L\|_F^2 = \frac{1}{2} \mathrm{Tr}(L + L^*)^2 = \frac{1}{2} \mathrm{Tr} T^2 - 2r\delta^2 \leq \frac{r}{2} (\epsilon_G^2 - 4\delta^2). \quad (\text{MD9})$$

The first equality for the Frobenius norm follows by expanding the square and using the two zero traces. The final inequality uses the r real eigenvalues of T , each bounded in modulus by ϵ_G . For the

original metric norm, $\|N_\rho\|_{G_\rho} = \|L\| \leq \|L\|_F$. Apply its ℓ -th power to the specified basis vector z^k . Since $N_\rho^\ell z^k = z^{k+\ell}$, the resulting squared norm inequality is exactly (MD7).

When $r \geq 2$, $N_\rho \neq 0$, so (MD9) makes the bracket strictly positive. Put $k = 0, \ell = r - 1$. The Rayleigh inequalities give $\kappa_2(G_\rho) \geq (G_\rho)_{0,0}/(G_\rho)_{r-1,r-1}$; combine with (MD7) to obtain (MD8). $\square$

All entries in (MD7) for the original sequence are explicitly $I_\rho^* U_\varepsilon^* K_N^{-1} U_\varepsilon I_\rho$. Thus this theorem supplies constraints on the actual moment calculation, with its original Taylor factors. At a critical root of order $r \geq 2$, a sequence with relative control approaching zero necessarily has the stated diverging Taylor-metric condition number. The surviving jets have not been erased; the inequalities calculate their relative metric scales. This is compatible with the complete graph and weighted-jet maps in Section 21, which retain those same coordinates.

14 Coherent packets, joint tensor layers, and their exact spectral cost

This section integrates the complete arithmetic interpolation construction of the 12 September coherent-interpolation delivery, whose source revision is 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, and proves its comparison with the preceding kernel and departure calculations. All zeros in a packet retain their complete multiplicities. A reflected packet is required only in statements explicitly using its reflection. The original spaces $V, \mathcal{B}$, map Θ , generator $D = -x\partial_x$, and Hilbert observation $L^2(dx)$ remain fixed.

14.1 The arithmetic unit and the coherent one-variable maps

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, write

$$d = \deg h, \quad E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \quad g = 2\xi. \quad (\text{CT.1})$$

Complete multiplicities make v_h invertible. Let $F_h \in \mathcal{B}$ be the original inverse satisfying $h(D)F_h = f_0$ and $\mathcal{M}F_h = g/h$, and put $\mathcal{T}_h P = P(D)F_h$. The integral construction of F_h , its estimates at both ends, and injectivity of the Mellin transform are proved in Section 6. They give

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (\text{CT.2})$$

Let $C_N : \mathcal{P}_N \rightarrow E_h$ be the full remainder map in the original monomial coordinates, let $M_{h,N}$ be the Gram matrix of $\mathcal{P}_N$ for $d\nu_h = |g(1/2 + it)/h(1/2 + it)|^2 dt/(2\pi)$, and use U_a for multiplication by a on E_h . The two kernel matrices occurring in the constructions have the exact types

$$K_N = C_N M_{h,N}^{-1} C_N^*, \quad J_{h,N} = U_{v_h} C_N, \quad \mathcal{C}_{h,N} = U_{v_h} K_N U_{v_h}^*. \quad (\text{CT.3})$$

Consequently their actual representative and metric coincide:

$$\begin{aligned} M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1} &= M_{h,N}^{-1} C_N^* K_N^{-1} U_{\varepsilon_h}, \\ G_{h,n} &= \mathcal{C}_{h,n+d}^{-1} = U_{\varepsilon_h}^* K_{n+d}^{-1} U_{\varepsilon_h}. \end{aligned} \quad (\text{CT.4})$$

Indeed invert the congruence in (CT.3) and cancel the adjacent inverse factors. The first line is exactly the polynomial numerator in (TB.7)–(TB.9). Its difference from $\text{rem}_h(\varepsilon_h u)$ is divisible by h and has quotient of degree at most n . Applying (CT.2) gives the same original theta correction. Thus the arithmetic unit remains in every coefficient map and every source representative.

For completeness, the minimum assertion follows directly without an existence assumption. For any positive Gram matrix M and surjective J , set $G = (JM^{-1}J^*)^{-1}$ and $L = M^{-1}J^*G$. Then $JL = I$, $L^*ML = G$, and for $Jz = 0$, $z^*ML = z^*J^*G = 0$. Every feasible vector is $Lu + z$, so

$$(Lu + z)^* M (Lu + z) = u^* G u + z^* M z. \quad (\text{CT.5})$$

Positivity makes this the unique minimum. In (CT.3), positivity holds because a nonzero polynomial cannot vanish almost everywhere on the original integration line; surjectivity follows from complete polynomial remainders. The kernel is exactly $h\mathcal{P}_n$ when $N = n + d$. Hence $R_{h,n} = (I - P_{L_n})s_h$, where $s_h = \mathcal{T}_h \text{rem}_h(\varepsilon_h \cdot)$ and $L_n = \Theta \text{span}(\phi_*, \dots, D^n \phi_*)$.

If $h \mid H$ are full packets, put $m = H/h$. The factors h, m are coprime. The original CRT injection $\iota : E_h \rightarrow E_H$ inserts zero jets at the additional centres, and satisfies $A_H \iota = \iota A_h$. Multiplication $P \mapsto mP$ is an isometry of the declared polynomial images, since $(g/H)mP = (g/h)P$; it satisfies

$$\mathcal{T}_H(mP) = \mathcal{T}_h P, \quad J_H \mathcal{T}_H(mP) = \iota J_h \mathcal{T}_h P. \quad (\text{CT.6})$$

At the added centres the second identity follows from the full zeros of g/h ; at the old centres it is the identity of entire functions. Both numerators for $s_H \iota = \mathcal{T}_H(m \text{rem}_h(\varepsilon_h \cdot))$ have degree below $\deg H$ and the same complete jets, proving equality. Since L_n is independent of the packet,

$$R_{H,n} \iota = R_{h,n}, \quad \iota^* G_{H,n} \iota = G_{h,n}, \quad \iota^* W_{H,n} \iota = W_{h,n}. \quad (\text{CT.7})$$

These are congruences, with no interchange of inversion and compression. Their split maps at a retained support λ are $(\lambda, u) \mapsto (\lambda, \iota u)$; all added zero jets are represented zeros at that support.

14.2 The original reflection and the exact diagonal coefficient

The original arithmetic density $|g(1/2 + it)|^2$ is even in t , since the real Taylor coefficients of g give $g(1/2 - it) = \overline{g(1/2 + it)}$. If Q_j is the monic orthogonal polynomial for that density in the variable s , the polynomial $(-1)^j Q_j(1 - s)$ is monic and orthogonal to all lower degrees, by the change $t \mapsto -t$. Uniqueness therefore gives

$$Q_j(1 - s) = (-1)^j Q_j(s), \quad \int t |Q_j(1/2 + it)|^2 |g(1/2 + it)|^2 \frac{dt}{2\pi} = 0. \quad (\text{CT.8})$$

The diagonal recurrence coefficient is thus exactly $\langle Q_j, sQ_j \rangle / \|Q_j\|^2 = 1/2$. For lower degrees the identity $D^* + D = I$ eliminates every coefficient except the adjacent one and gives its sign and actual norm ratio. More explicitly, sQ_j is orthogonal to Q_i for $i \leq j - 2$, because its inner product is the inner product of Q_j with $(1 - s)Q_i$. For $i = j - 1$ the leading term gives coefficient $-\|Q_j\|^2 / \|Q_{j-1}\|^2$. This proves, with the original vectors and norms,

$$Du_j = u_{j+1} + \frac{1}{2}u_j - \frac{\mathfrak{h}_j}{\mathfrak{h}_{j-1}}u_{j-1}, \quad u_j = Q_j(D)f_0, \quad \mathfrak{h}_j = \|u_j\|^2. \quad (\text{CT.9})$$

The final term is omitted at $j = 0$. The even density in (CT.8) is the additional reason the imaginary part of the diagonal coefficient vanishes; the adjoint identity by itself fixes only its real part.

For a packet stable under $\rho \mapsto 1 - \bar{\rho}$, including the same complete orders, the original maps are

$$\mathfrak{j}_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}, \quad \mathfrak{j}_{\mathcal{B}} F(x) = x^{-1} \overline{F(1/\bar{x})}.$$

The second is antiunitary for dx , and $D\mathfrak{j}_{\mathcal{B}} = \mathfrak{j}_{\mathcal{B}}(I - D)$. The identities $h^\dagger = (-1)^d h$ and $g^\dagger = g$ imply $(g/h)^\dagger = (-1)^d g/h$; they give $\mathfrak{j}_{\mathcal{B}} s_h = s_h \mathfrak{j}_h$ by the same complete jets and degree bounds as above. Equation (CT.9), or invariance of the polynomial space under $s \mapsto 1 - s$, shows $\mathfrak{j}_{\mathcal{B}} L_n = L_n$. The unique minimum (CT.5) then commutes with these maps. If $\mathfrak{j}_h u = C_h \bar{u}$,

$$C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}. \quad (\text{CT.10})$$

Substitution into a quadratic form proves that an upper bound and its reflected lower bound use this same $G_{h,n}$.

14.3 The joint total-degree source and its relation layer

Fix $k \geq 1$, $M \geq k(d-1)$, and set

$$E^{(k)} = E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{P}_M^{(k)} = \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad \mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}.$$

Here $\leq M$ means total degree. The full jet map is reduction by each $h(s_j)$, followed by multiplication by $v_h^{\otimes k}$, and its kernel is

$$\mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)). \quad (\text{CT.11})$$

Successive monic division in the declared order $s_1, \dots, s_k$ gives $P = \sum_j h(s_j) P_j$ in this kernel, with $\deg P_j \leq M - d$. Division lowers total degree at every replacement, so this bound holds at every division step. In the original complex $[V \rightarrow \mathcal{B}]$, placed in degrees 0, 1, the primitive for the j -th term uses $(-1)^{j-1} \phi_*$ in the j -th factor and F_h in the others, with the polynomial $P_j(D_1, \dots, D_k)$ applied to the whole tensor. The preceding $j-1$ factors have degree one, so the tensor differential supplies the further factor $(-1)^{j-1}$. Its boundary is exactly $+\mathcal{T}_h^{(k)}(h(s_j) P_j)$.

Write $\mathcal{B}_M = \mathcal{T}_h^{(k)} \mathcal{I}_{h,k,M}$ inside $\mathcal{H}_k = L^2(d^k x)$. The arithmetic moment matrix in the retained monomials is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s^{\alpha_j}} s^{\beta_j} d\nu_h, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (\text{CT.12})$$

Product integration proves the first formula. A polynomial vanishing almost everywhere for this product measure is zero, by successive one-variable uniqueness, proving positivity. The complete remainder basis has total degree at most $k(d-1)$, proving surjectivity. Thus (CT.5) proves

$$R_M = (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad J^{(k)} R_M = I, \quad q^{(k)} R_M = \sigma_h^{\otimes k}. \quad (\text{CT.13})$$

The kernel dimension is $\binom{M+k}{k} - d^k$. Since $\mathcal{T}_h^{(k)}$ is injective by its entire Mellin multiplier, the orthogonal relation layer $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ has exactly

$$\dim \mathcal{E}_M = \binom{M+k}{k-1}. \quad (\text{CT.14})$$

Define all of the following maps on the original $E^{(k)}$:

$$B_M = D^{(k)} R_M - R_M A^{(k)}, \quad Y_M = P_{\mathcal{E}_M} R_M, \quad C_M = (I - P_{\mathcal{B}_M}) B_M. \quad (\text{CT.15})$$

The complete jet of B_M is zero and its numerator has degree at most $M+1$, so $B_M(E^{(k)}) \subseteq \mathcal{B}_{M+1}$. Both Y_M, C_M therefore have target $\mathcal{E}_M$. Integration by parts on these rapidly decreasing functions gives $(D^{(k)})^* + D^{(k)} = kI$. Substitution into $R_M^*(D^{(k)} + (D^{(k)})^*) R_M = kG_M$, followed by orthogonality to $\mathcal{B}_M$, proves

$$\begin{aligned} W_M &= (A^{(k)})^* G_M + G_M A^{(k)} - kG_M = -(Y_M^* C_M + C_M^* Y_M), \\ \Delta_M &= G_M - G_{M+1} = Y_M^* Y_M \succeq 0. \end{aligned} \quad (\text{CT.16})$$

For the second identity use the orthogonal decomposition of $\mathcal{B}_{M+1}$ and $R_{M+1} = R_M - Y_M$. The actual quotient map $[z] \mapsto (I - P_{\mathcal{B}_M})z$ identifies $\mathcal{B}_{M+1}/\mathcal{B}_M$ with $\mathcal{E}_M$, with inverse the inclusion followed by quotient. A transported boundary is zero at its next support and remains represented there; this map sends no supported input to external τ .

For a reflection-stable packet, put $C_{h,k} = C_h^{\otimes k}$. The original product reflection acts on a numerator by $(-1)^{kd} \overline{P(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. This retains total degree and preserves the ideal in (CT.11), with

its displayed factor. Its action on $\mathcal{H}_k$ is antiunitary. The unique minimum (CT.5) therefore commutes with reflection and gives

$$C_{h,k}^* G_M C_{h,k} = \overline{G_M}, \quad A^{(k)} C_{h,k} = C_{h,k} (kI - \overline{A^{(k)}}), \quad C_{h,k}^* W_M C_{h,k} = -\overline{W_M}. \quad (\text{CT.16a})$$

If $W_M u = \lambda G_M u$, conjugation followed by $(C_{h,k}^*)^{-1}$ in the last identity gives $W_M C_{h,k} \bar{u} = -\lambda G_M C_{h,k} \bar{u}$. All generalized eigenvalues are real because $G_M > 0$. This invertible antilinear map pairs their full multiplicities. The source-layer maps themselves retain the reflection types. If $j_E = j_h^{\otimes k}$ and j_E is the restriction of the original function reflection to $\mathcal{E}_M$, then

$$j_E Y_M = Y_M j_E, \quad j_E C_M = -C_M j_E. \quad (\text{CT.16b})$$

Both boundary projections commute with reflection, proving the first identity. For the second use $D^{(k)} j_H = j_H (kI - D^{(k)})$, $A^{(k)} j_E = j_E (kI - A^{(k)})$, and $j_H R_M = R_M j_E$ in (CT.15). Expanding its two terms gives $j_H B_M = -B_M j_E$; the projection then proves (CT.16b).

14.4 Nested packets on the joint source: the additional projection

Let $h \mid H$, $q = \deg(H/h)$, $N = M + kq$, and $\iota_k = \iota_{hH}^{\otimes k}$. Multiplication of a numerator by $\prod_{j=1}^k (H/h)(s_j)$ gives a degree-preserving-with-shift map $\mathcal{P}_M^{(k)} \rightarrow \mathcal{P}_N^{(k)}$, with

$$\mathcal{T}_H^{(k)} (\prod_j (H/h)(s_j) P) = \mathcal{T}_h^{(k)} P, \quad J_H^{(k)} \mathcal{T}_h^{(k)} P = \iota_k J_h^{(k)} \mathcal{T}_h^{(k)} P. \quad (\text{CT.17})$$

The second equality follows on pure monomials from (CT.6) and then by linearity. In particular it maps the smaller relation space into the larger one in the same $\mathcal{H}_k$. Let

$$\mathcal{F} = \mathcal{B}_{H,k,N} \cap \mathcal{B}_{h,k,M}^\perp, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

Since $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$, the exact comparison is

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{CT.18})$$

Indeed nested orthogonal projections commute, and their difference is $P_{\mathcal{F}}$; apply them to the common section to prove the first line. Pythagoras proves the second; intertwining of $A^{(k)}$ and direct expansion prove the third. Also

$$\dim \mathcal{F} = \binom{N+k}{k} - (\deg H)^k - \binom{M+k}{k} + d^k. \quad (\text{CT.19})$$

This is the difference of the exact kernel dimensions. For $k = 1$, $M = n + d$, both boundary spaces are the original L_n , so $\mathcal{F} = 0$ and (CT.18) recovers (CT.7). For joint tensors (CT.18) retains every additional allowed boundary and calculates its effect on the control form.

14.5 Positive layer rank and the complete tensor spectral departure

List $\rho_1, \dots, \rho_d$ with all algebraic multiplicities, put $\delta_i = \text{Re } \rho_i - 1/2$, $\sigma_1 = \sum_i \delta_i$, $\sigma_2 = \sum_i \delta_i^2$, and define

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2}, \quad \epsilon_M = \|S_M\|_{\text{op}}, \quad r_M = \text{rank } \Delta_M, \\ \mathfrak{d}_M^{(k)} &= \text{Tr}(G_M^{-1} (A^{(k)})^* G_M A^{(k)}) - \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \rho_{i_j} \right|^2, \\ \mathcal{D}_k &= \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \delta_{i_j} \right|. \end{aligned} \quad (\text{CT.20})$$

All sums are ordered and retain repeated entries. The isometry $x \mapsto G_M^{1/2}x$ in this calculation leaves the original jet coordinates, source functions, metric and layer maps unchanged. For a map into $\mathcal{E}_M$, the Frobenius symbol below denotes its Hilbert–Schmidt norm for the layer’s inherited $L^2(d^kx)$ inner product.

Theorem 14.1 (Exact tensor departure and relative layer cost). *One has $\mathfrak{d}_M^{(k)} \geq 0$, and exactly*

$$\begin{aligned} \text{Tr } S_M &= 2kd^{k-1}\sigma_1, \\ \text{Tr}(S_M^2) &= 4\{kd^{k-1}\sigma_2 + k(k-1)d^{k-2}\sigma_1^2\} + 2\mathfrak{d}_M^{(k)}. \end{aligned} \quad (\text{CT.21})$$

The positive and negative inertias of W_M each obey

$$n_+(W_M), n_-(W_M) \leq \min(\text{rank } Y_M, \text{rank } C_M), \quad \text{rank } Y_M = r_M \leq \binom{M+k}{k-1}. \quad (\text{CT.22})$$

The complete ordered spectral cost satisfies

$$\boxed{\mathcal{D}_k \leq \frac{1}{2}\|S_M\|_1 \leq \|Y_M G_M^{-1/2}\|_F \|C_M G_M^{-1/2}\|_F}, \quad \mathcal{D}_k \leq r_M \epsilon_M. \quad (\text{CT.23})$$

In particular, the actual source matrices satisfy

$$\mathcal{D}_k^2 \leq \text{Tr}(G_M^{-1} \Delta_M) \text{Tr}(G_M^{-1} C_M^* C_M). \quad (\text{CT.24})$$

If the packet is reflection-stable, put $p_M = n_+(W_M) = n_-(W_M)$. Then

$$p_M \leq \min\{r_M, \text{rank } C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k \leq p_M \epsilon_M. \quad (\text{CT.24a})$$

In this case (CT.21) becomes $\text{Tr}(S_M^2) = 4kd^{k-1}\sigma_2 + 2\mathfrak{d}_M^{(k)}$. When $p_M > 0$, it implies

$$\epsilon_M^2 \geq \frac{2kd^{k-1}\sigma_2 + \mathfrak{d}_M^{(k)}}{p_M}. \quad (\text{CT.25})$$

Proof. Triangularize A_h over $\mathbb{C}$. The tensor product of this basis makes $A^{(k)}$ triangular, with diagonal entries $\sum_j \rho_{ij}$. Thus these are all its eigenvalues with algebraic multiplicity; its nilpotent entries have not been discarded. Now apply the unitary triangularization argument of (MD5) to $T = G_M^{1/2} A^{(k)} G_M^{-1/2}$. The sum of squared absolute values of its strictly upper triangular entries is exactly $\mathfrak{d}_M^{(k)}$, proving nonnegativity. Since $S_M = T + T^* - kI$, expansion of its squared trace gives

$$\text{Tr}(S_M^2) = 4 \sum_{i_1, \dots, i_k} \left(\sum_j \delta_{i_j} \right)^2 + 2\mathfrak{d}_M^{(k)}.$$

The k diagonal terms in this ordered sum contribute $kd^{k-1}\sigma_2$; the $k(k-1)$ ordered cross terms contribute $k(k-1)d^{k-2}\sigma_1^2$. The linear trace is computed in the same way. This proves (CT.21), including $k = 1$.

Write $\hat{Y} = Y_M G_M^{-1/2}$ and $\hat{C} = C_M G_M^{-1/2}$. Then $S_M = -(\hat{Y}^* \hat{C} + \hat{C}^* \hat{Y})$. Its form is zero on $\ker \hat{Y}$. A positive definite subspace of dimension greater than $\text{rank } \hat{Y}$ intersects this kernel nontrivially, a contradiction; the same argument holds for a negative definite subspace and with $\hat{C}$ in place of $\hat{Y}$. Sylvester inertia under the displayed invertible congruence proves the first part of (CT.22). The identity $\Delta_M = Y_M^* Y_M$ gives equality of their kernels, hence their ranks, proving the rest.

In the unitary triangular basis of T , the diagonal of S_M consists of $2 \sum_j \delta_{i_j}$. For any Hermitian matrix S , its spectral decomposition gives $|S_{ii}| \leq \sum_a |\lambda_a| |U_{ia}|^2$. Sum over i to obtain $\sum_i |S_{ii}| \leq \sum_a |\lambda_a| = \|S\|_1$. This proves the first inequality in (CT.23). For finite matrices B, C with a common row space, $\|B^* C\|_1 \leq \|B\|_F \|C\|_F$: choose a singular-value decomposition of $B^* C$, use its left and

right singular vectors to express its trace norm as $|\operatorname{Tr}(UB^*C)|$ for a unitary U , and apply Cauchy-Schwarz to the entries of BU^* and C . The triangle inequality now proves the second inequality of (CT.23). Finally (CT.22) bounds the number of nonzero eigenvalues of S_M by $2r_M$, so $\|S_M\|_1 \leq 2r_M\epsilon_M$. Squaring the middle bound and writing its two Frobenius norms as traces proves (CT.24).

For a reflected packet $\sigma_1 = 0$, with every multiplicity paired. The exact reflection (CT.16a) pairs the nonzero eigenvalues of S_M , giving equal positive and negative counts p_M . Equation (CT.22) proves the bounds on p_M , and there are exactly $2p_M$ nonzero eigenvalues of S_M , each of square at most ϵ_M^2 . Their trace norm is at most $2p_M\epsilon_M$, proving (CT.24a); their squared trace proves (CT.25). If $p_M = 0$, the pairing implies $W_M = 0$ directly, and (CT.21) states that all the nonnegative terms vanish; no division by p_M is made. $\square$

Every quantity in (CT.24) is attached to the same arithmetic moment matrix and its next original relation layer. In particular, $0 \preceq \Delta_M \prec G_M$, because $G_{M+1} > 0$, so all eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$ lie in $[0, 1)$, and its trace is strictly smaller than r_M when $r_M > 0$. Its actual value is retained in (CT.24). Equations (CT.21)–(CT.25) calculate necessary relative layer costs and the full nilpotent departure for the constructed tensor family. They assert no unproved upper bound on these arithmetic quantities.

There is an exact determinant refinement of the same source loss. For $r_M > 0$, put

$$\pi_M = \frac{\det G_{M+1}}{\det G_M}, \quad \tau_M = \operatorname{Tr}(G_M^{-1}\Delta_M), \quad \chi_M = \operatorname{Tr}(G_M^{-1}C_M^*C_M).$$

Then

$$1 - \pi_M \leq \tau_M \leq r_M(1 - \pi_M^{1/r_M}) < r_M, \quad \boxed{\mathcal{D}_k^2 \leq \tau_M \chi_M \leq r_M(1 - \pi_M^{1/r_M})\chi_M}. \quad (\text{CT.26})$$

To prove every factor, let $t_1, \dots, t_{r_M} \in (0, 1)$ be the positive eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$. The other eigenvalues are zero, so $\tau_M = \sum_i t_i$ and $\pi_M = \prod_i (1 - t_i)$. Induction on the number of factors proves $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$. For positive numbers with a fixed sum, their product is at most the product with all entries equal: a maximum exists on the closed nonnegative simplex, lies in its interior when the sum is positive, and any unequal pair can be replaced by its average to increase the product, since $((a+b)/2)^2 - ab = (a-b)^2/4$. Apply this fact to $1 - t_i$ to obtain $\pi_M \leq (1 - \tau_M/r_M)^{r_M}$, proving both bounds. The last two inequalities in (CT.26) now follow from (CT.24). For $r_M = 1$, $\tau_M = 1 - \pi_M$ exactly; for $r_M = 0$, $\Delta_M = 0$, $\pi_M = 1$, and $W_M = 0$, so this case requires no exponent $1/r_M$.

15 Density of the original joint boundaries and determinant dissipation

Retain the complete packet h , its degree $d \geq 1$, the arithmetic unit v_h , and every original map in Section 14. In particular

$$s_j = \frac{1}{2} + it_j, \quad v_h = g/h, \quad g = 2\xi, \quad d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad E^{(k)} = E_h^{\otimes k}, \quad \mathcal{H}_k = L^2(d^k x). \quad (\text{JD.1})$$

The one-variable cyclic-density argument in the coherent-interpolation source has a full joint-source extension. We prove it here, including the exact relation between the disappearing Hilbert representatives and their unchanged finite jets.

15.1 Polynomial density with the original weighted measures

Lemma 15.1. *Let μ be a finite positive Borel measure on $\mathbb{R}^k$ such that $\int e^{a \sum_j |t_j|} d\mu < \infty$ for some $a > 0$. The complex polynomials in $t_1, \dots, t_k$ are dense in $L^2(\mu)$.*

Proof. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial. Define the finite complex measure $d\eta = \bar{f} d\mu$. For $0 < b < a/2$, Cauchy–Schwarz gives

$$\int e^{b \sum_j |t_j|} d|\eta| \leq \|f\|_{L^2(\mu)} \left(\int e^{2b \sum_j |t_j|} d\mu \right)^{1/2} < \infty. \quad (\text{JD.2})$$

Thus $\Phi(z) = \int e^{i \sum_j z_j t_j} d\eta(t)$ is holomorphic in the connected tube $|\text{Im } z_j| < b$. On every smaller closed tube, each differentiated integrand is dominated by the integrable function in (JD.2), multiplied by a constant; the strict exponent margin absorbs each monomial. This justifies every derivative and the convergent Taylor series on small polydiscs. All derivatives at zero vanish by polynomial orthogonality. The Taylor series therefore vanishes near zero, and successive overlapping polydiscs along any path in the tube show $\Phi = 0$ throughout it. In particular the Fourier transform of η vanishes on $\mathbb{R}^k$.

For clarity, uniqueness here follows from an explicit approximate identity. Put

$$\gamma_\epsilon(x) = (4\pi\epsilon)^{-k/2} e^{-|x|^2/(4\epsilon)} = (2\pi)^{-k} \int e^{-\epsilon|u|^2} e^{iu \cdot x} du. \quad (\text{JD.3})$$

The second equality is the product Gaussian integral, obtained in one variable by differentiating its Fourier integral in x , integrating by parts in u , and evaluating its value at zero. The Gaussian integrability and finiteness of $|\eta|$ justify Fubini. Consequently $\eta * \gamma_\epsilon = 0$. For any continuous compactly supported φ , integration and Fubini give $\int (\varphi * \gamma_\epsilon) d\eta = 0$. These convolutions converge uniformly to φ : split the integral at a fixed small radius, use uniform continuity within it, and use the Gaussian tail and boundedness outside it. Hence $\int \varphi d\eta = 0$. Uniqueness of finite Borel measures from their continuous compactly supported integrals gives $\eta = 0$, and therefore $f = 0$ almost everywhere. The orthogonal complement of the polynomial closure is zero, proving the lemma. $\square$

The actual measure ν_h has every exponential moment of order less than $\pi/2$, with its full factor $dt/(2\pi)$, as proved in (GC.2). Multiplying it by any fixed polynomial in t preserves the same open range of exponential moments: choose a strictly larger exponent still below $\pi/2$ and absorb the polynomial into the difference. Product integration therefore verifies the lemma for both measures

$$\nu_h^{\otimes k}, \quad d\mu_{h,k}(t) = |h(s_1)|^2 \prod_{j=1}^k d\nu_h(t_j). \quad (\text{JD.4})$$

No factor h has been discarded. Its finitely many zeros on the first integration line have measure zero. The multiplication map

$$U_h : L^2(\mu_{h,k}) \longrightarrow L^2(\nu_h^{\otimes k}), \quad U_h P = h(s_1)P \quad (\text{JD.5})$$

is a unitary surjection: its norm identity follows directly from (JD.4), and its inverse is division by $h(s_1)$ away from that null set, with an arbitrary value on the null set. Polynomials in $s_j = 1/2 + it_j$ are precisely polynomials in t_j , since these coordinate changes are invertible. Thus (JD.5) and the lemma prove density of $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ in $L^2(\nu_h^{\otimes k})$. This dense space is contained in the original joint ideal $(h(s_1), \dots, h(s_k))$.

15.2 Strong limits of the actual source representatives

The Mellin Plancherel map, with its original sign and measure, is

$$(\mathcal{M}_k F)(t) = \int_{(0,\infty)^k} F(x) \prod_{j=1}^k x_j^{-1/2+it_j} d^k x, \quad \mathcal{M}_k : \mathcal{H}_k \longrightarrow L^2 \left(\frac{d^k t}{(2\pi)^k} \right). \quad (\text{JD.6})$$

Indeed the substitutions $x_j = e^{u_j}$ send F to $e^{\sum_j u_j/2} F(e^{u_1}, \dots, e^{u_k})$ isometrically in $L^2(d^k u)$, and (JD.6) becomes its Fourier transform with kernel $e^{it \cdot u}$. The inverse has kernel $e^{-it \cdot u}$ and factor $(2\pi)^{-k}$. Multiplication by $\prod_j v_h(s_j)$ is a unitary surjection from $L^2(\nu_h^{\otimes k})$ to the target in (JD.6). For surjectivity divide by this product almost everywhere: the entire nonzero function v_h has only isolated zeros on the line, and the product zero set is null. Composing with the inverse of (JD.6) gives the unitary extension

$$\mathcal{U}_{h,k} : L^2(\nu_h^{\otimes k}) \longrightarrow \mathcal{H}_k, \quad \mathcal{U}_{h,k} P = \mathcal{T}_h^{(k)} P \quad \text{for every polynomial } P. \quad (\text{JD.7})$$

The polynomial identity follows from the actual Mellin transform of F_h and D_j , so this is an extension of the original source map, not a change of representative.

Theorem 15.2 (Joint boundary density and exact finite-jet graph closure). *For each $k \geq 1$, the original total-degree boundary spaces $\mathcal{B}_M$ satisfy*

$$\overline{\bigcup_{M \geq k(d-1)} \mathcal{B}_M} = \mathcal{H}_k, \quad P_{\mathcal{B}_M} \longrightarrow I \quad \text{strongly.} \quad (\text{JD.8})$$

For the actual coherent minima of (CT.13), in the original fixed finite monomial tensor coordinates,

$$\|R_M\|_{E^{(k)} \rightarrow \mathcal{H}_k} \longrightarrow 0, \quad G_M \downarrow 0, \quad \det G_M \longrightarrow 0. \quad (\text{JD.9})$$

Here $G_M > 0$ at every finite M , and the middle limit is both decreasing in the positive-semidefinite order and convergent in matrix operator norm. Every finite jet is still retained: $J^{(k)} R_M = I$ for every M .

Define J_{pol} on the dense original source $\mathcal{D}_{h,k} = \mathcal{T}_h^{(k)} \mathbb{C}[s_1, \dots, s_k]$ by its complete arithmetic jets, including multiplication by $v_h^{\otimes k}$. Its graph has exactly the closure

$$\overline{\text{Graph}(J_{\text{pol}})} = \mathcal{H}_k \oplus E^{(k)}. \quad (\text{JD.10})$$

As a closed linear relation this has the split exact sequence

$$0 \longrightarrow E^{(k)} \xrightarrow{u \mapsto (0,u)} \overline{\text{Graph}(J_{\text{pol}})} \xrightarrow{(F,u) \mapsto F} \mathcal{H}_k \longrightarrow 0, \quad F \longmapsto (F, 0). \quad (\text{JD.11})$$

Proof. Every polynomial in $h(s_1) \mathbb{C}[s_1, \dots, s_k]$ belongs to one finite total-degree ideal $\mathcal{I}_{h,k,M}$. Its density proved in (JD.5), followed by the unitary map (JD.7), proves the first assertion in (JD.8). Given $F \in \mathcal{H}_k$ and $\epsilon > 0$, choose $b \in \mathcal{B}_N$ with $\|F - b\| < \epsilon$. For every $M \geq N$, the distance from F to $\mathcal{B}_M$ is at most that norm, so $\|(I - P_{\mathcal{B}_M})F\| < \epsilon$. This proves strong convergence.

Apply it to each of the d^k images of the fixed monomial basis under $s_h^{\otimes k}$. The identity $R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}$ makes each column tend to zero. For a vector with coordinate norm one, Cauchy-Schwarz bounds its image by the square root of the sum of the squared column norms, proving the first limit in (JD.9). Since $G_M = R_M^* R_M$ and $G_M - G_{M+1} = Y_M^* Y_M$, the norm limit and monotonicity follow. The determinant limit follows by continuity of the determinant in these same finite coordinates. Positivity and $J^{(k)} R_M = I$ are the proved finite interpolation identities; none of these finite algebraic equations is lost in taking the Hilbert norm limit.

The kernel of J_{pol} is exactly the union of the $\mathcal{B}_M$, so its graph closure contains $\mathcal{H}_k \oplus \{0\}$ by (JD.8). For any $u \in E^{(k)}$, the vectors $(R_M u, u)$ belong to its graph and converge to $(0, u)$ by (JD.9). The closure is a linear subspace, and therefore contains every (F, u) ; the reverse inclusion is its declared ambient space. This proves (JD.10), and the maps, exactness and section in (JD.11) follow immediately by evaluating their two coordinates. In particular the closure is a relation with vertical fibre $E^{(k)}$, and it cannot be the graph of a single-valued operator because $d^k > 0$. $\square$

There are corresponding exact limits for the original weight form and both interpolation kernels. Write $U = U_{v_h^{\otimes k}}$, let K_M be the remainder-only polynomial kernel on the joint source, and let $\mathcal{C}_M = G_M^{-1} = UK_M U^*$ be the complete arithmetic interpolation matrix. In the same fixed coordinates,

$$\begin{aligned} \|W_M\| &\leq (2\|A^{(k)}\| + k)\|G_M\| \longrightarrow 0, \\ \lambda_{\min}(\mathcal{C}_M) &= \|G_M\|^{-1} \longrightarrow \infty, \\ \lambda_{\min}(K_M) &\geq \frac{1}{\|U\|^2\|G_M\|} \longrightarrow \infty. \end{aligned} \tag{JD.9a}$$

The first line follows by the triangle inequality applied to the definition of W_M . Positivity of G_M proves the second. For the third, use $K_M = U^{-1}\mathcal{C}_M U^{-*}$ and $\|U^{-*}z\| \geq \|z\|/\|U\|$. Thus these coercive limits include every retained local jet, and the arithmetic unit remains in the comparison. The relative weight is related to its absolute form by the exact congruence

$$W_M = G_M^{1/2} S_M G_M^{1/2}, \quad \lambda_{\min}(G_M)\epsilon_M \leq \|W_M\| \leq \|G_M\|\epsilon_M. \tag{JD.9b}$$

The upper bound is submultiplicativity; applying it instead to $S_M = G_M^{-1/2} W_M G_M^{-1/2}$ gives the lower bound. Equations (JD.9a)–(JD.9b) retain the metric factors needed to pass from the proved absolute limits to the relative quantity.

The complementary quotient of the same closed relation is also explicit:

$$\frac{\overline{\text{Graph}(J_{\text{pol}})}}{\mathcal{H}_k \oplus \{0\}} \xrightarrow{\sim} E^{(k)}, \quad [(F, u)] \mapsto u, \quad u \mapsto [(0, u)]. \tag{JD.11a}$$

The kernel, well-definedness and both inverse identities follow by subtracting the two displayed pairs. With the product Hilbert norm the quotient norm is exactly $\|u\|$, since $\inf_{H \in \mathcal{H}_k} (\|F - H\|^2 + \|u\|^2)^{1/2} = \|u\|$, attained at $H = F$. Thus every completed-boundary class retains its original finite jet through an explicit isometric quotient.

This limit has an exact Split-Zero morphism. For a complex vector space X , use its one-support lift $\hat{X} = \{\tau_X\} \sqcup X^\bullet$, with τ_X the additive identity, vector addition within $X^\bullet$, and the original $G(\mathbb{C})$ -scalar action. In particular $0_X^\bullet \neq \tau_X$. Any linear map $T : X \rightarrow Y$ lifts to $\hat{T}(\tau_X) = \tau_Y$ and $\hat{T}(u^\bullet) = (Tu)^\bullet$; direct substitution checks addition and every scalar, including the supported scalar zero and external scalar zero. Give the supported fibre its norm topology and τ_X its isolated topology. Equation (JD.9) then says exactly

$$\hat{R}_M \longrightarrow \hat{0}, \quad \hat{0}(\tau_E) = \tau_{\mathcal{H}}, \quad \hat{0}(u^\bullet) = 0_{\mathcal{H}}^\bullet \quad (u \in E^{(k)}). \tag{JD.12}$$

Thus the limiting morphism retains the support of every input. Equation (JD.11) identifies all finite jets in the vertical fibre of the closed relation; it supplies the full typed relation between these supported Hilbert limits and the unchanged algebraic jets.

15.3 The accumulated determinant cost of the same boundary sequence

Continue to use the original ordered spectral sum and layer energy

$$\mathcal{D}_k = \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k (\text{Re } \rho_{i_j} - 1/2) \right|, \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M), \quad \tau_M = \text{Tr}(G_M^{-1} (G_M - G_{M+1})). \tag{JD.13}$$

The list $\rho_1, \dots, \rho_d$ includes every algebraic multiplicity. Define $\pi_M = \det G_{M+1} / \det G_M$. For integers $k(d-1) \leq a < b$, the following exact finite inequality holds whenever $\mathcal{D}_k > 0$:

$$\boxed{\mathcal{D}_k^2 \sum_{M=a}^{b-1} \frac{1}{\chi_M} \leq \sum_{M=a}^{b-1} \tau_M \leq \log \frac{\det G_a}{\det G_b}.} \tag{JD.14}$$

All denominators in this statement are proved positive. Indeed (CT.24) gives $\mathcal{D}_k^2 \leq \tau_M \chi_M$, with both factors nonnegative; strict positivity of the left side forces strict positivity of each factor. Let $t_{M,j} \in [0, 1)$ be the full eigenvalue list of $G_M^{-1/2}(G_M - G_{M+1})G_M^{-1/2}$, including zeros. The finite original matrices give exactly

$$\tau_M = \sum_j t_{M,j}, \quad \pi_M = \prod_j (1 - t_{M,j}), \quad -\log \pi_M - \tau_M = \sum_j \int_0^{t_{M,j}} \frac{u}{1-u} du \geq 0. \quad (\text{JD.15})$$

The first two identities are congruence and determinant identities; the last follows by integrating the derivative of $-\log(1-t) - t$. Sum (CT.24) divided by χ_M , then sum (JD.15). The product of the determinant ratios telescopes without a change of coordinates, proving (JD.14).

If $\mathcal{D}_k = 0$, every repeated tuple $(i, \dots, i)$ has zero summand, hence $\text{Re } \rho_i = 1/2$ for every retained zero. Conversely these equalities give $\mathcal{D}_k = 0$. The second inequality in (JD.14) and the exact remainder (JD.15) hold also in this case; no quotient $0/0$ is introduced. Equation (JD.9) proves that the right-hand side tends to infinity as $b \rightarrow \infty$ for each fixed a . Equations (JD.13)–(JD.15) retain the actual rate through χ_M , the complete relative Gram losses and their explicit logarithmic remainder. They impose no unproved upper estimate on that rate.

The same sequence supplies two further exact energy identities:

$$G_a = \sum_{M=a}^{\infty} Y_M^* Y_M, \quad \sum_{M=a}^{\infty} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M=a}^{\infty} \tau_M = \infty. \quad (\text{JD.16})$$

The first series converges in matrix operator norm, by telescoping $G_M - G_{M+1} = Y_M^* Y_M$ and the proved limit $G_M \rightarrow 0$. Taking traces of its finite partial sums proves the second identity; all summands are nonnegative. To prove the third, suppose instead that $\sum_M \tau_M$ were finite. Then beyond some index all $\tau_M \leq 1/2$, so every $0 \leq t_{M,j} \leq 1/2$. Integrating $(1-u)^{-1} \leq 2$ over $[0, t_{M,j}]$ gives $-\log(1-t_{M,j}) \leq 2t_{M,j}$. Hence the tail sum of $-\log \pi_M$ would be bounded by $2 \sum_M \tau_M < \infty$. The finite initial terms are finite because $G_M > 0$. Telescoping would therefore bound $\log(\det G_a / \det G_b)$ uniformly in b , contradicting $\det G_b \rightarrow 0$. This proves the divergence, including packets entirely on the critical line and every retained nilpotent chain. The equality $\tau_M = \|Y_M G_M^{-1/2}\|_{\text{HS}}^2$ identifies the relative energy in this divergence with the same layer map appearing in the convergent absolute energy series.

16 The actual kernel layer, its derivative image, and its quotient

This section integrates the complete kernel-layer delivery of 12 September 2026. Its archive hash and member hashes are recorded in `sources/web_kernel_layer_delivery/LOCAL_STAGING_PROVENANCE.json`. The source note's formulas (9)–(28) identify the interpolation kernel with an explicit basis of the next original theta relation layer. We first prove those maps in the retained coordinates and then calculate the exact derivative image and the entire quotient of that layer. The new quotient calculation retains the part of the layer not reached by the summed Euler derivative.

16.1 The original arithmetic source and its constrained minimum

Keep $g = 2\xi$, a nonempty full packet $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, and $d = \deg h$. Fix $k \geq 1$, and set

$$E = E_h^{\otimes k}, \quad E_h = \mathbb{C}[s]/(h), \quad A = \sum_{j=1}^k A_{h,j}, \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times. \quad (\text{KL.1})$$

All tensor coordinates use the original monomial remainders. The function F_h , with Mellin transform g/h , is the original test function satisfying $h(D)F_h = f_0$. The exact source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{M}_k \mathcal{T}(P) = P(s_1, \dots, s_k) \prod_{j=1}^k v_h(s_j), \quad s_j = \frac{1}{2} + it_j. \quad (\text{KL.2})$$

The Mellin multiplier proves injectivity on polynomials. Its Hilbert norm is the polynomial norm for the unchanged product measure

$$d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad \nu_h^{\otimes k}. \quad (\text{KL.3})$$

Let $\mathcal{P}_N$ be the polynomials of total degree at most N , and let $\mathcal{J}_N : \mathcal{P}_N \rightarrow E$ be full reduction by $h(s_1), \dots, h(s_k)$, followed by multiplication by the original unit $v_h^{\otimes k}$. For $N \geq k(d-1)$ this is onto, since the monomial remainder basis has at most that total degree. Fix an integer $N \geq k(d-1)$ for the remainder of this section; all subsequent levels $N+1, N+2$ satisfy the same threshold. Its kernel and its original theta image are

$$\mathcal{I}_N = \mathcal{P}_N \cap (h(s_1), \dots, h(s_k)), \quad \mathcal{L}_N = \mathcal{T}\mathcal{I}_N. \quad (\text{KL.4})$$

Let $p_j(s)$ be the actual monic orthogonal polynomial of ν_h , and retain $\kappa_j = \|p_j\|_{\nu_h}^2 > 0$. All moments exist by the proved arithmetic decay. The Gram is strictly positive because the density is positive except at a discrete set; this proves existence and uniqueness by finite orthogonal projection. For multi-indices set

$$p_\alpha = \prod_j p_{\alpha_j}, \quad \kappa_\alpha = \prod_j \kappa_{\alpha_j}, \quad a_\alpha = \bigotimes_j (v_h[p_{\alpha_j}]_h). \quad (\text{KL.5})$$

The monic triangular change preserves the full space of degree at most N , and product integration makes its Gram diagonal. It follows, by direct constrained minimization, that the original canonical representative and its metric are

$$\mathcal{K}_N = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha} > 0, \quad G_N = \mathcal{K}_N^{-1}, \quad R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (\text{KL.6})$$

Indeed the full jet of this vector is $\mathcal{K}_N G_N u = u$. If a coefficient vector has zero full jet, its inner product with the displayed coefficient vector is zero. Every other representative of that same jet is its sum with such a vector; its squared norm is therefore $u^* G_N u$ plus the squared norm of the added relation. This proves the unique minimum, its metric, and $R_N \perp \mathcal{L}_N$. These are the original maps $J^{(k)} R_N = I$ and $q^{(k)} R_N = \sigma$. The notation $\mathcal{K}_N$ here includes the local unit. To specify the comparison in the full tensor dimension, put

$$r_\alpha = \bigotimes_j [p_{\alpha_j}]_h, \quad K_N^{\text{rem}} = \sum_{|\alpha| \leq N} \frac{r_\alpha r_\alpha^*}{\kappa_\alpha}.$$

Then $\mathcal{K}_N = U_{v_h^{\otimes k}} K_N^{\text{rem}} U_{v_h^{\otimes k}}^*$, as an equality of operators on $E_h^{\otimes k}$. When $k = 1$, K_N^{rem} is exactly the earlier remainder-only kernel K_N .

16.2 The complete next-layer frame and its theta primitives

Write $\Sigma_n = \{\alpha \in \mathbb{N}^k : |\alpha| = n\}$ and $\mathbf{H}_n = \mathbb{C}^{\Sigma_n}$. The coordinate identification sends $x \in \mathbf{H}_n$ to the homogeneous polynomial $\sum_\alpha x_\alpha z^\alpha$; these z_j are coefficient variables and are not substituted for the original Mellin variables s_j . For $\beta \in \Sigma_{N+1}$, define the actual functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \mathcal{E}_N x = \sum_\beta x_\beta e_\beta, \quad \mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp. \quad (\text{KL.7})$$

The full jet of e_β is zero, hence it is in $\mathcal{L}_{N+1}$. The new-degree polynomial is orthogonal to all old polynomials and $R_N a_\beta \perp \mathcal{L}_N$, so $e_\beta \in \mathcal{E}_N$. The leading total-degree terms of the e_β are the distinct monomials s^β , proving independence by the injectivity of $\mathcal{T}$. For any element of $\mathcal{E}_N$, subtract its full degree- $N+1$ combination of the e_β . The remainder has zero jet and degree at most N , hence is in $\mathcal{L}_N$. It is also orthogonal to that space, so it is zero. This proves the exact isomorphism

$$\mathcal{E}_N : \mathcal{H}_{N+1} \xrightarrow{\sim} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1}/\mathcal{L}_N. \quad (\text{KL.8})$$

The inverse of the second arrow is $[f] \mapsto (I - P_{\mathcal{L}_N})f$.

Every vector in (KL.7) has a primitive in the original tensor complex. Its numerator polynomial is

$$p_\beta - \sum_{|\alpha| \leq N} p_\alpha \frac{a_\alpha^* G_N a_\beta}{\kappa_\alpha}. \quad (\text{KL.9})$$

Successive monic division in the ordered variables $s_1, \dots, s_k$ expresses this zero-jet polynomial as $\sum_i h(s_i) Q_{i,\beta}$, where $\deg Q_{i,\beta} \leq N+1-d$. At each replacement the leading power of the divided variable decreases and total degree does not increase, proving the bound. After the last division, the remainder has degree below d in every variable; its full jet is zero, so uniqueness of the complete remainder forces it to vanish. In the i -th tensor factor replace F_h by $(-1)^{i-1} \phi_*$, and apply $Q_{i,\beta}(D_1, \dots, D_k)$ to the whole tensor. The preceding $i-1$ degree-one factors give the differential sign $(-1)^{i-1}$. Together with $h(D)F_h = \Theta \phi_*$, the boundary of this primitive is exactly $+\mathcal{T}(h(s_i)Q_{i,\beta})$. Summing proves the asserted primitive with all ordered signs retained.

Put

$$\mathbf{U}_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad \mathbf{D}_{N+1} = \text{diag}(\kappa_\beta), \quad \mathbf{H}_N^{\text{lay}} = \mathcal{E}_N^* \mathcal{E}_N.$$

Expanding (KL.7) and using old/new polynomial orthogonality proves

$$\mathbf{H}_N^{\text{lay}} = \mathbf{D}_{N+1} + \mathbf{U}_N^* G_N \mathbf{U}_N > 0, \quad \mathcal{E}_N^* R_N = -\mathbf{U}_N^* G_N. \quad (\text{KL.10})$$

Define the projection loss $Y_N = P_{\mathcal{E}_N} R_N$. In particular its exact formula is $Y_N = -\mathcal{E}_N (\mathbf{H}_N^{\text{lay}})^{-1} \mathbf{U}_N^* G_N$. Its rank is exactly $\text{rank } \mathbf{U}_N$, because every other displayed factor is injective or invertible on its declared space.

16.3 The exact incidence map and the derivative-image rank

Define the actual old top-degree columns and their original norms by

$$\mathbf{A}_N = (a_\alpha)_{\alpha \in \Sigma_N} : \mathcal{H}_N \rightarrow E, \quad \mathbf{D}_N = \text{diag}(\kappa_\alpha)_{\Sigma_N}. \quad (\text{KL.11})$$

Define the coordinate incidence map by

$$\mathbf{T}_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}, \quad (\mathbf{T}_n x)_\beta = \sum_{i: \beta_i > 0} x_{\beta - \mathbf{e}_i}. \quad (\text{KL.12})$$

On the specified homogeneous coefficient polynomials this is multiplication by $\ell(z) = z_1 + \dots + z_k$. A product of two nonzero polynomials over $\mathbb{C}$ is nonzero: choose a monomial order compatible with products and multiply the two leading terms. Since $\ell \neq 0$, $\mathbf{T}_n$ is injective.

The full unprojected derivative defect and its projected layer map are

$$B_N = D^{(k)} R_N - R_N A, \quad C_N = (I - P_{\mathcal{L}_N}) B_N : E \rightarrow \mathcal{E}_N.$$

They have the exact coordinate factorization

$$\boxed{C_N = \mathcal{E}_N \mathbf{T}_N \mathbf{D}_N^{-1} \mathbf{A}_N^* G_N.} \quad (\text{KL.13})$$

To prove it, multiply the finite numerator in (KL.6) by $s_1 + \cdots + s_k$. Monicity makes its coefficient at the new top polynomial p_β equal to $\sum_{i:\beta_i>0} a_{\beta-\mathbf{e}_i}^* G_N u / \kappa_{\beta-\mathbf{e}_i}$, precisely the row on the right side of (KL.13). The numerator of $R_N A u$ has degree at most N . Subtract the right side of (KL.13) from $B_N u$: its numerator has degree at most N and its full jet is zero, so it is an original relation in $\mathcal{L}_N$. Orthogonal projection proves (KL.13), including the original quotient statement $[B_N u] = [C_N u]$ in $\mathcal{B}^{(k)} / \mathcal{L}_N$.

Theorem 16.1 (Exact derivative rank and its full kernel). *With every original jet column retained,*

$$\boxed{\text{rank } C_N = \text{rank } A_N, \quad \ker C_N = \mathcal{K}_N \ker A_N^*} \quad (\text{KL.14})$$

Define the original weight form $W_N = A^* G_N + G_N A - k G_N$. It obeys the refined bound

$$\text{rank } W_N \leq 2 \min\{\text{rank } A_N, \text{rank } U_N\}. \quad (\text{KL.15})$$

Proof. Both $\mathcal{E}_N$ and $\mathbf{T}_N$ are injective; D_N and G_N are invertible. Thus (KL.13) has rank $\text{rank } A_N^*$, which equals the rank of its conjugate transpose. Its kernel is $G_N^{-1} \ker A_N^* = \mathcal{K}_N \ker A_N^*$, with the displayed invertible coordinate map retained. The original Green identity and orthogonality give $W_N = -(Y_N^* C_N + C_N^* Y_N)$. Each summand has rank at most the lesser of the two map ranks; the rank of a sum is at most the sum of its ranks. Use (KL.10) and (KL.14) to obtain (KL.15). No nonzero homogeneous-leading-term assertion has been substituted for an actual column a_α . $\square$

16.4 The whole quotient of the next layer by the derivative image

For $k \geq 2$, let H'_n be the homogeneous polynomials of degree n in $z_1, \dots, z_{k-1}$. There is a split exact sequence of vector spaces

$$0 \longrightarrow H_n \xrightarrow{\mathbf{T}_n} H_{n+1} \xrightarrow{\rho_n} H'_{n+1} \longrightarrow 0, \quad \rho_n f = f(z_1, \dots, z_{k-1}, -\sum_{i < k} z_i). \quad (\text{KL.16})$$

For a complete proof and inverse, divide f by the monic linear polynomial $\ell = z_k + \sum_{i < k} z_i$ in the last variable. Every replacement retains total degree, giving uniquely

$$f = \ell q + \iota r, \quad q \in H_n, \quad r = \rho_n f \in H'_{n+1}, \quad (\text{KL.17})$$

where ι includes a polynomial independent of z_k . For uniqueness set $z_k = -\sum_{i < k} z_i$ first to obtain $r = 0$, then use injectivity of multiplication by ℓ . This also proves surjectivity, exactness and the specified section ι . The ordered last coordinate and the signs in the substitution remain part of the map.

Theorem 16.2 (Exact unused-layer quotient). *Set $Z_N = \text{im}(D_N^{-1} A_N^*) \subseteq H_N$. Then the actual quotient is*

$$\boxed{\mathcal{E}_N / \text{im } C_N \simeq (H_N / Z_N) \oplus H'_{N+1}} \quad (\text{KL.18})$$

For an original layer representative $e = \mathcal{E}_N f$, divide $f = \ell q + \iota r$ by (KL.17); the map sends $[e]$ to $([q], r)$. Its inverse sends $([q], r)$ to $[\mathcal{E}_N(\ell q + \iota r)]$.

Proof. The invertibility of G_N in (KL.13) gives $\text{im } C_N = \mathcal{E}_N \mathbf{T}_N Z_N$. Changing e by such an element changes exactly q by an element of Z_N , while leaving r unchanged. Thus the first map is well-defined. Changing the representative of $[q]$ has the same effect on the inverse, so the inverse is well-defined. The unique division (KL.17) proves both inverse identities. For $k = 1$, $\mathbf{T}_N$ is an isomorphism between the one-dimensional shell spaces and the same statement holds with $H'_{N+1} = 0$. $\square$

In particular $\dim H'_{N+1} = \binom{N+k-1}{k-2}$ for $k \geq 2$, by counting monomials; the first summand has dimension $\binom{N+k-1}{k-1} - \text{rank } A_N$. Their sum is exactly the original layer dimension minus $\text{rank } C_N$. The quotient therefore retains both the full-jet rank loss and the explicitly complementary relation directions. For every coordinate vector, (KL.9) supplies its original theta primitive. The quotient is of actual test-function relations, with those primitives, rather than only a quotient of dimensions.

16.5 The derivative between consecutive relation quotients

The original Euler sum has $D^{(k)}\mathcal{L}_N \subseteq \mathcal{L}_{N+1}$, because it multiplies the numerator polynomial by $s_1 + \cdots + s_k$ and preserves the ideal in (KL.4). Hence it induces the well-typed map

$$\partial_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [D^{(k)}f]. \quad (\text{KL.19})$$

Under the exact layer coordinates of (KL.8), its matrix is

$$\partial_N = \mathcal{E}_{N+1} \mathsf{T}_{N+1} \mathcal{E}_N^{-1}, \quad (\text{KL.20})$$

where each $\mathcal{E}$ is followed or preceded by its quotient isomorphism as in (KL.8). Write $e_\beta^{(N)}$ for (KL.7) at level N , and $e_{\beta+\mathbf{e}_i}^{(N+1)}$ for the next layer. The actual recurrence $sp_j = p_{j+1} + b_j p_j - (\kappa_j/\kappa_{j-1})p_{j-1}$ gives the complete retained difference

$$\begin{aligned} D^{(k)}e_\beta^{(N)} - \sum_i e_{\beta+\mathbf{e}_i}^{(N+1)} = \mathcal{T} \left[\left(\sum_i b_{\beta_i} \right) p_\beta - \sum_{i:\beta_i > 0} \frac{\kappa_{\beta_i}}{\kappa_{\beta_i-1}} p_{\beta-\mathbf{e}_i} \right] \\ - D^{(k)}R_N a_\beta + R_{N+1} \sum_i a_{\beta+\mathbf{e}_i}. \end{aligned}$$

The final term in the one-variable recurrence is omitted at $j = 0$. Every term on the right has numerator degree at most $N + 1$. Each term on the left has zero full jet, so their difference is in $\mathcal{L}_{N+1}$. This proves (KL.20) with all lower polynomial coefficients retained. Applying the explicit ordered division in (KL.9) to this displayed numerator supplies its original theta primitive.

It follows that ∂_N is injective and its cokernel is exactly H'_{N+2} , with maps (KL.16)–(KL.17) at index $N + 1$. Thus the original derivative connects consecutive relation quotient types by a proved injection; it has not been made an endomorphism of a fixed quotient. The ordinary relation-admission map has the same declared source and target:

$$\text{adm}_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [f] = 0.$$

Equation (KL.19) instead sends the representative to its next derivative class. These are explicitly different typed maps of the same original representative, related by (KL.20) and the inclusion of their actual numerator relations.

All maps here have their original supported lifts. At a fixed support label λ , a linear map T acts by $(\lambda, x) \mapsto (\lambda, Tx)$. The relation-admission map therefore sends a killed class to the receiving fibre's supported zero. The external zero τ remains external absence. Each displayed kernel and quotient records its full supported vector space and each derivative retains its consecutive source and target; no passage to the Hilbert closure is used.

16.6 The exact original-theta norm and volume bridge

Specialize to $k = 1$. We now distinguish the degree-indexed G_N of (KL.6) from the correction-indexed original metric by writing

$$\mathbf{G}_m = G_{d+m} \quad (m \geq -1), \quad \mathbf{R}_m = R_{d+m}, \quad r_m = \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{m-1}} \quad (m \geq 0). \quad (\text{KL.21})$$

At $m = -1$, $\mathcal{P}_{d-1}$ contains no nonzero multiple of h . Thus the original representative is exactly the fixed section

$$s_h u = \mathbf{R}_{-1} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u), \quad \varepsilon_h = v_h^{-1}, \quad \mathbf{G}_{-1} = s_h^* s_h.$$

This fixes the initial determinant in (KL.21), without introducing a negative-degree theta polynomial. Put $L_{-1} = 0$ and

$$f_j = D^j f_0, \quad L_j = \text{span}(f_0, \dots, f_j), \quad u_j = (I - P_{L_{j-1}})f_j, \quad \mathfrak{h}_j = \|u_j\|^2 > 0 \quad (j \geq 0). \quad (\text{KL.22})$$

These $u_j = Q_j(D)f_0$ are the monic orthogonal theta vectors for the original measure $|g(1/2 + it)|^2 dt/(2\pi)$. Their definition does not depend on h . The monic p_j and their norms κ_j continue to use the original packet-dependent measure $|g/h|^2 dt/(2\pi)$. Multiplication by the unchanged monic polynomial h is the exact isometry connecting their source polynomial spaces:

$$P \mapsto hP, \quad \|hP\|_{\nu_h}^2 = \|P\|_{|g|^2 dt/(2\pi)}^2, \quad \mathcal{T}_h(hP) = P(D)f_0. \quad (\text{KL.23})$$

This follows by cancellation of the pointwise factor $|h|^2$ in the integral; the identities extend across the discrete zeros by their entire expressions. It is an isometry onto the specified subspace of multiples of h , with its degree shifted by the full degree $d = \sum_\rho m_\rho$.

Theorem 16.3 (Original norm as an exact relative Gram volume). *For every $j \geq 0$, including the initial theta vector,*

$$\boxed{\kappa_{d+j} = \mathfrak{h}_j r_j}. \quad (\text{KL.24})$$

Consequently, for $m \geq 0$,

$$\boxed{\frac{\kappa_{d+m+1}}{\kappa_{d+m}} = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} \frac{r_{m+1}}{r_m}, \quad \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{-1}} = \prod_{j=0}^m \frac{\kappa_{d+j}}{\mathfrak{h}_j}}. \quad (\text{KL.25})$$

All ratios have positive denominators and all packet multiplicities are retained.

Proof. At the degree level $N = d + j - 1$, define $e_{d+j} = \mathcal{T}_h p_{d+j} - \mathbf{R}_{j-1} a_{d+j}$. Its numerator has zero full jet, is monic of degree $d + j$, and is divisible by the original monic h . By ordinary monic division its quotient is monic of degree j . Under (KL.23) this is f_j plus a vector in L_{j-1} . By (KL.7) it is orthogonal to that space. The defining orthogonal projection in (KL.22) therefore proves

$$e_{d+j} = u_j, \quad \mathfrak{h}_j = \kappa_{d+j} + a_{d+j}^* \mathbf{G}_{j-1} a_{d+j}. \quad (\text{KL.26})$$

When $j = 0$, the quotient of the numerator by h is the constant 1, so the same argument explicitly gives $e_d = f_0 = u_0$; no previous theta layer is required.

For the full unit cancellation put $v_l = [p_l]_h = p_l(A_h)[1]_h$, and $K_{l-1} = \sum_{i=0}^{l-1} v_i v_i^* / \kappa_i$, using the original monomial remainder coordinates. At $l = d + j$,

$$a_l = U_{v_h} v_l, \quad \mathbf{G}_{j-1} = U_{\varepsilon_h}^* K_{l-1}^{-1} U_{\varepsilon_h}, \quad a_l^* \mathbf{G}_{j-1} a_l = v_l^* K_{l-1}^{-1} v_l. \quad (\text{KL.27})$$

Both cancellations use $U_{\varepsilon_h} U_{v_h} = I$; hence they preserve every derivative of the original local unit at every full jet. The degree update $K_l = K_{l-1} + v_l v_l^* / \kappa_l$ has determinant ratio $1 + v_l^* K_{l-1}^{-1} v_l / \kappa_l$. To verify this scalar identity directly, expand $\det(I + xy^*)$ multilinearly in its columns: all terms with two updated columns vanish since those columns are multiples of x , leaving $1 + y^* x$. The fixed-unit congruences in (KL.27) then give

$$r_j = \frac{\kappa_l}{\kappa_l + v_l^* K_{l-1}^{-1} v_l} = \frac{\kappa_{d+j}}{\mathfrak{h}_j}.$$

This also proves $0 < r_j \leq 1$, including the case $v_l = 0$. Divide two consecutive identities and multiply the first $m + 1$ identities to obtain (KL.25). $\square$

For two full packets h, H , their original norm sequences are therefore connected at the same theta degree j by the exact scalar equality

$$\frac{\kappa_{\deg h+j}^{(h)}}{r_j^{(h)}} = \mathfrak{h}_j = \frac{\kappa_{\deg H+j}^{(H)}}{r_j^{(H)}}. \quad (\text{KL.28})$$

When $h \mid H$, the corresponding source isometry is $P \mapsto (H/h)P$, since $(g/H)(H/h)P = (g/h)P$. At the old centres this equality preserves every jet, and at each added centre g/h has the complete zero order, so it realizes exactly the CRT insertion of zero jets. The reference numerator has degree below $\deg H$ and is its unique full remainder. Both corrected representatives then subtract the same projection onto the packet-independent L_j , proving the coherent equality $\mathbf{R}_{H,j} \iota_{hH} = \mathbf{R}_{h,j}$, including $j = -1$. Thus (KL.28) connects precisely these original coherent source maps and their fixed-section metrics.

16.7 The complete scalar dictionary for the boundary energy

For $m \geq 0$, set $n = d + m + 1$ and retain the terminal notation

$$M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}, \quad \alpha_n = v_{n-1}^* M_n v_{n-1}, \quad \beta_n = v_n^* M_n v_n, \quad \gamma_n = v_n^* M_n v_{n-1}.$$

Define the original theta projection rows by

$$d_j = \mathfrak{h}_j^{-1} u_j^* s_h : E_h \rightarrow \mathbb{C}, \quad \mathbf{R}_m = s_h - \sum_{j=0}^m u_j d_j, \quad c_m = d_m \mathbf{G}_m^{-1} d_{m+1}^*. \quad (\text{KL.29})$$

Here an expression $u_j^* F$ denotes the Hilbert inner product $\langle u_j, F \rangle$, conjugate-linear in its first argument. The row d_j therefore retains the original unnormalized vector and the exact factor $\mathfrak{h}_j^{-1}$.

The complete row and scalar comparison is

$$\boxed{\begin{aligned} d_m &= -\frac{v_{n-1}^* K_{n-1}^{-1} U_{\varepsilon_h}}{\kappa_{n-1}}, \\ d_{m+1} &= -\frac{v_n^* K_{n-1}^{-1} U_{\varepsilon_h}}{\mathfrak{h}_{m+1}}, \\ d_m \mathbf{G}_m^{-1} d_m^* &= \frac{\alpha_n}{\kappa_{n-1}}, \\ d_{m+1} \mathbf{G}_m^{-1} d_{m+1}^* &= \frac{\kappa_{n-1} \beta_n}{\mathfrak{h}_{m+1}^2}, \\ \mathfrak{h}_{m+1} c_m &= \overline{\gamma_n}. \end{aligned}} \quad (\text{KL.30})$$

To prove the first row, the polynomial numerator of s_h has degree below d . In (KL.29), each u_j has numerator hQ_j , monic of degree $d+j$. Thus the coefficient of degree $d+m = n-1$ in the numerator of $\mathbf{R}_m$ is $-d_m$. The finite expansion (KL.6) gives the same coefficient as $a_{n-1}^* \mathbf{G}_m / \kappa_{n-1}$. Cancelling the same units as in (KL.27) proves the first row, also at $m = 0$. For the second row use $u_{m+1} = e_n$, so (KL.10) gives $u_{m+1}^* \mathbf{R}_m = -a_n^* \mathbf{G}_m$. The orthogonality of u_{m+1} to L_m also gives $u_{m+1}^* \mathbf{R}_m = u_{m+1}^* s_h = \mathfrak{h}_{m+1} d_{m+1}$. This proves its formula. Finally $\mathbf{G}_m^{-1} = U_{v_h} K_{n-1} U_{v_h}^*$; insert both rows, cancel both unit pairs and the adjacent kernel inverse. The three remaining products are exactly those in (KL.30), including the conjugation in the last line.

Theorem 16.4 (Exact original-norm formula for the reflected packet). *Suppose the original full packet is stable under $\rho \mapsto 1 - \bar{\rho}$. For its actual relative allowance ϵ_m ,*

$$\boxed{\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2.} \quad (\text{KL.31})$$

If the packet is also stable under ordinary conjugation, then $c_m = 0$, so its complete formula becomes

$$\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{KL.32})$$

Proof. The terminal identity (KT.4), with the original full multiplicities, gives $\text{Re } \gamma_n = \text{Re tr } A_h - d/2 = 0$: reflected centres contribute complementary real parts with equal orders. Its two possible nonzero generalized eigenvalues then yield $\epsilon_m^2 = \alpha_n \beta_n - |\gamma_n|^2$. The preceding and succeeding rank-one kernel updates, also proved directly in (KL.27), give

$$\alpha_n = 1 - r_m, \quad \beta_n = \frac{\kappa_n}{\kappa_{n-1}} \frac{1 - r_{m+1}}{r_{m+1}}.$$

Insert (KL.25) into the product $\alpha_n \beta_n$ and the last line of (KL.30) into $|\gamma_n|^2$. Every positive ratio is retained, and this is (KL.31).

For the final assertion, conjugation stability makes h real in the original monomial coordinates. The real Taylor coefficients of g give $(g/h)(1/2 - it) = \overline{(g/h)(1/2 + it)}$, so the actual ν_h is even in t . Its monomial Gram matrix is real: the integrand for entry $s^i s^j$ at $-t$ is the complex conjugate of its integrand at t . Therefore the unique solution of the real Gram linear equations for the monic p_l has real coefficients. Polynomial division by the real monic h makes each v_l real, and the finite sum for K_{n-1} is real. Its inverse and M_n are real as well. Consequently γ_n is real. The already proved reflection trace identity makes its real part zero, hence $\gamma_n = 0$. Equation (KL.30), with $\mathfrak{h}_{m+1} > 0$, proves $c_m = 0$. No division by a jet column, an off-line displacement, or a boundary energy has been used; all zero-column cases remain included. $\square$

The source PR15 denotes the relative determinant at correction m by $\delta_m = \det \mathbf{G}_{m+1} / \det \mathbf{G}_m$. Its exact index map is therefore $\delta_m = r_{m+1}$, with $\delta_{-1} = r_0$. Its cross-term row product is the c_m of (KL.29). Equations (KL.24)–(KL.31) prove the full map from its original-theta scalar formula to the terminal-kernel formula, including both monic norm families, the unit congruences, the conjugation, the fixed initial section, and the complete orders. These equalities calculate the same actual boundary energy at every finite stage. They do not supply an estimate tending to zero; that requires further information about these now explicitly related arithmetic quantities.

17 Original orthogonal shells and the exact projective-ratio algebra

The source's degree shell consists of its original orthogonal product polynomials. This section proves its full comparison with the homogeneous monomial shell, including every lower-degree coefficient. It also calculates the algebra generated by the ratio of the two original tensor coordinates. The arithmetic local unit and all multiplicities remain in the resulting linear maps.

17.1 Two sections of the same polynomial filtration

Retain a complete nonempty arithmetic packet

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(g/h), \quad g = 2\xi. \quad (\text{KPC.1})$$

The complete orders make v_h a unit. For $k \geq 1$, write $E^{(k)} = E_h^{\otimes k}$ in its original monomial tensor coordinates and use the original full-jet map

$$J(P) = U_{v_h^{\otimes k}}[P]_{(h(s_1), \dots, h(s_k))}, \quad J : \mathbb{C}[s_1, \dots, s_k] \longrightarrow E^{(k)}. \quad (\text{KPC.2})$$

Here U_a is multiplication by the stated element a . The integration variables remain $s_j = 1/2 + it_j$, with

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{KPC.3})$$

Let $p_j(s) = \sum_{r=0}^j p_{j,r} s^r$, with $p_{j,j} = 1$, be the original monic orthogonal polynomials for this measure. Retain $\kappa_j = \int |p_j(s)|^2 d\nu_h > 0$. For a multi-index α , put

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad a_\alpha = J(p_\alpha). \quad (\text{KPC.4})$$

All moment integrals in these definitions use the original density.

For $N \geq 0$, let $\mathcal{P}_{\leq N}$ be the polynomials of total degree at most N , let $\mathcal{P}_{\leq -1} = 0$, and let $\mathcal{H}_N$ be the homogeneous polynomials of degree N . The highest-degree map $\ell_N : \mathcal{P}_{\leq N} \rightarrow \mathcal{H}_N$ has kernel $\mathcal{P}_{\leq N-1}$. It therefore identifies $\mathcal{H}_N$ with the associated-graded space $\mathcal{P}_{\leq N}/\mathcal{P}_{\leq N-1}$. There are two specified sections:

$$\begin{aligned} i_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha s^\alpha, \\ O_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha p_\alpha. \end{aligned} \quad (\text{KPC.5})$$

Theorem 17.1 (The exact lower-degree correction). *Both maps in (KPC.5) are linear sections of ℓ_N . Their difference $L_N = O_N - i_N$ has target $\mathcal{P}_{\leq N-1}$ and is exactly*

$$L_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) = \sum_{|\gamma| < N} \left(\sum_{\substack{|\alpha|=N \\ \gamma_j \leq \alpha_j \ (1 \leq j \leq k)}} c_\alpha \prod_{j=1}^k p_{\alpha_j, \gamma_j} \right) s^\gamma. \quad (\text{KPC.6})$$

The image of O_N is the orthogonal complement of $\mathcal{P}_{\leq N-1}$ in $\mathcal{P}_{\leq N}$ for the original product measure, and its basis in (KPC.5) has Gram $\text{diag}(\kappa_\alpha)_{|\alpha|=N}$. Define the two full residue maps on the same $\mathcal{H}_N$ by

$$A_N^{\text{hom}} = J i_N, \quad A_N^{\text{orth}} = J O_N.$$

Then, with $J_{<N} = J|_{\mathcal{P}_{\leq N-1}}$,

$$\boxed{A_N^{\text{orth}} = A_N^{\text{hom}} + J_{<N} L_N.} \quad (\text{KPC.7})$$

In particular the original orthogonal-shell columns of A_N^{orth} are exactly a_α .

Proof. Expand each factor in p_α . Its coefficient of s^γ is $\prod_j p_{\alpha_j, \gamma_j}$ when $\gamma_j \leq \alpha_j$ for all j , and is zero otherwise. If $|\alpha| = |\gamma| = N$, these inequalities force $\gamma = \alpha$, whose coefficient is one. This proves both section identities and the complete expression (KPC.6). For a monomial s^γ of degree less than $N = |\alpha|$, at least one $\gamma_j < \alpha_j$. In the product integral for $\langle s^\gamma, p_\alpha \rangle$, that factor is zero by one-variable orthogonality. All product integrals are justified by their finite moments and Cauchy–Schwarz. Distinct product polynomials are mutually orthogonal, and their norms multiply to κ_α . Their leading monomials make them independent. Their number is $\dim \mathcal{P}_{\leq N} - \dim \mathcal{P}_{\leq N-1}$, so they exhaust the indicated orthogonal complement. Applying the unchanged linear map J to $O_N = i_N + L_N$ proves (KPC.7). This application includes the fixed multiplication by $v_h^{\otimes k}$. For $N = 0$ the sum in (KPC.6) is empty and $L_0 = 0$, so the same proof includes the initial shell. $\square$

The associated-graded target is explicit as well. Put $F_N E^{(k)} = J(\mathcal{P}_{\leq N})$ and $F_{-1} E^{(k)} = 0$. Projection onto $F_N E^{(k)} / F_{N-1} E^{(k)}$, composed with either map in (KPC.7), gives the same surjection

$$\overline{A}_N : \mathcal{H}_N \twoheadrightarrow F_N E^{(k)} / F_{N-1} E^{(k)}. \quad (\text{KPC.8})$$

Indeed their difference lands in $F_{N-1}E^{(k)}$, and $\mathcal{P}_{\leq N} = i_N \mathcal{H}_N + \mathcal{P}_{\leq N-1}$ proves surjectivity. If $Z_N = \ker \bar{A}_N$ and $K_N^a = \ker A_N^a$ for $a \in \{\text{hom}, \text{orth}\}$, then $K_N^a \subseteq Z_N$ and the exact sequence is

$$0 \longrightarrow Z_N/K_N^a \longrightarrow \mathcal{H}_N/K_N^a \longrightarrow \mathcal{H}_N/Z_N \longrightarrow 0. \quad (\text{KPC.9})$$

The middle space is identified with $\text{im } A_N^a$ by $[c] \mapsto A_N^a c$, the last space with the target of (KPC.8), and the first space with $\text{im } A_N^a \cap F_{N-1}E^{(k)}$. These identifications follow directly from their kernels and are bijective: for the first image, membership in F_{N-1} is exactly the condition $c \in Z_N$. Thus equality of the graded maps retains, rather than removes, the two potentially different lower-filtration image spaces. In particular, once $N-1 \geq k(d-1)$, complete monomial remainders show $F_{N-1}E^{(k)} = E^{(k)}$, so (KPC.8) has zero target; the full maps (KPC.7) still have their stated images and kernels.

17.2 The full correspondence between the two image spaces

Theorem 17.2 (Common quotient and exact graph correspondence). *Write $I_h = \text{im } A_N^{\text{hom}}$, $I_o = \text{im } A_N^{\text{orth}}$, $K_h = \ker A_N^{\text{hom}}$, and $K_o = \ker A_N^{\text{orth}}$. Define*

$$Q_N = \mathcal{H}_N/(K_h + K_o), \quad \Gamma_N = \{(A_N^{\text{hom}}c, A_N^{\text{orth}}c) : c \in \mathcal{H}_N\}. \quad (\text{KPC.10})$$

The maps $q_h : I_h \rightarrow Q_N$ and $q_o : I_o \rightarrow Q_N$, given respectively by $A_N^{\text{hom}}c \mapsto [c]$ and $A_N^{\text{orth}}c \mapsto [c]$, are well-defined surjections, with exact sequences

$$\begin{aligned} 0 \longrightarrow A_N^{\text{hom}}(K_o) \longrightarrow I_h \xrightarrow{q_h} Q_N \longrightarrow 0, \\ 0 \longrightarrow A_N^{\text{orth}}(K_h) \longrightarrow I_o \xrightarrow{q_o} Q_N \longrightarrow 0. \end{aligned} \quad (\text{KPC.11})$$

One has the exact vector-space identifications

$$\Gamma_N = I_h \times_{Q_N} I_o \cong \mathcal{H}_N/(K_h \cap K_o). \quad (\text{KPC.12})$$

The projection $\Gamma_N \rightarrow I_h$ has kernel $\{0\} \oplus A_N^{\text{orth}}(K_h)$; the projection onto I_o has kernel $A_N^{\text{hom}}(K_o) \oplus \{0\}$.

Proof. Changing a representative of an element of I_h changes c by K_h , hence leaves its class modulo $K_h + K_o$ unchanged. The same argument applies to I_o . The formulas give surjectivity. If $q_h(A_N^{\text{hom}}c) = 0$, write $c = c_h + c_o$ with $c_h \in K_h, c_o \in K_o$. Then $A_N^{\text{hom}}c = A_N^{\text{hom}}c_o$. Conversely every such vector maps to zero. This proves the first kernel formula in (KPC.11); interchanging the two maps proves the second.

The paired map from $\mathcal{H}_N$ onto Γ_N has kernel exactly $K_h \cap K_o$, proving the last identification in (KPC.12). Every paired image belongs to the fibre product. Conversely, suppose $(A_N^{\text{hom}}c, A_N^{\text{orth}}b)$ belongs to the fibre product. Then $c - b = c_h + c_o$ for some $c_h \in K_h, c_o \in K_o$. Put $e = c - c_h = b + c_o$. It has the two desired images, so the pair belongs to Γ_N . This proves equality with the full fibre product, not merely an inclusion. Taking the kernel of each coordinate projection gives the asserted remaining formulas. $\square$

The quotient in (KPC.10) also has its precise universal property. If linear maps $f_h : I_h \rightarrow Z$ and $f_o : I_o \rightarrow Z$ satisfy $f_h A_N^{\text{hom}} = f_o A_N^{\text{orth}}$, their common composite on $\mathcal{H}_N$ vanishes on both K_h and K_o , hence on their sum. It therefore factors uniquely through Q_N , and that factor gives both original maps after the surjections in (KPC.11). Likewise, a linear map $T : I_h \rightarrow I_o$ satisfying $T A_N^{\text{hom}} = A_N^{\text{orth}}$ exists exactly when $K_h \subseteq K_o$. Necessity follows by evaluating on K_h ; under that inclusion the formula $T(A_N^{\text{hom}}c) = A_N^{\text{orth}}c$ is well-defined and proves sufficiency. It is an isomorphism exactly when $K_h = K_o$. Thus (KPC.11)–(KPC.12) give the full typed correspondence even when no such isomorphism exists.

There is also an explicit polynomial deformation of the sections. For a complex parameter z , define, without division by z ,

$$p_j^{(z)}(s) = \sum_{r=0}^j p_{j,r} z^{j-r} s^r, \quad O_{N,z}(c) = \sum_{|\alpha|=N} c_\alpha \prod_j p_{\alpha_j}^{(z)}(s_j). \quad (\text{KPC.13})$$

For $|\gamma| < N$, its coefficient of s^γ is exactly the coefficient in (KPC.6) multiplied by $z^{N-|\gamma|}$; at degree N it is c_γ . Hence $O_{N,0} = i_N$, $O_{N,1} = O_N$, and $\ell_N O_{N,z} = I$ for every z . The maps $JO_{N,z}$ are matrices polynomial in the parameter, and (KPC.8) is unchanged throughout. A rank condition $\text{rank } JO_{N,z} \leq r$ is exactly the vanishing of every $(r+1)$ -minor. No constancy of their rank follows from this deformation. In particular (KPC.13) retains every original coefficient at $z = 1$; it does not replace the original measure or its orthogonal minimum.

17.3 The algebra generated by the original coordinate ratio

Now take $k = 2$ and retain $h(0) \neq 0$. Every original nontrivial zero in the arithmetic packet has $0 < \text{Re } \rho < 1$, so this domain includes that packet. Work in the original algebra

$$E^{(2)} = \mathbb{C}[s_1, s_2]/(h(s_1), h(s_2)), \quad \zeta = s_1 s_2^{-1}. \quad (\text{KPC.14})$$

The inverse is defined because s and $h(s)$ are coprime; the Euclidean identity for these polynomials supplies an inverse modulo h . All ordered pairs of centres and their multiplicities are retained.

Theorem 17.3 (Complete ratio algebra and homogeneous-shell rank). *For every distinct ratio $r = \rho/\sigma$, put*

$$M_r = \max_{\rho/\sigma=r} (m_\rho + m_\sigma - 1), \quad b = \sum_r M_r, \quad \mu_\zeta(Z) = \prod_r (Z - r)^{M_r}. \quad (\text{KPC.15})$$

Then evaluation at ζ has exactly kernel (μ_ζ) , and gives the unital algebra embedding

$$\mathbb{C}[Z]/(\mu_\zeta) \hookrightarrow E^{(2)}, \quad [P] \mapsto P(\zeta). \quad (\text{KPC.16})$$

Its image is the subalgebra $\mathbb{C}[\zeta]$. For every $N \geq 0$, the full arithmetic homogeneous-shell map is

$$\sum_{j=0}^N c_j s_1^j s_2^{N-j} \mapsto v_h^{\otimes 2} s_2^N \sum_{j=0}^N c_j \zeta^j. \quad (\text{KPC.17})$$

Its kernel consists exactly of the coefficient polynomials $\sum_{j=0}^N c_j Z^j$ divisible by μ_ζ , and

$$\boxed{\text{rank } A_N^{\text{hom}} = \min\{N+1, b\}.} \quad (\text{KPC.18})$$

The arithmetic unit and s_2^N in (KPC.17) are retained invertible multiplication maps. Formula (KPC.18) does not replace the orthogonal-shell map (KPC.7).

Proof. The factors $(s - \rho)^{m_\rho}$ of h are pairwise coprime. For each factor, a Euclidean identity with the product of the others gives a polynomial equal to one modulo that factor and zero modulo all others. Their residue classes are mutually orthogonal idempotents and sum to one: each of these assertions holds modulo every factor, hence modulo their product. They give the full Chinese-remainder isomorphism, with inverse obtained by multiplying each component representative by its idempotent and adding. Tensoring these explicit maps gives

$$E^{(2)} \cong \prod_{(\rho, \sigma)} \mathbb{C}[x, y]/(x^{m_\rho}, y^{m_\sigma}), \quad x = s_1 - \rho, \quad y = s_2 - \sigma. \quad (\text{KPC.19})$$

In each component the basis is the full set $x^a y^c$, $0 \leq a < m_\rho$, $0 \leq c < m_\sigma$. The inverse of $\sigma + y$ is explicitly $\sum_{j=0}^{m_\sigma-1} (-1)^j y^j / \sigma^{j+1}$, as multiplication and cancellation of consecutive powers verify.

Write $m = m_\rho$, $n = m_\sigma$. In this component the ratio retains the exact identity

$$\zeta - \frac{\rho}{\sigma} = \frac{\sigma x - \rho y}{\sigma(\sigma + y)}. \quad (\text{KPC.20})$$

Every monomial of total degree $m + n - 1$ vanishes, since one exponent must reach m or n . Thus the $(m + n - 1)$ -st power of (KPC.20) is zero. Its preceding power has the exact value

$$\left(\zeta - \frac{\rho}{\sigma}\right)^{m+n-2} = \binom{m+n-2}{m-1} \frac{\sigma^{m-1}(-\rho)^{n-1}}{\sigma^{2(m+n-2)}} x^{m-1} y^{n-1} \neq 0. \quad (\text{KPC.21})$$

Indeed only the displayed monomial survives in the numerator power; in the inverse-denominator power only its constant coefficient can multiply that monomial without making it zero. The displayed coefficient is nonzero over $\mathbb{C}$ because $\rho, \sigma \neq 0$. For $m = n = 1$ the preceding power is the zeroth power one, and the first power is zero, so this case is included. The local nilpotent index is therefore exactly $m + n - 1$.

For any nonzero polynomial $P(Z)$, factor at the local ratio r as $P(Z) = (Z - r)^e B(Z)$ with $B(r) \neq 0$. The element $B(\zeta)$ is invertible: after division by its nonzero constant $B(r)$, it is one plus a nilpotent polynomial in $\zeta - r$, whose finite geometric series is an inverse. Hence $P(\zeta) = 0$ in that component exactly when $e \geq m + n - 1$. Across all ordered components, the condition at each coincident ratio is its maximum exponent, and conditions at distinct ratios are coprime. This proves that the global evaluation kernel is precisely (μ_ζ) , including the zero polynomial. It proves (KPC.16) and its stated image.

Each homogeneous monomial satisfies $s_1^j s_2^{N-j} = s_2^N \zeta^j$ inside the original algebra. Applying the original arithmetic unit proves (KPC.17). Both multiplication factors there are invertible, so its kernel is exactly the asserted restricted polynomial kernel. For $N < b$, no nonzero polynomial of degree at most N is divisible by the monic μ_ζ . For $N \geq b$, all such multiples are $\mu_\zeta \mathbb{C}[Z]_{\leq N-b}$, which has dimension $N - b + 1$, since multiplication by the nonzero polynomial is injective. Subtracting this dimension from $N + 1$ gives (KPC.18) in both cases. $\square$

In the retained monomial tensor coordinates, set $S_1 = A_h \otimes I$, $S_2 = I \otimes A_h$, $c = [1] \otimes [1]$, and $Z_{\text{mat}} = S_1 S_2^{-1}$. Then (KPC.17) is exactly

$$(c_0, \dots, c_N) \mapsto U_{v_h^{\otimes 2}} S_2^N \sum_{j=0}^N c_j Z_{\text{mat}}^j c. \quad (\text{KPC.22})$$

This retains each complete Taylor block under (KPC.19) rather than replacing either coordinate by its eigenvalue. Multiplication by $v_h^{\otimes 2} s_2^N$ is a linear isomorphism from $\mathbb{C}[\zeta]$ to its displayed image, with inverse multiplication by $s_2^{-N} (v_h^{\otimes 2})^{-1}$ on that image. The image after this multiplication is not asserted to be a unital subalgebra.

The original summed generator also has an exact map between adjacent shells. Write $A^{(2)} = M_{s_1+s_2}$ on the retained algebra and $U = v_h^{\otimes 2}$. Then

$$A^{(2)}(U s_2^N f(\zeta)) = U s_2^{N+1} (1 + \zeta) f(\zeta). \quad (\text{KPC.22a})$$

This follows from $s_1 + s_2 = s_2(1 + \zeta)$ and commutativity of the original multiplication maps. Multiplication by $1 + Z$ sends degree at most N to degree at most $N + 1$, and sends a multiple of μ_ζ to another such multiple. Therefore (KPC.22a) is well-defined on the complete source quotients in (KPC.17). It is an adjacent-shell action; it does not assert that $\mathbb{C}[\zeta]$ alone is invariant under $A^{(2)}$. For example, if the packet has two distinct centres ρ, σ , the diagonal pairs (ρ, ρ) and (σ, σ) both have ratio one, while the constant terms of their summed generators are 2ρ and 2σ . A polynomial in the ratio has the same constant term at both pairs, so it cannot equal that summed coordinate. Formula (KPC.22a) retains the missing second-coordinate multiplication explicitly.

Theorem 17.4 (Exact algebra-retraction criterion). *In the local algebra $R_{\rho,\sigma} = \mathbb{C}[x, y]/(x^m, y^n)$, with $\rho, \sigma \neq 0$, the ratio-subalgebra inclusion $\mathbb{C}[\zeta] \hookrightarrow R_{\rho,\sigma}$ admits a unital complex-algebra retraction exactly when $\min(m, n) = 1$. In the full original tensor algebra $E_h^{\otimes 2}$, the inclusion (KPC.16) admits a unital complex-algebra retraction exactly when h is squarefree.*

Proof. First suppose $m = 1$, so $x = 0$ and $\zeta = \rho/(\sigma + y)$. This element is a unit, and $y = \rho\zeta^{-1} - \sigma$. Its inverse belongs to the ratio subalgebra: $\zeta = r + w$, with $r = \rho/\sigma \neq 0$ and $w^n = 0$, has inverse $\sum_{j=0}^{n-1} (-1)^j w^j / r^{j+1}$. Thus the ratio subalgebra contains y and is the whole local algebra. If $n = 1$, then $y = 0$ and $x = \sigma\zeta - \rho$, giving the same conclusion. In both cases the inclusion is an isomorphism and its inverse is the asserted retraction.

Now suppose $m, n \geq 2$, and put $L = m + n - 1$. The exact local index proved above identifies the ratio subalgebra with $B = \mathbb{C}[t]/(t^L)$, by $t \mapsto \zeta - \rho/\sigma$. In any putative retraction $f : R_{\rho,\sigma} \rightarrow B$, the images of x and y have zero constant term because their powers vanish. If the coefficient of t in $f(x)$ were a nonzero number a , the coefficient of t^m in $f(x)^m$ would be $a^m \neq 0$. Since $m < L$, that term survives in B , contradicting $x^m = 0$. The same argument using $n < L$ shows that $f(y)$ also has zero linear coefficient. The exact expression (KPC.20) then makes $f(\zeta - \rho/\sigma)$ have zero linear coefficient: its numerator lies in (t^2) and its denominator is a unit. A retraction would send that element to t . This contradiction proves the local converse.

If h is squarefree, (KPC.19) identifies the full tensor algebra with $\mathbb{C}^{\mathbb{Z} \times \mathbb{Z}}$. The ratio subalgebra is the algebra of functions constant on every fibre of $(\rho, \sigma) \mapsto \rho/\sigma$. Indeed (KPC.15) has exponent one at each ratio, and ordinary polynomial interpolation gives every assignment of values on the distinct ratios. Choose one ordered pair in each such fibre. Restriction of a function to those chosen pairs, then assigning that value to the entire corresponding fibre, is a unital algebra map back to this subalgebra. It is the identity on the subalgebra, proving existence of a retraction.

Conversely, let $M = \max_{\rho} m_{\rho} \geq 2$. The ratio-one component in (KPC.15) is

$$B_1 = \mathbb{C}[t]/(t^{2M-1}), \quad \zeta \mapsto 1 + t. \quad (\text{KPC.22b})$$

This is because $\rho/\sigma = 1$ for nonzero centres exactly when $\rho = \sigma$, and the maximum of $2m_{\rho} - 1$ is $2M - 1$. Compose a proposed global retraction with projection onto B_1 . A unital map from the product (KPC.19) into B_1 selects exactly one local factor. To justify that statement, an idempotent of B_1 has constant term zero or one. If it is zero, the idempotent lies in the nilpotent ideal (t) , and an idempotent which is nilpotent is zero, since all its positive powers equal itself. If the constant term is one, apply this argument to one minus the idempotent. Hence the only idempotents are zero and one. The mutually orthogonal source-factor idempotents sum to one, so exactly one maps to one and all others map to zero, proving the selection statement.

Suppose the selected factor is indexed by (ρ, σ) . Images of its nilpotent coordinates have zero constant term, so the constant term of its ratio image is ρ/σ . Equation (KPC.22b) forces this ratio to be one; thus $\rho = \sigma$, with common multiplicity $m = m_{\rho} \leq M < 2M - 1$. The relation $x^m = 0$ again forces the coefficient of t in the image of x to vanish, since a nonzero coefficient would produce a nonzero t^m term in the target. The same holds for y . Their ratio minus one therefore has zero linear coefficient by (KPC.20), contradicting its required image t in (KPC.22b). This proves the global converse while retaining all source-factor idempotents and the original ratio embedding. $\square$

17.4 A complete finite witness for unequal shell ranks

The following positive polynomial model tests the proposed passage from homogeneous rank to the original orthogonal rank. It is not a claim that $1/2$ is an arithmetic zero. Use the specified measure

$$\nu_{\text{ex}} = \delta_{1/2-i} + \delta_{1/2} + \delta_{1/2+i} \quad \text{on the variable } s, \quad h_{\text{ex}}(s) = s - \frac{1}{2}, \quad v_{\text{ex}} = 1. \quad (\text{KPC.23})$$

All three masses are exactly one. Through degree two the monic orthogonal polynomials and norms are

$$\begin{array}{c|c|c} j & p_j(s) & \kappa_j \\ \hline 0 & 1 & 3 \\ 1 & s - \frac{1}{2} & 2 \\ 2 & (s - \frac{1}{2})^2 + \frac{2}{3} & \frac{2}{3}. \end{array} \quad (\text{KPC.24})$$

To verify the complete Gram, their three value columns are

$$(1, 1, 1)^T, \quad (-i, 0, i)^T, \quad (-1/3, 2/3, -1/3)^T.$$

Their pairwise Hermitian inner products are zero, and their squared norms are respectively 3, $1 + 0 + 1 = 2$, and $1/9 + 4/9 + 1/9 = 2/3$. Thus the moment form through degree two is positive definite, as required for the source and next layer used below.

Take $k = 2$, $N = 1$. Here $d = 1$ and the threshold $N \geq k(d - 1) = 0$ holds. The original tensor residue algebra is $E_{\text{ex}}^{(2)} = \mathbb{C}$, with $s_1 = s_2 = 1/2$, and its full jet map is evaluation at that ordered pair. For $c = c_{10}s_1 + c_{01}s_2$, the two section and image maps are exactly

$$\begin{aligned} i_1(c) &= c_{10}s_1 + c_{01}s_2, \\ O_1(c) &= c_{10}(s_1 - \frac{1}{2}) + c_{01}(s_2 - \frac{1}{2}), \\ L_1(c) &= -\frac{1}{2}(c_{10} + c_{01}), \\ A_1^{\text{hom}}(c) &= \frac{1}{2}(c_{10} + c_{01}), \quad A_1^{\text{orth}}(c) = 0. \end{aligned} \quad (\text{KPC.25})$$

Thus the lower coefficient in (KPC.7) cancels the entire homogeneous image. The ranks are one and zero. The kernels are $K_h = \{(c_{10}, c_{01}) : c_{10} + c_{01} = 0\}$ and $K_o = \mathbb{C}^2$, so the common quotient in (KPC.10) is zero and the full correspondence is $\mathbb{C} \oplus \{0\}$. The ratio is $\zeta = 1$, its minimal polynomial is $Z - 1$, and (KPC.18) gives the correct homogeneous rank $\min(2, 1) = 1$. Equation (KPC.25) proves explicitly why it gives no equality with the orthogonal rank.

The actual finite minimum and next layer in this model can also be retained. In the source with product measure $\nu_{\text{ex}}^{\otimes 2}$, the degree-at-most-one orthogonal basis is p_0p_0, p_1p_0, p_0p_1 , with norms 9, 6, 6 and jets 1, 0, 0. Hence the full interpolation kernel is $K_1 = 1/9$, the metric is $G_1 = 9$, and its minimum is the constant polynomial $R_1u = u$. Multiplication by the original sum $s_1 + s_2$ acts on the packet by $A = 1$. Consequently

$$B_1u = (s_1 + s_2 - 1)u \in \mathcal{L}_1, \quad C_1 = (I - P_{\mathcal{L}_1})B_1 = 0, \quad W_1 = A^*G_1 + G_1A - 2G_1 = 0. \quad (\text{KPC.26})$$

Indeed $\mathcal{L}_1$ is spanned by $s_1 - 1/2, s_2 - 1/2$, the complete degree-one relation kernel. Thus the derivative rank zero is verified directly without presupposing the shell-rank formula.

At the next shell, ordered as $(2, 0), (1, 1), (0, 2)$, the original jets and norm diagonal are

$$U = \begin{pmatrix} 2/3 & 0 & 2/3 \end{pmatrix}, \quad D_{\text{next}} = \text{diag}(2, 4, 2).$$

The actual layer vectors are $e_{20} = (s_1 - 1/2)^2$, $e_{11} = (s_1 - 1/2)(s_2 - 1/2)$, and $e_{02} = (s_2 - 1/2)^2$. Each has zero jet and is orthogonal to $\mathcal{L}_1$, either by its product moments or by the layer construction. Their Gram is exactly

$$H = D_{\text{next}} + U^*G_1U = \begin{pmatrix} 6 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 6 \end{pmatrix}. \quad (\text{KPC.27})$$

For example the diagonal squared norm of e_{20} is $(1 + 0 + 1)(1 + 1 + 1) = 6$, and its pairing with e_{02} is $(-1 + 0 - 1)^2 = 4$; the odd cross terms vanish. The eigenvalues of the outer two-coordinate block are 10, 2, and the middle eigenvalue is 4, proving positivity. The next full kernel and metric therefore are

$$K_2 = \frac{1}{9} + \frac{(2/3)^2}{2} + \frac{(2/3)^2}{2} = \frac{5}{9}, \quad G_2 = \frac{9}{5}. \quad (\text{KPC.28})$$

The representative itself is the explicit polynomial

$$R_2 u = \left[1 + \frac{3}{5} \left((s_1 - \frac{1}{2})^2 + (s_2 - \frac{1}{2})^2 \right) \right] u. \quad (\text{KPC.28a})$$

For a direct verification, it has jet u . Its values for u equal to one are $-1/5$ at the four product nodes where both coordinates differ from $1/2$, $2/5$ at the four nodes where exactly one differs, and one at the central node. Its squared norm is therefore $4/25 + 16/25 + 1 = 9/5$. It is orthogonal to the five relation basis vectors $s_1 - 1/2, s_2 - 1/2, e_{20}, e_{11}, e_{02}$. The first two and e_{11} have an odd coordinate, so their pairings vanish by the explicit sign symmetry of the nine-node measure. For e_{20} , only the two noncentral first-coordinate values contribute, and for each of them the sum of the three values of $R_2(1)$ is $-1/5 + 2/5 - 1/5 = 0$. Its pairing is thus zero; the same computation with the coordinates interchanged treats e_{02} . These five independent zero-jet polynomials span the kernel in the six-dimensional degree-at-most-two source, whose jet map is onto the one-dimensional target. Every other feasible polynomial differs by a vector in this kernel. Pythagoras and positive definiteness therefore prove the unique-minimum assertion directly, including its actual norm. Thus the nonzero original relation layer and its positive metric loss are retained alongside the zero derivative layer. This complete finite calculation rejects only the proposed equality between homogeneous and orthogonal-shell ranks. The original arithmetic shell continues to be given by (KPC.4)–(KPC.7), and the exact graph and quotient maps relating its two presentations are (KPC.9)–(KPC.12).

Every linear map $T : V \rightarrow W$ in these comparisons has a support-preserving Split-Zero lift on the following specified carriers at a fixed support λ :

$$\widehat{V}_\lambda = \{\tau_V\} \sqcup (\{\lambda\} \times V), \quad \widehat{W}_\lambda = \{\tau_W\} \sqcup (\{\lambda\} \times W).$$

Addition has external τ as identity and uses vector addition inside the supported fibre. The scalar action is $r^\bullet(\lambda, v) = (\lambda, rv)$, while the external scalar zero sends every input to external absence and every scalar sends external absence to external absence. The lift maps τ_V to τ_W and (λ, v) to (λ, Tv) . Its addition identity follows from $T(v + w) = Tv + Tw$; its supported-scalar identity follows from $T(rv) = rTv$, including $r = 0$. The external cases follow from the definitions. Thus every displayed kernel vector maps to the represented zero at λ , and the graph, quotient and lower-degree maps retain this support even when the two ranks differ.

18 The original frontier energy, determinant growth, and the entire border fibre

The symmetric-frontier delivery of 12 September computes coordinates on the actual next relation layer. We retain these coordinates and prove their exact relation to the accumulated source energy of Section 15. The calculation also identifies the entire zero fibre of the delivery's smaller border operator, including its nilpotent extension.

18.1 Original moment coordinates and the relation graph

Keep a nonempty full-order packet h , its degree $d, g = 2\xi, E = \mathbb{C}[s]/h, v = j_h(g/h)$, and $A = M_s$. The positive measure remains

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}.$$

Let p_n be the monic orthogonal polynomial obtained from $1, s, \dots, s^n$ without dividing by its norm, and let $\omega_n = \|p_n\|_{\nu_h}^2 > 0$. The original generator has the exact recurrence

$$sp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \ (n \geq 1), \quad a_0 = 0, \quad b_n + \bar{b}_n = 1. \quad (\text{FE.1})$$

Here $p_{-1} = 0$. To verify the coefficients, multiplication by s has adjoint multiplication by $1 - s$ on the original line. Its pairing against degrees below $n - 1$ vanishes, the leading coefficient is one, and the pairing against p_{n-1} is $-\omega_n$. Its diagonal is $\int s|p_n|^2 d\nu_h/\omega_n$, which proves the last identity and retains its possibly nonzero imaginary part.

Fix $k \geq 1$ and $M \geq k(d - 1)$, and use $A_k = \sum_j A_j$ on the unchanged $E_k = E^{\otimes k}$. For $\alpha \in \mathbb{N}^k$, put

$$p_\alpha = \prod_j p_{\alpha_j}(s_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = v^{\otimes k}[p_\alpha] \in E_k.$$

All remainder coordinates, including every nilpotent jet and the entire local arithmetic unit, are retained in z_α . The original source map $\mathcal{T}^{(k)}$ is the map in (CT.11)–(CT.13). Its orthogonal polynomial images have norms ω_α . The constrained minimum gives exactly

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}, \quad R_M u = \sum_{|\alpha| \leq M} \mathcal{T}^{(k)} p_\alpha \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (\text{FE.2})$$

This is (CT.5) in the explicitly declared orthogonal source basis; no moment mass or polynomial norm has been changed.

Order the indices β of total degree $M + 1$, and put

$$r = \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\omega_\beta), \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \quad T_+ = [\mathcal{T}^{(k)} p_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k.$$

The two source maps have the same column ordering. Since the image of T_+ is orthogonal to the entire degree- M image,

$$b = T_+ - R_M F : \mathbb{C}^r \rightarrow \mathcal{E}_M, \quad H = b^* b = \Omega + F^* G_M F, \quad b^* R_M = -F^* G_M. \quad (\text{FE.3})$$

Here $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ is the original relation layer. The complete jets of b vanish; both of its terms are orthogonal to $\mathcal{B}_M$. Taking the highest orthogonal degree of bu recovers $T_+ u$, proving injectivity. For any boundary of degree $M + 1$, subtract the image under b of its highest coefficient. The difference lies in $\mathcal{B}_M$. This gives the inverse on the orthogonal layer and proves surjectivity. It also gives an original primitive: divide the numerator of each column of b in the fixed order by $h(s_1), \dots, h(s_k)$. The primitive in factor j has coefficient $(-1)^{j-1}$; its tensor differential contributes the same sign, giving the original positive boundary.

Define the second frontier matrix by

$$E_+ = [w_\beta], \quad w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}. \quad (\text{FE.4})$$

The notation E_+ denotes a matrix $\mathbb{C}^r \rightarrow E_k$. From (FE.2), the highest coefficient of $D^{(k)} R_M - R_M A_k$ is $\Omega^{-1} E_+^* G_M$. Projection onto the graph (FE.3) therefore proves, with the original types and signs,

$$Y_M = -b H^{-1} F^* G_M, \quad C_M = b \Omega^{-1} E_+^* G_M, \quad \Delta_M = Y_M^* Y_M = G_M F H^{-1} F^* G_M. \quad (\text{FE.5})$$

The notation C_M is the original derivative-layer map in (CT.15), not the inverse interpolation matrix $\mathcal{C}_M$.

Applying (FE.1) to (FE.2) cancels every internal neighbouring pair, because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonals give exactly $k C_M$. The remaining pairs have top degree $M + 1$, proving

$$\mathcal{Z}_M = A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{FE.6})$$

Alternatively (FE.5) inserted into $W_M = -(Y_M^* C_M + C_M^* Y_M)$ gives exactly (FE.6). Thus both the polynomial cancellation and its actual boundary primitive yield the same original form.

18.2 Exact relative loss and the source derivative energy

For the next formulas only, use the displayed finite isometries from the specified metric coordinate spaces and define

$$U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}, \quad P = U U^*, \quad K_U = U^* U. \quad (\text{FE.7})$$

The original F, E_+, Ω, G_M are still given by (FE.1)–(FE.5), and can be recovered by the inverse coordinate maps in (FE.7). In particular no polynomial is assigned norm one and no arithmetic coefficient is suppressed.

Theorem 18.1 (Frontier energy and every relative Gram eigenvalue). *In these explicitly related coordinates,*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = U V^* + V U^*, \\ T_M &= G_M^{-1/2} \Delta_M G_M^{-1/2} = U (I + K_U)^{-1} U^* = P (I + P)^{-1}, \\ G_M^{-1/2} G_{M+1} G_M^{-1/2} &= (I + P)^{-1}. \end{aligned} \quad (\text{FE.8})$$

If $\lambda_1, \dots, \lambda_q > 0$ are all positive eigenvalues of P , including multiplicity, then

$$\begin{aligned} q &= \text{rank } F = \text{rank } Y_M = \text{rank } \Delta_M, \\ \tau_M &= \text{Tr}(G_M^{-1} \Delta_M) = \sum_{i=1}^q \frac{\lambda_i}{1 + \lambda_i}, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M} = \prod_{i=1}^q (1 + \lambda_i)^{-1}, \\ \chi_M &= \text{Tr}(G_M^{-1} C_M^* C_M) = \|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{FE.9})$$

The empty product equals one if $q = 0$. In addition $\text{rank } C_M = \text{rank } E_+$.

Proof. Equation (FE.6) proves the first line of (FE.8). The identity $H = \Omega^{1/2}(I + U^* U) \Omega^{1/2}$ substituted into (FE.5) proves the second line's first expression. The equality $(I + U U^*) U = U (I + U^* U)$, multiplied by the indicated invertible factors, identifies it with $P (I + P)^{-1}$. Subtract it from I to get the third line. The positive matrices $I + K_U$ and $I + P$ are invertible directly from their quadratic forms.

The graph map b is injective, and H, Ω, G_M are invertible, so (FE.5) proves both asserted ranks. Equivalently, the formula for T_M identifies its positive eigenvalues with $\lambda_i / (1 + \lambda_i)$, retaining zeros on $\ker P$. The trace and determinant formulas now follow from the finite spectral decomposition in these same coordinates. Finally

$$C_M G_M^{-1/2} = b \Omega^{-1/2} V^*, \quad (b \Omega^{-1/2})^* (b \Omega^{-1/2}) = I + U^* U.$$

Its squared Hilbert–Schmidt norm is $\text{Tr}(V(I + U^* U)V^*)$, which expands to the last line of (FE.9). This is the energy of the original source-layer map for its inherited $L^2(d^k x)$ inner product. $\square$

The exact remainder in the accumulated determinant calculation is therefore also a frontier formula:

$$-\log \pi_M - \tau_M = \sum_{i=1}^q \left(\log(1 + \lambda_i) - \frac{\lambda_i}{1 + \lambda_i} \right) = \sum_{i=1}^q \int_0^{\lambda_i} \frac{x}{(1+x)^2} dx. \quad (\text{FE.10})$$

Differentiate each displayed scalar expression and evaluate at zero to prove the equality, including every eigenvalue and its sign. Thus (JD.14) uses the exact original derivative cost $\|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2$, and (JD.16) is precisely the pair of identities

$$\sum_{M \geq a} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M \geq a} \sum_i \frac{\lambda_{M,i}}{1 + \lambda_{M,i}} = \infty. \quad (\text{FE.11})$$

Both use the already proved actual boundary density; the second retains its full inverse-interpolation cost.

18.3 The entire small border operator, including its zero extension

Let $n = d^k$, and define

$$B = [U, V] : \mathbb{C}^{2r} \rightarrow \mathbb{C}^n, \quad K = B^*B, \quad J_0 = \begin{pmatrix} 0 & I_r \\ I_r & 0 \end{pmatrix}, \quad A_0 = J_0 K, \quad H_0 = K^{1/2} J_0 K^{1/2}. \quad (\text{FE.12})$$

The symbol A_0 here is a finite border operator, not the arithmetic generator A or A_k . This distinction is accompanied by the exact maps below. The source's unscaled border operator is

$$J[F, E_+]^* G_M[F, E_+], \quad J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

With $D_\Omega = \text{diag}(\Omega^{1/2}, \Omega^{1/2})$, it is exactly $D_\Omega^{-1} A_0 D_\Omega$: substitution of (FE.7) gives

$$[F, E_+]^* G_M[F, E_+] = D_\Omega K D_\Omega.$$

This states the full coordinate similarity, including zero.

Put $N = \ker K = \ker B$ and $\mathcal{R} = \text{ran } K$. They are orthogonal complements since K is Hermitian, and $K^{1/2}$ restricts to an invertible positive map on $\mathcal{R}$. There is an exact sequence of spaces with their specified operators:

$$0 \longrightarrow (N, 0) \longrightarrow (\mathbb{C}^{2r}, A_0) \xrightarrow{K^{1/2}} (\mathcal{R}, H_0|_{\mathcal{R}}) \longrightarrow 0. \quad (\text{FE.13})$$

Indeed $K^{1/2} A_0 = H_0 K^{1/2}$, its kernel is N , and its image is $\mathcal{R}$. On N , $A_0 = 0$. The map

$$Q|_{\mathcal{R}} = B(K^{1/2}|_{\mathcal{R}})^{-1}, \quad Q|_N = 0 \quad (\text{FE.14})$$

is a partial isometry onto $\text{ran } B$: for $x, y \in \mathcal{R}$, substitution of $B^*B = K$ gives $\langle Qx, Qy \rangle = \langle x, y \rangle$. It satisfies $B = QK^{1/2}$ and

$$S_M = B J_0 B^* = Q H_0 Q^*, \quad S_M|_{(\text{ran } B)^\perp} = 0. \quad (\text{FE.15})$$

Thus the restriction of H_0 to $\mathcal{R}$ and that of S_M to $\text{ran } B$ are unitarily intertwined by an explicitly proved map; both remaining orthogonal complements and their zero actions are specified.

Theorem 18.2 (Exact zero fibre of the border comparison). *Let $Z_0 = \ker(H_0|_{\mathcal{R}})$ and $q_0 = \dim Z_0$. The complete generalized zero eigenspace of A_0 is*

$$\mathcal{N}_0 = (K^{1/2})^{-1}(Z_0), \quad A_0^2|_{\mathcal{N}_0} = 0. \quad (\text{FE.16})$$

Define $L = (K^{1/2}|_{\mathcal{R}})^{-1} : \mathcal{R} \rightarrow \mathcal{R}$. For its restriction to $z \in Z_0$, set

$$Lz = (K^{1/2}|_{\mathcal{R}})^{-1}z, \quad N_0z = J_0 K^{1/2}z \in N. \quad (\text{FE.17})$$

The map $N_0 : Z_0 \rightarrow N$ is injective, and the exact vector space decomposition $\mathcal{N}_0 = N \oplus LZ_0$ gives

$$A_0(n + Lz) = N_0z. \quad (\text{FE.18})$$

Consequently its zero Jordan data consist of exactly q_0 blocks of size two and $\dim N - q_0$ blocks of size one. Its nonzero Jordan blocks have size one and coincide, with full multiplicities, with the nonzero eigenvalues of S_M .

Proof. If $A_0^m x = 0$, (FE.13) gives $H_0^m K^{1/2} x = 0$. Hermitian spectral decomposition of $H_0|_{\mathcal{R}}$ identifies its generalized zero space with Z_0 , proving that x belongs to the preimage in (FE.16). Conversely, if $K^{1/2} x \in Z_0$, then $K^{1/2} A_0 x = 0$, so $A_0 x \in N$ and $A_0^2 x = 0$. This proves (FE.16) exactly.

For $z \in Z_0$, one has $K^{1/2}J_0K^{1/2}z = H_0z = 0$, proving $N_0z \in N$. Since J_0 is invertible and $K^{1/2}$ is injective on $\mathcal{R}$, N_0 is injective. Equation $K^{1/2}Lz = z$ gives the direct decomposition, and $A_0Lz = J_0KLz = J_0K^{1/2}z$ proves (FE.18). Choose a basis of Z_0 , use its injective image as part of a basis of N , and extend it in N . Each pair Lz, N_0z is a size-two chain, and the remaining basis vectors of N are size-one chains. This also proves $q_0 \leq \dim N$ without a rank assumption.

For a nonzero eigenvalue λ , take an eigenvector $z \in \mathcal{R}$ of H_0 . If $x = Lz$, then $n = A_0x - \lambda x \in N$, so $x + n/\lambda$ is an eigenvector of A_0 lifting z . The lift is unique because A_0 vanishes on N and $\lambda \neq 0$. This supplies every eigenvector of the Hermitian quotient and leaves no nontrivial Jordan chain at a nonzero eigenvalue: applying (FE.13) to such a chain would give one for the Hermitian quotient, and a residual chain contained in N is impossible at nonzero λ . Finally (FE.15) identifies those eigenvalues with the actual relative form. All claims include their algebraic multiplicities. $\square$

In particular,

$$\dim \ker S_M = n - \text{rank } B + q_0, \quad \dim \ker A_0 = 2r - \text{rank } B, \quad \text{mult}_0(A_0) = 2r - \text{rank } B + q_0. \quad (\text{FE.19})$$

The first equality is (FE.15), the second uses the invertibility of J_0 , and the third counts (FE.18). Thus replacing the source's border calculation by a Hermitian one has an exact kernel, quotient, complement and zero extension, all explicitly retained here. At any fixed retained support λ , each displayed linear map has its Split-Zero lift $(\lambda, x) \mapsto (\lambda, Tx)$, with external $\tau \mapsto \tau$. A size-two chain in (FE.18) is therefore the exact supported chain

$$(\lambda, Lz) \mapsto (\lambda, N_0z) \mapsto (\lambda, 0). \quad (\text{FE.19a})$$

The last value is the represented zero in that fibre. For $z \neq 0$, injectivity of N_0 proves that the intermediate value has nonzero amplitude; neither map sends it to external absence.

18.4 A quantified link between the frontier bound and determinant cost

Let $\lambda_M = \|UU^*\|_{\text{op}}$, and retain the original positive recurrence quantity

$$\Gamma_M = \max_{|\alpha|=M} \sum_{j=1}^k \{a_{\alpha_j+1} + (k-1)a_{\alpha_j}\} > 0. \quad (\text{FE.20})$$

The strict inequality follows from $a_{\alpha_j+1} > 0$. For clarity, its upper estimate uses an exact matrix comparison. Let $Z = [z_\alpha]_{|\alpha|=M}$, $\Omega_M = \text{diag}(\omega_\alpha)$, and let $L_{\alpha\beta} = a_{\beta_j}$ for $\alpha = \beta - e_j$, summed over those j . Then $E_+ = ZL$, and $\Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is the displayed sum in (FE.20): $\omega_\alpha a_{\alpha_j+1} / \omega_{\alpha+e_j} = 1$ for each outgoing edge, and the other predecessors contribute $\sum_{l \neq j} a_{\alpha_l}$. For a matrix with nonnegative entries, an eigenvector component of largest absolute value bounds the eigenvalue's absolute value by its largest row sum. The matrix is similar to the positive Hermitian matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$, so

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_M Z \Omega_M^{-1} Z^* \preceq \Gamma_M \mathcal{C}_M.$$

It follows that $VV^* \preceq \Gamma_M I$. For any vector x , Cauchy–Schwarz applied to (FE.8) gives $|x^* S_M x| \leq 2\|U^* x\| \|V^* x\|$. Thus

$$\epsilon_M \leq 2\sqrt{\Gamma_M \lambda_M}. \quad (\text{FE.21})$$

Every actual eigenline survives in the full tensor family. If $\delta_* = \max_{\rho \in Z} |\text{Re } \rho - 1/2|$, its repeated tensor gives $\epsilon_M \geq 2k\delta_*$. Combining this with (FE.9) and (FE.21) proves, at every finite degree,

$$\lambda_M \geq \frac{k^2 \delta_*^2}{\Gamma_M}, \quad \frac{\det G_M}{\det G_{M+1}} \geq 1 + \frac{k^2 \delta_*^2}{\Gamma_M}. \quad (\text{FE.22})$$

Indeed the determinant inverse is $\prod_i (1 + \lambda_i)$, which is at least $1 + \lambda_M$. For integers $k(d-1) \leq a < b$, multiplication, followed by the increasing logarithm, gives the explicit source constraint

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{k^2 \delta_*^2}{\Gamma_M} \right). \quad (\text{FE.23})$$

When $\delta_* = 0$, this remains a valid inequality with zero right side and no division by a zero spectral quantity. It is an exact necessary growth relation for the constructed arithmetic matrices, not an assumed upper estimate on that growth.

The original derivative energy supplies a second exact growth bound without the recurrence maximum. By (FE.8), $\|Y_M G_M^{-1/2}\|_{\text{op}}^2 = \lambda_M / (1 + \lambda_M)$. By (FE.9), $\|C_M G_M^{-1/2}\|_{\text{op}}^2 \leq \chi_M$. The original cross-pairing formula, the operator triangle inequality, and submultiplicativity therefore give

$$\epsilon_M^2 \leq 4 \frac{\lambda_M}{1 + \lambda_M} \chi_M. \quad (\text{FE.24})$$

If $\delta_* > 0$, combine it with the retained eigenline $\epsilon_M \geq 2k\delta_*$. The finite number $\lambda_M / (1 + \lambda_M)$ is strictly below one, so

$$\chi_M > k^2 \delta_*^2, \quad \lambda_M \geq \frac{k^2 \delta_*^2}{\chi_M - k^2 \delta_*^2}, \quad \log \frac{\det G_M}{\det G_{M+1}} \geq -\log \left(1 - \frac{k^2 \delta_*^2}{\chi_M} \right). \quad (\text{FE.25})$$

Every denominator and logarithm in (FE.25) has just been proved valid. The middle inequality follows by rearranging (FE.24); the last uses $\det G_M / \det G_{M+1} \geq 1 + \lambda_M$. Summing it over any $k(d-1) \leq a < b$ gives the corresponding finite accumulated determinant inequality. For $\delta_* = 0$, (FE.24) remains valid, and (FE.25) is left un-divided; no 0/0 value is assigned to an original zero-energy layer.

19 The signed symmetric summand and its complete spectral cost

This section proves the cochain, coordinate, and spectral comparisons for the symmetric summand in the arithmetic frontier delivery. The original packet is $E = \mathbb{C}[t]/h$, of dimension $d \geq 1$, and its operator is $A = M_t$. The tensor degree is $k \geq 1$. All orders in $h = \prod_\rho (t - \rho)^{m_\rho}$ remain present. The original joint representative, metric, and layer maps of (CT.12)–(CT.16) are denoted by R_M, G_M, Y_M, C_M , with $M \geq k(d-1)$. The source's symmetric construction is used through the maps proved below; its orbit sums keep their literal factorials.

19.1 The chain projector and its completed and finite targets

The completion in this subsection is specifically the completed projective tensor product $\widehat{\otimes}_\pi$ of the strong-Schwartz locally convex spaces. The global-retraction source, equations (16), (19), and (24), constructs continuous maps

$$\Lambda : \mathcal{B} \longrightarrow V, \quad \Lambda\Theta = 1, \quad s : Q \longrightarrow \mathcal{B}, \quad s[F] = (1 - \Theta\Lambda)F, \quad qs = 1. \quad (\text{SS.1})$$

Its specified cutoffs and moment projection occur in $\Lambda = P_V(1 - A_\alpha B_\beta)^{-1} Y_{\alpha,\beta}$. The convergence and continuity of this formula are proved in that source; no Hilbert-space replacement of these topologies is used here. Put $C = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $P = sq$ as a cochain map with degree-zero part zero, and $H^1 = \Lambda$. Then in the two degrees respectively, $\Lambda\Theta = 1$ and $\Theta\Lambda = 1 - sq$ prove

$$dH + Hd = 1 - P. \quad (\text{SS.2})$$

For homogeneous inputs define the chain permutation by

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i| |x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (\text{SS.3})$$

Here the degrees in the exponent belong to the original inputs. An adjacent interchange contributes $(-1)^{ab}$ for degrees a, b . Interchanging twice contributes $(-1)^{2ab} = 1$; disjoint interchanges commute; and the two three-interchange braid words both contribute $(-1)^{ab+ac+bc}$. Thus (SS.3) is a representation of S_k . On two adjacent factors, $d(x \otimes y) = dx \otimes y + (-1)^a x \otimes dy$. After interchange the coefficients of $dy \otimes x$ and $y \otimes dx$ are respectively $(-1)^{ab}$ and $(-1)^{ab+b}$, on either side of the cochain equation. This proves $dT_\pi = T_\pi d$, first for generators and then for every permutation. Each map is continuous and therefore extends to $C^{\otimes_\pi k}$.

The signed average and its complementary map are

$$S_k = \frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi, \quad 1 - S_k. \quad (\text{SS.4})$$

For each γ , exactly $k!$ pairs (π, σ) satisfy $\pi\sigma = \gamma$, and their signed coefficient is $\text{sgn}(\gamma)/(k!)^2$. Hence $S_k^2 = S_k$. In top degree every factor has degree one, so $T_\pi = \text{sgn}(\pi) P_\pi$ and S_k is the ordinary symmetric projection there. The unsigned chain average has top part $k!^{-1} \sum_{\pi} \text{sgn}(\pi) P_\pi$; this exact equation specifies its alternating target.

To verify the completed cohomology directly, extend the finite sum

$$H_k = \sum_{j=1}^k P^{\otimes_\pi(j-1)} \hat{\otimes}_\pi H \hat{\otimes}_\pi 1^{\otimes_\pi(k-j)}. \quad (\text{SS.5})$$

The degree -1 factor contributes the Koszul sign $(-1)^{\sum_{i < j} |x_i|}$. Expanding $dH_k + H_k d$, the terms in which the differential and homotopy occupy different slots cancel in pairs. The term in slot j is $P^{\otimes(j-1)} \otimes (1 - P) \otimes 1^{\otimes(k-j)}$ by (SS.2). Summing telescopes to $1 - P^{\otimes k}$. Continuity extends this identity from the dense algebraic tensor product. Replace H_k by the explicit equivariant homotopy

$$\overline{H}_k = \frac{1}{k!} \sum_{\pi} T_\pi H_k T_\pi^{-1}. \quad (\text{SS.6})$$

The operator $P^{\otimes k}$ commutes with every permutation, so (SS.6) has the same homotopy identity and commutes with S_k . The maps $q^{\otimes_\pi k}$ and $s^{\otimes_\pi k}$, restricted to the images of (SS.4), now give continuous inverse cohomology maps. Thus

$$H^r(\text{im } S_k) = \begin{cases} (Q^{\otimes_\pi k})^{S_k}, & r = k, \\ 0, & r \neq k. \end{cases} \quad (\text{SS.7})$$

The complementary cohomology is the image of $1 - \frac{1}{k!} \sum_{\pi} P_\pi$ on the same completed Q -tensor. The full cochain isomorphism and its inverse are $x \mapsto (S_k x, (1 - S_k)x)$ and $(u, v) \mapsto u + v$. Equations (SS.5)–(SS.7) prove the completion statement using the supplied continuous contraction; they require no exactness assertion for arbitrary completed tensor products.

Here is the finite-rank cochain map that supports the finite traces. The full packet jet $J_h : \mathcal{B} \rightarrow E$ annihilates ΘV , and descends to a continuous $j_h : Q \rightarrow E$. The original packet inclusion $\sigma_h : E \rightarrow Q$ satisfies $j_h \sigma_h = 1$, as $J_h R_M = 1$ and $q R_M = \sigma_h$ in one variable. Set $\iota_h = s \sigma_h : E \rightarrow \mathcal{B}$ and define

$$\Pi_h^0 = 0, \quad \Pi_h^1 = \iota_h J_h, \quad J_h \iota_h = 1. \quad (\text{SS.8})$$

This is a continuous finite-rank cochain projector: $\Pi_h^1 \Theta = 0$ and $(\Pi_h^1)^2 = \Pi_h^1$. It commutes with permutations after taking identical tensor powers. Consequently $\Pi_h^{\otimes k} S_k$ and $\Pi_h^{\otimes k} (1 - S_k)$ are the finite cochain projectors with ranks $\binom{k+d-1}{d-1}$ and $d^k - \binom{k+d-1}{d-1}$. The projector S_k on the full infinite complex

itself has not been assigned either finite rank. The finite inclusion and retraction on cohomology are precisely $\sigma_h^{\otimes k}$ and $j_h^{\otimes k}$, restricted to the respective permutation summands. Their composite on the packet is the identity.

For $U(a) = e^{(\log a)A}$, $a > 0$, the arithmetic action obeys $J_h U_{\mathcal{B}}(a) = U(a) J_h$. Thus $J_h U_{\mathcal{B}}(a) \iota_h = U(a)$. If $L : F \rightarrow X$, $R : X \rightarrow F$, and F is finite dimensional, the trace of the finite-rank map LR is $\text{Tr}_F(RL)$: expanding LRx in a basis of F expresses both as the sum of the same coefficient functionals evaluated on the same image vectors. Apply this identity to (SS.8) and its tensor powers. Expanding a permutation trace in a tensor basis contracts one matrix index around each cycle, proving

$$\text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|}). \quad (\text{SS.9})$$

This proof uses all entries of $U(a)$ and applies to its nonsemisimple action. The cochain supertrace carries $(-1)^k$. The complementary finite supertrace is exactly $(-1)^k$ times $(\text{Tr } U(a))^k$ minus the trace in (SS.9). No infinite trace is assigned by these equations.

At a fixed permutation-invariant support fibre, every displayed linear map lifts by $(\lambda, x) \mapsto (\lambda, Tx)$. For arbitrary masks the support of (SS.4) is the specified orbit join $c(\lambda) = \bigvee_{\pi} \pi\lambda$, and the amplitude maps include the original transports into that fibre. The two projection maps then decompose the transported vector in the fibre over $c(\lambda)$; their sum is that transport. Their sum has not been declared the identity on a different original fibre. The upstream kernel is exactly the kernel of its specified transport $V_{\lambda} \rightarrow V_{c(\lambda)}$. At invariant masks the transport is the identity and the decomposition above applies. Every vanishing projected amplitude remains the supported zero in its target fibre; external absence is sent to external absence.

19.2 Orbit coordinates, their Gram, and all local nilpotents

Let $e_1, \dots, e_d$ denote the original power basis with indices shifted by one. For $n \in \mathbb{N}^d$, $|n| = k$, let b_n be the sum of the distinct basis tensors having occupancy n . Define

$$N = \begin{pmatrix} k+d-1 \\ d-1 \end{pmatrix}, \quad I : \mathbb{C}^N \longrightarrow E^{\otimes k}, \quad Ie_n = b_n, \quad D_o = I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, \quad L = D_o^{-1} I^*. \quad (\text{SS.10})$$

Distinct tensor basis vectors are orthonormal for this coordinate calculation; hence D_o records exactly their orbit sizes. It is separate from the arithmetic Gram. One has $LI = 1$ and $IL = P_{\text{sym}} = k!^{-1} \sum P_{\pi}$, since either map replaces every coefficient on an orbit by its average. Here $\text{Sym}^k E$ denotes this invariant image. Its comparison with the symmetric tensor quotient is the quotient map restricted to that image: it sends b_n to $(k! / \prod_i n_i!) e_1^{n_1} \cdots e_d^{n_d}$. The inverse sends this quotient monomial to $(\prod_i n_i! / k!) b_n$. These formulas prove the isomorphism and retain the exact diagonal factors between the two presentations.

Permutation invariance of the arithmetic constraints and measure makes the unique constrained minimum satisfy $R_M P_{\pi} = P_{\pi} R_M$; applying the inverse permutation to a feasible minimizer gives a feasible vector of the same norm, and uniqueness identifies it. Thus G_M and G_M^{-1} commute with P_{sym} . Retain the matrices

$$G_M^s = I^* G_M I, \quad W_M^s = I^* W_M I, \quad A^s = L A^{(k)} I, \quad A^{(k)} I = I A^s, \quad (G_M^s)^{-1} = L G_M^{-1} L^*. \quad (\text{SS.11})$$

The last equality includes both D_o^{-1} factors in L . Indeed its product with $I^* G_M I$ is

$$L G_M^{-1} (L^* I^*) G_M I = L G_M^{-1} P_{\text{sym}} G_M I = LI = 1.$$

This proves the inverse formula without exchanging arbitrary compression and inversion.

In these unscaled coordinates, if $Ae_j = \sum_i A_{ij}e_i$, the full generator is

$$A^{(k)}b_n = \left(\sum_i n_i A_{ii} \right) b_n + \sum_{\substack{i \neq j \\ n_j > 0}} (n_i + 1) A_{ij} b_{n+e_i-e_j}. \quad (\text{SS.12})$$

For a specified target tensor of occupancy $n + e_i - e_j$, each of its $n_i + 1$ entries equal to e_i can have come from an e_j in a source tensor, and these are exactly the contributing preimages. The diagonal term counts its n_i unchanged positions. This proves every coefficient in (SS.12). Using the original quotient's complete CRT isomorphism, set $E_\rho = \mathbb{C}[x_\rho]/x_\rho^{m_\rho}$, $t = \rho + x_\rho$. The explicit CRT map is reduction modulo each $(t - \rho)^{m_\rho}$; its inverse is the unique degree $< d$ polynomial with the prescribed full jets. Applying its tensor power and restriction through I, L is the exact change from the original orbit coordinates to the following direct sum:

$$\text{Sym}^k E \simeq \bigoplus_{\substack{\ell_\rho \geq 0 \\ \sum_\rho \ell_\rho = k}} \bigotimes_{\rho} \text{Sym}^{\ell_\rho} E_\rho, \quad w_\ell = \prod_{\rho} \binom{\ell_\rho + m_\rho - 1}{m_\rho - 1}. \quad (\text{SS.13})$$

The isomorphism groups the unscaled tensor orbit sums by the number of factors in each CRT summand; these grouped basis vectors exhaust the full basis, which proves the direct sum and its dimensions.

On the summand ℓ , the generator is $\lambda_\ell + N_\ell$, where $\lambda_\ell = \sum_\rho \ell_\rho \rho$ and N_ℓ is the sum of the multiplications by the x_ρ in their original slots. Put $h_\ell = \sum_\rho \ell_\rho (m_\rho - 1)$. Every application raises the total local exponent by one, so $N_\ell^{h_\ell+1} = 0$. On the tensor of local constants,

$$N_\ell^{h_\ell} \left(\bigotimes_{\rho} 1^{\otimes \ell_\rho} \right) = \frac{h_\ell!}{\prod_{\rho} ((m_\rho - 1)!)^{\ell_\rho}} \bigotimes_{\rho} (x_\rho^{m_\rho-1})^{\otimes \ell_\rho} \neq 0. \quad (\text{SS.14})$$

The coefficient counts the ordered choices of increments in each slot, each of which must be incremented exactly $m_\rho - 1$ times. Therefore the nilpotent index is exactly $h_\ell + 1$. Even when different ℓ have the same λ_ℓ , their direct summands in (SS.13) are retained separately. Traces may kill individual nilpotent diagonal contributions, while the complete operators (SS.12)–(SS.14) retain them.

19.3 Complete symmetric spectral sums and their exact departure

List $\rho_1, \dots, \rho_d$ with every algebraic multiplicity and put

$$\delta_i = \text{Re } \rho_i - \frac{1}{2}, \quad \sigma_1 = \sum_i \delta_i, \quad \sigma_2 = \sum_i \delta_i^2, \quad a_k = \binom{k+d-1}{d}, \quad b_k = \binom{k+d-1}{d+1}. \quad (\text{SS.15})$$

Binomial coefficients with lower index exceeding the upper index are zero. Thus these formulas include $k = 1$ and $d = 1$. Triangularize A . The induced ordered orbit basis is also triangular under (SS.12), with diagonal $\lambda_n = \sum_i n_i \rho_i$, once for each $n \in \mathbb{N}^d$, $|n| = k$. For example, order the monomials compatibly with the triangular basis's index weight; each off-diagonal substitution in (SS.12) decreases that weight. Consequently

$$\det(z - A^s) = \prod_{|n|=k} (z - \lambda_n), \quad \sum_{k \geq 0} q^k \text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \prod_{i=1}^d (1 - qa^{\rho_i})^{-1}. \quad (\text{SS.16})$$

The second identity is an identity of formal power series: expand each geometric series and select its occupancy. On grouping repeated roots it is $\prod_{\rho} (1 - qa^{\rho})^{-m_\rho}$, and the coefficient for grouped occupancy ℓ is precisely w_ℓ from (SS.13). This gives the exact multiplicity dictionary between the cycle trace (SS.9) and the full local jets.

Define the symmetric absolute spectral cost and metric departure by

$$\begin{aligned}\mathcal{D}_k^s &= \sum_{|n|=k} \left| \sum_i n_i \delta_i \right| = \sum_{|\ell|=k} w_\ell \left| \sum_\rho \ell_\rho (\operatorname{Re} \rho - \tfrac{1}{2}) \right|, \\ \mathfrak{d}_M^s &= \operatorname{Tr}((G_M^s)^{-1} (A^s)^* G_M^s A^s) - \sum_{|n|=k} |\lambda_n|^2, \\ S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2}, \quad \epsilon_M^s = \|S_M^s\|_{\text{op}}.\end{aligned}\tag{SS.17}$$

The square root is used only as the displayed isometry for computing the original Hermitian pencil; it changes none of the orbit vectors, source functions, arithmetic constants, or entries of G_M^s .

Theorem 19.1 (Exact occupancy moments and symmetric departure). *The full unweighted occupancies satisfy*

$$\begin{aligned}\sum_{|n|=k} \sum_i n_i \delta_i &= a_k \sigma_1, \\ \sum_{|n|=k} \left(\sum_i n_i \delta_i \right)^2 &= (a_k + b_k) \sigma_2 + b_k \sigma_1^2.\end{aligned}\tag{SS.18}$$

In the original arithmetic orbit Gram,

$$\begin{aligned}\mathfrak{d}_M^s &\geq 0, \quad \operatorname{Tr} S_M^s = 2a_k \sigma_1, \\ \operatorname{Tr}((S_M^s)^2) &= 4\{(a_k + b_k) \sigma_2 + b_k \sigma_1^2\} + 2\mathfrak{d}_M^s.\end{aligned}\tag{SS.19}$$

In particular, if $\bar{\delta} = \sigma_1/d$, the exact mean-square spread of the occupancy offsets is

$$\sum_{|n|=k} \left(\sum_i n_i \delta_i - k\bar{\delta} \right)^2 = \frac{Nk(k+d)}{d(d+1)} \left(\sigma_2 - \frac{\sigma_1^2}{d} \right).\tag{SS.20}$$

Proof. The product $\prod_i (1 - q_i)^{-1}$ enumerates every occupancy. Apply $q_i \partial_{q_i}$ and then set all $q_i = q$. Its coefficient of q^k gives $\sum n_i = [q^k] q(1 - q)^{-d-1} = a_k$. For two distinct indices, applying the two derivatives gives $\sum n_i n_j = [q^k] q^2 (1 - q)^{-d-2} = b_k$. Applying $q_i^2 \partial_{q_i}^2$ gives $\sum n_i (n_i - 1) = 2b_k$. These coefficient identities follow by multiplying geometric series, or by induction from $(1 - q)^{-r} = \sum_{m \geq 0} \binom{m+r-1}{r-1} q^m$. Therefore $\sum n_i^2 = a_k + 2b_k$. Expanding the square, the cross terms have coefficient b_k ; collecting the diagonal terms gives (SS.18). The identities $a_k = Nk/d$ and $a_k + b_k = Nk(k+d)/(d(d+1))$ follow directly from the factorials, including the stated edge cases. Expanding the centered square proves (SS.20).

Let $T = (G_M^s)^{1/2} A^s (G_M^s)^{-1/2}$. A unitary upper triangularization exists by induction: choose a unit eigenvector as the first basis vector, complete it to an orthonormal basis, and repeat on the remaining diagonal block. If its strictly upper triangular part is N_T , its Hilbert–Schmidt squared norm is exactly $\mathfrak{d}_M^s$; the diagonal entries are the complete λ_n . This proves nonnegativity while retaining every nonzero off-diagonal entry. Since $S_M^s = T + T^* - kI$, its diagonal is $2 \sum_i n_i \delta_i$; each strictly upper triangular entry and its adjoint contribute twice their squared absolute value to the squared trace. Thus $\operatorname{Tr}((S_M^s)^2) = 4 \sum_n (n \cdot \delta)^2 + 2\mathfrak{d}_M^s$. Equation (SS.18) proves (SS.19). $\square$

For comparison with (CT.20)–(CT.21), the ordered tensor has an occupancy n exactly $k! / \prod_i n_i!$ times. The permutation complement therefore has that multiplicity minus one. Its characteristic polynomial, trace, and absolute spectral cost are obtained with these nonnegative integer multiplicities. In particular

$$\mathcal{D}_k = \mathcal{D}_k^s + \sum_{|n|=k} \left(\frac{k!}{\prod_i n_i!} - 1 \right) |n \cdot \delta|.\tag{SS.21}$$

This equality retains every contribution omitted by the symmetric inclusion. Moreover the permutation projection is orthogonal also for G_M , because it commutes with that Gram. Its image and kernel are invariant under $A^{(k)}$. Thus the full metric operator is the orthogonal direct sum of its symmetric and complementary restrictions; the squared Hilbert–Schmidt norms and the eigenvalue-square sums add separately. The full departure in (CT.20) is exactly $\mathfrak{d}_M^s$ plus the corresponding complementary departure, each nonnegative. This proves the full operator comparison as well as the scalar identity (SS.21).

19.4 The cost in the original symmetric relation layer

Restrict the original source maps without changing their Hilbert targets:

$$Y_M^s = Y_M I, \quad C_M^s = C_M I, \quad \Delta_M^s = G_M^s - G_{M+1}^s = (Y_M^s)^* Y_M^s, \quad W_M^s = -((Y_M^s)^* C_M^s + (C_M^s)^* Y_M^s). \quad (\text{SS.22})$$

Their images lie in the invariant original relation layer, whose dimension is $r_{\text{lay}} = p_{\leq k}(M+1)$. Indeed an unscaled symmetric polynomial basis in exact degree $M+1$ is indexed by its partitions into at most k parts. The invariant full-jet map is onto at every $M \geq k(d-1)$: average a complete tensor remainder to lift each invariant jet. The difference of the two kernel dimensions is therefore the number of the new degree $M+1$ basis vectors. The injective polynomial realization and orthogonal layer quotient transport this dimension to the actual source layer, proving the assertion with its original relation ideal. Put

$$r_M^s = \text{rank } Y_M^s, \quad \tau_M^s = \text{Tr}((G_M^s)^{-1} \Delta_M^s), \quad \chi_M^s = \text{Tr}((G_M^s)^{-1} (C_M^s)^* C_M^s). \quad (\text{SS.23})$$

Then the precise relative source costs are

$$n_{\pm}(W_M^s) \leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}\}, \quad \mathcal{D}_k^s \leq \frac{1}{2} \|S_M^s\|_1 \leq \sqrt{\tau_M^s \chi_M^s}, \quad \mathcal{D}_k^s \leq r_M^s \epsilon_M^s. \quad (\text{SS.24})$$

To prove the inertia bound, the Hermitian form in (SS.22) vanishes on $\ker Y_M^s$. Any strictly positive or strictly negative subspace of dimension greater than $\text{rank } Y_M^s$ intersects that kernel nontrivially, a contradiction. Apply the same argument to C_M^s . In the triangular basis used in (SS.19), the absolute diagonal sum of S_M^s is $2\mathcal{D}_k^s$. Its spectral expansion gives $|S_{ii}| \leq \sum_j |\lambda_j| |U_{ij}|^2$; summing over i bounds this sum by the trace norm. Finally, for any two finite-domain Hilbert-space maps B, C , their singular-vector decomposition and entrywise Cauchy–Schwarz prove $\|B^* C\|_1 \leq \|B\|_{\text{HS}} \|C\|_{\text{HS}}$. Apply it to $B = Y_M^s (G_M^s)^{-1/2}$ and $C = C_M^s (G_M^s)^{-1/2}$ in (SS.22). The squared norms are τ_M^s, χ_M^s . The inertia bound gives at most $2r_M^s$ nonzero eigenvalues, proving the last assertion.

If the packet is stable, with full orders, under $\rho \mapsto 1 - \bar{\rho}$, its original antiunitary reflection commutes with permutations. Restricting (CT.16a) to the invariant space pairs every nonzero eigenvalue of the Hermitian pencil with its negative. Thus $\sigma_1 = 0$, and with $p_M^s = n_+(W_M^s) = n_-(W_M^s)$,

$$\begin{aligned} p_M^s &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}, \lfloor N/2 \rfloor\}, \\ \mathcal{D}_k^s &\leq p_M^s \epsilon_M^s, \\ p_M^s (\epsilon_M^s)^2 &\geq \frac{2Nk(k+d)}{d(d+1)} \sigma_2 + \mathfrak{d}_M^s. \end{aligned} \quad (\text{SS.25})$$

There are $2p_M^s$ nonzero eigenvalues, each of absolute value at most ϵ_M^s . Bounding their absolute and squared sums and using (SS.19) proves both inequalities. They involve no division by p_M^s ; when it vanishes, (SS.19) forces every displayed nonnegative term to vanish.

Let $\delta_* = \max_i |\delta_i|$. If it is positive, $|n \cdot \delta| \leq k\delta_*$ yields the additional explicit lower cost

$$\mathcal{D}_k^s \geq \frac{(a_k + b_k) \sigma_2 + b_k \sigma_1^2}{k\delta_*}. \quad (\text{SS.26})$$

Indeed square each occupancy offset, bound it by $k\delta_*|n \cdot \delta|$, and sum using (SS.18). For a reflected packet this becomes $\mathcal{D}_k^s \geq N(k+d)\sigma_2/(d(d+1)\delta_*)$. Each repeated eigenvector $v^{\otimes k}$ is nonzero and invariant, and its exact original Rayleigh quotient is $2k\delta_i$. Hence $\epsilon_M^s \geq 2k\delta_*$. Equations (SS.25)–(SS.26) additionally measure the cumulative cost of every occupancy, with all multiplicities and original layer ranks. If any $m_\rho > 1$, (SS.14) gives a nonzero nilpotent on the all- ρ summand. An operator with that Jordan block cannot be unitarily diagonal, so the sum of squares defining $\mathfrak{d}_M^s$ is strictly positive for every positive G_M^s . This retains a further finite-metric contribution even when all $\delta_i = 0$.

Finally the density proved in (JD.8) passes to the invariant source by the bounded permutation projection. The canonical representatives are the restrictions $R_M I$; hence $G_M^s \downarrow 0$ and $\det G_M^s \rightarrow 0$, while the full jets remain the identity at every finite M . With $\pi_M^s = \det G_{M+1}^s / \det G_M^s$, the positive eigenvalues t_j of $(G_M^s)^{-1/2} \Delta_M^s (G_M^s)^{-1/2}$ lie in $(0, 1)$, their number is r_M^s , and $\tau_M^s = \sum t_j$, $\pi_M^s = \prod (1 - t_j)$. The arithmetic mean–geometric mean inequality therefore gives

$$(\mathcal{D}_k^s)^2 \leq \tau_M^s \chi_M^s \leq r_M^s (1 - (\pi_M^s)^{1/r_M^s}) \chi_M^s \quad (r_M^s > 0). \quad (\text{SS.27})$$

For $r_M^s = 0$, (SS.22) gives $\Delta_M^s = W_M^s = 0$, without using an exponent $1/r_M^s$. Since $-\log(1-t) - t = \int_0^t u/(1-u) du$, the same finite source sequence also obeys, for integers $k(d-1) \leq a < b$ and $\mathcal{D}_k^s > 0$,

$$(\mathcal{D}_k^s)^2 \sum_{M=a}^{b-1} (\chi_M^s)^{-1} \leq \sum_{M=a}^{b-1} \tau_M^s \leq \log \frac{\det G_a^s}{\det G_b^s}. \quad (\text{SS.28})$$

Positivity of every denominator follows from (SS.24). Telescoping proves the last inequality. Moreover $\sum_{M=a}^\infty \tau_M^s = \infty$: otherwise the tail has $t_j \leq \tau_M^s \leq 1/2$, and $-\log(1-t_j) \leq 2t_j$ would keep the telescoped logarithm bounded, contrary to $\det G_M^s \rightarrow 0$. These identities transfer the entire determinant and relative-energy calculation to the source's smaller invariant family through its exact inclusion; they supply no assumed arithmetic upper estimate.

20 Arithmetic quartet parity and the exact positive frontier operator

Let h be a nonempty packet of actual zeros with their full orders, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Their composition also preserves the packet. Retain $d = \deg h$, $g = 2\xi$, $v_h = g/h$, $v = j_h v_h$, and the actual measure $d\nu_h = |v_h(1/2 + it)|^2 dt/(2\pi)$. Throughout this section $k \geq 1$, $M \geq k(d-1)$, and

$$E_k = (\mathbb{C}[s]/h)^{\otimes k}, \quad A_k = \sum_{j=1}^k A_{h,j}, \quad C_M = G_M^{-1}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{AP.1})$$

The metric G_M is the actual total-degree minimum of (CT.12)–(CT.13). All coordinates below remain its original monomial tensor coordinates.

20.1 The actual two frontier matrices

Let q_j be the original monic orthogonal polynomials for ν_h , and $\kappa_j = \int |q_j|^2 d\nu_h$. Thus κ_0 retains the mass of the original measure. For a multi-index α , put

$$q_\alpha = \prod_{j=1}^k q_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad z_\alpha = U_{v^{\otimes k}}[q_\alpha] \in E_k. \quad (\text{AP.2})$$

The source's notation p_j, ω_j denotes these same q_j, κ_j : their monic leading terms and identical lower-degree orthogonality determine the polynomial uniquely, and then determine its norm.

No polynomial or measure is divided by its norm. The triangular polynomial change of basis and product integration in (CT.12) give exactly

$$\mathcal{C}_M = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\kappa_\alpha}. \quad (\text{AP.3})$$

Indeed this basis has Gram diagonal κ_α , and its full interpolation matrix has columns z_α ; substitution in $JM^{-1}J^*$ proves (AP.3), retaining $v^{\otimes k}$ in every column.

Write $a_j = \kappa_j / \kappa_{j-1}$ for $j \geq 1$, $a_0 = 0$, and retain the ordered frontier $|\beta| = M + 1$. Its size is $r = \binom{M+k}{k-1}$. Define maps with the same declared source $\mathbb{C}^r$:

$$\begin{aligned} \Omega &= \text{diag}(\kappa_\beta)_{|\beta|=M+1}, & F &= [z_\beta]_{|\beta|=M+1}, \\ w_\beta &= \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, & E_+ &= [w_\beta]_{|\beta|=M+1}. \end{aligned} \quad (\text{AP.4})$$

The original recurrence $sq_j = q_{j+1} + b_j q_j - a_j q_{j-1}$, with $b_j + \bar{b}_j = 1$, applied to (AP.3) gives

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{AP.5})$$

To check the cancellation completely, an adjacent pair $\alpha, \alpha + e_j$ both of degree at most M has coefficients κ_α^{-1} and $-a_{\alpha_j+1} \kappa_{\alpha+e_j}^{-1}$, whose sum is zero. Its transposed conjugate cancels in the same way. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, which cancels the last term. The remaining frontier pair has $\beta = \alpha + e_j$ and coefficient $\kappa_\alpha^{-1} = a_{\beta_j} / \kappa_\beta$; summing these pairs is exactly (AP.4)–(AP.5). The same sum gives the full inverse-metric update

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (\text{AP.6})$$

20.2 The original linear reflection separates the two frontiers

Conjugation closure makes h real, and reflection closure then gives $h(1-s) = (-1)^d h(s)$. The original functional equation gives

$$v_h(1-s) = (-1)^d v_h(s), \quad q_j(1-s) = (-1)^j q_j(s). \quad (\text{AP.7})$$

For the second identity, the actual density is even in t . The polynomial $(-1)^j q_j(1-s)$ is monic, and substitution $t \mapsto -t$ verifies its orthogonality to every lower degree. The difference from q_j has degree less than j and is orthogonal to itself, so positivity proves it is zero. This proves (AP.7) with the original leading coefficient.

Let $\mathfrak{c} : E_k \rightarrow E_k$ be the complex-linear map substituting $s_j \mapsto 1 - s_j$ in every factor. It is well-defined because the ideal $(h(s_1), \dots, h(s_k))$ is preserved, and $\mathfrak{c}^2 = I$. Let C denote its matrix in the retained original basis. The original function map

$$(\mathfrak{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}) \quad (\text{AP.8})$$

is a linear unitary involution for $d^k x$, by the change of variables $y_j = x_j^{-1}$. Its Mellin transform replaces s_j by $1 - s_j$. Consequently, on an original numerator,

$$\mathfrak{t}_k \mathcal{T}_h^{(k)} P = (-1)^{kd} \mathcal{T}_h^{(k)} P(1 - s_1, \dots, 1 - s_k). \quad (\text{AP.9})$$

Equations (AP.7) and (AP.2) show that its full jet is $\mathfrak{c} \mathcal{J} P$, with exactly the displayed unit sign. Total degree and the original relation ideal are preserved. Applying $\mathfrak{t}_k$ to the unique minimum for a

prescribed jet therefore gives the unique minimum for its $\mathfrak{c}$ -image. Uniqueness follows from the orthogonal decomposition (CT.5). Hence

$$\mathfrak{t}_k R_M = R_M C, \quad C^* G_M C = G_M, \quad C z_\alpha = (-1)^{kd+|\alpha|} z_\alpha. \quad (\text{AP.10})$$

In particular, with $\eta = (-1)^{kd+M+1}$,

$$CF = \eta F, \quad CE_+ = -\eta E_+. \quad (\text{AP.11})$$

Every term defining w_β has degree M , which proves the second identity without suppressing any coefficient.

Declare the following maps between the original positive spaces and their coordinate Hilbert spaces:

$$P = G_M^{1/2} C G_M^{-1/2}, \quad U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}. \quad (\text{AP.12})$$

Here $G_M^{1/2}$ and $\Omega^{1/2}$ are the explicit isometries from the declared original metric spaces to Euclidean coordinate spaces, with their displayed inverses. Every expression in (AP.12) retains both original matrices. From (AP.10), P is unitary and $P^2 = I$; hence $P^* = P$. Equations (AP.11) imply $PU = \eta U$, $PV = -\eta V$, and thus

$$U^* V = (PU)^* (PV) = -U^* V = 0. \quad (\text{AP.13})$$

20.3 The complete positive operator and the sharper arithmetic bound

Put $A = U^* U$, $B = V^* V$, retaining their source type $\mathbb{C}^r$. Equivalently,

$$A = \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, \quad B = \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}. \quad (\text{AP.14})$$

These positive-semidefinite matrices are built from the two original frontiers; A is distinct from the arithmetic generator A_k .

Theorem 20.1 (Exact positive frontier operator). *The actual relative weight satisfies*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = UV^* + VU^*, \\ \boxed{\epsilon_{k,M} &= \|UV^*\|, \quad \epsilon_{k,M}^2 = \lambda_{\max}(A^{1/2} B A^{1/2}),} \quad p_M = \text{rank}(A^{1/2} B^{1/2}), \\ \text{Tr}(S_M^2) &= 2 \text{Tr}(AB). \end{aligned} \quad (\text{AP.15})$$

Its positive and negative inertias both equal p_M . Define the original incidence bound and volume ratio by

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad \lambda_{k,M} = \|U\|^2, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M}. \quad (\text{AP.16})$$

Then

$$\boxed{\epsilon_{k,M}^2 \leq \Gamma_{k,M} \lambda_{k,M} \leq \Gamma_{k,M} (\pi_M^{-1} - 1).} \quad (\text{AP.17})$$

All formulas include the zero-rank cases and every full arithmetic jet. No growth estimate for $\Gamma_{k,M}$ or π_M is assumed.

Proof. Equation (AP.5) proves the first equality in (AP.15). Decompose the retained coordinate Hilbert space into the two orthogonal eigenspaces of the explicit Hermitian involution P . Relative to the order $(\eta, -\eta)$, the operator has blocks

$$S_M = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = UV^*|_{\ker(P+\eta I)}. \quad (\text{AP.18})$$

Indeed U^* vanishes on the opposite eigenspace and V^* vanishes on the first, by (AP.11). Squaring gives diagonal blocks TT^*, T^*T . For each nonzero singular value σ of T , choose unit singular vectors x, y with $Ty = \sigma x$, $T^*x = \sigma y$; then (x, y) and $(x, -y)$ have eigenvalues $\sigma, -\sigma$. The remaining eigenspaces are $\ker T^* \oplus \ker T$, with eigenvalue zero. These vectors form the full orthogonal decomposition by the spectral theorem for T^*T , including all its zero vectors. It follows that $\epsilon = \|T\| = \|UV^*\|$ and both inertias equal $\text{rank}(UV^*)$.

Now $(UV^*)(UV^*)^* = UBU^*$. Its nonzero eigenvalues and rank are those of $B^{1/2}AB^{1/2}$: the maps $UB^{1/2}$ and its adjoint transfer each nonzero eigenvector, with inverse divided by that eigenvalue. The latter matrix is $T_0^*T_0$, with $T_0 = A^{1/2}B^{1/2}$, while $T_0T_0^* = A^{1/2}BA^{1/2}$. Their full positive spectra agree by the same maps, and their ranks equal $\text{rank } T_0$. This proves the norm and rank formulas, including rank zero. Finally (AP.13) kills $(UV^*)^2$ and $(VU^*)^2$. Taking the trace of the remaining two products gives the last line of (AP.15).

For the incidence estimate, let $Z = [z_\alpha]_{|\alpha|=M}$ and $\Omega_M = \text{diag}(\kappa_\alpha)$. Define $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, and zero otherwise; summation applies if more than one specified incidence contributes. Then $E_+ = ZL$. The matrix $N = \Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is exactly

$$\sum_{j=1}^k \left(a_{\alpha_j+1} + \sum_{l \neq j} a_{\alpha_l} \right) = \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (\text{AP.19})$$

To verify each factor, its j -th contributing frontier is $\beta = \alpha + e_j$, and $\kappa_\alpha a_{\beta_j} / \kappa_\beta = 1$. The predecessors of this β supply $\sum_l a_{\beta_l}$, which is precisely the displayed summand. Every eigenvalue of N has modulus at most the largest row sum: in an eigenvector choose a coordinate of largest absolute value and apply that row equation. The matrix N is similar, by $\Omega_M^{1/2}$, to the Hermitian positive matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$. Its eigenvalues therefore lie in $[0, \Gamma_{k,M}]$. Undoing the congruence and using (AP.3) proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z \Omega_M^{-1} Z^* \preceq \Gamma_{k,M} \mathcal{C}_M, \quad VV^* \preceq \Gamma_{k,M} I. \quad (\text{AP.20})$$

Consequently $\epsilon^2 = \|UV^*\|^2 \leq \|U\|^2 \|V\|^2 \leq \Gamma_{k,M} \lambda_{k,M}$. Finally (AP.6) gives

$$\pi_M^{-1} = \det(I + UU^*) = \prod_j (1 + \lambda_j(UU^*)) \geq 1 + \lambda_{k,M}. \quad (\text{AP.21})$$

All eigenvalues here are nonnegative, proving the second bound in (AP.17). If $U = 0$ or $V = 0$, (AP.15) gives zero weight directly; all determinant denominators remain positive. $\square$

The arithmetic departure has an exact expression in these same two frontier Grams. List the original ρ_i with every algebraic multiplicity and put $\delta_i = \text{Re } \rho_i - 1/2$. Comparing (AP.15) with the fully proved tensor trace (CT.21) gives

$$\mathfrak{d}_M^{(k)} = \text{Tr}(AB) - 2 \sum_{i_1, \dots, i_k} (\delta_{i_1} + \dots + \delta_{i_k})^2. \quad (\text{AP.22})$$

This retains the complete nonsemisimple generator in the left-hand definition of (CT.20). For $k = 1$, a four-distinct-point tesserine orbit of common full order μ has the further exact chain, with $M = d + m$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_{1,M}^2 = \lambda_{\max}(A^{1/2}BA^{1/2}) \leq \Gamma_{1,M}(\pi_M^{-1} - 1). \quad (\text{AP.23})$$

The first inequality, original character, multiplicity and critical supported-zero case are proved in (R9); (AP.15)–(AP.17) prove the rest. Thus the original character observation is related to an explicit positive frontier operator through the original metric, full interpolation unit and declared isometries.

20.4 The same positive frontier on the actual symmetric summand

The cochain projector used here is exactly $S_k = (k!)^{-1} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, with the original Koszul permutations T_π . Its top-degree action is the ordinary permutation projection, because T_π there equals $\text{sgn}(\pi) P_\pi$. Equations (SS.1)–(SS.9) prove its continuous cochain maps, completed projective-tensor cohomology, and finite packet projector, including the complementary summand. We now use its finite invariant image with the original total-degree representatives R_M .

Write $e_1, \dots, e_d$ for the original power basis of E , with indices shifted by one, and retain each occupancy $n \in \mathbb{N}^d$, $|n| = k$. Let b_n be the literal sum of the distinct basis tensors of occupancy n . Define the maps and dimensions by

$$\begin{aligned} N_s &= \binom{k+d-1}{d-1}, & I : \mathbb{C}^{N_s} &\longrightarrow E_k, & Ie_n &= b_n, \\ D_o &= I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, & L &= D_o^{-1} I^*, & P_s &= IL = \frac{1}{k!} \sum_{\pi} P_\pi, & LI &= I_{N_s}. \end{aligned} \quad (\text{AP.24})$$

Indeed the distinct original tensor basis vectors are orthonormal for this coordinate calculation; hence their disjoint orbit sums have precisely the displayed diagonal Gram D_o . The map L averages coefficients on each orbit, proving all the compositions in (AP.24). In particular $P_s^* = P_s$ and $P_s^2 = P_s$. The factorials in D_o are retained separately from the arithmetic Gram.

Permutation invariance of the original product measure, relation ideal, total degree and full interpolation unit gives $R_M P_\pi = P_\pi R_M$, by applying the unitary inverse permutation to each constrained minimum. Thus G_M and G_M^{-1} commute with P_s . Put

$$\begin{aligned} R_M^s &= R_M I, & G_M^s &= I^* G_M I, & W_M^s &= I^* W_M I, \\ A_k^s &= L A_k I, & A_k I &= I A_k^s, & W_M^s &= (A_k^s)^* G_M^s + G_M^s A_k^s - k G_M^s, \\ I^* G_M &= G_M^s L, & G_M I &= L^* G_M^s, & \mathcal{K}_M^s &:= (G_M^s)^{-1} = L G_M^{-1} L^*. \end{aligned} \quad (\text{AP.25})$$

These are the same representatives and matrices as (SS.10)–(SS.11). For completeness, $I^* G_M = I^* P_s G_M = I^* G_M P_s = I^* G_M IL$, proving the first rectangular identity and then its adjoint. The generator commutes with every factor permutation, so it preserves the image of I , proving its intertwining identity and the displayed weight formula. Finally

$$(L G_M^{-1} L^*)(I^* G_M I) = L G_M^{-1} P_s G_M I = LI = I_{N_s},$$

which proves the inverse formula with both factors D_o^{-1} present. Positivity follows since I is injective and $G_M \succ 0$. The notation $\mathcal{K}_M^s$ refers only to this inverse kernel; it will not denote a reflection.

Keep the complete, possibly redundant frontier of (AP.4), with its unchanged domain $\mathbb{C}^r$ and norm matrix Ω . Its symmetric target maps are

$$\begin{aligned} F_s &= L F : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, & E_s &= L E_+ : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, \\ \mathcal{Z}_M^s &= F_s \Omega^{-1} E_s^* + E_s \Omega^{-1} F_s^*, & W_M^s &= G_M^s \mathcal{Z}_M^s G_M^s, \\ \mathcal{K}_{M+1}^s &= \mathcal{K}_M^s + F_s \Omega^{-1} F_s^*. \end{aligned} \quad (\text{AP.26})$$

The rectangular identities (AP.25), applied to $I^* G_M \mathcal{Z}_M G_M I$, prove the weight formula. Applying L and L^* to (AP.6) proves the inverse update. It uses the same orbit maps at both source degrees. Thus the update concerns the actual next symmetric minimum, not a newly selected positive form.

The original linear reflection C commutes with every factor permutation, since it substitutes the same $1 - s_j$ in each factor. Its invariant-coordinate matrix is

$$\begin{aligned} C_s &= L C I, & C I &= I C_s, & C_s L &= L C, & C_s^2 &= I_{N_s}, \\ C_s^* G_M^s C_s &= G_M^s, & C_s F_s &= \eta F_s, & C_s E_s &= -\eta E_s, & \eta &= (-1)^{kd+M+1}. \end{aligned} \quad (\text{AP.27})$$

For example $CI = P_s CI = ILCI$, and $C_s L = LCP_s = LP_s C = LC$; these prove the first intertwiners and then the involution. The metric equation follows by substituting $IC_s = CI$ into $I^* G_M I$ and using (AP.10). Applying L to (AP.11) proves both parities with the full arithmetic unit and its original sign.

Define the explicit isometries and the two retained frontier Grams by

$$\begin{aligned} P_M^s &= (G_M^s)^{1/2} C_s (G_M^s)^{-1/2}, \\ U_s &= (G_M^s)^{1/2} F_s \Omega^{-1/2}, \quad V_s = (G_M^s)^{1/2} E_s \Omega^{-1/2}, \\ A_s &= U_s^* U_s, \quad B_s = V_s^* V_s. \end{aligned} \tag{AP.28}$$

The superscript on P_M^s distinguishes this transported reflection from the fixed permutation projection P_s in (AP.24). Equation (AP.27) proves that P_M^s is a Hermitian involution and that U_s, V_s have opposite eigenvalues $\eta, -\eta$. Their ranges are consequently orthogonal, so $U_s^* V_s = 0$.

Theorem 20.2 (Exact symmetric arithmetic frontier). *For the actual signed symmetric cohomology summand, the original Hermitian pencil has*

$$\begin{aligned} S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2} = U_s V_s^* + V_s U_s^*, \\ \boxed{\epsilon_M^s &= \|U_s V_s^*\|, \quad (\epsilon_M^s)^2 = \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}),} \\ p_M^s &= \text{rank}(A_s^{1/2} B_s^{1/2}), \quad n_+(W_M^s) = n_-(W_M^s) = p_M^s, \\ \text{Tr}((S_M^s)^2) &= 2 \text{Tr}(A_s B_s). \end{aligned} \tag{AP.29}$$

With the original $\Gamma_{k,M}$ of (AP.16), put

$$\lambda_M^s = \|U_s\|^2, \quad \pi_M^s = \frac{\det G_{M+1}^s}{\det G_M^s}.$$

Then the same source sequence satisfies

$$\boxed{(\epsilon_M^s)^2 \leq \Gamma_{k,M} \lambda_M^s \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), \quad \lambda_M^s \leq \lambda_{k,M}.} \tag{AP.30}$$

All formulas retain every occupancy and nilpotent direction in (SS.12)–(SS.14), as well as the ungrouped frontier multiplicities.

Proof. Equation (AP.26) gives the first line of (AP.29). Decomposing $\mathbb{C}^{N_s}$ into the two orthogonal eigenspaces of P_M^s gives exactly the block operator (AP.18), now with $T_s = U_s V_s^*$ between those two spaces. The singular-vector maps and their inverses proved after (AP.18) apply to these declared matrices without any rank assumption. They give opposite eigenvalues for each positive singular value, $\text{rank rank}(A_s^{1/2} B_s^{1/2})$, and all remaining eigenvalues zero. The same two explicit composites $U_s B_s^{1/2}$ and its adjoint identify the positive eigenvalues with those of $A_s^{1/2} B_s A_s^{1/2}$. Finally $U_s^* V_s = 0$ kills the two square cross factors; the traces of the other two products both equal $\text{Tr}(A_s B_s)$. This proves every assertion in (AP.29).

Applying L and L^* to the original inequality (AP.20), and using (AP.25), proves

$$E_s \Omega^{-1} E_s^* \preceq \Gamma_{k,M} \mathcal{K}_M^s, \quad V_s V_s^* \preceq \Gamma_{k,M} I_{N_s}. \tag{AP.31}$$

Thus $\|U_s V_s^*\|^2 \leq \|U_s\|^2 \|V_s\|^2 \leq \Gamma_{k,M} \lambda_M^s$. Taking determinants of the exact inverse update in (AP.26) gives

$$(\pi_M^s)^{-1} = \det(I_{N_s} + U_s U_s^*) = \det(I_r + A_s) \geq 1 + \lambda_M^s, \quad 0 < \pi_M^s \leq 1. \tag{AP.32}$$

The determinant equality follows either by the two rectangular composites and their nonzero spectra, or by multiplying the two block factors for $\det(I + XY) = \det(I + YX)$. The inequality multiplies the nonnegative factors $1 + \lambda_j(A_s)$, exactly as in (AP.21). This proves the first two bounds in (AP.30).

For the final bound, the literal orbit maps give

$$\begin{aligned} L^* G_M^s L &= P_s G_M P_s \preceq G_M, \\ A_s &= \Omega^{-1/2} F^* P_s G_M P_s F \Omega^{-1/2} \preceq \Omega^{-1/2} F^* G_M F \Omega^{-1/2} = A. \end{aligned} \quad (\text{AP.33})$$

Indeed G_M commutes with the orthogonal projection P_s , so their omitted cross terms vanish and $G_M - P_s G_M P_s = (I - P_s) G_M (I - P_s) \succeq 0$. Taking the largest Rayleigh quotient of the two positive matrices in (AP.33) proves $\lambda_M^s \leq \lambda_{k,M}$. No comparison of arbitrary compressed inverse matrices is used. $\square$

There is also an exact comparison of the two relative operators. The map

$$Q_M = G_M^{1/2} I (G_M^s)^{-1/2} : \mathbb{C}^{N_s} \longrightarrow \mathbb{C}^{d^k}$$

has $Q_M^* Q_M = I_{N_s}$, and the inverse formula in (AP.25) gives $Q_M Q_M^* = P_s$. Direct multiplication therefore gives

$$S_M^s = Q_M^* S_M Q_M, \quad U_s = Q_M^* U, \quad V_s = Q_M^* V, \quad \epsilon_M^s \leq \epsilon_{k,M}. \quad (\text{AP.34})$$

Here $G_M^{1/2}$ commutes with P_s by its spectral decomposition. The full S_M also commutes with P_s , since both A_k and G_M do. Consequently (AP.34) is its invariant orthogonal restriction; the complementary operator remains on $\ker P_s$. The norm inequality follows directly by applying S_M to the unit vector $Q_M u$ and then the contraction Q_M^* .

If desired, the source frontier can be grouped with every factorial retained. For an orbit $\mathcal{O}$ of degree- $M + 1$ multi-indices, let $m_j(\beta) = |\{l : \beta_l = j\}|$ and

$$n_{\mathcal{O}} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!}, \quad \hat{f}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lz_{\beta}, \quad \hat{e}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lw_{\beta}, \quad \hat{\omega}_{\mathcal{O}} = n_{\mathcal{O}} \kappa_{\beta}. \quad (\text{AP.35})$$

The product norm κ_{β} is constant on an orbit. Equivariance of the full unit, polynomials and incidence sums implies that Lz_{β} and Lw_{β} are individually constant on the same orbit. Hence, for example,

$$\sum_{\beta \in \mathcal{O}} \frac{(Lz_{\beta})(Lw_{\beta})^*}{\kappa_{\beta}} = \frac{\hat{f}_{\mathcal{O}} \hat{e}_{\mathcal{O}}^*}{\hat{\omega}_{\mathcal{O}}}. \quad (\text{AP.36})$$

The same calculation with both columns of the first or second type proves the corresponding kernel and incidence identities. Equations (AP.35)–(AP.36) are the exact map to the source's orbit-factorial presentation. The ungrouped matrices in (AP.26)–(AP.34) continue to retain all their columns.

All zero-rank cases follow without inversion of A_s or B_s . If their square-root product vanishes, then $p_M^s = 0$, $W_M^s = 0$, and $\epsilon_M^s = 0$, even if the kernel update is nonzero. If $\pi_M^s = 1$, (AP.32) forces $U_s = 0$, with the same conclusion. The dimension N_s is always at least one; if it is one, a Hermitian involution has only one nonzero eigenspace, so one of U_s, V_s vanishes and the weight is zero. This argument also covers a zero-dimensional reflection eigenspace in a larger target. It neither removes the retained quotient nor asserts existence of a particular actual packet. For $k = 1$, every orbit cardinality is one and $I = L = I_d$; all the displayed formulas recover (AP.15)–(AP.21) exactly.

Finally the symmetric source is still $R_M I$, and its two actual boundary maps are $Y_M I$ and $C_M I$ in (SS.22). The original relation-graph map, followed by the bounded permutation projection on its Hilbert target, realizes the redundant frontier used here. Grouping it by (AP.35) gives the invariant relation layer of dimension $p_{\leq k}(M + 1)$, as proved in (SS.22)–(SS.24). Its primitive is the original fixed-order division primitive followed by S_k ; the cochain identity in (SS.3)–(SS.4) preserves its boundary and all signs. At invariant support fibres the same maps lift with that support; vanishing amplitudes remain the corresponding supported zero. For other masks the original orbit-joined transport and its kernel remain as stated after (SS.9). Thus (AP.30) is a proved bound for the same finite arithmetic cohomology summand, with no assumed asymptotic estimate for either $\Gamma_{k,M}$ or π_M^s .

21 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space, the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

21.1 Logarithmic coordinates and the original measure

Let $\mathcal{B}, V, \Theta$ have the definitions in Section 6. Put $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathcal{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \frac{1}{2} - \partial_y, \quad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y) e^{(\rho-1/2)y} dy. \quad (\text{JT1})$$

On $\mathcal{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathcal{H}^{(r)}$ be the domain of the first r powers of D in $\mathcal{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^k F\|_{\mathcal{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 21.1 (Finite differential graph norms and the complete jet graph). *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathcal{B} \rightarrow E_Z$ the complete Mellin-jet map. For every finite r ,*

$$\overline{\text{span}\{D^j f_0 : j \geq 0\}}^{\mathcal{H}^{(r)}} = \mathcal{H}^{(r)}, \quad f_0 = \Theta\phi_*, \quad (\text{JT2})$$

$$\overline{\{(F, J_Z F) : F \in \mathcal{B}\}}^{\mathcal{H}^{(r)} \oplus E_Z} = \mathcal{H}^{(r)} \oplus E_Z. \quad (\text{JT3})$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathcal{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_c e^{-c|t|}$. Indeed $e^{z/2}f_0(e^z)$ is holomorphic in $|\text{Im } z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\text{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $Jf_0 = f_0$, so analytic continuation gives $e^{z/2}f_0(e^z) = e^{-z/2}f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2 + it)|^2 dt/(2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \bar{u} d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2 , since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\text{TV}} \|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$ is an isometry onto $L^2(w_r dt/(2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2 + it)^j g(1/2 + it)$, proving (JT2).

Every theta polynomial has zero J_Z -jet. Thus the closure in (JT3) contains $(h, 0)$ for every h . The actual packet section $R_{\text{ref}} : E_Z \rightarrow \mathcal{B}$ satisfies $J_Z R_{\text{ref}} = 1$. Approximate $R_{\text{ref}} u$ in $\mathcal{H}^{(r)}$ by theta polynomials $p_n(D)f_0$. Then $(R_{\text{ref}} u - p_n(D)f_0, u) \rightarrow (0, u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. $\square$

Corollary 21.2 (Differential on the completed graph). *The continuous map*

$$(D, A) : \mathcal{H}^{(r+1)} \oplus E_Z \longrightarrow \mathcal{H}^{(r)} \oplus E_Z, \quad (F, u) \mapsto (DF, Au) \quad (\text{JT4})$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. *On the quotient by the boundary closure it is exactly* $A = M_t$.

Proof. The norm definitions bound $D : \mathcal{H}^{(r+1)} \rightarrow \mathcal{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). $\square$

Its split observation records $(e_{\mathcal{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathcal{H}}$ and to $u^\bullet$; they do not identify it with (τ, τ) .

21.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathcal{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|} dx), \quad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|} dy. \quad (\text{JT5})$$

The identity inclusions $\mathcal{B} \hookrightarrow \mathcal{H}_a \hookrightarrow \mathcal{H}_0$ are continuous. Source seminorms with powers exceeding $a + 1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 21.3 (Exact threshold, including the boundary). *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \text{Re } \rho - 1/2$, with $k \geq 0$. The functional on $\mathcal{B}$ extends continuously to $\mathcal{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2} \left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}} \right). \quad (\text{JT6})$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\bar{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2} \int y^{2k} e^{2by-2a|y|} dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. $\square$

Theorem 21.4 (Complete confluent Cauchy Gram). *For distinct centres ρ_i with $|\text{Re } \rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathcal{H}_a \rightarrow \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a^* J_a$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!l!} \left(\frac{1}{(2a - z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a + z_{ij})^{k+l+1}} \right), \quad z_{ij} = \rho_i + \bar{\rho}_j - 1. \quad (\text{JT7})$$

It is positive definite. The exact right inverse $S_a = J_a^ C_a^{-1}$ satisfies*

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1} (J_a F). \quad (\text{JT8})$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l} \rangle_a$; integrating $y^{k+l} e^{z_{ij}y-2a|y|} / (k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y) e^{\bar{\rho}_i y} = 0$. For each i , apply $\prod_{j \neq i} (\partial_y - \bar{\rho}_j)^{m_j}$. On the i -th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto $\text{range } J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathcal{H}_a/\overline{\Theta V}^{\mathcal{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \quad \ker = (\ker J_a)/\overline{\Theta V}^{\mathcal{H}_a}. \quad (\text{JT9})$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \quad T[p] = (p^{(k)}(\rho_i)/k!)_{i,k}.$$

Then $J_a|_{\mathcal{B}} = TJ_Z$, and the exact representative difference is $R_{\text{ref}}T^{-1} - S_a : \mathbb{C}^d \rightarrow \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathcal{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1}C_a(T^{-1})^*$ and the right inverse is S_aT .

21.3 The energy term carried by the weighted topology

Theorem 21.5 (Weighted boundary identity on the actual representatives). *On $\mathcal{B}$, with $(\Sigma H)(x) = \text{sgn}(\log x)H(x)$,*

$$\langle DF, H \rangle_a + \langle F, DH \rangle_a - \langle F, H \rangle_a = 2a \langle F, \Sigma H \rangle_a. \quad (\text{JT10})$$

*For an actual packet representative $R : E_Z \rightarrow \mathcal{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^*R$, $T_a = R^*\Sigma R$. Adjoints use exactly $\mathcal{H}_a$. Then*

$$A^*G_a + G_aA - G_a = 2aT_a - (B^*R + R^*B), \quad -G_a \leq T_a \leq G_a. \quad (\text{JT11})$$

For $Au = \rho u \neq 0$,

$$2 \operatorname{Re} \rho - 1 = 2a \frac{u^*T_a u}{u^*G_a u} - 2 \operatorname{Re} \frac{\langle Ru, Bu \rangle_a}{\|Ru\|_a^2}. \quad (\text{JT12})$$

Proof. In log coordinates integrate $-(\bar{f}h)'e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a \operatorname{sgn}(y)e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

21.4 The tessarine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi \exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \quad \frac{u^*(A^*G + GA - G)u}{u^*G u} = \frac{c_{-+}}{2}, \quad |b| < a \iff |c_{-+}| < 4a. \quad (\text{JT13})$$

The first equality is exactly $z = \psi p^{\rho-1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

22 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$ -module cohomology Q , a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion. The subsequent continuation now supplies certified entries of the original arithmetic moment matrix, the exact finite-compression and infinite-tail resolvent comparison, and a convergence rate for the initial resolvent including its complete off-line jets.

22.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\text{ref}}} \mathcal{B} \xrightarrow{q} Q, \quad J_Z R_{\text{ref}} = 1, \quad DR_{\text{ref}} - R_{\text{ref}}A = \Theta\phi_*\ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m . The complete calculation is

$$\begin{aligned} d\nu_Z(t) &= \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad N = d + m, \quad C_N P = j_h P, \quad K_N = C_N (H_N^\nu)^{-1} C_N^*, \\ p_{N,u} &= (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad R_m u = R_{\text{ref}} u + \Theta(B_{N,u}(D)\phi_*), \\ G_m &= U_\varepsilon^* K_N^{-1} U_\varepsilon, \quad W_m = U_\varepsilon^* K_N^{-1} (A K_N + K_N A^* - K_N) K_N^{-1} U_\varepsilon. \end{aligned} \quad (\text{R1})$$

The middle line evaluates the specified polynomial at the original D ; every source vector lies in V .

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\operatorname{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\operatorname{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \quad (\text{R2})$$

The proofs in Section 9 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right|. \quad (\text{R3})$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 8; those results do not identify its arbitrary member with R_m .

The full two-row source observation in Section 9 is injective after the original A -iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

22.2 Topology, duality and the two sides of a weight estimate

The actual source moments now have the absolutely convergent incomplete-gamma expression and explicit double-tail bound of Section 7. Its certified six-by-six Gram matrix concerns the original $|2\xi|^2 dt/(2\pi)$ measure. The exact identity (AM.19) supplies the original monic recurrence norms through degree five, with interval enclosures.

For the $|g/h|^2$ kernel, Section 10 proves a universal polynomial quadrature formula, including centres that coincide with a quadrature node. On the precise nonintersection domain its alternate expression is

$$K_{n-1} = \kappa_{n-1}^{-1} q_n(A) S_n q_n(A)^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*, \quad r_n = F_n(A)c.$$

The exact Schur comparison (KR.15)–(KR.20) then joins the initial and terminal observations to the actual infinite-dimensional tail. Section 11 proves the explicit initial-resolvent rate (GC.8) and its complete finite-algebra bound (GC.23). The projection to that off-line domain retains the critical summand as its stated kernel. All factors in the Schur identity remain in converting the initial estimate to the terminal form that determines ϵ_m .

In particular, Section 13 identifies exactly the two contributions in a reflection-stable packet:

$$\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).$$

Its full-jet inequalities also quantify the required metric ratios along each retained nilpotent chain. These identities do not infer a relative estimate from absolute matrix shrinkage.

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathcal{H}^{(r)} \oplus E$, with quotient E , not a quotient of $\mathcal{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re} \rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 21.3. The original differential then has the energy term $2a \operatorname{sgn}(\log x)$ in Theorem 21.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z , reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad Z^\dagger = \{1 - \bar{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z , so that this map is an involution on E_Z . The arithmetic residue form is the source form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g} ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f, g'u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{R4})$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \dots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)} ds, \quad 0 \leq i, j < d, \quad J_g = M_{jhg'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g . Its exact comparison to the positive source metric is therefore $G_m^{-1} SJ_g$, since $f^* G_m (G_m^{-1} SJ_g) u = f^* SJ_g u$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

22.3 Provenance and mathematical status

The coherent interpolation now has the full coefficient identity $J_{h,N} = U_{v_h} C_N$, with $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$. Equations (CT.3)–(CT.7) prove equality with (R1), including the original representative, and compatibility with the complete CRT injections between finite packets. The actual reflection preserves the unique minimum and proves the same-metric signed pairing of (CT.10) and (CT.16a).

On the original tensor complex, the complete calculation is

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A^{(k)} &= \sum_{j=1}^k A_{h,j}, & \mathcal{I}_{h,k,M} &= \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)), \\ R_M &= (I - P_{B_M}) s_h^{\otimes k}, & G_M &= (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}, \\ W_M &= -(Y_M^* C_M + C_M^* Y_M), & G_M - G_{M+1} &= Y_M^* Y_M, & Y_M, C_M &: E^{(k)} \longrightarrow \mathcal{E}_M, \\ \dim \mathcal{E}_M &= \binom{M+k}{k-1}. \end{aligned} \tag{R5}$$

Section 14 proves the original ordered division, both cochain signs, every layer map, and the extra orthogonal projection required when tensor packets are enlarged. Its formula (CT.18) calculates the resulting change in both G and W .

For a reflection-stable packet, let p_M be either signed inertia of W_M , and let $\mathfrak{d}_M^{(k)}$ be its complete tensor departure from (CT.20). The same source layers give

$$\begin{aligned} \mathcal{D}_k &\leq p_M \epsilon_M, & p_M \epsilon_M^2 &\geq 2kd^{k-1} \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_M^{(k)}, \\ p_M &\leq \min\{\operatorname{rank} Y_M, \operatorname{rank} C_M, \lfloor d^k/2 \rfloor\}, & \mathcal{D}_k^2 &\leq \operatorname{Tr}(G_M^{-1}(G_M - G_{M+1})) \operatorname{Tr}(G_M^{-1} C_M^* C_M). \end{aligned} \tag{R6}$$

The zero-inertia case is proved directly in (CT.25). The full ordered sum $\mathcal{D}_k$ and all multiplicities are retained in (CT.20). The joint density and exact graph closure in Section 15 show $G_M \downarrow 0$ for this actual total-degree sequence and identify every retained jet in the vertical fibre of (JD.11). The same section proves the coercive inverse-interpolation limit with its arithmetic unit, the supported-zero limit morphism, and the accumulated determinant cost (JD.14)–(JD.15). These formulas connect the relative layer energy to the exact rate of determinant loss through the same original matrices.

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) orthogonal boundary control (PR 13), and coherent interpolation with joint tensor boundaries (PR 14). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. The next tranche adds the certified original theta-moment series and its full truncation bound, the confluent quadrature and finite/terminal/tail resolvent identities, the explicit initial convergence rate, and the metric-departure and nilpotent-chain identities. The recovered June 24 historical

packet calculation is reproduced with a full-jet comparison in Section 2. Its exact A -invariant saturation is every off-axis local factor; the quotient, involutions, source-metric transport and split-zero endomorphism descent are proved there. The original transcript excerpt and its response locators are retained in the source package. This is an explicitly recorded historical recovery, with its new arithmetic saturation proof distinguished from the original permutation calculation. The subsequent historical reconstruction in Section 3 proves the inverse map from a complete local Hurwitz zero-cluster germ to every moment of its original compact probability measure, followed by the explicit Bernstein recovery of that measure. The monic cluster polynomial retains arbitrary zero multiplicity. The same section proves the complete Laurent period maps and their collision kernels. For two distinct primes, the joint Laurent map is surjective onto the full finite spectral algebra; Theorem 3.1 gives every primary kernel and the inverse original spectral coordinate, including its truncated logarithm on each nilpotent factor. The factorwise Split-Zero lifts and their exact comparison with the synchronized global lift are proved in (S4)–(S5). Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation includes (R2) on the actual $|g/h|^2$ -moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 21.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue. The next estimate must propagate (GC.8) through (KR.16), retaining q_n^2 , κ_n , a_n , the terminal ratio F_n and the actual tail denominator. The arithmetic moment formulas give an executable source for those finite data. The exact departure identity and the full-jet metric bounds specify what a resulting relative estimate would control in the original packet; no component is removed by changing observation.

The next step of that propagation is now calculated in Section 12. For $u_n = q_n(A)c$, $w_n = q_{n-1}(A)c$, and $M_n = \kappa_{n-1}^{-1}K_{n-1}^{-1}$, let $\beta_n = u_n^* M_n u_n$, and let P_n be the exact M_n -orthogonal projection with kernel $\mathbb{C}u_n$ when $\beta_n > 0$. Put $p_n = P_n w_n$, $z_n = P_n A u_n$. The universal step is

$$\begin{aligned} M_{n+1} &= a_n^{-1} \left(M_n - \frac{M_n u_n u_n^* M_n}{a_n + \beta_n} \right), & \frac{\det G_{m+1}}{\det G_m} &= \frac{a_n}{a_n + \beta_n}, \\ \Delta_{n+1} &= \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}, & \epsilon^{(n)} &= |gz| + \sqrt{g_Z^2 + \Delta_n}, \quad gz = \operatorname{Re} \operatorname{tr} A - d/2. \end{aligned} \quad (\text{R7})$$

Here $m = n - d - 1$. The zero-vector exceptional step is given completely in (KT.14), so critical centres and every polynomial coincidence remain in (R7)'s source calculation. The alternate terminal Gram follows by the exact isometry (KT.15) and recurrence (KT.16). The full scalar-to-matrix extraction uses the doubled off-line algebra and the explicit sequence (KT.23); at a reflected pair its matrix coefficient is exactly $(-1)^l F_n^{[k+l+1]}(\rho)$. Finally (KT.24)–(KT.25) recover these coefficients from the original initial error and tail, with both reciprocal series and all polynomial/norm factors displayed. The remaining energy estimate is therefore attached to the three explicit terms of (KT.11), including their actual cross pairing, on the same arithmetic moment sequence.

For a packet closed under both conjugation and reflection, the original real-coefficient calculation in (KT.28) closes this step in terms of consecutive determinant ratios. With the fixed section metric G_{-1} retained, put $r_m = \det G_m / \det G_{m-1}$ for $m \geq 0$. Then the actual arithmetic allowance obeys exactly

$$\boxed{\epsilon_m^2 = a_{d+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}.} \quad (\text{R8})$$

The coefficient reality and full reflection trace prove $\gamma_n = 0$ in the original coordinates; (KT.7)–(KT.8) then prove (R8), including zero-step cases. This places the relative quartet control directly on the same monic norm ratios and consecutive Gram determinants as the certified moment problem.

This also gives a direct arithmetic bound for the original tesserine violation channel. Retain the original quartet element

$$Q(\rho) = \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa, \quad \rho = \beta + i\gamma,$$

and its character $c_{-+}(\rho) = 2(2\beta - 1)$, as calculated in the carrier section. If $\gamma \neq 0$, $\beta \neq 1/2$, and Z is this entire four-point zero orbit with common full order μ , then $d = 4\mu$ and for every $m \geq 0$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = a_{4\mu+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{R9})$$

To prove every coefficient, conjugation and the functional equation preserve the full order, and the four real displacements are $\beta - 1/2, \beta - 1/2, -(\beta - 1/2), -(\beta - 1/2)$. The proved aggregate bound (R3) is therefore $\epsilon_m \geq 4\mu|\beta - 1/2| = \mu|2(2\beta - 1)|$. Squaring and using (R8) gives (R9). The sector morphism on the original supported quartet is precisely

$$G(\chi_{-+}) : G(\mathbb{C}[K_4]) \longrightarrow G(\mathbb{C}), \quad Q(\rho)^\bullet \longmapsto c_{-+}(\rho)^\bullet, \quad \tau \longmapsto \tau.$$

Here $\chi_{-+} = \text{ev}_{-1,+1}$ is the original group character extended linearly; it preserves multiplication because the character does, and hence its displayed G -lift is a semiring morphism. At $\beta = 1/2$ this morphism takes the supported value e ; the same original quartet has phase channel $c_{--} = 4i\gamma$. The orbit then has two distinct points when $\gamma \neq 0$, so its degree is 2μ and (R8) applies with that exact degree; the left-hand side of the sector bound is zero. Thus the arithmetic volume comparison and the supported critical value use the same typed character map, with the phase coordinate retained.

22.4 The original theta norm and the positive arithmetic frontier

The two newly delivered source notes, reproduced completely in the appendices, admit the following exact common formula with (R8). Keep $u_j = (I - P_{L_{j-1}})D^j f_0$, $\mathfrak{h}_j = \|u_j\|^2$, and the original measure $|g|^2 dt / (2\pi)$. The polynomial division and original minimum in (KL.23)–(KL.27) prove, at every full packet,

$$\kappa_{d+j} = \mathfrak{h}_j r_j \quad (j \geq 0), \quad \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2 \quad (\text{R10})$$

for reflection-stable packets and $m \geq 0$, where $c_m = d_m G_m^{-1} d_{m+1}^*$ uses the original theta projection rows. The first identity includes $j = 0$ and the fixed section G_{-1} : the actual new-layer vector at that endpoint is exactly f_0 . The row dictionary (KL.30) retains its separate domain $m \geq 0$; its cross term is $\mathfrak{h}_{m+1} c_m = \overline{\gamma_n}$, with $n = d + m + 1$. Both signs and the conjugation are obtained by multiplying those rows against the original inverse metric, including both local-unit matrices. Equation (KL.25) substitutes the first identity in (R10) into the complete terminal energy, proving its second identity. Under conjugation closure the original coefficients are real, so γ_n is real; the reflection trace makes its real part zero. Thus $c_m = 0$ in this case, by precisely the same calculation, with no discarded cross term.

For the full quartet in (R9), this gives the proved chain

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{R11})$$

Indeed the order-preserving symmetries establish the hypotheses of (R10) and its vanishing cross term; the lower bound was proved with every multiplicity in (R9). The norms $\mathfrak{h}_j$ are independent of the packet. For $h \mid H$, the original polynomial map $P \mapsto (H/h)P$ is an isometry because

$(g/H)(H/h)P = (g/h)P$. Its jets are the old complete jets and zero jets at each added centre. Its fixed representative and all its corrected representatives agree under this insertion, as proved in (KL.28). Thus the common theta sequence in (R11) connects the actual nested-packet maps; the packet dependence remains in the two displayed determinant ratios.

For the tensor frontier, retain the original full column maps $F, E_+ : \mathbb{C}^r \rightarrow E_h^{\otimes k}$ and its original norm diagonal Ω in (AP.4). With G_M the actual total-degree minimum, define

$$\begin{aligned} \mathbf{A} &= \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, & \mathbf{B} &= \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}, \\ \Gamma_{k,M} &= \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), & \pi_M &= \frac{\det G_{M+1}}{\det G_M}. \end{aligned} \quad (\text{R12})$$

All complete conjugation-and-reflection packets satisfy

$$\epsilon_{k,M}^2 = \lambda_{\max}(\mathbf{A}^{1/2} \mathbf{B} \mathbf{A}^{1/2}) \leq \Gamma_{k,M} \lambda_{\max}(\mathbf{A}) \leq \Gamma_{k,M} (\pi_M^{-1} - 1). \quad (\text{R13})$$

Here the square roots are the explicit metric isometries, with their inverses in (AP.12); the original matrices in (R12) remain part of the formula. The full proof in (AP.7)–(AP.21) starts from the actual linear reflection of the original Mellin source. It places the two frontiers in opposite orthogonal eigenspaces and gives $U^*V = 0$. The relative weight is consequently the two off-diagonal blocks UV^* and its adjoint. Its complete singular-vector decomposition identifies its norm squared with the positive operator in (R13), including all zero vectors. The explicit incidence row sum gives $VV^* \preceq \Gamma I$; the inverse-metric update gives $\pi_M^{-1} = \det(I + UU^*)$. Multiplication of its nonnegative eigenvalue factors proves the two inequalities. For $k = 1, M = d + m$, (R13), (R11), and (AP.23) calculate the same original tesserine bound through its norm, volume, and positive-operator presentations.

22.5 The symmetric summand, the actual relation layer, and support

Every factorial in the symmetric comparison is retained. The literal orbit-sum injection I , its diagonal $D_o = I^*I$, and $L = D_o^{-1}I^*$ in (AP.24) give $G_M^s = I^*G_M I$ and the exact identity $(G_M^s)^{-1} = L G_M^{-1} L^*$. The proof uses permutation invariance of the actual minimum, so that its metric commutes with the projection IL . Hence the original redundant frontier has symmetric maps $F_s = LF, E_s = LE_+$ with the unchanged Ω , and

$$\begin{aligned} \mathbf{A}_s &= \Omega^{-1/2} F_s^* G_M^s F_s \Omega^{-1/2}, & \mathbf{B}_s &= \Omega^{-1/2} E_s^* G_M^s E_s \Omega^{-1/2}, \\ (\epsilon_M^s)^2 &= \lambda_{\max}(\mathbf{A}_s^{1/2} \mathbf{B}_s \mathbf{A}_s^{1/2}) \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), & \pi_M^s &= \frac{\det G_{M+1}^s}{\det G_M^s}. \end{aligned} \quad (\text{R14})$$

Equations (AP.25)–(AP.36) prove all these maps, the full positive and zero spectra, the opposite reflection parities, and the orbit-factorial grouping. In particular the isometry $Q_M = G_M^{1/2} I (G_M^s)^{-1/2}$ satisfies $Q_M^* Q_M = I$ and $Q_M Q_M^* = IL$; it carries the symmetric relative operator to the invariant restriction of the original one, retaining the complementary operator on $\ker L$. The continuous cochain projector in (SS.1)–(SS.9) has exactly this cohomology image. Its finite packet version uses the original section and full jet, and all Koszul signs in that construction remain in its primitive.

The actual next relation layer underlying these matrices is proved in (KL.7)–(KL.20). Its complete graph frame, Gram, fixed-order division primitive, derivative image, and derivative cokernel are given there. The comparison with a homogeneous shell is the exact polynomial identity (KPC.7), with every lower coefficient in (KPC.6). Its full graph is the fibre product (KPC.12), and its two projection kernels are specified in (KPC.11). For two tensor coordinates the original ratio algebra embeds by (KPC.16), with exponent $\max_{\rho/\sigma=r} (m_\rho + m_\sigma - 1)$ at each coincident ratio. Its adjacent-shell generator action is (KPC.22a); the missing second-coordinate multiplier remains in that formula. The finite example (KPC.23)–(KPC.28a) calculates both original shell ranks, all nine node values, and the actual minimum, establishing the exact scope of that comparison.

For every linear map in these graph, kernel, quotient and cochain comparisons, the fixed-support lift at the end of the projective-comparison section maps (λ, v) to (λ, Tv) and maps external absence to absence. Its scalar and additive identities are proved there, including supported zero. For measured support, the historical branch chapter proves the separate full connection from ordinary branches to frame points through $\text{Idl}(L) \rightarrow L$, with its exact kernel congruence and descent criterion. Its L^∞ correspondence identifies every Boolean branch with a multiplicative state and its normal branches with actual measure atoms. The factor $\mu(a)^{-1} \int_a f d\mu$ is retained. These explicit maps specify the support observations available to the analytic construction and preserve the nonatomic support component alongside its atomic quotient.

The analytic continuation now has the common packet-independent theta norm sequence in (R10)–(R11), the exact positive operators in (R13)–(R14), and the full relation-layer and support maps just proved. Their finite identities determine the quantities to be estimated in the original arithmetic moment problem. The reader does not assert a vanishing asymptotic for those quantities.

Part II

Complete source proof notes and chronology

A PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, [Zeta research repository](#).

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

A.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

A.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \rightarrow \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_h E_h, \quad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \rightarrow E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q , not only quotients of Q , and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

A.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx$. Poisson and the two moment conditions put ΘV in $\mathcal{B}$. Also $D\Theta = \Theta D$, and D preserves V .

Set

$$A = \mathbb{C}[t], \quad t \cdot F = DF, \quad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1, \quad Q = \mathcal{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathcal{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\psi, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = (1 - D)\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

On the open strip $\mathcal{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \quad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

A.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathcal{B} : \mathcal{M}H(\rho) = 0\}$

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt.$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho)S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re \rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathcal{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$(D - \rho)\mathcal{B} = \ker(F \mapsto \mathcal{M}F(\rho)).$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathcal{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x) dx = \int_{\mathbb{R}} \phi(x) dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathcal{S}).$$

At the level of Taylor coefficients $\phi(x) = \sum_{j \geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j \geq 1} a_{2j} x^{2j} / (\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints 0, 1 are not included in this source range theorem.

A.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \dots, \rho_r \in \mathcal{S}$ and positive integers m_i . Define the particular polynomial

$$h(t) = \prod_{i=1}^r (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \quad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1} . In particular $g_h = j_h g$ retains the original local units of g .

Iterate Section 2 through the factors of h . Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = \text{id}.$$

The exact range identities are

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k)y} \frac{y^{j+l}}{j!l!} dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\bar{\theta}} & \mathcal{B}/h(D)\mathcal{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}$$

are actual A -module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \quad \beta_h[F] = j_h \mathcal{M}F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathcal{B}$; the class is independent of that choice by the range theorem.

A.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathcal{B}$ are torsion-free A -modules: every factor $D - \rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A , they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$C_+ \otimes_A^{\mathbf{L}} E_h \simeq [E_h \xrightarrow{\times g_h} E_h]$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \rightarrow V \xrightarrow{\Theta} \mathcal{B} \rightarrow Q \rightarrow 0$ also gives

$$0 \rightarrow Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \rightarrow Q/h(D)Q \rightarrow 0.$$

Here $Q[h(D)] = \ker(h(D) : Q \rightarrow Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M}F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \rightarrow E_h)$, choose its polynomial representative of degree less than d and set

$$\partial_h(P) = [S_h \Theta(P(D)\phi_*)] \in Q[h(D)].$$

The last expression is defined because its theta input has zero jets modulo h . Replacing P by $P + hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D , with $U_a F(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h , the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho + z) = u(z)z^r$, where $u(0) \neq 0$ (and $r = 0$ is allowed). For test order m , the two cohomology modules have dimension $\min(m, r)$. When $r = 0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q . The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

A.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ . Put

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z . Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z . This inverse retains the actual Taylor coefficients of g/h_Z ; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \quad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q , followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}(S_{h_Z} \Theta(P(D)\phi_*)) = \frac{gP}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z q_Z \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}}^{-1} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \quad J_Z[F] = j_{h_Z} \mathcal{M}F.}$$

Then

$$\boxed{J_Z \sigma_Z = \text{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \text{im } \Pi_Z = Q[h_Z(D)], \quad \ker \Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathcal{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z} \mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z} \Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ , this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho} \int_x^\infty \Theta \phi_*(t) t^{\rho-1} dt \right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q , proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z \Pi_W = \Pi_{Z \cap W}, \quad \Pi_{Z \cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \emptyset).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D . Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

A.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\text{sp}} : M^\tau \rightarrow (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semi-modules.

Consequently the finite observation has the split-linear equivalence

$$(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_h E_h), \quad [F]^\bullet \mapsto (j_h \mathcal{M} F \bmod g_h E_h)^\bullet.$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\tilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q . At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\tilde{\Pi}_Z \tilde{\Pi}_W = \varepsilon, \quad \varepsilon(\lambda, x) = (\lambda, 0).$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\emptyset = 0_Q$ and lift it to ε . The scalar representation

$$G(\text{End}_{\mathbb{C}} Q) \longrightarrow \text{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant- τ map. It is a unital semiring morphism and sends the supported scalar zero to ε .

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathcal{B}/\Theta W \longrightarrow Q_{W_D} := \mathcal{B}/\Theta W_D, \quad \ker = \Theta W_D/\Theta W.$$

For a D -stable W , define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$Q_W/h(D)Q_W \cong E_h/(g_h J_{W,h}).$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h / (g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

A.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathcal{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV , hence also its closure. They therefore descend to

$$Q^H = \mathcal{B} / \overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^H$. Indeed

$$\ker J_Z = h_Z(D)\mathcal{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^H = \sigma_Z(E_Z) \oplus h_Z(D)Q^H.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z . No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \rightarrow \prod_\rho \mathcal{O}_\rho / (g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathcal{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathcal{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \quad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [bMF].$$

Let $\text{tors}_{\mathcal{O}}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\text{tors}} : \text{tors}_{\mathcal{O}}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker \beta$ has a canonical $\mathcal{K}$ -module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \quad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathcal{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q . Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \text{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D - \rho)^r] \oplus (D - \rho)^r Q.$$

On the second summand, $(D - \rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D - \rho)^r Q = (D - \rho)^{2r} Q$$

follows by applying $(D - \rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r \mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x , so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β . Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$ -module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ , reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \quad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains

in the structural coefficient square throughout; x is the analytic image of the operator $t - \rho$, not a replacement for the scalar e or τ .

The theorem does not assert $\mathcal{N} = 0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

A.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s) = 1 - \bar{s}$. For germs define

$$f^\dagger(s) = \overline{f(1 - \bar{s})}.$$

The chosen monic polynomial satisfies $h^\dagger = (-1)^d h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f, u) = \sum_i \text{Res}_{s=\rho_i} \frac{f^\dagger(s)u(s)}{h(s)} ds.$$

It satisfies $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$. No phase is inserted to change this sign. Since $g^\dagger = g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\overline{\text{coker } g_h} \otimes \ker g_h \longrightarrow \mathbb{C}.$$

Well-definedness is the identity $R_h(ga, u) = R_h(a, gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+ \otimes_A^{\mathbf{L}} E_h$.

For an exact packet h_Z , set $v_{Z,h} = j_h(g/h_Z)$ as above and

$$j_Z : A_Z \rightarrow \ker g_h = E_Z, \quad [f] \mapsto [j_h(h_Z/g)f].$$

Here $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g) \cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$R_h(f, j_Z k) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger k}{g} ds =: R_Z(f, k).$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k] = [g'k]$. Its contraction is

$$R_h(f, j_Z J_g k) = R_Z(f, J_g k) = \sum_{\rho \in Z} r_\rho \overline{f(1 - \bar{\rho})} k(\rho) = \text{Tr}(M_{f^\dagger k} | A_Z).$$

In particular the factor 2 in $g = 2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ .

A.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A -action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathbf{A}_\tau(\mathcal{T}_+ \otimes_A^{\mathbf{L}} E_h) \xrightarrow{\sim} \mathbf{A}_\tau(\mathcal{T}_+) \otimes_A^{\mathbf{L}} E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H_\tau^1 = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k ,

$$H_k F(x) = \int_0^\infty k(a) F(x/a) \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r -fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \dots + \rho_r} \exp \left((\log a) \sum_{j=1}^r N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1) \log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

A.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

B PR 9: Chain descent

Source attribution. Web contribution, pull request #9, [Zeta research repository](#).

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. workbenches/tau-chain-descent/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

B.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

B.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide `Recovery.recoverEquiv`, `Relations.coequalizer_universal`, the internal-homology coequalizer, `ChainMap.homologyKernelEquiv`, and `homology_int_ertwines`. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

B.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad |V| = 0, \quad |\mathcal{B}| = 1, \quad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathcal{B}], \quad (\phi, \psi) \mapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1 - D)$; the inclusion $C_+ \rightarrow C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ . Set

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g . Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \quad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathcal{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V . For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

The stated source inverse uses $0 < \Re \rho < 1$ and the zero moment. Iteration supplies S_h . Define the actual maps

$$J : \mathcal{B} \rightarrow E, \quad JF = j_h \mathcal{M}F,$$

$$s : E \rightarrow \mathcal{B}, \quad s(u) = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \quad (1)$$

These equations fix the representative map, including all derivatives of g/h . On Q , the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

B.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathcal{B}_Z = \{F \in \mathcal{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathcal{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces,}$$

with maps

$$(\phi, u) \mapsto \Theta\phi + s(u), \quad F \mapsto (\Theta^{-1}(F - sJF), JF). \quad (2)$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathcal{B}_Z$, and there is a short exact sequence of A -modules

$$\boxed{0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J} E \longrightarrow 0.} \quad (3)$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\text{Ext}_A^1(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$\boxed{[\mathcal{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times.} \quad (4)$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H -jet is ε_Z .

In particular, (3) has no A -linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathcal{B}$: an A -linear map $E \rightarrow \mathcal{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

B.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \quad k_P(u) = q_P(u)(D)\phi_* \in V. \quad (5)$$

The numerator is divisible by h , since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathcal{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \quad (6)$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \quad (7)$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d . Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*} \quad (8)$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathcal{B}_Z} = \begin{pmatrix} D_V & \phi_*\ell_Z \\ 0 & M_t \end{pmatrix}.} \quad (9)$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1} dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)} \Theta\phi_*} \quad (10)$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E . Define

$$k_a(u) = S_h^V (U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*). \quad (11)$$

The source of S_h^V has zero H -jets because $a^s \varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a . The exact identities are

$$U_a s - s A_a = \Theta k_a, \quad k_{ab} = U_a k_b + k_a A_b, \quad k_1 = 0. \quad (12)$$

The same construction applies to a smooth compactly supported multiplicative test f : replace U_a by $H_f = \int f(a) U_a da/a$ and A_a by $M_{j_h \mathcal{M} f}$. The source integral belongs to V ; its H -jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

B.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathcal{B} \rightarrow \mathcal{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathfrak{p}_Z = (0, P_Z) : C_+ \rightarrow C_+ \quad (13)$$

is a cochain idempotent. Its image is the two-term complex $[0 \rightarrow s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+ , define a degree-minus-one map with sole component

$$\mathbf{H}_P^1 = k_P J : \mathcal{B} \rightarrow V.$$

Equation (6) gives

$$\boxed{P(D)\mathfrak{p}_Z - \mathfrak{p}_Z P(D) = d\mathbf{H}_P + \mathbf{H}_P d.} \quad (14)$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathbf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A -linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathcal{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \quad \pi : C_Z \rightarrow E[-1], \quad \pi^0 = 0, \pi^1 = J \quad (15)$$

are strictly A -linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \xleftarrow[\simeq]{\pi} C_Z \xrightarrow{\iota} C_+.} \quad (16)$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathcal{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A -linear; the A -linear replacement is the roof (16).

The strictly A -linear map $j : C_+ \rightarrow E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A -linear section into $\mathcal{B}$.

On the two-leg complex use $\mathfrak{p}_Z^0 = 0$ on $V \oplus V$, $\mathfrak{p}_Z^1 = P_Z$, and $\mathbf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \rightarrow C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

B.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathcal{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda, \lambda \vee \mu} x + \rho_{\mu, \lambda \vee \mu} y),$$

$$e(\lambda, x) = (\lambda, 0), \quad \tau(\lambda, x) = (\perp, 0). \quad (17)$$

The cochain projector $\tilde{p}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$H_{a,+}^1(F) = b, \quad H_{a,-}^1(F) = -\hat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\hat{b})$. The difference is the explicit cycle

$$\boxed{(b, \hat{b}) \in \ker d_{+-} = H^0(C_{+-}).} \quad (18)$$

For nonzero k_a , there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\hat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \quad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L \lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$\boxed{d\mathfrak{r}^0 = \mathfrak{r}^1 d, \quad \mathfrak{r}^2 = \mathfrak{r}.} \quad (19)$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-} . On degree-one homology, the map is

$$(\lambda, [F]) \mapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q . The original mask is retained through

$$x \mapsto (\text{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L . Define

$$\tilde{H}_a^{\text{left},1}(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\tilde{H}_a^{\text{right},1}(\lambda, F) = (\{+, -\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\tau(\tilde{U}_a \tilde{p}_Z + (-1)^{\bullet} \tilde{p}_Z \tilde{U}_a) = \tilde{d} \tilde{H}_a + \tilde{H}_a \tilde{d}. \quad (20)$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \rightarrow C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\} / B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D -stable admitted source $W \subset V$, the map $Q_W \rightarrow Q$ has kernel $\Theta V / \Theta W$. A homotopy value may not already lie in W . The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V , where every displayed homotopy is defined.

B.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) u(s)}{h(s)} ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \rightarrow E.$$

Substituting (4) into the residue formula gives

$$R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds = R_Z(f, u). \quad (21)$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho) = \text{Tr}(M_{f^\dagger u} \mid E). \quad (22)$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ . Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$j_B F(x) = x^{-1} \overline{F(1/x)}, \quad j_V \phi = \widehat{\phi}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta j_V = j_B \Theta$ and $J j_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $j_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$j_B S_h^{\mathcal{B}} = (-1)^d S_h^{\mathcal{B}} j_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$j_B s = s j_E, \quad j_B P_Z = P_Z j_B. \quad (23)$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$j_V k_a = a k_{1/a} j_E. \quad (24)$$

Indeed, apply j_B to (12), use $j_B U_a = a U_{1/a} j_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar{r}^\bullet$, which fixes both e and τ .

B.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+ . The cochain endomorphism

$$p_Z U_f = (0, s J H_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\text{Str}(p_Z U_f \mid C_+) = -\text{Tr}(s J H_f \mid \mathcal{B}) = -\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (25)$$

The finite-rank identity $\text{Tr}(AB) = \text{Tr}(BA)$ reduces the middle expression to $\text{Tr}(J H_f s \mid E)$; this is multiplication by $j_h \mathcal{M}f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n -fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathfrak{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \dots, f_n$,

$$\text{Str} \left(\bigotimes_j (\mathfrak{p}_Z \cup_{f_j}) \right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \quad (26)$$

Let $A = U_a \mathfrak{p}_Z$ and $B = \mathfrak{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^n B^{\otimes(j-1)} \otimes H_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

B.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θ_{k_a} on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

B.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and

original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, `FINITE_OPERATOR_COMPARISON.md`, sections 1–10; `TAU_BASE_MODEL.md` is its already-published model.
- Owner’s `SplitZero_Derived_Mathematics_2026-09-12(1).md`, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- `formal/splitzero/SplitZeroHomology.lean`, lines 175–end, read at the same revision; blob `7b9e5c49a494dd0c751133e0234b3e01048f7b67`.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, `arXiv:math/0412277`, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner’s programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

C PR 10: Global retraction

Source attribution. Web contribution, pull request #10, [Zeta research repository](#).

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. workbenches/tau-global-retraction/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

C.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

C.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\mathfrak{b}_\tau = \text{Spec } \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathcal{B} \longrightarrow V, \quad \boxed{\Lambda_{\alpha,\beta}\Theta = \text{id}_V}. \quad (2)$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathcal{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

C.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \quad (3)$$

These seminorms give $\mathcal{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y . Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\begin{aligned} \Theta\phi(x) &= \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{F}\phi(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x \xi} d\xi, \\ JF(x) &= x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (4)$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \quad J^2 = 1, \quad DJ = J(1 - D). \quad (5)$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\hat{\phi}$. This proves continuity $\Theta : V \rightarrow \mathcal{B}$. Also J is continuous on $\mathcal{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F)$, $0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, over $A = \mathbb{C}[t]$ with t acting as D . The two-leg tau chart retains $C_{+-} = [V^2 \rightarrow \mathcal{B}]$ with differential $\Theta(\phi - \hat{\psi})$ and source action $(D, 1 - D)$.

C.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathcal{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|)}. \quad (6)$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F .

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (7)$$

Indeed each summand is bounded by $C_{N,j}(F) n^{-N} x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \quad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \quad (8)$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x . The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

C.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \quad B_\beta = \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad T_{\alpha,\beta} = A_\alpha B_\beta. \quad (9)$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f , so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k \geq 0} T_{\alpha,\beta}^k, \quad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \quad (10)$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1 - \alpha)\mathcal{U}F, \quad b_F = (1 - \beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \quad (11)$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F . Therefore $Y : \mathcal{B} \rightarrow \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \quad (12)$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \leq C_q\|u\|_2$, then

$$q(\Phi F) \leq q(YF) + \frac{C_q}{1 - c_{\alpha,\beta}}\|YF\|_2. \quad (13)$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1 - A_\alpha)\phi + A_\alpha(1 - B_\beta)\phi = (1 - T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \quad (14)$$

C.1.5 4. Keep both original moment conditions

For arbitrary F , (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V .

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int \phi)$ are $(1, 1)$ and $(0, 1)$. Define

$$\mathbf{m}(\phi) = (\phi(0), \int_{\mathbb{R}} \phi), \quad j_0(a, b) = a\gamma_0 + (b - a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathbf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int \phi - \phi(0)\right)\gamma_2. \quad (15)$$

Then $\mathbf{m}j_0 = 1$, $P_V^2 = P_V$, $\text{im } P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta} : \mathcal{B} \rightarrow V.} \quad (16)$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $m\Phi : \mathcal{B} \rightarrow \mathbb{C}^2$ and the explicit correction $j_0 m\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

C.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \quad K = 1_{\mathcal{B}} - \Theta\Lambda. \quad (17)$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \text{im } K = \ker \Lambda. \quad (18)$$

Since K is continuous, its kernel is closed. Thus $Q = \mathcal{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathcal{B} \rightarrow Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \quad (19)$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathcal{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \quad (20)$$

For a convergent sequence $\Theta\phi_n \rightarrow F$ in $\mathcal{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \rightarrow \Lambda F$ in V , so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathcal{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathcal{B}/\overline{\Theta V}, \quad [F] \mapsto [F], \quad (21)$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \rightarrow \prod_{\rho} \mathcal{O}_{\rho}/(g), \quad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_{\infty}$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathcal{B}/\Theta W \longrightarrow Q, \quad \ker = \Theta V/\Theta W \cong V/W. \quad (22)$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W ; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V .

C.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\tilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \tilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \tilde{K}(\lambda, F) = (\lambda, KF). \quad (23)$$

They preserve external bottom and commute with scalar reflection. In particular

$$\tilde{K}\tilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathcal{B}}) = z_{V, \mathcal{B}}(\lambda, \phi).$$

Here $z_{V, \mathcal{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \xrightleftharpoons[s]{q} Q[-1], \quad H^1 = \Lambda,$$

$$\boxed{qs = 1, \quad 1_{C_+} - sq = dH + Hd.} \quad (24)$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \quad (25)$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \widehat{\phi}, \quad p_R^1 = q, \quad H_R^1(F) = (0, -\widehat{\Lambda F}). \quad (26)$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \quad (27)$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\text{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \quad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\boxed{\mathfrak{r} - \tilde{i}\tilde{p}_L\mathfrak{r} = \tilde{d}\tilde{H}_L + \tilde{H}_L\tilde{d},} \quad (28)$$

where $\tilde{H}_L(\lambda, F) = (\{+, -\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\text{supp } x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ .

C.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_a F(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$\boxed{k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a.} \quad (29)$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (30)$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$\boxed{U_a|_{\mathcal{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \quad D|_{\mathcal{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}.} \quad (31)$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \rightarrow V$. Then

$$s_1 - s_0 = \Theta b_{10}, \quad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10} \overline{U}_a. \quad (32)$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \rightarrow \mathcal{B}$, $\sigma_Z = q s_Z : E_Z \rightarrow Q$, and $J_Z : \mathcal{B} \rightarrow E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s \sigma_Z = K s_Z = s_Z - \Theta b_Z. \quad (33)$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \quad (34)$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \quad g = 2\xi. \quad (35)$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z} ; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_{h_Z}g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z \Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (36)$$

C.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \rightarrow Q'_b \xrightarrow{q'} \mathcal{B}'_b \xrightarrow{\Theta'} V'_b \rightarrow 0, \quad \Theta' \Lambda' = 1.} \quad (37)$$

The continuous cochain-dual complex is $[\mathcal{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0 . The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto (u \mapsto R_Z(f, j_Z u)), \quad (38)$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z . It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+ , the homotopy

$$H_n = \sum_{j=1}^n P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \quad (39)$$

includes $(-1)^{\sum_{i < j} |x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \quad (40)$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

C.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \rightarrow \mathcal{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g \rangle = a \langle f, g \rangle$:

$$\boxed{\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a \mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle.} \quad (41)$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e .

For a reduced eigenline of the actual dilation representation with exponent ρ , the left side at $u = v$ is $(a^{2\rho} - a) \mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathcal{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta \phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \quad W F(y) = e^{y/2} F(e^y).$$

The Fourier transform in y , with kernel $e^{-2\pi i \nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i \nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of $W b_*$ is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $W U_a b_* = a^{1/2} (W b_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \quad (42)$$

Its kernel is all of Q . Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

C.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

C.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

D PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, [Zeta research repository](#).

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. workbenches/tau-homotopy-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

D.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

D.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \quad d(v, w) = \Theta(v - \mathcal{F}w), \quad Q = B/\Theta V,$$

where Θ is injective and $\mathcal{F}$ is an R -linear automorphism. For a fixed $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, every solution of BOTH equations

$$dH = \Theta \kappa, \quad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b), \quad \alpha : Q \rightarrow_R V.$$

Proof: writing $H = (u, v)$, injectivity gives $u - \mathcal{F}v = \kappa b$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$. Hence $\mathcal{F}v$ factors uniquely through q , and supplies α . Conversely substitution proves both equations. The reference solution is $H_0 = (\kappa, 0)$; the family is an affine space under $\text{Hom}_R(Q, V)$, relative to that reference.

Let $\gamma(v) = (v, \mathcal{F}^{-1} v)$. Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map $H_1(C) \rightarrow H_0(C)$. There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate $R = \mathbb{C}$. With the quotient topology on Q , the continuous version follows: $\alpha q = \mathcal{F} \text{pr}_2 H$ is continuous, so the quotient property makes α continuous. Conversely the displayed formula sends continuous α to continuous H .

For the polynomial action $A_0 = \mathbb{C}[t]$, Fourier has the exact type

$$\mathcal{F} : V^j \xrightarrow{\sim} V, \quad t \cdot_{V^j} v = (1 - D)v, \quad j(t) = 1 - t,$$

because $F D = (1 - D)F$. Thus d is A_0 -linear from $V + V^\wedge j$ to B . A complex-linear homotopy is not automatically A_0 -linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

D.1.2 2. Original split support on every homotopy

Retain the base b_τ and the square $G(Z) \rightarrow G(C)$ over $Z > C$. The quotient $G(Z) \rightarrow Z$ remains infinite. Both lifted coefficient maps preserve τ and the supported scalar e separately.

Use the original support lattice $L = \{\text{bottom}, +, -, J\}$, $J = \{+, -\}$. Degree-zero fibres are $0, V, V, V + V$ with the coordinate inclusions. Degree-one active fibres are all B , with identity transports. Their differentials are Θ , $-\Theta$, F and d . Retain zero modules in the other degrees; their reconstructed zero map is $z(\lambda, x) = (\lambda, 0)$.

Literal multiplication by $E_{\text{sync}} = (1, e)$ induces

$$r^0(a, b) = (a + eb, b + ea), \quad r_L(\perp) = \perp, \quad r_L(\lambda) = J(\lambda \neq \perp).$$

Every homotopy lifts by

$$\tilde{H}_\alpha(\lambda, b) = (J, (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b)), \quad \lambda \neq \perp; \quad \tilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J , and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives $(J, (0, 0))$; external τ gives the global zero. Therefore

$$\tilde{d} \tilde{H}_\alpha + \tilde{H}_\alpha \tilde{d} = r \tilde{\Delta}.$$

The difference is $(J, \gamma(\alpha - \beta)qb)$, retaining its degree-zero class. For this chart, the pair $(\text{support}(x), r(x))$ is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W , keep $Q \rightarrow W$ with kernel $\Theta V / \Theta W$; q in the formulas means the displayed quotient by the full admitted V .

D.1.3 3. Coherence for all dilations

Let $S_a = U_a$ on V , $T_a = U_a$ on B , $U_a F(x) = F(x/a)$, and T_{bar}_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1} S_a \mathcal{F}) = (U_a, a U_{1/a}), \quad \mathsf{T}_a^1 = T_a.$$

It satisfies $T_a^0 \gamma = \gamma S_a$ and $q T_a = T_{\text{bar}}_a q$. With the PR #10 section s , take $P = (0, sq)$, and write

$$[\mathsf{T}_a, P]^1 = \Theta \kappa_a, \quad \kappa_a = k_a q, \quad T_a s - s \overline{T}_a = \Theta k_a.$$

The commutator product identity gives $\kappa_{ab} = S_a \kappa_b + \kappa_a T_b$. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\alpha_{ab} = S_a \alpha_b + \alpha_a \bar{T}_b.$$

Proof: the reference family $(\kappa_a, 0)$ is coherent. Subtract it and use injectivity of γ and surjectivity of q . No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with $L = (d/dt)|_0 \alpha(e^t)$, the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L \bar{T}_{e^r} dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t . Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular $\alpha_a = S_a b - b \bar{T}_a$ has infinitesimal parameter $D_V b - b D_Q$.

D.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

Use the existing $R: E_Z \rightarrow B$ and $J_Z: B \rightarrow E_Z$ with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z \Theta = 0, \quad J_Z D = A J_Z, \quad DR - RA = \Theta k.$$

The cochain projector $P_R = (0, R|_Z)$ has commutator $(0, \Theta k|_Z)$, so the reviewed theorem applies with $\kappa = k|_Z$. The quotient $\text{jet } J_{\text{bar}}_Z$ satisfies $J_{\text{bar}}_Z \sigma_Z = 1$, hence any $\alpha_E: E_Z \rightarrow V$ extends to $\alpha_E J_{\text{bar}}_Z$ on Q . Restriction to R gives

$$H_{\alpha_E} R = (k + \alpha_E, \mathcal{F}^{-1} \alpha_E), \quad dH_{\alpha_E} R = \Theta k.$$

Take $H = L^2(R_+, dx)$ with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^* R, \quad B_R = \Theta k, \quad W_R = -(R^* B_R + B_R^* R).$$

The composite of actual maps

$$\text{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \text{Hom}(E_Z, V^2) \xrightarrow{d \circ (-)} \text{Hom}(E_Z, H) \xrightarrow{B \mapsto -(R^* B + B^* R)} \text{Herm}(E_Z)$$

is constant at W_R . Its last map is real-linear. The retained H^0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$A^* G_R + G_R A - G_R = W_R.$$

The boundary term $[x \text{ conjugate}(F)H]_0^\infty$ vanishes by the stated decay. No claim that a θ boundary is orthogonal to a representative is used.

D.1.5 5. The actual representative-change family

Every representative of the same inclusion σ_Z is uniquely

$$R_b = R + \Theta b, \quad b : E_Z \rightarrow_{\mathbb{C}} V.$$

Indeed $q(R'-R)=0$ gives values in ΘV , and the PR #10 retraction gives the unique $b=\Lambda(R'-R)$. Finite source dimension gives continuity. Global continuous sections have the identical formula with $b:Q \rightarrow V$ continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is $U_b(v,u)=(v+bu,u)$, and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre $(V+E_Z)^\tau$. For the original masks, first keep the pair $(\text{support}(x), r(x))$; apply the shear on the joint fibre, and recover the original amplitudes by U_b^{-1} and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey $P_{(R_b)} - P_R = \text{partial } L_b$ with $L_b^{-1}(F)=(b|_Z F, 0)$. Their finite cochain traces agree because

$$\text{Tr}(\Theta b J_Z H_f) = \text{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by $h_Z(D)b(1)$, so its class in $V/h_Z(D)V$ remains $\epsilon_Z = j_h(h_Z/(2\xi))$. The residue/Jacobian trace form is consequently unchanged.

D.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain $\phi_j = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$, $\phi_j = D^j \phi_0$, and

$$\Phi_m : \mathbb{C}^{m+1} \rightarrow V, \quad (c_j) \mapsto \sum_{j=0}^m c_j \phi_j.$$

Their actual Mellin identities are $M \Theta \phi_j(s) = 2\xi(s)s^j$. They remain in the original two-moment source and are linearly independent. The source section $R_0 = s_Z$ has the exact rank-one defect $DR_0 - R_0 A = \Theta \phi^* \ell_Z$, with the unchanged arithmetic remainder functional ℓ_Z .

Define maps, including the target enlargement,

$$I_m e_j = e_j, \quad S_m e_j = e_{j+1}, \quad I_m, S_m : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^{m+2}, \quad D\Phi_m = \Phi_{m+1} S_m.$$

For B in $\text{Mat}_((m+1) \times d)(\mathbb{C})$, $d = \dim E_Z$, put

$$R_B = R_0 + \Theta \Phi_m B, \quad K_B = e_0 \ell_Z + S_m B - I_m B A.$$

Then

$$DR_B - R_B A = \Theta \Phi_{m+1} K_B, \quad q R_B = \sigma_Z, \quad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \rightarrow L^2(\mathbb{R}_+, dx), \quad (u, c) \mapsto R_0 u + \Theta \Phi_{m+1} c, \quad M_m = \mathcal{T}_m^* \mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives $u=0$; Mellin then gives $c=0$. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \quad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \quad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

D.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures $||M\text{-}\hat{M}|| \leq \eta_M$, $||X\text{-}\hat{X}|| \leq \eta_X$, $||Y\text{-}\hat{Y}|| \leq \eta_Y$, put $x = ||\hat{X}||$, $y = ||\hat{Y}||$, $m_0 = ||\hat{M}||$. Product estimates give

$$\eta_G = \eta_M (x + \eta_X)^2 + m_0 (2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M (x + \eta_X)(y + \eta_Y) + m_0 (x\eta_Y + y\eta_X + \eta_X \eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for $\hat{G}\text{-}\eta_G \geq 0$ and

$$\epsilon \hat{G} \pm \hat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies $S^* = -S$ and $A^* S + S A = S$. Its exact dual transport is

$$G_B^D = S^* G_B^{-1} S, \quad W_B^D = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays $S \hat{J}_g$, with metric comparison $G_B^{-1} S \hat{J}_g$; no positivity is assigned by choosing G_B .

For tensor degree n , retain $A_n = \sum_j 1$ tensor A tensor 1 and $G_{(B,n)} = G_B^{\wedge}(\text{tensor } n)$. The product control is $\sum_j G_B$ tensor W_B tensor G_B . Changing a homotopy by $\gamma \alpha q$ changes

the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n} dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of $G_{B,n}$ now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration $D_{\text{amb}} = \text{diag}(1/2+i, 1/2+2i)$, $\Theta(v) = (v, 0)$, $A = 1/2+2i$, $R(u) = (u, u)$ has boundary $\Theta(-i)$, Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map $B \mapsto (R^*B + B^*R)$ explains which nonzero boundaries can have zero control.

Conversely every eigenvector $Av = \rho v$ obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re \rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

D.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

E PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, [Zeta research repository](#).

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

E.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

E.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective $\Theta: V \rightarrow B$, an R -linear automorphism F , $Q = B/\Theta V$, and $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, all homotopies satisfying BOTH $dH = \Theta \kappa$ and $Hd = 0$, for $d(v, w) = \Theta(v - Fw)$, are

$$H_\alpha(b) = (\kappa b + \alpha q_b, F^{-1} \alpha q_b), \quad \alpha: Q \rightarrow R V.$$

Writing $H = (u, v)$ gives $u - Fv = \kappa$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$; hence Fv descends uniquely to α . Substitution proves the converse. The difference is $\gamma(\alpha - \beta)q$ with $\gamma(v) = (v, F^{-1}v)$. The solution set is affine relative to H_0 , not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomotopy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over $C[t]$, support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$, with $p: G(Z) \rightarrow Z$ still infinite, $e = \text{ofR}(0)$, and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R_+: sup x^b |D^k F(x)| < infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*= (4 pi^2 x^4 - 6 pi x^2) exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```

The Hilbert observation is $H=L^2(\mathbb{R}_+, dx)$, conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let $E=C[t]/h_Z$, $A=M_t$. The previous analytic constructions supply

$R_\theta: E \rightarrow B$, $J_Z: B \rightarrow E$, $qR_\theta = \sigma_Z$, $J_{ZR_\theta} = 1$, $J_Z \theta = 0$,
 $DR_\theta - R_\theta A = f_\theta \in \ell_Z$.

The exact ℓ_Z comes from $\epsilon_Z = j_h(h/(2\xi))$; at a simple zero its value on 1 is $1/(2\xi'(\rho))$. These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

E.1.2 2. An actual minimum over the admitted theta relations

For $m \geq 0$ define

$W_m = \text{span}(\phi_0, \dots, \phi_m)$;
 $\Phi_m: C^{(m+1)} \rightarrow V$, $c \mapsto \sum c_j \phi_j$;
 $F_m = \theta \Phi_m: C^{(m+1)} \rightarrow H$;
 $H_m = F_m^* F_m$; $C_m = F_m^* R_\theta$; $G_\theta = R_\theta^* R_\theta$.

The ϕ_j are independent: Mellin sends a relation to $g(s)P(s)=0$. Thus H_m is positive definite. Here C_m is a cross-Gram matrix $E \rightarrow C^{(m+1)}$, not the complex of that name in the endpoint workbench.

Set

$B_m = -H_m^{-1} C_m$;
 $\Pi_m = F_m H_m^{-1} F_m^*$;
 $R_m = R_\theta + \theta \Phi_m B_m = (1 - \Pi_m) R_\theta$.

The correction is a finite original theta boundary, so R_m lands in B , $qR_m = \sigma_Z$, and $J_{ZR_m} = 1$ at every m . Direct multiplication proves

$F_m^* R_m = 0$;
 $G_m = R_m^* R_m = G_\theta - C_m^* H_m^{-1} C_m \succ 0$.

Positive definiteness follows from $J_{ZR_m} = 1$. For any $B: E \rightarrow C^{(m+1)}$,

$(R_\theta + F_m B)^* (R_\theta + F_m B)$
 $= G_\theta + (B - B_m)^* H_m (B - B_m)$.

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also $(1 - \Pi_m)(R_\theta + F_m B') = R_m$: changing the initial representative by a relation already in W_m leaves the result unchanged.

E.1.3 3. The complete control form has rank at most two

Let $I_m, S_m: C^{(m+1)} \rightarrow C^{(m+2)}$ be $I_m e_j = e_j$ and $S_m e_j = e_{(j+1)}$. Define

$K_m = e_\theta \in \ell_Z + S_m B_m - I_m B_m A$.

Then

$DR_m - R_m A = F_{(m+1)} K_m = \theta \Phi_{(m+1)} K_m$.

No boundary component is discarded in this identity. Put $b_m = e_m^* B_m: E \rightarrow C$, and $z_m = R_m^* f_{(m+1)}: C \rightarrow E$. Integration by parts on the stated source functions gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

with vanishing boundary term $[x \text{ conjugate}(F)H]_0^\infty$. Hence

$$\begin{aligned} W_m &= A^* G_m + G_m A - G_m \\ &= -(R_m^* F_{(m+1)} K_m + K_m^* F_{(m+1)} R_m) \\ &= -z_m b_m - b_m^* z_m^*. \end{aligned}$$

The last equality uses $F_m^* R_m = 0$ and the exact last row of K_m , which is b_m . Thus $\text{rank}(W_m) \leq 2$ for every packet dimension and every finite m . The pairings with the first $m+1$ theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

$$\begin{aligned} u_m &= z_m^* G_m^{-1} z_m \geq 0; \\ v_m &= b_m^* G_m^{-1} b_m \geq 0; \\ c_m &= b_m^* G_m^{-1} z_m. \end{aligned}$$

The nonzero spectrum of $G_m^{-1} W_m$ is the nonzero spectrum of

$$-\begin{bmatrix} c_m & v_m \\ u_m & \text{conjugate}(c_m) \end{bmatrix}.$$

This follows by factoring through the two-column map $[z_m, b_m^*]$ and using the equality of nonzero spectra of the two compositions. Its trace is $-2 \text{Re}(c_m)$, determinant $|c_m|^2 - u_m v_m$. Therefore the least symmetric control allowance is

$$\begin{aligned} \epsilon_m &= |\text{Re}(c_m)| + \sqrt{u_m v_m - (\text{Im}(c_m))^2}, \\ -\epsilon_m G_m &\leq W_m \leq \epsilon_m G_m. \end{aligned}$$

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

$$\text{trace}(G_m^{-1} W_m) = 2 \text{Re}(\text{trace } A) - \dim E = 0,$$

so $\text{Re}(c_m) = 0$. This is an exact trace identity, not a choice of coordinates. For every eigenvector $Av = \rho v$,

$$(v^* W_m v) / (v^* G_m v) = 2 \text{Re}(\rho) - 1.$$

Thus the relative control still detects precisely the original spectral weight.

E.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

$$\begin{aligned} a_{(m+1)} &= (1 - \Pi_m) f_{(m+1)} = \Theta \psi_{(m+1)}, \\ \psi_{(m+1)} &= \phi_{(m+1)} - \Pi_m H_m^{-1} F_m^* f_{(m+1)}, \\ \eta_{(m+1)} &= \|a_{(m+1)}\|^2 > 0. \end{aligned}$$

Then $R_m^* a_{(m+1)} = z_m$, and orthogonal decomposition gives

$$\begin{aligned} \Pi_{(m+1)} &= \Pi_m + a_{(m+1)} \eta_{(m+1)}^{-1} a_{(m+1)}^*; \\ R_{(m+1)} &= R_m - a_{(m+1)} \eta_{(m+1)}^{-1} z_m^*; \\ G_{(m+1)} &= G_m - \eta_{(m+1)}^{-1} z_m z_m^*. \end{aligned}$$

The control update is obtained by applying the same real-linear operator $X \mapsto A^*X + XA - X$ to the last identity.

The original quotient transport is

$$Q_W = B / \Theta W_m \rightarrow Q_{W(m+1)} = B / \Theta W_{(m+1)},$$

$$\text{kernel} = \Theta W_{(m+1)} / \Theta W_m \simeq W_{(m+1)} / W_m.$$

That kernel is one-dimensional. The square $E \rightarrow Q_W$ by R_m and $E \rightarrow Q_{W(m+1)}$ by $R_{(m+1)}$ commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

$$H_{\text{cal}_m}([F], [G]) = \langle (1 - \Pi_m)F, (1 - \Pi_m)G \rangle.$$

It is positive definite since the kernel of $1 - \Pi_m$ in B is exactly ΘW_m . Let $\lambda_{(m+1)}[F] = \langle a_{(m+1)}, F \rangle$. This is well defined on Q_W and

$$H_{\text{cal}_m}(x, y) = H_{\text{cal}_{(m+1)}}(qx, qy) + \eta_{(m+1)}^{-1} \text{conjugate}(\lambda(x)) \lambda(y).$$

The map $(q, \lambda): Q_W \rightarrow Q_{W(m+1)}$ plus C is a linear isomorphism. Its inverse is

$$([F], c) \mapsto [F - a \eta^{-1} \langle a, F \rangle + a \eta^{-1} c]_W.$$

Changing F by a relation at level $m+1$ changes this expression only by a relation at level m . This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join $\max(m, n)$ and the above transports. The scalar e sends $(m, [F])$ to $(m, [0])$; τ sends it to external absence. The two-leg primitives are $(k_m u, 0)$ and $(0, \text{Fourier}^{-1} k_m u)$, where $k_m = \Phi_{(m+1)} K_m$. Synchronization by the original $E_{\text{sync}} = (1, e)$ supplies their common joint support. Their difference remains the H_0 cycle $(k_m u, \text{Fourier}^{-1} k_m u)$. The other session's $\text{Hom}(Q, H_0)$ freedom leaves their differential and hence W_m unchanged.

E.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since $JF(x) = x^{-1} F(1/x)$ is unitary on $L_2(dx)$, $Jf_0 = f_0$ and $JD = (1 - D)J$,

$$H_{ij} = \int_{-\infty}^{\infty} [\text{conjugate}(f_i) f_j + \text{conjugate}(f_i^{\text{ref}}) f_j^{\text{ref}}] dx,$$

$$f_j^{\text{ref}} = (1 - D)^j f_0.$$

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on $x \geq 1$. Write

$$\phi_j(x) = P_j(x) \exp(-\pi x^2),$$

$$P_0 = 4 \pi^2 x^4 - 6 \pi x^2,$$

$$P_{(j+1)} = 2 \pi x^2 P_j - x P_j',$$

$$P_j = \sum_r c_{(j, r)} x^{(2r)},$$

$$c_{(j+1, r)} = 2 \pi c_{(j, r-1)} - 2r c_{(j, r)}.$$

For $f_j^{\text{N}} = 2 \sum_{(n=1)}^N P_j(nx) \exp(-\pi n^2 x^2)$, put

$$C_j = \sum_r |c_j(r)| \sqrt{\Gamma(2r+1/2) / (2 \pi^{(2r+1/2)})}.$$

Then

$$\begin{aligned} & \|f_j - f_j^N\|_{L^2([1, \infty))} \\ & \leq 2 C_j (N+1)^{-1/2} \exp(-\pi(N+1)^2/2) \\ & \quad / [1 - \exp(-\pi(2N+3)/2)]. \end{aligned}$$

Proof: on $x \geq 1$, $2 \pi n^2 x^2 \geq \pi n^2 + \pi n^2 x^2$. Integrating each monomial's squared bound gives $n^{-1/2} \exp(-\pi n^2/2)$ times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

$$\begin{aligned} & \int_1^\infty x^{(2k)} \exp(-c x^2) dx \\ & = \Gamma(k+1/2, c) / (2 c^{(k+1/2)}), \quad c = \pi(n^2 + n'^2). \end{aligned}$$

For truncation tail norms δ_i, δ_j , each inner-product error is bounded by $\delta_i \|f_j^N\| + \|f_i^N\| \delta_j + \delta_i \delta_j$, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is $H_{(i+1,j)} + H_{(i,j+1)} = H_{ij}$. Cross-Gram columns $F_m * R_0$ remain their actual source integrals and are not replaced by arbitrary matrices.

E.1.6 6. The same source proves a strong topology comparison

For this actual f_0 ,

$$\text{closure span}\{D^j f_0 : j \geq 0\} \text{ in } L^2(\mathbb{R}_+, dx) = H.$$

Proof. The explicit unitary Mellin-line map sends F to $M F(1/2+it)$ in $L^2(\mathbb{R}, dt/(2\pi))$. It sends D to $1/2+it$ and f_j to $(1/2+it)^j g(1/2+it)$. The log-space function $h(z) = \exp(z/2) f_0(\exp z)$ is holomorphic in $|\text{Im } z| < \pi/4$ by its Gaussian series. For $|\text{Im } z| \leq a < \pi/4$ it decays superexponentially as $\text{Re } z$ tends to positive infinity. Poisson gives $h(z) = h(-z)$, so the same holds at the other end. Fourier contour displacement therefore gives $|g(1/2+it)| \leq C_a \exp(-a|t|)$.

The measure $d\mu = |g(1/2+it)|^2 dt/(2\pi)$ has a positive exponential moment. If an $L^2(\mu)$ function is orthogonal to all polynomials, multiplying μ by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in $L^2(\mu)$. Multiplication by g is unitary from this weighted space onto $L^2(dt/(2\pi))$, because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

$$\begin{aligned} & R_m \rightarrow 0 \text{ in } \text{Hom}(E, H), \quad G_m \rightarrow 0 \text{ and } W_m \rightarrow 0, \\ & \text{while } q_{R_m} = \sigma_Z \text{ and } J_{ZR_m} = 1 \text{ for every } m. \end{aligned}$$

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

$$\begin{aligned} & \text{closure}\{(F, J_{ZF}) : F \text{ in } B\} = H \text{ plus } E; \\ & \text{closure}\{(\Theta \phi, 0) : \phi \text{ in } V\} = H \text{ plus } 0. \end{aligned}$$

Indeed finite theta polynomials give every $(h,0)$ in the closure, and the sequence $(R_m u, u)$ gives $(0, u)$. For a specified positive metric on E , completion in $\|F\|_2^2 + \|J_Z F\|_E^2$ is this graph closure, and the quotient by the boundary closure is E by $[(h, u)] \rightarrow u$. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

$$((R_m u)^{\bullet}, u^{\bullet}) \rightarrow (e_H, u^{\bullet})$$

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (τ, τ) .

E.1.7 7. Duality, trace and scope

The original residue map has matrix S with $S^* = -S$ and $A^*S + SA = S$. Its pullback of the dual metric is

$$\begin{aligned} G_m^* D &= S^* G_m^{-1} S, \\ W_m^* D &= -S^* G_m^{-1} W_m G_m^{-1} S. \end{aligned}$$

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays $S J_g$, with explicit metric comparison $G_m^{-1} S J_g$. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class $j_h(h/(2\xi))$ is unchanged; the induced finite cochain projectors differ by the homotopy $(b)_Z, 0$. The trace difference is zero because $J_Z \Theta = 0$. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance $n \epsilon_m$; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied `Tau_Deligne_Control` and `Tau_Global_Comparison` source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

F Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. formal/splitzero/DERIVED_MATHEMATICS.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

F.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

F.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

F.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R -modules V_i with linear transition maps

$$\begin{aligned} \text{rho_ij} &: V_i \rightarrow V_j \quad (i \leq j), \\ \text{rho_ii} &= \text{id}, \quad \text{rho_jk} \circ \text{rho_ij} = \text{rho_ik}. \end{aligned}$$

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

$$\text{Total}(D) = \text{disjoint union of } V_i \text{ over } i \text{ in } L.$$

The global zero is $(\text{bottom}, 0)$, and

$$\begin{aligned} (i, x) + (j, y) &= (i \text{ join } j, \text{rho_i, i-join-j}(x) + \text{rho_j, i-join-j}(y)). \\ \text{tau} \cdot (i, x) &= (\text{bottom}, 0). \\ \text{ofR}(r) \cdot (i, x) &= (i, r \cdot x). \end{aligned}$$

`LinearDiagram.totalAddCommMonoid` and `LinearDiagram.totalModule` construct the actual instances and prove every law. For associativity, transport both sides to $i \text{ join } j \text{ join } k$. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R -module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to $(i,0)$.

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual $G(R)$ -linear maps between reconstructed totals. Identity and composition are proved. More substantially, `Recovery.recoverEquiv` reconstructs the intrinsic support fibres of **any** existing $G(R)$ -semimodule M and gives a full $G(R)$ -linear equivalence back to M :

$$(1,x) \mapsto x, \quad m \mapsto (e\ m, m).$$

Its addition proof uses the original absorption identity $m + e\ m = m$. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

F.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i . The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

$$q : \text{Total}(D) \rightarrow \text{Total}(D/B), \quad q(i,x) = (i,[x])$$

is a surjective $G(R)$ -linear map. The code proves the exact equality criterion

$$q(i,x)=q(i,y) \text{ iff } x-y \text{ is in } B_i.$$

Different labels cannot become equal under q . In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** $G(R)$ -semimodule, not assumed additively cancellative, and let $f:\text{Total}(D)\rightarrow N$ be split-linear. If

$$f(i,b)=f(i,0) \text{ for every } b \text{ in } B_i,$$

then there exists a unique split-linear $g:\text{Total}(D/B)\rightarrow N$ with $g\circ q=f$. The proof does not subtract elements in N . If $x-y$ is in B_i , use the identity $y+(x-y)=x$ inside V_i and then apply additivity of f . This proves constancy on q -fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

`Relations.coequalizer_universal` states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

F.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

$C^{(i-1)} \xrightarrow{d_{\text{Prev}}} C^i \xrightarrow{d_{\text{Next}}} C^{(i+1)}, \quad d_{\text{Next}} d_{\text{Prev}} = 0.$

Window.boundaryToCycles constructs the incoming map into $\ker(d_{\text{Next}})$. Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map $f:C \rightarrow D$, the code constructs

$f_Z : Z(C) \rightarrow Z(D),$
 $H(f) : Z(C)/B(C) \rightarrow Z(D)/B(D).$

It proves identity, composition, and

$H(f)([z]) = 0 \iff \text{there exists } y \text{ with } d_{\text{Prev}_D}(y) = f_i(z).$

Define the module of killed representatives and its original relation submodule:

$\text{Krep} = \{z \text{ in } Z(C) : f_Z(z) \text{ is in } B(D)\},$
 $\text{Bold} = B(C), \text{ viewed as a submodule of } \text{Krep}.$

There is a constructed linear equivalence

$\text{Krep}/\text{Bold} \simeq \ker H(f).$

Forward map: the class of z goes to the kernel element $[z]$. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying $f \circ a = b \circ f$, homology_intertwines proves $H(f) \circ H(a) = H(b) \circ H(f)$. kernelAction constructs the restriction of $H(a)$ to $\ker H(f)$. Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

F.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

$R_L(d_{\text{Next}}) \circ R_L(d_{\text{Prev}}) = z_L : R_L(C^{(i-1)}) \rightarrow R_L(C^{(i+1)}),$
 $z_L(1, x) = (1, 0).$

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation $d(x) = e.d(x)$.

The internal homology projection

$q : R_L(Z(C)) \rightarrow R_L(H(C))$

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. `homology_coequalizes` and `homology_coequalizer` establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition $i \leq j$,

$\text{rho}^{\wedge} H_{ij}([z]) = 0$ iff $\text{rho}_{ij}(z)$ is a boundary at j .

If z is not a boundary at i , its retained class $(i, [z])$ is not $(i, 0)$, regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

F.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k . In the displayed degree let

$C: k \xrightarrow{-\theta} k \xrightarrow{-\theta} k,$
 $D: k \xrightarrow{-a} (a, \theta) \xrightarrow{-} k^2 \xrightarrow{-\theta} k.$

Use zero maps in the outside degrees. The two middle chain maps are

$f(a) = (a, \theta),$ $g(a) = (\theta, a).$

The code proves that the source class $[1]$ is nonzero. Its f -image is the **nonzero vector** $(1, 0)$, but that vector is a boundary in D , so $H(f)([1]) = 0$. In fact $H(f) = 0$ on every source class. The g -image $(0, 1)$ cannot be a boundary, because every boundary has second coordinate zero. Hence $H(g)([1])$ is nonzero and $H(f)$ is not $H(g)$.

The ordinary calculation gives $H(C) = k$ and $H(D) = k$ (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

F.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a $k[t]$ -algebra structure on B . Let $u: k[t] \rightarrow B$ denote that map and

$I_{-}(\text{rho}, m) = ((t - \text{rho})^{\wedge} m), \quad m \geq 0.$

There is an explicit B -algebra equivalence

$B \otimes_{k[t]} (k[t] / I_{-}(\text{rho}, m)) \simeq B / ((u(t) - u(\text{rho}))^{\wedge} m).$

It maps $b \otimes [p]$ to $[u(p)b]$, with inverse $[b]$ to $b \otimes 1$. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem `Algebra.tensorProduct.quotIdealMapEquivTensorQuot`, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves τ and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any $k[t]$ -module V and v in V , the exact balancing calculation is

$$(1-u(t)) \text{ tensor } (t.v) - u(t) \text{ tensor } (v-t.v) = 0.$$

Expanding gives $1 \text{ tensor } (t.v) - u(t) \text{ tensor } v$, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice $u(t)=s$ and $t.v=Dv$ is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

F.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison β to the spectral sheaf $\mathcal{O}/(2\xi)$. Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c , their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of $H^1(c)$. Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter $a>1$, an eigenvalue a^ρ has modulus $a^{(\text{Re } \rho)}$. A proved weight-one modulus $a^{(1/2)}$, for the representation identified with every relevant spectral zero, then forces $\text{Re } \rho=1/2$. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

F.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256

e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathematiques de l'IHES*, 52, 137-252. <https://doi.org/10.1007/BF02684780>

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, LinearAlgebra/Quotient/{Defs,Basic}.lean, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

F.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪ SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.

G PR 14: Coherent arithmetic interpolation and joint tensor boundary control

Source attribution. Complete TeX witness delivered with [web pull request #14](#); the associated Git revision is recorded in its delivery receipt.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Coherent_Interpolation_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 01a4fa76072ca4f3d8266c9fdbf61c816e53d09c673f8b8776b981b736c09e26

Original archive. Tau_Coherent_Interpolation_Control_2026-09-12.zip

Archive SHA-256. a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26

G.1 Coherent arithmetic interpolation and joint tensor boundary control

Owner-directed continuation, 12 September 2026. Written mathematical arguments for review; the new analytic and matrix results in this note have not been Lean-checked.

G.1.1 0. Accepted inputs and the change in the continuation

The new formalization receipt reports—and the repository metadata and workflow step summaries confirm—a successful check at 5d2772ea0a16d177ac3e01ff70394b90ba95a232 in PR #12. That certificate covers the specified retraction’s algebraic/topological consequences, the complete coefficient homotopy family, and the stated finite-chart dual maps. It does not certify the analytic construction of the theta retraction, support-changing synchronization homotopies, or the matrix estimates developed here. Its branch is left untouched.

The second attachment proves, for an unrestricted finite correction matrix B , the exact formula

$$G_B = G_{\perp,m} + (B + H_m^{-1}C_m)^* H_m (B + H_m^{-1}C_m). \quad (0.1)$$

Once $m+1 \geq \dim E_Z$, every positive metric ray is realized by an actual theta-boundary correction, with the arithmetic jets unchanged. Its further conclusion

$$\inf_B \epsilon_A(G_B) = \max_{\rho \in Z} |2\Re \rho - 1| \quad (0.2)$$

therefore has to be retained. This note does not count unrestricted metric optimization as a new arithmetic estimate. It instead uses the unique least-norm representative for a fixed, increasing space of the original arithmetic boundaries. It proves compatibility of these representatives for all finite packets, an exact full-jet moment-matrix formula, and a joint total-degree tensor construction.

All conclusions are derived on the existing characteristic-zero theta complex over the marked tau-base. None is asserted to be an application of finite-field purity to a space merely renamed tau.

G.1.2 1. Fixed scalar, cochain, and arithmetic maps

Retain the original scalar construction

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the entire square

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}} \quad (1.1)$$

The target of $p_{\mathbb{Z}}$ is infinite. This quotient remains part of the marked geometry over $\mathfrak{b}_\tau$. Coefficient maps below fix external absence and carry a represented scalar zero to the represented scalar zero of the target.

Use the unchanged spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \forall b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{m \neq 0} \phi(mx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Write $q : \mathcal{B} \rightarrow Q$. Its action is by the original D , not a shifted operator. Let

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}f_0 = g. \quad (1.2)$$

The established range identities are used with these exact constants. They give $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire. In particular every finite jet map used below annihilates ΘV .

For $n \geq 0$, retain

$$V_n = \text{span}\{\phi_*, D\phi_*, \dots, D^n\phi_*\}, \quad L_n = \Theta V_n,$$

$$Q_n = \mathcal{B}/L_n \longrightarrow Q_{n+1}, \quad \ker = L_{n+1}/L_n. \quad (1.3)$$

Their split reconstructions have labels V_n . The transport of a class in L_{n+1}/L_n is the supported zero at V_{n+1} ; it is not external absence. The derivative has type $Q_n \rightarrow Q_{n+1}$ because $DL_n \subseteq L_{n+1}$.

G.1.3 2. A polynomial presentation of the actual finite representatives

Take a nonempty finite set Z of actual zeros of g , with their complete orders m_ρ , and put

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[t]/(h), \quad v_h(s) = g(s)/h(s). \quad (2.1)$$

The polynomial variable t acts on the source by D and is evaluated at s by Mellin transformation. Full orders make v_h nonzero at all selected centres. Let j_h be the complete Hermite-jet/Chinese-remainder map into E_h and retain

$$v_h = j_h v_h \in E_h^\times, \quad \varepsilon_h = v_h^{-1}. \quad (2.2)$$

The unit ε_h is not the scalar e . Its split scalar is $\varepsilon_h^\bullet$.

There is an actual $F_h \in \mathcal{B}$ with $h(D)F_h = f_0$ and $\mathcal{M}F_h = v_h$. It can be obtained successively from

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad (2.3)$$

whenever the required Mellin value vanishes. At zero, replace the integral by its negative from 0 to x , using that vanishing; at infinity use the displayed integral. The defining estimates of $\mathcal{B}$ prove rapid decay of every scaling derivative at both ends. Repeated zero jets permit iteration. Uniqueness follows from injectivity of Mellin and $h\mathcal{M}F = 0 \Rightarrow F = 0$.

Define the linear map

$$\boxed{\mathcal{T}_h : \mathbb{C}[t] \longrightarrow \mathcal{B}, \quad P \longmapsto P(D)F_h.} \quad (2.4)$$

It is injective and satisfies

$$\mathcal{M}\mathcal{T}_h(P) = \frac{g}{h}P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (2.5)$$

Here $J_h F = j_h \mathcal{M}F$. The first equation retains the whole completed function, not its values only on the critical line.

For $N = n + d$, let $\mathcal{P}_N = \mathbb{C}[t]_{\leq N}$, and define

$$\mathcal{J}_{h,N} : \mathcal{P}_N \twoheadrightarrow E_h, \quad P \longmapsto v_h[P]_h.$$

Its exact sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_n \xrightarrow{\times h} \mathcal{P}_{n+d} \xrightarrow{\mathcal{J}_{h,n+d}} E_h \longrightarrow 0.} \quad (2.6)$$

Applying $\mathcal{T}_h$ sends the left map to the original theta relations L_n . Thus its split internal quotient has exactly the relation $P - P' \in h\mathcal{P}_n$, with the support label retained.

The original source section is

$$s_h(u) = \mathcal{T}_h(\text{rem}_h(\varepsilon_h u)), \quad J_h s_h = 1, \quad q s_h = \sigma_h. \quad (2.7)$$

No arbitrary metric or section has entered (2.6).

G.1.4 3. Exact arithmetic moment matrices and the full-jet interpolation formula

The Hilbert observation is $\mathcal{H} = L^2((0, \infty), dx)$. The precise transform map is

$$\mathcal{H} \xrightarrow{\mathcal{L}} L^2(\mathbb{R}, dy) \xrightarrow{\mathcal{F}_+} L^2(\mathbb{R}, dt/(2\pi)),$$

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{F}_+\psi(t) = \int_{\mathbb{R}} \psi(y)e^{ity}dy, \quad (\mathcal{F}_+\mathcal{L}F)(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (3.1)$$

Both arrows are unitary with these measures. The inverse Fourier factor is $1/(2\pi)$; no mass of a measure below is set equal to one. The same map carries D to multiplication by $s_t = 1/2 + it$.

Define the finite positive measure

$$d\nu_h(t) = \left| \frac{g(\frac{1}{2} + it)}{h(\frac{1}{2} + it)} \right|^2 \frac{dt}{2\pi}. \quad (3.2)$$

Any apparent singularity at a selected critical-line zero is removed by the entire function v_h . All moments are finite. In the fixed basis $1, t, \dots, t^N$ of $\mathcal{P}_N$, put

$$(M_{h,N})_{ij} = \int_{\mathbb{R}} \overline{s_t^i} s_t^j d\nu_h(t), \quad 0 \leq i, j \leq N. \quad (3.3)$$

In the integrand t is the real integration variable; the polynomial basis is evaluated at s_t . This prevents the polynomial variable from being confused with the spectral-height coordinate.

The matrix $M_{h,N}$ is positive definite. A polynomial with zero norm vanishes almost everywhere on a line on which v_h is nonzero almost everywhere, hence is zero. Write $J_{h,N}$ for the matrix of $\mathcal{J}_{h,N}$ in the fixed quotient basis $1, t, \dots, t^{d-1}$. It includes multiplication by the actual unit v_h .

Set

$$\mathcal{C}_{h,N} = J_{h,N} M_{h,N}^{-1} J_{h,N}^*, \quad C_{h,N} = M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1}. \quad (3.4)$$

These are finite matrices on declared coordinate spaces. $J_{h,N} C_{h,N} = 1$. The actual test-function map is

$$R_{h,n} = \mathcal{T}_h C_{h,n+d} : E_h \longrightarrow \mathcal{B}. \quad (3.5)$$

Theorem. This is precisely the preceding orthogonal representative $(1 - P_n)s_h$, where P_n is the $L^2(dx)$ orthogonal projection onto L_n . In particular,

$$qR_{h,n} = \sigma_h, \quad J_h R_{h,n} = 1, \quad G_{h,n} := R_{h,n}^* R_{h,n} = \mathcal{C}_{h,n+d}^{-1}. \quad (3.6)$$

Proof. Every polynomial with jet u is $C_{h,N}u + z$ with $J_{h,N}z = 0$. The cross term vanishes because

$$z^* M_{h,N} C_{h,N} u = z^* J_{h,N}^* \mathcal{C}_{h,N}^{-1} u = 0.$$

Its norm is therefore $u^* \mathcal{C}_{h,N}^{-1} u + z^* M_{h,N} z$. This proves the unique minimizer. By (2.6), its difference from the degree-below- d lift (2.7) is exactly a member of L_n . It must be $(1 - P_n)s_h$. This also proves (3.6).

Thus the subtraction-of-nearby-Grams formula of the previous note has the exact alternative (3.6): invert the full-jet interpolation matrix. This is a matrix-valued Christoffel construction with the arithmetic local unit retained. The finite least-norm identity itself is standard linear algebra; its application here identifies the actual original representative, source relation, and completed multiplier.

For one simple zero ρ , the same formula is

$$G_{\rho,n} = \frac{1}{|g'(\rho)|^2 \mathcal{K}_{\nu_{\rho,n+1}}(\rho, \rho)}, \quad \mathcal{K}_{\nu,N}(z, w) = (1, z, \dots, z^N) M_N^{-1} (1, w, \dots, w^N)^*. \quad (3.7)$$

The complex conjugate in the second evaluation is part of the displayed kernel. Higher orders use (3.4), not just point evaluations. The factor $g' = 2\xi'$ is retained.

G.1.5 4. All finite packets fit the same source filtration

Let $h \mid H$ be polynomials for nested packets with full orders and set $m = H/h$. Since the newly added centres are disjoint, h and m are coprime. There is a $\mathbb{C}[t]$ -linear CRT injection

$$\iota_{hH} : E_h \hookrightarrow E_H,$$

which inserts zero jets at the new centres. It is multiplication by the appropriate CRT idempotent after lifting a residue; it sends the unit of E_h to that idempotent, so it is not being asserted to be a unital ring map.

The polynomial map $P \mapsto mP$ has the exact properties

$$\mathcal{T}_H(mP) = \mathcal{T}_h(P), \quad J_H \mathcal{T}_h(P) = \iota_{hH} \mathcal{J}_h(P), \quad (4.1)$$

and is isometric from ν_h to ν_H because $|v_H mP| = |v_h P|$. The CRT equations and degree bounds prove

$$s_H \iota_{hH} = s_h.$$

The original L_n is independent of the packet, so

$$\boxed{R_{H,n} \iota_{hH} = R_{h,n}, \quad \iota_{hH}^* G_{H,n} \iota_{hH} = G_{h,n}, \quad \iota_{hH}^* W_{H,n} \iota_{hH} = W_{h,n}.} \quad (4.2)$$

The action also intertwines: $A_H \iota_{hH} = \iota_{hH} A_h$, where $A_h = M_t$. Thus a certificate on the larger packet restricts by this actual congruence to the smaller packet. The inverse interpolation matrices satisfy the corresponding inverse-compression formula; inverse and compression are not commuted.

The finite sections consequently glue on the algebraic direct limit of the actual packet injections. This statement does not assert convergence of an infinite sum of packet projectors or identify the whole Q with that direct limit.

At every support label, lift these maps as $(\lambda, u) \mapsto (\lambda, \iota_{hH} u)$ and $(\lambda, u) \mapsto (\lambda, R_{h,n} u)$. Added zero jets remain represented zeros. The label map in (1.3) is still the map that admits one more source relation.

G.1.6 5. A global density result, with its precise cohomological consequence

The single derivative family $\{D^n f_0 : n \geq 0\}$ is already dense in $\mathcal{H}$.

Proof. Put $d\nu(t) = |g(1/2 + it)|^2 dt / (2\pi)$. It has an exponential moment of every order $c < \pi/2$. To verify this, the alternating-series formula and partial summation give

$$|\zeta(\tfrac{1}{2} + it)| \leq \frac{2|\tfrac{1}{2} + it|}{\sqrt{2} - 1}.$$

The gamma vertical-line asymptotic gives, for sufficiently large $|t|$,

$$|g(\tfrac{1}{2} + it)|^2 \leq C(1 + |t|)^{11/2} e^{-\pi|t|/2}.$$

The constant C is a fixed bound, not a numerical enclosure claimed here. The gamma fact used is NIST DLMF 5.11.9. The same exponential-moment property holds for ν_h , since v_h is entire at the finitely many possible denominator zeros and has polynomially modified tails.

If $f \in L^2(\nu)$ is orthogonal to every polynomial, the finite measure $\bar{f} d\nu$ has a holomorphic Fourier transform in a nonempty horizontal strip, by Cauchy–Schwarz and the exponential moment. All

derivatives at the origin vanish. Analytic uniqueness makes its transform zero; Fourier uniqueness for finite measures makes the measure zero. Thus polynomials are dense in $L^2(\nu)$.

Multiplication by the actual function $g(1/2 + it)$ is a unitary map $L^2(\nu) \rightarrow L^2(dt/(2\pi))$: its inverse divides by g away from its measure-zero set of real-line zeros. The inverse is a map of weighted L^2 spaces, not holomorphic division through an arithmetic zero. Combining it with (3.1) sends polynomials to $P(D)f_0$. This proves the assertion.

It follows for each fixed packet that

$$\boxed{G_{h,n} \longrightarrow 0 \quad \text{in every matrix entry and in operator norm.}} \quad (5.1)$$

Indeed $P_n \rightarrow 1$ strongly on $\mathcal{H}$, and the finite number of columns of s_h is fixed. Since each $G_{h,n}$ is positive definite, the least eigenvalue of its inverse tends to infinity. Every nonzero arithmetic row functional $\ell : E_h \rightarrow \mathbb{C}$ therefore has

$$\boxed{\|\ell\|_{G_{h,n}^\vee}^2 = \ell \mathcal{C}_{h,n+d} \ell^* \longrightarrow \infty.} \quad (5.2)$$

These facts do not kill a cohomology class. The actual maps are

$$\mathcal{B}/L_n \longrightarrow \mathcal{H}/L_n, \quad Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{H}/\overline{\Theta V}^{\mathcal{H}} = 0. \quad (5.3)$$

The first map is injective because L_n is finite-dimensional and consists of the same smooth functions in both spaces. The second map forgets the stronger topology and has the whole Q as kernel. Its split lift lands at the represented zero in $G(0)$, not at τ . $G(0)$ retains both elements. An active class may therefore tend to zero under the L^2 observation while remaining fixed by $qR_{h,n} = \sigma_h$ and $J_h R_{h,n} = 1$.

The constructive consequence is to estimate relative boundary flux against the complete interpolation cost (3.4), not to use the absolute smallness of $G_{h,n}$ as a weight bound.

G.1.7 6. Tail representation and same-metric reflection control

Retain the unscaled orthogonal vectors from the previous calculation:

$$u_0 = f_0, \quad u_n = (1 - P_{n-1})D^n f_0, \quad \mathfrak{h}_n = \|u_n\|^2, \quad (6.1)$$

$$Du_n = u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1).$$

The diagonal $1/2$ and the negative last coefficient follow from $D^* = 1 - D$ on these functions. Neither the u_n nor their norms are rescaled. Define $r_n = \mathfrak{h}_n^{-1}u_n^* s_h$. The new density result makes the entire tail formula exact:

$$\boxed{G_{h,n} = \sum_{j>n} \mathfrak{h}_j r_j^* r_j.} \quad (6.2)$$

This series converges in finite matrix norm. Its source map is $u \mapsto (r_j u)_{j>n}$ into the weighted sequence space with norm $\sum_{j>n} \mathfrak{h}_j |c_j|^2$. The map retains every coefficient; it is an isometry for $G_{h,n}$.

The previous boundary calculation, or telescoping (6.1), gives

$$\boxed{W_{h,n} = A_h^* G_{h,n} + G_{h,n} A_h - G_{h,n} = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}).} \quad (6.3)$$

It pairs two explicit maps into L_{n+1}/L_n . At the original support V_n the derivative defect is $-[u_{n+1}]r_n$; at V_{n+1} it becomes the supported zero. This is also the input to the now-checked coefficient homotopy family, with $\kappa_n = K_n J_h$ and $\kappa_n \Theta = 0$.

For a reflection-stable packet define the actual antilinear involution

$$j_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}.$$

On functions use $j_B F(x) = x^{-1} \overline{F(1/x)}$. It is antiunitary on $\mathcal{H}$ and satisfies $Dj_B = j_B(1 - D)$. The specific section (2.7) satisfies $j_B s_h = s_h j_h$: since $h^\dagger = (-1)^d h$, both the unit and the numerator have the corresponding $(-1)^d$ factor. Also $j_B L_n = L_n$, so uniqueness of orthogonal projection gives $j_B R_{h,n} = R_{h,n} j_h$.

Write $j_h u = C_h \bar{u}$ in the retained coordinates. Then

$$\boxed{C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}.} \quad (6.4)$$

Consequently the generalized spectrum of $(W_{h,n}, G_{h,n})$ is symmetric under $\lambda \mapsto -\lambda$ **for this same canonical metric**, with full multiplicities retained. An upper inequality $W_{h,n} \preceq \epsilon G_{h,n}$ therefore implies the lower inequality $W_{h,n} \succeq -\epsilon G_{h,n}$ by applying j_h to the input vector.

The residue duality is still the separate explicit morphism with matrix S_h , and its pulled-back metric is $S_h^* G_{h,n}^{-1} S_h$. The relation to the current metric is the linear map $G_{h,n}^{-1} S_h$ into its antilinear dual coordinates. The arithmetic Weil form remains $S_h J_g$; no positivity is supplied merely by (6.4).

For completeness, retain the full rank-two formula

$$a_n = r_n G_{h,n}^{-1} r_n^*, \quad b_n = r_{n+1} G_{h,n}^{-1} r_{n+1}^*, \quad c_n = r_n G_{h,n}^{-1} r_{n+1}^*.$$

The least control excess is

$$\epsilon_{h,n} = \mathfrak{h}_{n+1} \left(|\Re c_n| + \sqrt{a_n b_n - (\Im c_n)^2} \right). \quad (6.5)$$

For reflection-stable Z , the additional proved identity is $\Re c_n = 0$: take the trace of $G_{h,n}^{-1} W_{h,n}$, which is $2\Re \operatorname{Tr} A_h - d = 0$. Thus (6.5) also equals $\mathfrak{h}_{n+1} \sqrt{a_n b_n - (\Im c_n)^2}$. Both equations are retained. Equation (3.6) supplies every occurrence of $G_{h,n}^{-1}$ without a difference of nearly equal Gram matrices.

G.1.8 7. Joint total-degree control on the actual product complex

The tensor degree will be denoted k , independently of the relation degree n and the packet multiplicities. Put

$$E_h^{(k)} = E_h^{\otimes k}, \quad A_h^{(k)} = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}.$$

The existing product complex over $\mathfrak{b}_\tau$ has differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k.$$

Define $\mathcal{P}_M^{(k)} = \mathbb{C}[t_1, \dots, t_k]_{\text{total degree} \leq M}$ for $M \geq k(d-1)$ and

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) (F_h^{\otimes k}). \quad (7.1)$$

Its Mellin transform is $P(s_1, \dots, s_k) \prod_j g(s_j)/h(s_j)$. Its full jet map $J_{k,M}$ is reduction modulo each $h(t_j)$ followed by multiplication by $v_h^{\otimes k}$. The target is $E_h^{(k)}$ and this map is surjective. The tensor remainder basis has total degree at most $k(d-1)$.

The exact relation space is

$$\mathcal{I}_{k,M} = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)). \quad (7.2)$$

Polynomial division in the fixed order $t_1, \dots, t_k$ represents each member as $\sum_j h(t_j)P_j$, with $\deg P_j \leq M-d$. It terminates because every h is monic and replacing a leading power lowers total degree. Under (7.1), each term is an actual top cochain boundary: in the j -th source factor use ϕ_* in place of F_h and give its primitive the factor $(-1)^{j-1}$. The differential contributes the second $(-1)^{j-1}$. The resulting boundary is $+h(t_j)P_j$ with no removed sign.

Thus these joint corrections are maps in the original product complex, not freely chosen positive tensor metrics. Set $\mathcal{B}_{k,M} = \mathcal{T}_h^{(k)} \mathcal{I}_{k,M}$.

The moment matrix on the monomials of total degree at most M is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s_t^{\alpha_j}} s_t^{\beta_j} d\nu_h(t). \quad (7.3)$$

Its positive definiteness follows because a polynomial cannot vanish almost everywhere on the full real product unless it is zero. It is a principal submatrix of the stated product moment matrix, not a declaration that the total-degree space is a tensor product of smaller univariate spaces.

The actual joint least-norm representative and its metric are

$$R_{k,M} = \mathcal{T}_h^{(k)} M_{k,M}^{-1} J_{k,M}^* (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1},$$

$$G_{k,M} = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (7.4)$$

The proof is the same explicit constrained-minimum calculation as (3.6), now with the relation ideal (7.2). In particular its arithmetic class is exactly $\sigma_h^{\otimes k}$ and every jet is fixed.

For a rectangular degree budget N one has the actual inclusion

$$\mathcal{P}_N^{\otimes k} \hookrightarrow \mathcal{P}_{kN}^{(k)}.$$

It carries the old product representative into the feasible set of the new problem. Therefore

$$G_{k,kN} \preceq G_{h,N-d}^{\otimes k} \quad (N \geq d). \quad (7.5)$$

This is an inequality from an inclusion of admissible original boundaries. It is not asserted to imply the same ordering of control forms, whose dependence on A is retained separately.

G.1.9 8. The product error is one new total-degree layer

Let $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, and let $P_{\mathcal{B}_{k,M}}$ be its orthogonal projection onto the finite-dimensional actual boundary space. With the fixed low-degree section $s_h^{\otimes k}$,

$$R_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) s_h^{\otimes k}.$$

Put

$$\mathcal{E}_{k,M} = (1 - P_{\mathcal{B}_{k,M}})\mathcal{B}_{k,M+1}.$$

Because the jet map is already onto at both degrees,

$$\dim \mathcal{E}_{k,M} = \binom{M+k}{k-1}. \quad (8.1)$$

This is $\dim \mathcal{P}_{M+1}^{(k)} - \dim \mathcal{P}_M^{(k)}$; the d^k jet constraints cancel from the dimension difference.

Let $D^{(k)} = \sum_j D_j$. Its induced action on jets is $A_h^{(k)}$. The actual derivative defect lies in $\mathcal{B}_{k,M+1}$:

$$B_{k,M} = D^{(k)} R_{k,M} - R_{k,M} A_h^{(k)}.$$

Define the two explicitly typed maps into the same new relation layer

$$Y_{k,M} = P_{\mathcal{E}_{k,M}} R_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M},$$

$$C_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) B_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M}.$$

Integration by parts gives $(D^{(k)})^* + D^{(k)} = kI$ on these functions. Orthogonality then proves

$$W_{k,M} := (A_h^{(k)})^* G_{k,M} + G_{k,M} A_h^{(k)} - k G_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad (8.2)$$

$$G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}, \quad \text{rank } W_{k,M} \leq 2 \binom{M+k}{k-1}. \quad (8.3)$$

For (8.3), use the orthogonal decomposition $\mathcal{B}_{k,M+1} = \mathcal{B}_{k,M} \oplus \mathcal{E}_{k,M}$. Its projection sends $R_{k,M}$ to $Y_{k,M}$ and the new representative is $R_{k,M} - Y_{k,M}$.

The source quotient map is the same degree transport as before:

$$\mathcal{B}_{k,M+1}/\mathcal{B}_{k,M} \xrightarrow{\sim} \mathcal{E}_{k,M}, \quad [z] \mapsto (1 - P_{\mathcal{B}_{k,M}})z.$$

A derivative class in this kernel becomes the supported zero only at the next total-degree label. The primitive retains which tensor factor supplied the theta boundary and its cochain sign. Synchronization by $(1, e)$ in each required two-leg factor supplies the common joint support, while the original mask remains recorded.

Thus the product estimate need not simply copy k identical one-factor errors: (8.2) gives a new jointly optimized boundary-layer expression with completely specified moment data and maps. Its rank bound alone does not bound its generalized eigenvalues.

G.1.10 9. Same-metric reflection and residue duality on the product

For a reflection-stable packet, $j_h^{\otimes k}$ is the antilinear involution on $E_h^{(k)}$, defined on pure tensors by applying j_h to every factor. The corresponding function reflection is $F(x_1, \dots, x_k) \mapsto \prod_j x_j^{-1} \overline{F(x_1^{-1}, \dots, x_k^{-1})}$.

It preserves both the total-degree numerator space and its relation ideal. Since $(g/h)^\dagger = (-1)^d (g/h)$, its action on a numerator is $(-1)^{kd} P^\dagger$; this factor is retained. Uniqueness of the minimum in (7.4) proves reflection compatibility of $R_{k,M}$. Therefore the analogue of (6.4) gives symmetry of the generalized spectrum of $(W_{k,M}, G_{k,M})$ on the same metric.

Residue duality remains the explicitly specified map with matrix $S_h^{\otimes k}$ and target line $\mathcal{L}_k$, on which dilation acts by a^k . It satisfies

$$(A_h^{(k)})^* S_h^{\otimes k} + S_h^{\otimes k} A_h^{(k)} = k S_h^{\otimes k}.$$

Any fixed degree-orientation scalar in the cochain evaluation is retained separately; the ordered product of the scalar residue pairings here has matrix exactly $S_h^{\otimes k}$.

Its dual metric and control are

$$\begin{aligned} G_{k,M}^D &= (S_h^{\otimes k})^* G_{k,M}^{-1} S_h^{\otimes k}, \\ W_{k,M}^D &= -(S_h^{\otimes k})^* G_{k,M}^{-1} W_{k,M} G_{k,M}^{-1} S_h^{\otimes k}. \end{aligned} \quad (9.1)$$

The Jacobian contraction is still the original multiplication trace, with every multiplicity, and is unchanged by the source boundaries used to construct (7.4). Neither (9.1) nor reflection alone establishes an arithmetic upper bound.

G.1.11 10. Quantitative certificate and the Deligne comparison

For one or several variables write M for the positive moment matrix, J for its fixed full-jet map, and $\mathcal{C} = JM^{-1}J^*$. These actual matrices now replace an unrestricted metric search.

A useful deterministic enclosure is as follows. Suppose a Hermitian approximation $\widehat{M}$ has a verified bound $\|M - \widehat{M}\| \leq \eta$ and $\widehat{M} \succeq \mu I$ for a retained $\mu > \eta$. Then

$$\|M^{-1} - \widehat{M}^{-1}\| \leq \frac{\eta}{\mu(\mu - \eta)}, \quad \|\mathcal{C} - J\widehat{M}^{-1}J^*\| \leq \|J\|^2 \frac{\eta}{\mu(\mu - \eta)}. \quad (10.1)$$

This is the resolvent identity for inverses. The positive lower bound on $\mathcal{C}$ must also be retained before a second inversion. With $\widehat{G}$ and $\widehat{W}$ and verified errors η_G, η_W , the sufficient matrix certificate is

$$\widehat{G} - \eta_G I \succ 0, \quad \epsilon \widehat{G} \pm \widehat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0. \quad (10.2)$$

No numerical interval enclosure for the actual infinite arithmetic moments is claimed merely from a floating-point calculation. The previous exterior-theta sums give another exact representation of their one-variable inputs; all tails and roundoff must be supplied to (10.1).

Deligne's retained Corollaries 3.3.4–3.3.6 place upper and dual lower bounds on the same compact-to-ordinary image; the tensor step in 3.2.13 improves an estimate only after additional geometric control. This note supplies a canonical coherent family, its exact source relation ideal on products, and the same-metric reflection/dual operations with which such an upper estimate can now be tested. It does not import finite-field hypotheses into the tau chart.

For an actual eigenvector $A_h v = \rho v$, every constructed tensor metric still has the exact diagnostic

$$\frac{(v^{\otimes k})^* W_{k,M} v^{\otimes k}}{(v^{\otimes k})^* G_{k,M} v^{\otimes k}} = k(2\Re \rho - 1). \quad (10.3)$$

The quantity to bound is therefore the generalized boundary form (8.2) relative to (7.4). A dimension/rank estimate, a decrease of a Gram matrix, or unrestricted metric realizability is not substituted for that relative estimate. The new constructive route is the exact finite full-jet moment problem together with its next-layer flux, not another homotopy choice.

G.1.12 11. Reading, provenance, and validation

Newly read: both supplied formalization/control notes in full; the preceding local `Tau_Homotopy_Boundary_Control/NOTE.tex`, especially sections 4–10; the retained Deligne English TeX excerpt 3.3.4–3.3.6 and French 6.2.4–6.2.7. This is not a new audit of every page of Weil II. No OCR or PDF extraction was used. Source hashes are in the package reading record.

External context checked: NIST DLMF 5.11.9 for the gamma bound; Connes–Consani–Moscovici, arXiv:2403.01247, abstract, for prior moment/Jacobi constructions from squared absolute local factors; Romik, arXiv:1902.06330, abstract, for prior orthogonal-polynomial expansions of Ξ ; Lasserre–Pauwels, arXiv:1606.03858, abstract, for inverse-moment/Christoffel context. Only the gamma asymptotic is used as an external estimate; the other papers are priority/context references, not sources of an unread theorem. The arguments above prove the specific maps used here.

The accompanying exact checker uses declared finite discrete calibration measures and rational complex polynomial data. It checks the general finite matrix identities, joint relation ranks, signs, full repeated-root reductions, reflection, naturality, and comparison metrics. Its model polynomial is never labelled an arithmetic zeta counterexample. Analytic density, asymptotic gamma bounds, and the original theta range identities are written arguments, not certified by those finite tests. No new Lean execution is claimed. No RH or GRH conclusion is claimed.

H Full-jet interpolation kernels and the actual next relation layer

Source attribution. Complete TeX witness from the dated kernel-layer supplement, delivered 12 September 2026. Its handoff identifies PR 14 as the arithmetic source and the existing PR 15 revision as its formal baseline. The supplement’s add-only patch was verified locally; its receipt records no remote branch modification. These NOTE bytes are identified by the source and archive hashes below; neither baseline revision is represented as their Git head.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Kernel_Layer_Integration/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Arithmetic source baseline PR 14

Cited existing PR 15 formal baseline. 21970bbf4760d0bbca512a4a7996a2e38f968947

Source SHA-256. be8485471fc7dac41d97e4a9fd52413da14e83fbc42c6c55c01a5f0908caa06c

Original archive. Tau_Kernel_Layer_Integration_2026-09-12(1).zip

Archive SHA-256. ebd927c8a9afa90600f5320bd4189ecb48085889835205057b1bfafdb68d23d2

H.1 Full-jet interpolation kernels and the actual next relation layer

H.1.1 Integration supplement — 12 September 2026

This supplement reviews the supplied **Coherent arithmetic interpolation and joint tensor boundary control** package and derives an additional finite-stage bridge. The source’s canonical representative, original theta boundaries, full multiplicities, and support transport are retained. The new formulas below have written proofs and exact finite regression tests; they are **not new Lean certificates**.

The supplied ZIP has SHA-256 a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26. All 29 entries of its manifest verified. Its note, handoff, and checker match the published Git blobs at PR #14, commit 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc. The source checker was independently rerun: nineteen tests passed normally and under optimized Python, with identical result records; the intentionally false twentieth test failed in both modes. See SOURCE_REVIEW_RECEIPT.json.

The comparison with PR #15 is limited precisely: its existing five-module receipt certifies its stated support-changing, relation-layer, finite-Krylov and rank-two interfaces. This supplement does not extend that receipt to the source’s theta analytics, all-degree orthogonal-polynomial construction, or the new tensor-kernel calculations.

H.1.2 1. Source results retained, rather than reconstructed differently

Write the source polynomial degree as N , to distinguish it from a moment matrix. Fix tensor degree $k \geq 1$, a nonempty packet polynomial h of degree d , and $N \geq k(d - 1)$. The packet includes the complete orders of the selected actual zeros. Retain

$$g = 2\xi, \quad E = E_h^{\otimes k}, \quad A = A_h^{(k)}, \quad \mathcal{J}_N : \mathcal{P}_N^{(k)} \twoheadrightarrow E. \quad (1)$$

The last map is reduction modulo the original ideal $(h(t_1), \dots, h(t_k))$, followed by multiplication by the original local unit $v_h^{\otimes k}$. Its kernel is $\mathcal{I}_{k,N}$. The source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k) F_h^{\otimes k}, \quad \mathcal{MT}(P) = P(s_1, \dots, s_k) \prod_i \frac{g(s_i)}{h(s_i)}. \quad (2)$$

Let $\mathcal{L}_N = \mathcal{TI}_{k,N}$, $P_N = P_{\mathcal{L}_N}$, and let $\sigma : E \hookrightarrow Q^{\otimes k}$ be the source arithmetic inclusion. Its canonical representative and forms are

$$R_N = (I - P_N) s_h^{\otimes k}, \quad G_N = R_N^* R_N \succ 0, \quad q^{(k)} R_N = \sigma, \quad J^{(k)} R_N = I, \quad (3)$$

$$B_N = D^{(k)} R_N - R_N A, \quad W_N = A^* G_N + G_N A - k G_N. \quad (4)$$

The supplied note proves $B_N(E) \subseteq \mathcal{L}_{N+1}$ and, with $\mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp$,

$$Y_N = P_{\mathcal{E}_N} R_N, \quad C_N = (I - P_N) B_N, \quad W_N = -(Y_N^* C_N + C_N^* Y_N). \quad (5)$$

These are source results (source note, §§2–4 and 7–9), not claimed as discoveries of this supplement. Their finite-dimensional derivations check: the constrained minimum is over precisely the original ideal, the derivative increases total degree by one, and orthogonal projection removes only the admitted relation space. The source's exact sequence and packet naturality require full orders. Its total-degree statements require the threshold in (1). Its analytic function-space identities remain the stated analytic inputs, not consequences of a finite fixture.

The original coefficient square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, with projections $p_{\mathbb{Z}}, p_{\mathbb{C}}$, is unchanged. A vector in an admitted relation maps to the receiving fibre's supported zero, never to external absence τ .

H.1.3 2. Calculate the inverse interpolation metric by full-jet columns

Use exactly the source measure

$$d\nu_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}. \quad (6)$$

Let $p_j^{(h)}(s)$ be its monic orthogonal polynomials, with the polynomial variable evaluated at $s = 1/2 + it$, and retain their positive squared norms

$$\kappa_j^{(h)} = \int |p_j^{(h)}(1/2 + it)|^2 d\nu_h(t) > 0. \quad (7)$$

These are **not** the earlier theta-Krylov vectors u_j , nor their norms $\mathfrak{h}_j$. In particular they depend on h ; they are not assigned norm one. Existence through any finite degree follows from the source measure's positive-definite moment matrices. No limiting assertion is needed here.

Suppress h in notation. For a multi-index α , define

$$p_\alpha = \prod_i p_{\alpha_i}, \quad \kappa_\alpha = \prod_i \kappa_{\alpha_i}, \quad a_j = v_h[p_j]_h \in E_h, \quad a_\alpha = \bigotimes_i a_{\alpha_i} \in E. \quad (8)$$

Every a_j is the full jet, including the local unit. Product orthogonality gives $\langle \mathcal{T} p_\alpha, \mathcal{T} p_\gamma \rangle = \kappa_\alpha \delta_{\alpha\gamma}$. These polynomials form a basis of $\mathcal{P}_N^{(k)}$, because the monic triangular basis change preserves total degree.

Set

$$K_N := G_N^{-1} = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha}. \quad (9)$$

Proof. In this basis the source moment matrix is $\text{diag}(\kappa_\alpha)$ and the full-jet matrix has columns a_α . Substitute into the source identity $G_N^{-1} = J_N M_N^{-1} J_N^*$. A change of polynomial basis does not change that product. This proves (9) for the same canonical metric, not a newly selected metric.

Its representative has the explicit coefficient formula

$$R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (10)$$

In particular $R_N K_N$ is an actual map from finite antidual coordinates into the original test-function space. It satisfies $J^{(k)} R_N K_N = K_N$ and $q^{(k)} R_N K_N = \sigma K_N$. The invertible coordinate change K_N is retained, not identified with the identity.

H.1.4 3. An explicit basis of the original next relation layer

Let $\Sigma_{N+1} = \{\beta \in \mathbb{N}^k : |\beta| = N + 1\}$. Define actual test functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \beta \in \Sigma_{N+1}. \quad (11)$$

They are in $\mathcal{E}_N$. First, their full jets vanish by (3), so their numerator polynomials lie in the original ideal at degree at most $N + 1$; thus $e_\beta \in \mathcal{L}_{N+1}$. Second, $\mathcal{T}p_\beta$ is orthogonal to all degree-at-most- N polynomials and $R_N a_\beta$ is orthogonal to $\mathcal{L}_N$. Hence $e_\beta \perp \mathcal{L}_N$.

Define

$$\mathcal{E}_N : \mathbb{C}^{\Sigma_{N+1}} \longrightarrow \mathcal{E}_N, \quad c \longmapsto \sum_{\beta} c_\beta e_\beta, \quad (12)$$

$$U_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad D_N = \text{diag}(\kappa_\beta)_{\beta \in \Sigma_{N+1}}. \quad (13)$$

The map $\mathcal{E}_N$ is an isomorphism. Independence follows by taking the degree- $N + 1$ polynomial part of (11). For spanning, subtract from a relation its degree- $N + 1$ combination of the e_β . The remainder is a relation of degree at most N , hence belongs to $\mathcal{L}_N$. An element of $\mathcal{E}_N$ has no such remaining component.

Its Gram matrix and pairing with the representative are exactly

$$H_N := \mathcal{E}_N^* \mathcal{E}_N = D_N + U_N^* G_N U_N, \quad \mathcal{E}_N^* R_N = -U_N^* G_N. \quad (14)$$

The cross terms vanish because the new-degree polynomials are orthogonal to the old polynomial space. In particular $H_N \succ 0$, and no original norm has been suppressed.

This identifies the existing quotient equivalence in coordinates:

$$\mathbb{C}^{\Sigma_{N+1}} \xrightarrow{\mathcal{E}_N} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1} / \mathcal{L}_N. \quad (15)$$

The inverse quotient-to-layer map is the source's $[z] \mapsto (I - P_N)z$. For each nonzero coordinate vector, its class in this relative quotient is nonzero, and its image in the next admitted quotient is zero. At the split level this remains the next supported zero.

For an explicit theta primitive, form $p_\beta - \sum_{|\alpha| \leq N} p_\alpha a_\alpha^* G_N a_\beta / \kappa_\alpha$. Its full jet is zero. Fixed-order monic division expresses it as $\sum_i h(t_i) Q_{i,\beta}$, with $\deg Q_{i,\beta} \leq N + 1 - d$. In the i -th cochain factor use the original ϕ_* in place of F_h , and multiply its primitive by $(-1)^{i-1}$. The differential supplies the same sign, producing the displayed relation with positive sign. This supplies the actual primitive, not just a dimension count.

H.1.5 4. Both boundary maps now have explicit coordinates

Define the matrix $F_N : E \rightarrow \mathbb{C}^{\Sigma_{N+1}}$ by its rows

$$(F_N)_\beta = \sum_{i: \beta_i > 0} \frac{a_{\beta - \mathbf{e}_i}^*}{\kappa_{\beta - \mathbf{e}_i}}. \quad (16)$$

Then

$$\boxed{Y_N = -\mathcal{E}_N H_N^{-1} U_N^* G_N, \quad C_N = \mathcal{E}_N F_N G_N.} \quad (17)$$

The first equation is the actual orthogonal projection formula using (14). For the second, multiply the finite expansion (10) by $s_1 + \cdots + s_k$. Monicity implies that its degree- $N + 1$ coefficient at p_β is exactly $(F_N)_\beta G_N u$. The term $R_N A u$ has degree at most N . Subtract $\mathcal{E}_N F_N G_N u$ from $B_N u$: the result has degree at most N and zero full jets, hence is in $\mathcal{L}_N$. Projecting by $I - P_N$ proves the equation.

Consequently the old quotient records

$$\boxed{[B_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N} = [\mathcal{E}_N F_N G_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N}.} \quad (18)$$

It becomes the supported zero under the next relation transition. The symbol $\mathcal{B}^{(k)}$ denotes the original product test-function space used by the source; no derivative on all of its L^2 completion is defined here.

H.1.6 5. The control form is a top-degree full-jet endpoint

Define

$$\boxed{V_N := A K_N + K_N A^* - k K_N.} \quad (19)$$

Substituting (17) into the source boundary identity (5), and using $\mathcal{E}_N^* \mathcal{E}_N = H_N$, gives

$$\boxed{W_N = G_N (U_N F_N + F_N^* U_N^*) G_N.} \quad (20)$$

Meanwhile direct multiplication of (4) by K_N on both sides gives $K_N W_N K_N = V_N$. Thus

$$\boxed{V_N = U_N F_N + F_N^* U_N^* = \sum_{|\alpha|=N} \sum_{i=1}^k \frac{a_{\alpha + \mathbf{e}_i} a_\alpha^* + a_\alpha a_{\alpha + \mathbf{e}_i}^*}{\kappa_\alpha}.} \quad (21)$$

This is a finite endpoint identity derived through the actual theta relation layer. It does not assume a small boundary, discard mixed terms, or replace the packet operator. It can also be obtained by finite Christoffel–Darboux telescoping. The standard scalar orthogonal-polynomial kernel and telescoping identities are recorded by NIST (n.d., §18.2(v)); the full-jet, original-layer identification (11)–(21) is proved here rather than imported from that scalar formula.

For this alternative telescoping proof, multiplication satisfies $sp_j = p_{j+1} + \beta_j p_j - (\kappa_j / \kappa_{j-1}) p_{j-1}$, with $\operatorname{Re} \beta_j = 1/2$. The imaginary part need not vanish for the packet-dependent measure. Only $\beta_j + \overline{\beta_j} = 1$ is needed. Interior adjacent terms cancel; (21) is the uncanceled total-degree boundary. The new tests include an asymmetric measure with a nonzero imaginary diagonal coefficient.

Regroup by α and put $b_\alpha = \sum_i a_{\alpha + \mathbf{e}_i}$. Equation (21) factors through the two sets of columns $(a_\alpha)_{|\alpha|=N}$ and $(b_\alpha)_{|\alpha|=N}$. Therefore

$$\boxed{\text{rank } W_N = \text{rank } V_N \leq \min \left\{ d^k, 2 \binom{N+k-1}{k-1} \right\}.} \quad (22)$$

This sharpens the source's general next-layer bound $2 \binom{N+k}{k-1}$ for this product-polynomial construction. It does **not** change the actual dimension of $\mathcal{E}_N$, which remains $\binom{N+k}{k-1}$. The sharpening comes from the scalar summed derivative and regrouping its top-degree columns. For $k = 2, d = 4, N = 6$, the two structural bounds are 14 and 16 on a 16-dimensional packet; the exact regression tests cover this nonvacuous case. Neither rank bound is a small-eigenvalue estimate.

H.1.7 6. Estimate the same form without a second inversion

Since K_N is Hermitian positive definite,

$$\boxed{-\epsilon G_N \preceq W_N \preceq \epsilon G_N \iff -\epsilon K_N \preceq V_N \preceq \epsilon K_N.} \quad (23)$$

This is congruence by the specified invertible map K_N . More explicitly, for $u = K_N \lambda$, $u^* G_N u = \lambda^* K_N \lambda$ and $u^* W_N u = \lambda^* V_N \lambda$. This identifies the generalized Rayleigh quotients, including multiplicities of generalized eigenvalues. The original generator A is not replaced.

The source's same-metric reflection therefore still transfers an upper estimate to a lower estimate for precisely the original G_N . Under the natural weight- k antidual action $A^D = kI - A^*$, the defect of the antidual metric K_N is $-V_N$, not V_N . The dual sign and weight are both retained.

The reflection map itself can be retained explicitly. If the source involution is $j(u) = C\bar{u}$, its antidual involution is $j^\#(\lambda) = C^T \bar{\lambda}$. From $C\bar{C} = I$ and $C^* G_N C = \overline{G_N}$, inversion gives $(C^T)^* K_N C^T = \overline{K_N}$. The control identity transports to $(C^T)^* V_N C^T = -\overline{V_N}$. Thus the two-sided equivalence retains the actual involution, not only spectral symmetry.

Equation (23) avoids the *second* inversion $G_N = K_N^{-1}$ in the final inequality test. It does not eliminate the need to enclose moments, full-jet coefficients, or the conditioning of the first construction of K_N .

A deterministic finite certificate can therefore work directly with $\hat{K}$, $\hat{A}$ and error bounds $\|K_N - \hat{K}\| \leq \eta_K$, $\|A - \hat{A}\| \leq \eta_A$, where $\hat{K}$ is Hermitian. Put

$$\hat{V} = \hat{A}\hat{K} + \hat{K}\hat{A}^* - k\hat{K}, \quad \eta_V = (2\|\hat{A}\| + k)\eta_K + 2\eta_A(\|\hat{K}\| + \eta_K). \quad (24)$$

Expanding both perturbations proves $\|V_N - \hat{V}\| \leq \eta_V$. Sufficient conditions, for $\epsilon \geq 0$, are

$$\hat{K} - \eta_K I \succ 0, \quad \epsilon \hat{K} \pm \hat{V} - (\epsilon \eta_K + \eta_V) I \succeq 0. \quad (25)$$

No enclosure of the actual infinite arithmetic moments is asserted in this supplement.

H.1.8 7. The metric update and the univariate connection

The positive update from (9) and its inverse are

$$\boxed{K_{N+1} = K_N + U_N D_N^{-1} U_N^*,} \quad (26)$$

$$\boxed{G_{N+1} = G_N - G_N U_N (D_N + U_N^* G_N U_N)^{-1} U_N^* G_N.} \quad (27)$$

Equation (27) follows either by direct multiplication with (26), or from $R_{N+1} = R_N - Y_N$ and the orthogonal decomposition. The middle matrix is exactly the actual layer Gram matrix H_N in (14), not an unrelated auxiliary matrix. The lost Gram form is $Y_N^* Y_N$.

For $k = 1$ and $N = n + d$, (11) recovers the earlier monic theta vector:

$$\boxed{e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1}.} \quad (28)$$

Indeed the numerator of (11) is monic of degree $n + d + 1$, divisible by monic h , and its quotient has degree $n + 1$ and leading coefficient one. Applying $\mathcal{T}_h$ gives f_{n+1} plus an element of L_n , orthogonal to L_n . This uniquely identifies u_{n+1} . Thus the packet-dependent polynomial construction is connected to the packet-independent original theta norm by an exact identity; those two norm families were not conflated.

H.1.9 8. Degree recursion and the remaining estimate

For $A_j = a_j a_j^* / \kappa_j$, define finite matrix sums

$$Q_{k,m} = \sum_{|\alpha|=m} \bigotimes_i A_{\alpha_i}, \quad Q_{k,m} = \sum_{j=0}^m A_j \otimes Q_{k-1,m-j}, \quad K_{k,N} = \sum_{m=0}^N Q_{k,m}. \quad (29)$$

These are finite degree convolutions; they do not assert convergence of an infinite generating series. They compute the product full-jet interpolation matrix using one-variable polynomial norms and jets, without inverting a total-degree moment matrix of dimension $\binom{N+k}{k}$. The target dimension remains d^k ; no polynomial-time complexity or unconditional numerical stability is claimed.

In univariate packet inclusions, the source's exact equality $R_{H,n} \iota_{hH} = R_{h,n}$ remains in force. The packet-dependent polynomial bases used here need not themselves commute under inclusion. No extension of this univariate equality to different joint total-degree budgets is presumed.

Finally, on $Av = k\rho v$, with $v = v_0^{\otimes k}$, and $\lambda = G_N v$,

$$\frac{\lambda^* V_N \lambda}{\lambda^* K_N \lambda} = k(2 \operatorname{Re} \rho - 1). \quad (30)$$

The additional work is therefore the explicit layer basis, derivative-coordinate map, endpoint identity, and direct relative certificate for the same source-controlled object. It is not a proof of a uniform sublinear excess. No arithmetic block or nilpotent jet has been removed.

H.1.10 Verification and literature scope

The independent new checker contains eighteen exact test methods. They cover the kernel basis change, repeated jets, two-variable layers, actual derivative-class coordinates, Gram updates, the original univariate theta vector, endpoint factorization, a nonvacuous improved rank bound, antidual signs, generalized characteristic polynomials, nonzero off-line *polynomial model* diagnostics, nonreal recurrence diagonals, fixed-order theta primitives, and deterministic matrix perturbation certificates. Normal and optimized result records agree. Both intentional-failure modes return failure.

These are finite regression tests, not proofs of the general analytic inputs and not Lean execution. The general finite identities have the written proofs above. The source's nineteen tests and this supplement's eighteen tests are separate suites, not counts of novel theorems.

The matrix-kernel and Christoffel–Darboux methods have established antecedents. NIST's general orthogonal-polynomial formulas supply the standard scalar comparison. Lasserre and Pauwels (2016) connect inverse moment matrices with orthogonal polynomials and the Christoffel function; only their abstract was examined here, so no theorem from that paper is used as a premise. No global priority claim is made for standard kernel or inverse-update identities.

H.1.11 References

Coherent arithmetic interpolation and joint tensor boundary control. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #14, commit 3bff1b2bd87dfa2592f65c5cc231f6ee9541dfc. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/14>

Formalize support-changing relation layers and the Krylov boundary-control calculation. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #15, verified source commit 21970bbf4760d0bbca512a4a7996a2e38f968947. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/15>

Lasserre, J.-B., & Pauwels, E. (2016). *Sorting out typicality with the inverse moment matrix SOS polynomial* [Preprint]. arXiv. <https://arxiv.org/abs/1606.03858>

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, §18.2(v), Christoffel–Darboux formula. <https://dlmf.nist.gov/18.2#v>

I Arithmetic frontier estimates and symmetric tensor control

Source attribution. Complete TeX witness from the dated symmetric-frontier continuation, delivered 12 September 2026. Its delivery receipt records an add-only local patch based on PR 14, with no remote write. The exact revision below is that patch base; it is not a new remote head for this NOTE. The source and archive hashes identify the complete delivered witness.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Symmetric_Frontier_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Recorded add-only patch base, PR 14

Source SHA-256. 734b6c935892ebd9d18b506f6b592f1ca3489afcb95d5c669a489f4ca04656cf

Original archive. Tau_Symmetric_Frontier_Control_2026-09-12(1).zip

Archive SHA-256. a024ee4bbb4df9555bc6f94c2d74a2e90c077533ad2ce312863342b758539796

I.1 Arithmetic frontier estimates and symmetric tensor control

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. New written deductions from the precise arithmetic source of PR #14, submitted for mathematical review. The supplied Lean receipt certifies its stated coefficient, quotient, homotopy, and topology interfaces; it does not certify the analytic inputs or the results in this note. No purity estimate is assumed. No claim of global novelty is made for the underlying orthogonal-polynomial, finite-group, or matrix identities.

I.1.1 0. Source and purpose

The source is PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, research-note Git blob 4ec40c15397128552bc276ec903fe59c89657d4e. Its actual total-degree representative, full-jet map, and theta relation ideal are used without changing them.

The source obtained two maps into the next relation layer, with

$$W_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}. \quad (0.1)$$

This note constructs coordinates on that *actual relation layer*, computes the relative control in them, proves a finite arithmetic upper bound, and constructs a smaller symmetric tensor cohomology summand that still contains every repeated eigenline used in amplification. Its exact complement and its trace are retained.

The decisive new coordinate map is

$$b_M(q) = \mathcal{T}_h^{(k)} q - R_{k,M} \mathcal{J} q. \quad (0.2)$$

It converts the highest orthogonal polynomial degree into an original theta boundary and has a calculated inverse and Gram matrix. A highest-degree polynomial alone need not be a boundary; its full-jet value is subtracted by the specific source representative in (0.2).

I.1.2 1. Unchanged split base, arithmetic source, and quotients

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.1)$$

The base is the marked absolute point $\mathfrak{b}_\tau$. The first arithmetic target in (1.1) is still infinite. Every fixed-label linear map below is reconstructed by $(\lambda, x) \mapsto (\lambda, Tx)$; its value on external absence is external absence. A vanishing amplitude at an active label remains that label's supported zero.

Use the original $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, where

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= \sum_{n \neq 0} \phi(nx), \quad q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V. \end{aligned}$$

The input is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_*(s) = g(s) = 2\xi(s).$$

Fix a nonempty finite packet of actual nontrivial zeros of g , with their full orders, and define

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[t]/(h), \quad v_h = g/h, \quad v = j_h v_h \in E^\times. \quad (1.2)$$

No squarefree assumption is made. The source constructs $F_h \in \mathcal{B}$ with $h(D)F_h = \Theta\phi_*$ and $\mathcal{M}F_h = v_h$. It supplies

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{M}\mathcal{T}_h(P) = v_h P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v[P]_h. \quad (1.3)$$

For tensor degree k , put $E_k = E^{\otimes k}$ and $A_k = \sum_{j=1}^k A_j$, where $A = M_t$. Let

$$\begin{aligned} \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, \quad M \geq k(d-1), \\ \mathcal{T}^{(k)}P &= P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{J}P = v^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \quad (1.4)$$

The relation space and its realization are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)}\mathcal{I}_M. \quad (1.5)$$

The source's least-norm representative $R_M : E_k \rightarrow \mathcal{B}^{\otimes k} \subset \mathcal{H}_k$, with $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, satisfies

$$J^{(k)}R_M = 1, \quad q^{\otimes k}R_M = \sigma_h^{\otimes k}, \quad R_M(E_k) \perp \mathcal{B}_M. \quad (1.6)$$

Its finite Gram is $G_M = R_M^* R_M \succ 0$. Every adjoint here is of a finite-domain map into the declared Hilbert observation; no bounded Hilbert-space extension of the full theta map is assumed.

1.1.3 2. Unscaled polynomial coordinates with every moment retained

The arithmetic measure is exactly

$$d\nu_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 \frac{dy}{2\pi}. \quad (2.1)$$

The source's log-coordinate and Fourier maps give $\|\mathcal{T}_h P\|_{L^2(dx)}^2 = \int |P(\tfrac{1}{2} + iy)|^2 d\nu_h(y)$. The mass of this measure is not set to one.

Starting from $1, t, t^2, \dots$, subtract its orthogonal projections onto the preceding span:

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathcal{P}_{n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (2.2)$$

No polynomial is divided by its norm. Let T_N be the triangular matrix with columns $p_0, \dots, p_N$ in the original power basis. The exact coordinate relation is

$$T_N^* M_{h,N} T_N = \text{diag}(\omega_0, \dots, \omega_N). \quad (2.3)$$

The earlier seed-boundary vectors $u_n = P_n(D)\Theta\phi_*$ are related to these coordinates by $\mathcal{T}_h(hP_n) = u_n$. Their measure is $|g|^2 dy/(2\pi)$; equation (1.3) is the map to the present $|g/h|^2$ measure. The two orthogonalizations have not been interchanged.

Recurrence. Define $a_0 = 0$ and $a_n = \omega_n/\omega_{n-1}$ for $n \geq 1$. Then

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad b_n = \frac{\int (\tfrac{1}{2} + iy) |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \tfrac{1}{2}. \quad (2.4)$$

Here $p_{-1} = 0$. The possibly nonzero $\Im b_n$ remains. The packet measure need not be even in the height coordinate y .

Proof. Multiplication by t has adjoint multiplication by $1 - t$ for this polynomial inner product. Pairing tp_n with $p_j, j < n-1$, gives zero. Its coefficient of p_{n+1} is one by its leading term; pairing with p_{n-1} gives $-\omega_n/\omega_{n-1}$. The diagonal is the displayed integral. This proves every coefficient. These are the usual orthogonal-polynomial operations, applied through the specified coordinate map rather than changing the arithmetic generator.

For $\alpha \in \mathbb{N}^k$, define

$$p_\alpha = \prod_j p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}p_\alpha \in E_k. \quad (2.5)$$

The columns z_α contain the complete unit $v^{\otimes k}$ and all nilpotent jets.

Inverse-metric formula. In the fixed original basis of E_k ,

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}. \quad (2.6)$$

Indeed (2.3) applied to the tensor polynomial basis transforms $JM^{-1}J^*$ exactly into this sum. The representative coefficient on p_α is

$$c_\alpha(u) = \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (2.7)$$

These formulas avoid inversion of a large raw monomial moment matrix, without imposing a new metric.

I.1.4 3. The highest polynomial degree is isomorphic to the actual new relation layer

Let $\mathcal{U}_M = \text{span}\{p_\beta : |\beta| = M + 1\}$ and retain its ordered basis. Put

$$\Omega = \text{diag}(\omega_\beta)_{|\beta|=M+1}, \quad F = [z_\beta]_{|\beta|=M+1} : \mathbb{C}^r \rightarrow E_k, \quad r = \binom{M+k}{k-1}. \quad (3.1)$$

Write $T_+ : \mathbb{C}^r \rightarrow \mathcal{H}_k$ for the map with columns $\mathcal{T}^{(k)} p_\beta$. Its image is orthogonal to the entire degree- M polynomial image, including $R_M(E_k)$.

The prior actual new relation layer is

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}.$$

Theorem: relation-graph isomorphism. The map

$$b_M = T_+ - R_M F : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M \quad (3.2)$$

is an isomorphism and has

$$H_M = b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (3.3)$$

Proof. Its full jet is $F - F = 0$, so (1.5) makes it an actual boundary of degree at most $M + 1$. Both terms are orthogonal to $\mathcal{B}_M$. If $b_M u = 0$, its component in the highest orthogonal polynomial degree is $T_+ u = 0$, so $u = 0$. The source calculation gives $\dim \mathcal{E}_M = r$, proving surjectivity. Alternatively every degree- $M + 1$ boundary has a unique highest orthogonal component; subtract its b_M image and the result lies in $\mathcal{B}_M$. This also constructs the inverse. Orthogonality $T_+^* R_M = 0$ proves (3.3).

A primitive is fully specified: form the numerator $p_\beta - c_M(z_\beta)$, where c_M is (2.7), divide it in the fixed variable order by $h(t_1), \dots, h(t_k)$, and use the original theta primitive in the factor where $h(t_j)$ appears. Give that primitive coefficient $(-1)^{j-1}$; the tensor differential supplies the same sign. No relation is inferred solely from a dimension count.

The comparison with the original internal quotient is

$$\mathbb{C}^r \xrightarrow{b_M} \mathcal{E}_M \xrightarrow{v \mapsto [v]} \mathcal{B}_{M+1}/\mathcal{B}_M \hookrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_M \longrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_{M+1}. \quad (3.4)$$

The last composite is zero. Its split lift sends (λ_M, u) to $(\lambda_{M+1}, 0)$, not to τ . At λ_M , $u \neq 0$ represents a nonzero kernel class. The map b_M describes the entire class that is about to become a supported zero.

Representative update. Orthogonal projection onto $b_M(\mathbb{C}^r)$ gives

$$Y_M = -b_M H_M^{-1} F^* G_M, \quad R_{M+1} = R_M + b_M H_M^{-1} F^* G_M. \quad (3.5)$$

Consequently

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (3.6)$$

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (3.7)$$

The complete volume identity is

$$\frac{\det \mathcal{C}_{M+1}}{\det \mathcal{C}_M} = \frac{\det(\Omega + F^* G_M F)}{\det \Omega}. \quad (3.8)$$

The full matrices in (3.6)–(3.7) remain available; their determinant in (3.8) is an additional observation, not their replacement.

I.1.5 4. Relative control is a highest-degree Christoffel–Darboux identity

For each $|\beta| = M + 1$, put

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (4.1)$$

The name E_+ denotes this finite matrix, not the scalar e or the arithmetic quotient E_h ; its type is $\mathbb{C}^r \rightarrow E_k$.

Theorem: exact displacement.

$$\boxed{Z_M := A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*}. \quad (4.2)$$

Proof. Apply (2.4) in each factor of (2.6). A pair of adjacent indices inside $|\alpha| \leq M$ cancels because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, precisely the term subtracted in (4.2). The remaining ordered pairs have $|\beta| = M + 1$ and predecessor $\beta - e_j$; collecting them gives (4.1)–(4.2). The imaginary diagonal coefficients cancel with their conjugates here; they were retained in the recurrence and are not assigned zero.

This identity is the tensor total-degree version of the Christoffel–Darboux cancellation. The standard one-variable formula is background; the maps to the full arithmetic quotient and its source boundaries are specified above.

The derivative defect also has an exact expression in the same relation-graph coordinates:

$$B_M = D^{(k)} R_M - R_M A_k, \quad \boxed{C_M^{\text{rel}} := (1 - P_{\mathcal{B}_M}) B_M = b_M \Omega^{-1} E_+^* G_M}. \quad (4.3)$$

Proof. The coefficient of p_β in $D^{(k)} R_M u$ is $\sum_j \omega_{\beta-e_j}^{-1} z_{\beta-e_j}^* G_M u = \omega_\beta^{-1} w_\beta^* G_M u$. The defect is an original boundary. The inverse of (3.2), followed by projection away from $\mathcal{B}_M$, therefore gives (4.3).

Substituting (3.5) and (4.3) into the previous cross-pairing formula proves

$$\boxed{W_M = A_k^* G_M + G_M A_k - k G_M = G_M Z_M G_M = -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M)}. \quad (4.4)$$

Thus the new highest-degree computation is connected to both maps in the preceding note, including their minus sign and their support-valued boundary primitive.

I.1.6 4.1 Exact finite certificate and a proved upper estimate

For every real ϵ , congruence by the actual invertible G_M gives

$$\boxed{-\epsilon G_M \preceq W_M \preceq \epsilon G_M \iff -\epsilon C_M \preceq Z_M \preceq \epsilon C_M}. \quad (4.5)$$

This transports the inequality; no eigenvalue or generator has been rescaled.

Put $\mathcal{B} = [F, E_+]$ and

$$J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

The nonzero generalized eigenvalues of (Z_M, C_M) are the nonzero eigenvalues of

$$\boxed{J \mathcal{B}^* G_M \mathcal{B}}. \quad (4.6)$$

The proof is the pair of maps $\mathcal{B} : \mathbb{C}^{2r} \rightarrow E_k$ and $J\mathcal{B}^* G_M : E_k \rightarrow \mathbb{C}^{2r}$ and the equality of nonzero spectra of their two composites. These eigenvalues are real because the first composite is the operator of the Hermitian pencil in (4.5). The $2r$ -matrix itself need not be Hermitian in its displayed coordinates.

There is also a direct upper bound. Let

$$\lambda_{k,M} = \max_{v \neq 0} \frac{v^* F \Omega^{-1} F^* v}{v^* \mathcal{C}_M v},$$

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (4.7)$$

Then the actual arithmetic control excess satisfies

$$\boxed{\epsilon_{k,M} \leq 2\sqrt{\Gamma_{k,M} \lambda_{k,M}}}. \quad (4.8)$$

Proof. Let Z_M be the columns z_α , $|\alpha| = M$, and $\Omega_M = \text{diag}(\omega_\alpha)$. The lowering-incidence matrix L has $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, summed over matching j . Then $E_+ = Z_M L$. The matrix $\Omega_M L \Omega^{-1} L^*$ is similar to a positive semidefinite matrix and has nonnegative entries. Its row sum at α is exactly the sum in (4.7), so its spectral radius is at most $\Gamma_{k,M}$. This proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z_M \Omega_M^{-1} Z_M^* \preceq \Gamma_{k,M} \mathcal{C}_M.$$

Cauchy–Schwarz in the same Ω^{-1} pairing now gives

$$|v^* \mathcal{Z}_M v| \leq 2\sqrt{v^* F \Omega^{-1} F^* v} \sqrt{v^* E_+ \Omega^{-1} E_+^* v} \leq 2\sqrt{\lambda_{k,M} \Gamma_{k,M}} v^* \mathcal{C}_M v.$$

This proves (4.8). The phase-sensitive formula (4.6) is retained for a sharper estimate. Equation (4.8) is not claimed sublinear in k ; its two arithmetic factors are now fully specified.

1.1.7 5. A symmetric cohomology summand that retains the amplification witnesses

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$ with its original tensor differential. For $\pi \in S_k$, the chain permutation is

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i| |x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (5.1)$$

The exponent uses the original inputs crossed by π . Adjacent transpositions verify the cochain equation and the permutation relations. In top degree every input has degree one, so $T_\pi = \text{sgn}(\pi) P_\pi$, where P_π is ordinary factor permutation.

Define the actual idempotent

$$\boxed{S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi}. \quad (5.2)$$

The unscaled numerator squares to $k!$ times itself. The displayed $1/k!$ is necessary to make the specified morphism idempotent and is not absorbed into a pairing or an arithmetic function. In degree k the two signs multiply to $+1$, hence

$$\boxed{H^k(S_k \mathcal{C}_k) \cong (Q^{\widehat{\otimes} k})^{S_k}}. \quad (5.3)$$

On a finite packet this contains $(E^{\otimes k})^{S_k}$ with its entire nilpotent action. If $Av = \rho v$, then $v^{\otimes k}$ is nonzero, lies in this summand, and has generator eigenvalue $k\rho$. Using the unsigned chain average instead would give the alternating top-degree summand and would kill a repeated vector for $k > 1$; equation (5.2) is the precise replacement.

The complement is retained through the actual cochain decomposition

$$\mathcal{C}_k \xrightarrow{x \mapsto (S_k x, (1-S_k)x)} \text{im } S_k \oplus \text{im}(1 - S_k), \quad (u, v) \mapsto u + v. \quad (5.4)$$

At an invariant joint support this is reconstructed fibrewise. On all support masks the lift of (5.2) has the specified support map $\lambda \mapsto \bigvee_{\pi} \pi \lambda$. Its formula is the sum of the actual supported scalars $(\text{sgn } \pi / k!) \bullet T_{\pi}$. A vanishing projected amplitude lands at the orbit-joined supported zero, not at τ . Original transports into that joint fibre remain recorded. No injectivity of arbitrary support transports is assumed.

1.1.8 5.1 The canonical representatives commute with this operation

Permutation of variables preserves the total-degree numerator, its relation ideal, the product arithmetic measure, and the full unit $v^{\otimes k}$. Uniqueness of the constrained minimum gives

$$R_M P_{\pi} = P_{\pi} R_M. \quad (5.5)$$

Consequently R_M restricts to the invariant arithmetic module

$$E_k^{\text{sym}} = (E^{\otimes k})^{S_k}, \quad \boxed{\dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}}. \quad (5.6)$$

This is not a change of the spectral packet. The inclusion $I_k : E_k^{\text{sym}} \hookrightarrow E_k$ gives

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad W_M^{\text{sym}} = I_k^* W_M I_k, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (5.7)$$

All repeated eigenlines remain, so this smaller family still contains every witness needed by the tensor-amplification upper estimate. For example, $d = 3, k = 8$ gives dimension 45 rather than $3^8 = 6561$. This is a dimension reduction, not a bound on W_M^{sym} .

1.1.9 6. Orbit sums give the symmetric matrices with all factorials retained

For a nondecreasing tuple $\lambda \in \mathbb{N}^k$, let $\text{Orb } \lambda$ denote its distinct permutations and

$$n_{\lambda} = \frac{k!}{\prod_{j \geq 0} m_j(\lambda)!}, \quad p_{[\lambda]} = \sum_{\alpha \in \text{Orb } \lambda} p_{\alpha}, \quad \omega_{[\lambda]} = n_{\lambda} \prod_j \omega_{\lambda_j}. \quad (6.1)$$

Distinct product basis vectors are orthogonal, proving the last formula directly. The orbit sum is not divided by its norm or its orbit size.

The target basis is likewise the unscaled orbit sums of $e_{a_1} \otimes \cdots \otimes e_{a_k}$, $0 \leq a_j < d$, where $e_i = [t^i]$. Let I_k be their column inclusion and

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_i n_i!} \right), \quad L_k = D_k^{-1} I_k^*. \quad (6.2)$$

The explicit L_k extracts the common coefficient on an orbit. For each orbit polynomial, define

$$z_{[\lambda]} = L_k \mathcal{J} p_{[\lambda]}.$$

Then

$$\boxed{\mathcal{C}_M^{\text{sym}} = \sum_{|\lambda| \leq M} \frac{z[\lambda] z[\lambda]^*}{\omega[\lambda]}, \quad G_M^{\text{sym}} = (\mathcal{C}_M^{\text{sym}})^{-1}.} \quad (6.3)$$

The comparison with full coordinates is

$$\boxed{\mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad G_M^{\text{sym}} = I_k^* G_M I_k.} \quad (6.4)$$

Inversion and compression have not been commuted without the displayed D_k factors. Equation (6.4) follows because the full form commutes with the orthogonal permutation projection and by substituting $L_k = D_k^{-1} I_k^*$.

The symmetric polynomial numerator has one basis vector per partition of its degree into at most k parts. For $M \geq k(d-1)$ the jet map is onto, because a tensor remainder has total degree at most $k(d-1)$ and averaging makes a symmetric lift of every invariant jet. Thus

$$\boxed{\dim(\mathcal{E}_M^{S_k}) = p_{\leq k}(M+1),} \quad (6.5)$$

where $p_{\leq k}(n)$ is the number of partitions of n into at most k parts. The maps $b_M, Y_M, C_M^{\text{rel}}$ are permutation-equivariant and therefore restrict to that very subspace. Fixed-order polynomial division yields a primitive; applying S_k to it gives a symmetric cochain primitive because $dS_k = S_k d$.

The highest-layer formulas in sections 3–4 hold on the invariant coordinates by grouping F, E_+ into orbit sums, applying L_k , and using the diagonal entries $\omega_{[\lambda]}$. This computes the symmetric bound directly, instead of building a d^k -dimensional matrix first. The upper-bound row-sum estimate can still use the fully retained $\Gamma_{k,M}$; the exact grouped incidence matrix can give a smaller constant.

I.1.10 6.1 Reflection and the original trace

For a reflection-stable packet the source involution j_h induces $j_h^{\otimes k}$ on both summands in (5.4), commutes with permutations, and preserves the canonical representatives. Hence on the symmetric metric an upper control implies the opposite lower control by the same involution. The residue pairing and its Jacobian map are still transported through their specified inclusions; no auxiliary positive Gram is declared to be the Weil form.

For a finite packet action $U(a) = \exp((\log a)A)$, the exact trace on the invariant summand is

$$\boxed{\text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|} | E).} \quad (6.6)$$

The proof expands the permutation operator in the tensor basis; each cycle forces the corresponding contracted indices and gives the indicated trace. It holds for nonsemisimple U . The cohomological supertrace is $(-1)^k$ times (6.6). The complement trace remains

$$(-1)^k \left(\text{Tr}(U(a) | E)^k - \text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) \right). \quad (6.7)$$

These are traces of finite-rank cochain projectors on the original complex. An infinite packet sum and its trace topology are not supplied by these formulas.

I.1.11 7. An exact joint-control calculation with the nilpotent direction retained

This section is a declared algebraic calibration, not a packet of zeta zeros. Take the polynomial quotient

$$E = \mathbb{C}[s]/((s - \frac{1}{2})^2), \quad d\nu(y) = e^{-y^2/2} dy, \quad c = \sqrt{2\pi}.$$

The measure has its literal mass c , and the unit in the calibration jet map is 1. The polynomial basis comparison $x = s - 1/2$ is given by

$$B : \text{span}(1, x) \xrightarrow{\sim} \text{span}(1, s), \quad B = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}. \quad (7.1)$$

The operator remains multiplication by $s = 1/2 + x$. In the symmetric basis

$$1 \otimes 1, \quad 1 \otimes x + x \otimes 1, \quad x \otimes x,$$

its two-fold generator is

$$A_2^{\text{sym}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}. \quad (7.2)$$

The two-fold total-degree-4 calculation gives

$$\boxed{G_{2,4}^{\text{sym}} = c^2 \text{diag}(1/3, 2/3, 1/4),} \quad (7.3)$$

$$\boxed{W_{2,4}^{\text{sym}} = c^2 \begin{pmatrix} 0 & 2/3 & 0 \\ 2/3 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{pmatrix}.} \quad (7.4)$$

Its generalized characteristic polynomial is

$$\frac{\det(zG_{2,4}^{\text{sym}} - W_{2,4}^{\text{sym}})}{\det G_{2,4}^{\text{sym}}} = z(z^2 - 7/2),$$

so the exact two-sided excess is $\sqrt{7/2}$. The independent rectangular degree-2-per-factor construction gives $\sqrt{6}$ at the same containing total degree. This proves an actual improvement for the specified joint construction in this calibration; the nilpotent operator in (7.2) is unchanged.

For comparison, the smaller joint total degree $M = 2$ has excess $\sqrt{6}$, whereas the rectangular degree-1 construction contained in it has excess 2. Consequently a larger correction space alone does not monotonically improve the control constant. The typed comparison is inclusion of the two polynomial numerators and the resulting Gram compression; the direct formula (4.2) supplies the actual control value. Both calibrations are retained so that the first improvement is not promoted into a false monotonicity theorem.

I.1.12 8. Exact target, counterfactuals, and handoff

The next estimate is now an arithmetic highest-layer estimate, either by the phase-sensitive pencil (4.5)–(4.6), or by bounding the two factors in (4.8). On the symmetric cohomology family it concerns the same G_M^{sym} and W_M^{sym} , with all relation primitives retained. The source of every matrix entry is the actual measure $|2\xi/h|^2 dy/(2\pi)$, not a fitted metric.

A sublinear excess in tensor degree has not been proved. For every actual eigenvector $Av = \rho v$, the retained witness in the symmetric summand satisfies

$$\frac{(v^{\otimes k})^* W_M(v^{\otimes k})}{(v^{\otimes k})^* G_M(v^{\otimes k})} = k(2\Re \rho - 1). \quad (8.1)$$

The symmetric reduction therefore retains the witness needed by Deligne-style amplification; it does not assume its disappearance. The input Deligne passages use tensoring *and* a separate geometric estimate. Our new work supplies a computable smaller arithmetic target and a proved finite bound, not that geometric estimate by renaming a category.

Constructive continuations are recorded with their exact maps:

- Raw moment inversion: use (2.3)–(2.7), retaining every ω_j and the full jet unit.
- Highest polynomials that are not relations: use $b_M = T_+ - R_M F$, not an assertion that high degree is zero.
- Absolute Gram decay: use the congruence (4.5), retaining $\mathcal{C}_M = G_M^{-1}$ and the whole highest-layer update.
- Excessive tensor dimension: use the chain projector (5.2) and the symmetric numerator/quotient (6.1)–(6.5), while retaining the complementary cohomology and trace.
- A Cauchy–Schwarz bound too large: use the exact signed matrix (4.6); its phases and cross terms have not been replaced by their absolute values in the stored data.
- Ordinary zero observations: reconstruct the maps and kernels in (3.4). A new relation is *e*-zero at the next label; the original class and the full kernel remain upstream.

The supplied quotient/homotopy Lean results can formalize the finite algebra of (3.2)–(4.6) and the chain-permutation calculation. The original analytic range theorem, arithmetic moment bounds, and any uniform tensor estimate remain separate written-analysis inputs.

I.1.13 References and evidence scope

The current input was read in the mounted NOTE.tex and checked against the pinned public mark-down (especially sections 7–9). The prior formalization receipt was read at its recorded scope: its section-change and internal-quotient statements are used, without treating its analytic assumptions as proved Lean theorems.

The classical orthogonal-polynomial recurrence and Christoffel–Darboux formula are documented in NIST DLMF sections 18.2(iv)–(v), <https://dlmf.nist.gov/18.2>. The tensor complex and its signed commutativity are documented in the Stacks Project, Tag 0GWN, <https://stacks.math.columbia.edu/tag/0GWN>. These are background mechanisms; sections 3–6 above prove their specific maps on this arithmetic source.

The retained Deligne reading for the intended amplification is the supplied account of Weil II 3.2.13 and 3.3.4–3.3.6. No new full rereading of Weil II is claimed in this turn. No external source corpus is redistributed in this package.

The checker uses exact finite Gaussian moment fixtures, including repeated-root quotients, non-constant units, a nonzero imaginary recurrence diagonal, and both signed chain degrees. They are not arithmetic measurements. All derived infinite-dimensional statements rest on the written arguments and the explicitly named prior analytic inputs, not on the finite tests.